

A Course On Systematic Trading With RMA

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1st of June 2025

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QUANT FINANCE LTD
Working Paper
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1st of June 2025

Revised on the 10th of October 2025

Version : 1.0.5

Abstract

Systematic trading operates at the intersection of quantitative analysis and financial uncertainty, where market prices evolve not deterministically, but as outcomes of complex, non-stationary processes. This article presents a rigorous framework for developing trading strategies that are robust to structural breaks, regime shifts, and unpredictable events such as price jumps, fakeouts, and reversal traps. We begin by examining foundational models, momentum and mean-reversion, and assess their limitations in dynamic environments. We then address the design of systems capable of navigating uncertainty, incorporating probabilistic reasoning, adaptive estimators, and real-time risk controls.

At the core of our contribution is the Relative Moving Average (RMA) framework, a novel indicator that fuses statistical structure with contextual awareness. By constructing a dynamic price distribution around a moving average, RMA captures local disequilibria and enables responsive strategy deployment without relying on assumptions of stationarity. Through analytical discussion and empirical testing, we demonstrate how RMA outperforms conventional techniques in volatile and discontinuous markets.

Ultimately, we argue that the success of a trading strategy hinges not on predictive accuracy, but on disciplined decision-making under uncertainty. This requires models that respond intelligently to evolving conditions and systems that embed robustness at every level, from signal detection to risk management. This article provides the tools and principles to build such strategies.

Systematic Trading, Quantitative Strategy, Risk Management, Relative Moving Average

1 Introduction

Financial markets can be understood as dynamic systems driven by the continuous arrival of new information, which alters supply and demand and leads to persistent price fluctuations (see Dolfin et al. [2024], Ataei [2025]). These changes are not deterministic but stochastic in nature, making them more appropriately characterised by evolving probability distributions rather than fixed equations. The role of systematic trading is to detect and respond to these dislocations, extract meaningful signals from noise, and express those insights through disciplined, repeatable strategies that incorporate robust risk management (see Narang [2009], Chan [2021]).

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†All mistakes are ours.

Unlike physical systems governed by immutable laws, financial markets are non-stationary and structurally adaptive. They evolve in response to shifting economic conditions, regulatory developments, central bank policies, geopolitical shocks, and investor sentiment. This inherent instability manifests as regime shifts, structural breaks, volatility clustering, and extreme events, phenomena that challenge the core assumptions of classical statistics and traditional forecasting approaches.

Early pioneers such as Benoit Mandelbrot [1963] [1967] and Mandelbrot et al. [1968] [1997] laid the groundwork for understanding markets as multifractal systems, questioning the validity of Gaussian assumptions. He described the financial market as a system with fat tails, stable distributions and persistence (long-range memory). His insights were later reinforced by models such as ARCH and GARCH (see Engle [1982], Bollerslev [1986]), which captured volatility's time-varying nature. Later, Peters [1994] proposed the Fractal Market Hypothesis (FMH) to model investor behaviour and market price movements. Those hypothesis state that the different investment horizons value information differently, leading to uneven diffusion of information, so that at any one time, prices only reflect the information important to that investment horizon.

This unpredictable complexity of market behaviour has exposed the fundamental limitations of any framework, be it statistical or machine-learning based, that aims to predict market movements with precision. Indeed, forecasting in financial markets is not only difficult, it is, in a deep sense, impossible (see Bloch [2025]). Each market outcome is a new draw from a high-dimensional, partially observable distribution. Traditional indicators and AI models alike fail when confronted with stop-runs, fakeouts, reversal traps, or sudden jumps. These phenomena are not rare exceptions but rather integral features of market dynamics. They frequently invalidate even well-calibrated strategies by breaking assumed patterns and exposing latent vulnerabilities.

In this environment, the objective of quantitative trading is not to predict precise future prices, but to formulate probabilistic strategies that remain robust under structural uncertainty. Recent literature, including Manesoonthorn et al. [2024], Guo et al. [2025], Buehler et al. [2025], strongly supports the view that quantitative trading in modern financial markets is better served by probabilistic frameworks capable of accommodating structural uncertainty, rather than by deterministic price forecasts. Achieving this requires models that account for both long-term structural tendencies and short-term fluctuations, supported by a risk management framework designed to preserve capital in the face of unexpected events. Crucially, we must recognise that markets rarely provide clean, unambiguous signals. More often, they present conflicting cues, where local and global information diverge, forcing systematic strategies to navigate trade-offs and adapt in real time.

Systematic trading, defined as the execution of trades based on predefined rules with minimal discretionary intervention, is increasingly being enhanced by real-time data and machine learning models to uncover alpha-generating signals ahead of market consensus (see Wang et al. [2024], Macri et al. [2024], Zhou et al. [2025]). However, integrating such adaptive systems demands a fundamental shift in traditional risk frameworks, ensuring that strategies remain disciplined, explainable, and resilient under evolving market conditions. True automation requires not only technological sophistication but also a cultural transformation within firms to align innovation with robust risk management and regulatory obligations. This transformation is part of a broader organisational shift known as digital transformation, characterised by the convergence of technology, process reengineering, and cultural adaptation. Within this framework, systematic trading faces three core, interrelated challenges: detecting signals in real time, embedding live risk controls, and developing AI-driven execution algorithms capable of operating safely and effectively in fast-moving, uncertain environments. However, integrating these components is far from trivial, it demands a structured, multi-stage mapping process that systematically translates external signals into actionable trading decisions.

We begin in Section (2) by introducing quantitative and systematic trading, which stands at the crossroads of timeless principles and cutting-edge innovation. From trend following and probabilistic rule-based strategies to tools like the Relative Moving Average (RMA) that adapt to shifting market dynamics, today's approaches are built for complexity and change. Futuretesting expands the toolkit further by enabling rigorous evaluation in synthetic yet statistically

grounded environments. Now, a new wave is taking hold: AI-driven trading intelligence. By uncovering hidden patterns in vast datasets and adapting in real time, machine learning models are not just refining the edge, they are redefining it.

We continue in Section (3) by examining the foundational building blocks of quantitative and systematic trading. We introduce the two dominant paradigms, momentum and mean-reversion, and discuss how historical data and statistical rules can be used to reduce discretionary bias and support scalable, algorithmic decision-making. The section then turns to the critical role of risk management, before concluding with a discussion of estimation in non-stationary environments, where we explore adaptive tools such as shrinkage estimators and exponential weighting. Finally, we highlight how intraday systematic trading demands a distinct portfolio allocation approach that accounts for high-frequency dynamics, short horizons, and evolving market microstructure, departing fundamentally from traditional multi-day portfolio theory.

In Section (4), we examine how modern systematic trading organisations integrate expertise across trading, quantitative research, and machine learning. The coordination between these functions defines how efficiently market insights are transformed into executable strategies. The Head of Trading ensures that alpha ideas are operationally viable and cost-efficient in real markets; the Head of Quantitative Research focuses on developing economically grounded models and risk frameworks; and the Head of Machine Learning drives innovation in adaptive modelling, data processing, and predictive infrastructure. Together, these roles form a tightly coupled ecosystem in which theoretical signal development, algorithmic execution, and technological scalability reinforce one another. Understanding this cross-functional structure is essential for appreciating how successful systematic firms sustain both research innovation and trading performance in dynamic market environments.

Section (5) introduces a reassessment of classical statistical tools and model evaluation practices in financial time series analysis. Traditional moment-based measures and error metrics like MSE assume stationarity and i.i.d. returns, assumptions routinely violated in practice. Financial data exhibit jumps, volatility clustering, and structural breaks, making such tools unreliable. We also rethink how models should be evaluated in finance, asking what metrics better reflect economic relevance. In particular, we recommend alternatives tailored to one-step-ahead trajectory forecasting in trading contexts.

Section (6) explores the core principles of digital transformation and examines how they shape the evolution of modern systematic trading systems. We present five key dimensions that define digital transformation, emphasising the integration of technology, process reengineering, and cultural change. Building on this foundation, we focus on three interrelated areas critical to systematic trading in today's markets: (1) real-time signal detection, (2) the live risk management frameworks required to support it, and (3) AI-driven trade execution algorithms. Together, these elements form the basis for adaptive, compliant, and competitive trading architectures in an increasingly complex regulatory and technological landscape.

Intraday systematic trading using deep learning models, such as deep neural networks (DNNs) and large language models (LLMs), has opened new avenues for exploiting short-term market inefficiencies, with threshold-based execution strategies providing a practical framework by triggering trades when the model's estimated probability of a directional move crosses a predefined threshold. Yet despite their appeal, these strategies face significant challenges in live, volatile markets, where sudden price jumps can distort rolling inputs, causing prediction lags, premature exits, and costly reversals-effects that are compounded by execution costs and limited trade frequency. In Section (7), we examine the core challenge of reliably estimating near-term market direction, contrasting two distinct approaches: data-driven AI methodologies leveraging real-time features and calibration techniques, and stochastic models linking probability forecasts to underlying risk premia and variance, both ultimately constrained by the inherent unpredictability of short-term price action.

Section (8) confronts the practical reality that uncertainty is not an inconvenience but a structural condition of financial markets. Here, we examine how to build systems that are not designed to eliminate uncertainty but to function within

it. Through a case study, we explore how strategies can respond to price anomalies and structural breaks by embedding protective mechanisms, fallback logic, and real-time heuristics into their design.

Section (9) introduces an innovative technical indicator, the Relative Moving Average (RMA), which addresses a blind spot shared by traditional momentum and mean-reversion models: their purely reactive nature. RMA integrates statistical rigour with price action sensitivity, forming a dynamic distribution around a sliding moving average. This distribution evolves in real time, enabling the strategy to identify and exploit local disequilibria. RMA is calibrated to handle non-stationary environments, supports both trend-following and mean-reverting behaviour, and naturally extends to multi-asset applications such as pairs trading. We formally define the methodology, outline its components, introduce the Relative Extremum Ratios for detecting market regimes and their transition phases, and demonstrate how it can be used to construct robust and responsive trading strategies.

In Section (10), we analyse the intrinsic boundaries of quantitative approaches. Using a detailed case study, we explore how momentum and mean-reversion strategies each respond to a sequence of deceptive market events, such as reversal traps and fakeouts. Despite initial success, both models can underperform when faced with abrupt structural shifts. These cases highlight the danger of assuming stationarity and reinforce the importance of incorporating regime awareness and risk containment directly into strategy logic.

Section (11) presents empirical tests of the RMA-based strategies introduced earlier. We evaluate their performance across a range of market conditions and compare them with standard benchmarks, including traditional moving average systems and static mean-reversion setups. Key performance metrics such as Sharpe ratio, drawdown resilience, and trade consistency are analysed to demonstrate the adaptability and robustness of the RMA framework.

In summary, successful systematic trading does not hinge on predictive precision but on disciplined adaptability. It demands robust statistical thinking, continuous recalibration, and a deep respect for uncertainty. Knowing the signal is not enough, what matters is how we map it to action, and how we manage the risk when the signal fails. This course is designed to train you in exactly that: systematic decision-making under uncertainty.

2 Foundations and innovations: An Overview

This section introduces the foundational principles and modern innovations that underpin systematic trading. It begins with a review of traditional market mechanics, outlining how price action reflects a complex interplay of information flow, order dynamics, and investor sentiment. Classic strategies such as trend following, technical analysis, and price action trading are explored as practical tools for navigating this uncertainty. These approaches are complemented by quantitative models, which formalise trading decisions through systematic rules, statistical learning, and risk management protocols calibrated to evolving market conditions.

At the heart of quantitative trading lies a shift from prediction to probabilistic response: rather than forecasting exact outcomes, modern strategies adapt to unfolding patterns and volatility regimes using dynamic tools like trailing stops, relative indicators, and distribution-aware thresholds. The inherent limits of predictability, rooted in non-stationarity and regime shifts, are addressed through innovations such as the Relative Moving Average (RMA), which provides a localised, data-driven framework for assessing price deviations from dynamic equilibrium.

To test and refine strategies beyond the constraints of historical data, the Futuretesting paradigm introduces statistically grounded synthetic market environments. This enables rigorous evaluation of strategy robustness under diverse and extreme scenarios, advancing beyond the narrow lens of backtesting.

Finally, the rise of AI-driven trading intelligence represents a new frontier, integrating deep learning and data-driven modelling to uncover latent structures and nonlinear dependencies in financial data. By learning directly from high-dimensional inputs, ranging from market microstructure to alternative data, AI systems offer the potential to generate

adaptive signals, enhance execution efficiency, and optimise strategies in real time. This convergence of human-designed frameworks and machine-learned intelligence exemplifies the future of adaptive, robust, and scalable trading.

2.1 Foundations of trading: From market mechanics to strategy design

Trading involves buying and selling financial assets, stocks, bonds, derivatives, with the objective of turning a profit. Traders aim to capitalise on price fluctuations by going long (buy low, sell high) or short (sell high, buy low), often over short- to medium-term horizons (see Harris [2002]). In contrast, investors typically adopt a longer-term view, seeking to buy undervalued shares and profit as those prices rise over time. This distinction is visible in the long-term trajectory of the *S&P 500* Index displayed in Figure (1), which reflects both secular trends and cycles of boom and bust.

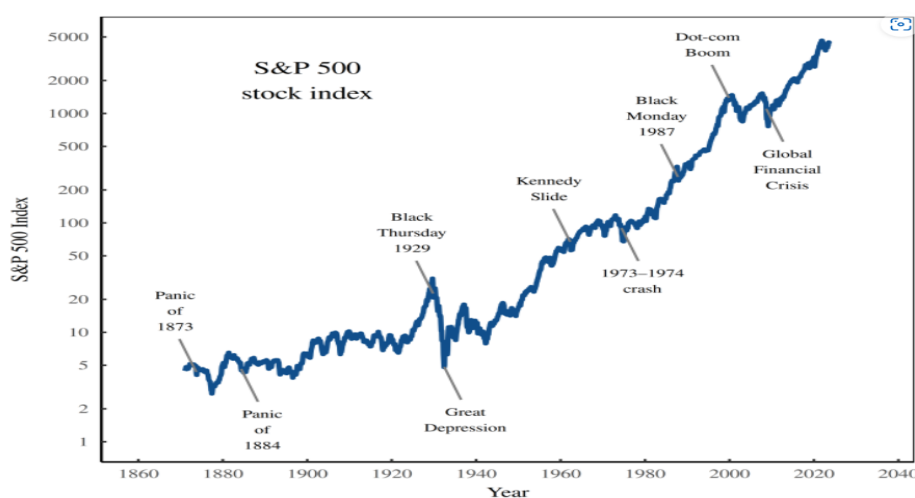


Figure 1: Long-term history of the SP500 Index.

At the heart of trading lies a fundamental question: Where are prices headed next? Market prices fluctuate in response to a complex interplay of supply and demand, macroeconomic indicators, investor psychology, and real-time information flow, ranging from corporate earnings to geopolitical news. Though driven by fundamentals, price action is ultimately a manifestation of order flow and sentiment, creating patterns that skilled traders aim to decipher and exploit.

Quantitative trading (see Velu et al. [2020], Chan [2021]) leverages mathematical models to forecast market direction, magnitude, and pattern formation. These predictions rely on two core approaches:

1. Data-driven models, which analyse historical price movements using charts, indicators, and statistical methods.
2. Process-driven models, which model asset dynamics using stochastic processes or econometric frameworks.

Systematic trading refers to executing trades based on a predefined set of rules, without discretionary intervention during live trading. It prioritises automation, discipline, and consistency (see Treleaven et al. [2013], Pereira [2021]). In contrast, quantitative trading focuses on how strategies are designed, using data, statistical models, and mathematical analysis. While quantitative trading drives the development of a strategy, systematic trading ensures its mechanical and repeatable execution.

Forecasting the future trajectory of asset prices is never about predicting a single outcome, it is about navigating a spectrum of plausible scenarios representing the full spread of potential returns. Quantitative and systematic trading, at

its core, is an exercise in estimating the probabilities of these near-term directional moves. But financial markets don't play by stationary rules (see Benoit Mandelbrot [1963], Peters [1994]). They evolve, shaped by shifting volatility, sudden structural breaks, and unpredictable regime changes. These dynamics routinely break the assumptions behind conventional statistical models. The harsh reality is that any informational edge from probabilistic forecasts is fleeting, local in time, transient in value, and constantly eroded by market drift. In other words, the very probabilities we rely on for trade decisions are themselves stochastic, undermining the dream of consistent, systematic profits across assets, strategies, or timeframes.

Rather than aiming to predict exact outcomes, quantitative trading focuses on responding intelligently to evolving market conditions. It leverages structured tools such as Technical Indicators and Price Action patterns to translate raw market behaviour into systematic, rule-based decisions. A key concept in Technical Analysis (TA) is trend identification (see Neely et al. [2014]). Markets move in three directions: up, down, or sideways. Uptrends form higher highs and higher lows, indicating bullish momentum. Downtrends create lower highs and lows, signalling bearish sentiment. Sideways markets exhibit price stability within a range. Reversals mark the transition between these states and are central to timing trades effectively.

To visualise price action, traders use candlestick charts, which encapsulate the open, high, low, and close (OHLC) over fixed intervals (see Figure (2)). These charts not only reveal price movement but also reflect market sentiment and volatility. Embedded within them are recurring price patterns, head-and-shoulders, triangles, flags, that help forecast future movements. Trendlines, drawn along peaks or troughs, identify key support and resistance levels, where price tends to stall, reverse, or break out.

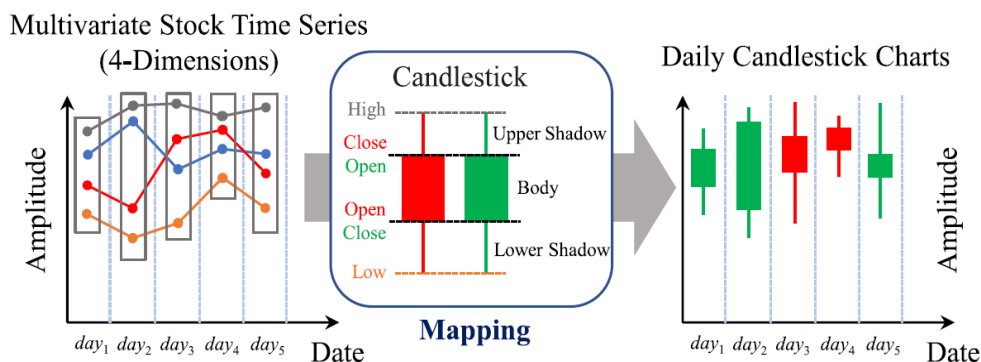


Figure 2: Daily candlestick charts.

Traders adopt strategies aligned with their time horizon, risk tolerance, and market outlook: No one-size-fits-all approach, styles range from milliseconds to multi-year positions. Common trading strategies include:

- **News Trading:** Reacts to macroeconomic releases, earnings reports, and real-time news flow.
- **End-of-Day Trading:** Executes decisions at market close, leveraging peak liquidity and daily settlement dynamics.
- **Swing Trading:** Targets short- to medium-term momentum between support and resistance levels.
- **Day Trading:** Opens and closes positions within the same session; no overnight exposure.
- **Trend Trading:** Identifies and rides sustained directional moves confirmed by technical indicators.
- **Scalping:** Executes high-frequency trades to profit from small price fluctuations and bid-ask spreads.

- **Position Trading:** Maintains trades over longer horizons (weeks to years), driven by macro or fundamental views.

Each style demands specific tools, discipline, and risk management: Choosing the right approach depends on trader profile and market conditions.

Regardless of the strategy, risk management is non-negotiable. Recent market turbulence and heightened volatility highlight the importance of protecting capital. Two key tools for managing individual trades dominate this arena:

- Stop-loss orders, which cap potential downside by exiting losing positions;
- Take-profit orders, which lock in gains when prices hit predefined targets.

Combining these tools yields effective risk/reward setups. Beyond these individual trades, advanced capital preservation techniques, such as diversification, use of inverse correlations, and protective puts, help mitigate drawdowns and secure portfolio integrity. Effective risk management ensures strategy durability: it preserves capital, stabilises returns, and supports compounding over time.

2.2 Quantitative trading and dynamic risk management

Quantitative trading operates at the crossroads of mathematics, statistics, and real-time financial decision-making (see Murphy [1999], Chan [2021]). In fast-moving markets, it offers a structured way to uncover and exploit inefficiencies. By modelling asset return distributions, including the sharp, unexpected spikes seen in daily returns (see Figure (3)), quant strategies aim to tilt the odds in favour of profitable trades.

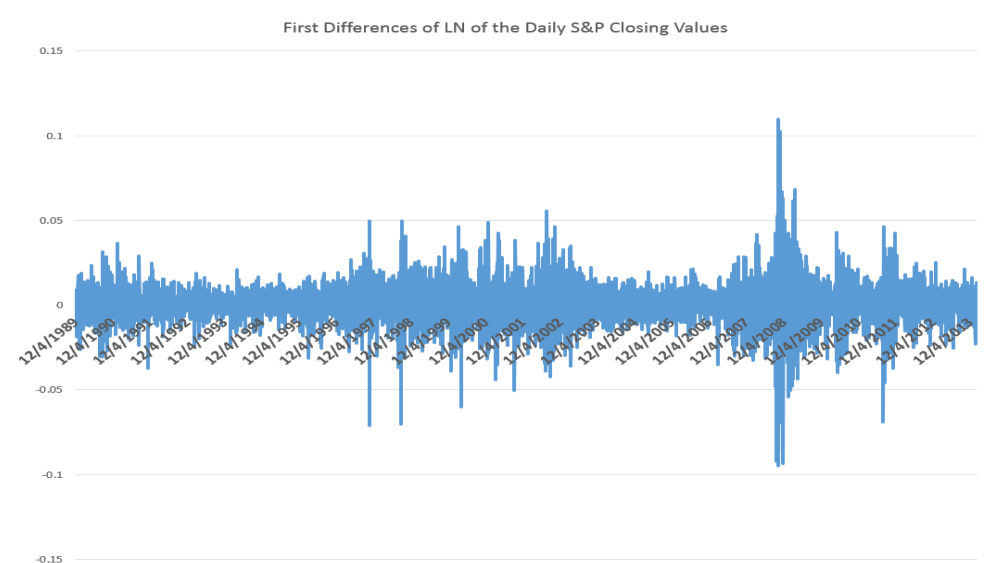


Figure 3: Daily market returns

At the core of any quant strategy lies a distinction between Alpha and the strategy itself. Alpha represents the expected excess return derived from information. The trading strategy is the systematic rule set, algorithmic and risk-aware, that converts this insight into positions via carefully defined entry and exit criteria, including stop-loss and take-profit levels.

These strategies vary in form, econometric, fundamental, or purely mathematical, but share a common foundation: actionable rules, typically tuned through historical backtesting. Yet, this tuning brings risk. Backtest overfitting, where models are too tightly fitted to past data, can erode performance in live markets. Successful strategies strike a balance between historical accuracy and generalization to future conditions.

The quantitative trading workflow is a full-stack process, integrating:

1. Market signal generation: Identify predictive signals or alpha factors from data (price, volume, fundamentals, alternative data, etc.).
2. Entry/exit rule formulation: Translate signals into actionable trading rules, often including thresholds, time windows, or conditions for entering/exiting trades.
3. Initial risk management and constraint definition: Define position sizing, stop-loss/take-profit limits, exposure constraints, and capital allocation rules. This informs portfolio construction.
4. Portfolio construction: Optimise position weights based on signals, constraints, and risk preferences (e.g., through mean-variance optimisation, risk parity, etc.).
5. Trade execution: Send orders to the market, possibly using execution algorithms that minimise slippage and market impact.
6. Ongoing risk monitoring and post-trade evaluation: Continuously monitor positions, PnL, exposures, and risk metrics (VaR, drawdown, etc.) and adjust as needed in real time.

Despite the sophistication of tools, such as stochastic models, technical indicators, and option theory, predicting financial time series remains inherently difficult. Market returns are noisy, non-Gaussian, and often exhibit extreme values and long memory, complicating efforts to forecast price dynamics with precision (see Peters [1991-96], Mandelbrot et al. [1997]). This uncertainty makes dynamic risk/reward calibration essential. We illustrate the distribution dynamics in Figure (4).

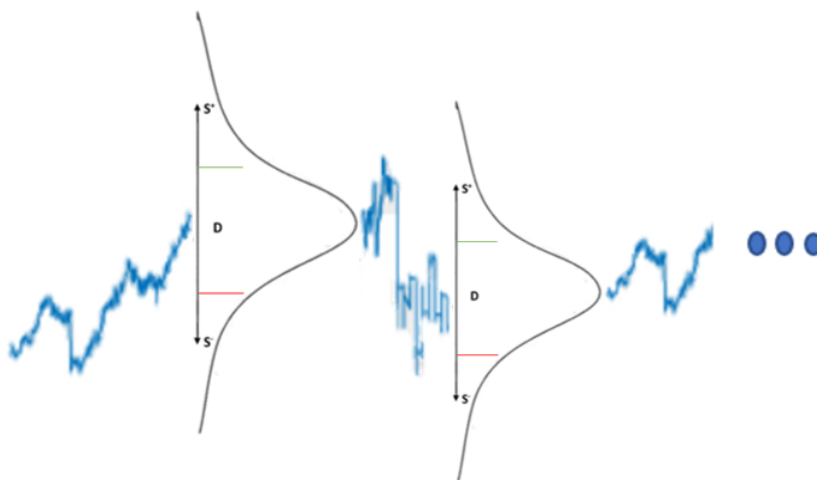


Figure 4: Distribution dynamics

A trader's edge often lies not in knowing where the market is going, but in managing what to do if it doesn't. The risk/reward ratio, the expected gain versus the potential loss, is the cornerstone of trade viability. In a long trade, a stop-loss defines downside risk, while a take-profit target defines the upside. The 'first hitting time', when price reaches one of these levels, dictates execution. Risk targets can be estimated using:

- Data-driven tools (e.g., support/resistance levels)
- Process-driven models (e.g., return distributions at maturity)

More refined tools like trailing stops help capture gains during favourable price moves while providing downside protection. But precision is key: overly tight stops can cut trades prematurely; overly loose ones may lead to drawdowns. As such, stop-loss and take-profit boundaries must adapt with market conditions.

One widely used adaptive technique is the ATR-based trailing stop (see Appendix (16)). It adjusts thresholds based on recent volatility, effectively creating a 'cone of diffusion' that evolves as market conditions shift (see Figure (5)). Bollinger Bands (see Bollinger [1992]) and Keltner Channels offer similar volatility-sensitive bands for dynamic thresholding.

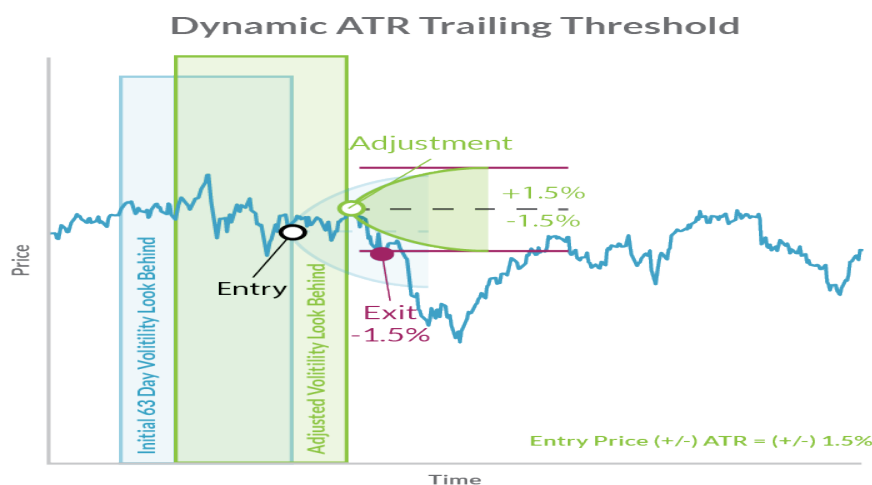


Figure 5: Dynamic ATR trailing threshold.

While Gaussian models often assume symmetric price movement, real-world returns exhibit asymmetry, especially in the tails. To improve robustness, prices can be clustered over sliding time windows, building empirical return distributions that better reflect actual market behaviour, avoiding bias when prices sit in the extremes. An example of such price clustering is the Relative Moving Average presented in Section (9).

Ultimately, dynamic risk management is not an option, it is a necessity. In a world where volatility is constant and certainty is rare, the ability to adjust and manage positions intelligently defines sustainable trading success.

2.3 The limits of predictability in financial markets

2.3.1 The non-stationary nature of financial markets

Financial markets are inherently non-stationary. Their behaviour evolves in response to a constantly shifting landscape of economic policies, geopolitical events, regulatory interventions, and investor sentiment. Unlike physical systems governed by stable laws, markets adapt and reorganise dynamically. This results in price behaviour characterised by time-varying volatility, structural breaks, and regime shifts, features that violate the assumptions underpinning many conventional statistical and mathematical models.

Benoit Mandelbrot [1963] [1967] was among the first to challenge the Gaussian and stationary assumptions in financial theory. His work introduced fractal and multifractal models to capture the irregular and self-similar nature of

asset prices, highlighting properties such as heavy tails and long memory. Subsequent developments by Engle [1982] and Bollerslev [1986] through ARCH and GARCH models formalised the notion of volatility clustering, reinforcing the view that markets cannot be described by simple, time-invariant distributions. Recently, Marcondes et al. [2023] presented financial markets as metastable systems¹ and showed that, under this assumption, financial time series evolve as Hidden Markov Models.

At a deeper level, financial time series are best understood as realisations of an underlying stochastic process. These processes are fully described by their finite-dimensional distributions (FDDs). If the process were stationary, these FDDs would remain invariant under time shifts, allowing past data to inform future outcomes reliably. However, financial markets do not exhibit this property. The FDDs depend on absolute time points rather than relative differences, meaning that observations from one period do not generalize to another.

This point is underscored by Kolmogorov’s extension theorem: since the entire stochastic process is characterised by its FDDs, a model trained on historical data approximates only those specific past distributions. It cannot extrapolate with fidelity to future regimes, especially when the underlying dynamics evolve. As a result, mathematical models built solely on historical trajectories are fundamentally limited in their predictive capacity. No data-driven method can fully account for the complex, time-varying structure of financial markets, making reliable long-term forecasting, in a strict theoretical sense, unattainable.

2.3.2 Addressing uncertainty in time series forecasting

When forecasting the future trajectory of asset prices based on past data, there is rarely a single ‘correct’ prediction. Instead, there exists a spectrum of plausible future scenarios. In the context of diffusion-based models, this uncertainty is often visualised as a diffusion cone, spanning the range $[-\alpha\sigma\sqrt{T-t}, \alpha\sigma\sqrt{T-t}]$ for some constant α , volatility σ , and time to maturity $T-t$ (see Black et al. [1973], Cox et al. [1976]). This cone represents the spread of potential future returns under the assumed stochastic dynamics. Consequently, the realistic goal of forecasting is not to provide a single deterministic trajectory, but rather to estimate a distribution over possible outcomes, ideally accompanied by their associated probabilities.

In practice, autoregressive neural networks (ANNs) are widely used for forecasting high-dimensional time series, drawing on both historical patterns and exogenous features (see Benidis et al. [2020]). To quantify predictive uncertainty, these models often impose an explicit parametric distribution on their outputs and apply maximum likelihood estimation. However, this reliance on parametric forms introduces sensitivity to the choice of distribution family, potentially limiting the model’s expressiveness, especially in capturing non-standard or asymmetric uncertainty structures.

More recently, the advent of conditional diffusion models and other general-purpose generative frameworks (see Ho et al. [2020], Rasul et al. [2021]) has driven substantial progress in probabilistic forecasting of complex, high-dimensional time series. These models can implicitly capture rich uncertainty structures, but they typically come with significant computational burdens during inference, especially when real-time generation of multiple ‘what-if’ scenarios (hypotheses) is required. Moreover, standard generative models often lack explicit mechanisms to enforce diversity across generated hypotheses, leading to redundancies or narrow coverage of the outcome space.

To address these limitations, Cortes et al. [2025] proposed TimeMCL, a method specifically designed to produce multiple diverse and plausible future trajectories for time series forecasting. TimeMCL builds on the framework of Multiple Choice Learning (MCL), which encourages a model to generate a set of distinct predictions rather than a single best guess. By explicitly optimising for diversity among forecasts, TimeMCL provides a principled approach to scenario generation, improving robustness and informativeness in applications like market trajectory forecasting where multiple future paths must be evaluated in parallel.

¹Metastability is a phenomenon observed in stochastic systems which stay in a false-equilibrium within a region of its state space until the occurrence of a sequence of rare events that leads to an abrupt transition to a different region.

2.3.3 The illusion of edge: Probabilistic trading in non-stationary markets

Threshold-based execution strategies (see Section (7.1)) provide a compelling framework for systematic trading, triggering positions when model-estimated probabilities of near-term market direction exceed predefined thresholds. The central challenge in this approach lies in reliably estimating those probabilities over short horizons, typically from a few minutes to an hour, depending on market structure and strategy design. Two fundamentally distinct methodologies dominate this task: data-driven, AI-enhanced forecasting models, and stochastic modelling frameworks rooted in option pricing theory. Yet, regardless of technique, forecasting short-term market direction remains inherently subjective, favouring those with faster, sharper access to relevant information.

While probabilistic forecasting offers a disciplined framework for making risk-informed trade decisions, the non-stationary nature of financial markets breaks the classical argument that repeated trades will reliably converge to positive expected returns. In reality, any informational edge derived from probability estimates is fleeting, local in time, transient in value, and demands continuous recalibration to stay aligned with ever-evolving market conditions.

For example, Marcondes et al. [2023] demonstrated how regime shifts invalidate standard assumptions of stationarity, and highlights that probabilistic edges are inherently local and transient. Liu et al. [2024] proposed TimeBridge to explicitly tackle the dual challenges posed by non-stationarity at both short and long horizons. It segments time series into patches and uses attention mechanisms to mitigate short-term non-stationarity while preserving long-term dependencies. Both papers show that statistical edges erode without adaptive recalibration, reinforcing the statement that probabilistic forecasts are local and transient. This highlights the necessity of continuous recalibration in forecasting models when faced with temporal instability.

2.3.4 From prediction to reaction

As discussed above, given the inherent unpredictability of financial markets, the traditional goal of forecasting future prices with precision must be reconsidered. Rather than attempting to predict exact outcomes, quantitative trading shifts the objective toward reacting intelligently to observable market behaviour. This shift reflects a practical and theoretically grounded approach: when reliable prediction is unattainable, the next best alternative is to design strategies that adapt to and exploit market structure as it unfolds in real time.

Quantitative trading relies on tools such as Technical Indicators (TIs) (see Murphy [1999], Lo et al. [2000], Wang et al. [2007]) and Price Action (PA) (see Thomas [2000]) to identify patterns, assess trends, and make data-driven decisions without relying on forecasts of future prices. These tools serve as a bridge between raw price data and trading decisions by extracting relative information from recent price movements, volatility, momentum, and trend reversals.

While technical indicators, such as moving averages, oscillators, and volatility bands, are often criticized for their lag due to their reliance on historical data, they offer systematic frameworks for evaluating market conditions. When calibrated properly, they can identify regions of expansion and contraction, approximate support and resistance zones, and quantify market momentum. This allows traders to respond consistently to changing conditions, even if the direction of future prices remains uncertain.

In contrast, price action trading emphasises the interpretation of short-term market structure, including candlestick patterns, trading ranges, and breakout formations. Though traditionally more qualitative, price action methods can be formalised and incorporated into quantitative frameworks to enable objective decision-making. For example, a reversal pattern or a breakout from a trading range can be identified algorithmically and used as a signal to enter or exit a position.

Taken together, these approaches do not claim to predict market movements. Rather, they aim to respond to market behaviour as it emerges, exploiting statistical regularities and market inefficiencies when they arise. This reactive

paradigm aligns with the theoretical constraints discussed previously: since the future statistical structure of financial time series cannot be known in advance, strategies must remain adaptive, local, and probabilistic rather than deterministic.

Quantitative trading, therefore, represents a paradigm shift, from prediction to pattern recognition, from certainty to conditional probability, and from static models to dynamic, sliding-window frameworks. It provides a mathematically consistent and practically effective way to engage with the market without violating the theoretical limitations of forecastability.

2.4 Introducing Relative Moving Average

Traditional methods such as Technical Analysis (TA), Price Action Trading (PAT), Stochastic Models, and Option Theory use a combination of indicators and mathematical tools to estimate alpha and define trading rules. While effective in certain market conditions, these approaches face well-known limitations. Technical Indicators (TIs), for instance, often lag due to their dependence on historical data, resulting in delayed signals and, at times, conflicting information. PAT offers valuable insights by interpreting price behaviour in real time, yet remains inherently subjective, different traders may draw different conclusions from identical data.

Moving Averages (MAs) help identify trends and support/resistance zones, but also suffer from lag. PAT, by contrast, emphasises real-time responsiveness but struggles to characterise evolving stochastic structures. Traditional statistical moments, such as kurtosis and skewness, capture the shape of distributions but do not localize these patterns meaningfully within the time series.

To address these challenges, Bloch [2024a] introduced the Relative Moving Moments (RMM) framework, which normalises sample moments relative to specific reference points within a sliding window. Of particular relevance is the first-order RMM, known as the Relative Moving Average (RMA), which blends the strengths of MAs and PAT. By referencing asset prices to statistical anchors, such as segment extrema or quantiles, within a dynamic window, RMA enables a more adaptive and informative profile of market behaviour.

The RMA indicator quantifies the deviation of price from its dynamic equilibrium (the sliding MA), projecting these deviations into a distribution that evolves over time. This evolving distribution provides actionable trading signals while reflecting market cycles of expansion and contraction. Figure (9) illustrates how RMA tracks such deviations, offering a real-time view of price displacement from equilibrium, which can be interpreted as a proxy for profit potential.

This framework proves particularly effective in high-frequency and intraday contexts. The use of normalised rolling moments supports precise calibration of entry and exit points, as well as integrated risk management via stop-loss orders, profit targets, and trailing stops. The approach is versatile and complements established trading strategies, including momentum, mean-reversion, crossovers, flags, pullbacks, and breakouts, while offering a unified, data-driven methodology across various timeframes and asset classes.

Furthermore, the RMA structure extends naturally to advanced strategies such as Pairs Trading and Correlation Trading. It captures both expansions and contractions in relative price movements, making it particularly valuable in identifying regime changes and breakout opportunities. These insights can be translated into clear and actionable strategies that respond dynamically to shifting market conditions.

At its core, RMA addresses a fundamental limitation of forecasting in finance: we observe only the sequential evolution of asset prices in real time, while predictions necessarily involve uncertain distributions over future states. The RMA bridges this gap by continuously measuring the distance between observed price trajectories and a dynamically updated MA, thereby constructing a localised probabilistic structure for strategic decision-making.

2.5 Futuretesting: A new paradigm for strategy evaluation and risk analysis

Once trading rules have been identified, the critical challenge becomes estimating the strategy parameters that are most likely to yield consistent profitability. Yet, the future evolution of market prices is inherently uncertain, making the precise prediction of strategy performance infeasible. Traditional approaches, such as backtesting and forward testing, rely on limited historical trajectories, introducing overfitting risks and providing a narrow lens on strategy robustness.

When calibrating a strategy, it is essential to consider the most challenging conditions for making predictions. To this end, we apply the algorithm to a synthetic dataset that mirrors the structure of the real data but consists of randomly generated, non-stationary values. A forecast is only repeatable if it arises from a process that is not purely random. Berdondini [2021] quantified the true uncertainty of statistical outcomes by evaluating the probability of achieving an equal or better result by chance. This probability decreases as the result becomes more consistently reproducible. In this framework, a high probability implies that the data is more likely to be random and, therefore, less informative. Conversely, for a result to be deemed useful, the likelihood of achieving it by random chance must be very low. In this context, overfitting, is understood as exploiting seemingly useful patterns that, in fact, stem from randomness.

To address these shortcomings, Bloch [2023] introduced Futuretesting, an innovative framework that leverages realistic synthetic market data to rigorously test and optimise trading strategies, perform market risk analysis, and evaluate financial models. Unlike conventional methods that calibrate strategies on a single historical path, Futuretesting operates across a diverse ensemble of statistically grounded or data-driven synthetic trajectories. The aim is to model the complex, stochastic dynamics of financial markets and enable robust inference under varied conditions, including market regimes not previously observed.

Conventional backtesting engines evaluate strategies on historical data, estimating how they would have performed in the past. However, these engines rest on two problematic assumptions: that historical price patterns will persist, and that the strategy being tested would not have influenced those prices. Furthermore, real-world financial markets offer only a single realization of asset price paths, limiting the statistical power and leading to fragile strategies that fail under alternative scenarios.

Forward testing, or paper trading, simulates live trading based on a strategy's logic without executing real trades. While this adds a layer of out-of-sample evaluation, it still relies on a single market path, failing to capture the full distribution of possible futures. Both methods overlook the need to stress test strategies against a wide range of potential outcomes.

Futuretesting solves this by generating statistically coherent synthetic data that reflects real market behaviour while exploring new, unobserved scenarios. This is achieved through several approaches:

- Statistical modelling using parametric processes (e.g., GARCH and its extensions) to simulate price dynamics calibrated on historical data.
- Monte Carlo simulations for high-dimensional or portfolio-level analysis, capturing joint distributions and tail dependencies.
- Stress testing, where synthetic data is tailored to mimic extreme but plausible market conditions, such as bubbles, crashes, or regime shifts.
- Data-driven generation, employing deep learning (e.g., GANs, variational autoencoders) and forecasting models (e.g., neural networks, DNNs) to create market scenarios based on predicted price movements or market features.

This synthetic data empowers us to test trading strategies and risk models across thousands of plausible market paths, uncovering vulnerabilities that would be invisible in a single historical trajectory. Moreover, it enables the development of resilient quantitative strategies, optimised for out-of-sample performance rather than past success.

The Futuretesting engine formalises this approach. It allows for the definition of financial or economic performance criteria, such as mean-variance efficiency in incomplete markets, and uses synthetic data to search for strategies that perform robustly across a pre-defined space of market conditions. All strategies and models are then validated out-of-sample using the full span of real historical data, significantly reducing overfitting and enhancing generalisability.

By unifying statistical, data-driven, and scenario-based simulation methods, Futuretesting provides a versatile and forward-looking toolkit for discovering robust trading strategies and performing comprehensive risk diagnostics. It marks a shift from merely analysing the past to systematically exploring the space of possible futures.

2.6 AI-driven trading intelligence

2.6.1 Empowering autonomous trading with advanced machine learning models

With rapid advances in computer science, automated trading programs now replicate the decision-making of technical traders but with far greater autonomy (see Marti [2022]). To accurately forecast financial returns, particularly in time series exhibiting long-range dependence (LRD), machine learning (ML) models require specialised modifications (see Dung et al. [2024], Zhang et al. [2024]). For instance, Recurrent Neural Networks (RNNs) can be adapted to capture LRD properties, enabling them to identify key technical patterns.

Predicting financial time series is notoriously complex due to their multifractal nature, non-Gaussian distributions, extreme values, and persistent dynamics. Trend forecasting, which targets directional movement instead of exact prices, reduces model complexity while offering substantial advantages for investors. By combining trend accuracy with return metrics that assess the magnitude of trends, we can enhance overall performance. Our framework further refines precision by forecasting both price levels and directional movements, delivering more robust insights.

ML model success hinges on choosing input variables, such as features, correlated assets, and technical indicators, that best explain future asset variations. As computing power has increased, so has the scope for experimenting with diverse input sets, shifting the focus from identifying a single statistical model to identifying the most informative inputs. This data-driven approach allows the model to, speak for itself, revealing hidden relationships. Research backs this method, highlighting how both fundamental and technical factors can align to forecast market prices effectively.

In designing neural networks, prioritising relevant inputs optimises accuracy while reducing complexity. Through the Relative Moving Average (RMA), we address the statistical properties of financial time series. Using generative adversarial networks (GANs), our framework forecasts both market prices and directional movements by optimally combining technical indicators, trading rules, and strategies, leading to a more dynamic and adaptive trading system.

2.6.2 Maximising gains through optimal timing and reinforced risk management

For maximising future returns, investors need to determine the ideal timing for opening and closing positions while managing downside risk, an approach grounded in solving an optimal stopping problem. When holding a long position, the goal is to sell at the optimal moment while using a stop-loss strategy for protection against downside risks. Liquidation challenges, particularly those with constraints, are highly sensitive to corporate events and sudden price jumps, making it challenging to accurately model such gaps in stock prices.

Reinforcement learning techniques, such as Temporal Difference Backward Propagation (TDBP), provide a robust, model-independent approach to addressing these complexities. TDBP handles corporate events effectively and enables precise pricing for contingent claims, such as European and barrier options. In this approach, liquidation constraints are viewed as an optimal stopping problem with a barrier, specifically an American Claim on a Down-and-Out (AC-DOC) barrier contract, which is then valued using the OST-TDBP framework (see Bloch [2024b] [2024c]). This innovative solution supports investors in navigating the risks and timing decisions critical to realising gains.

2.7 The complexity of real-time systematic trading

Systematic trading increasingly relies on the integration of real-time data with machine learning models designed to detect alpha-generating signals intraday (see Ndikum et al. [2024]). With sufficient historical data, a model can be trained to probabilistically classify events, estimate the likelihood of various outcomes, and act based on the most probable scenario. Further, it can anticipate order book dynamics and liquidity shifts before they become apparent to other market participants (see Nagy et al. [2023], Vytelingum et al. [2025]). By doing so, firms can identify and exploit short-lived market opportunities, providing a potential edge in increasingly competitive environments.

However, the adoption of real-time, adaptive systems introduces new layers of complexity, particularly regarding model explainability, operational risk, and compliance. Sophisticated risk explainability frameworks are required before actionable signals can be executed, ensuring that trades are not only economically justified but also ethically and legally defensible.

Traditional model risk frameworks are insufficient for this environment. They must be adapted specifically for real-time, adaptive machine learning systems. Crucially, real-time signal detection engines must include robust safeguards such as kill switches and circuit breakers to prevent cascading trades triggered by false positives or unexpected feedback loops. In this paradigm, risk management is no longer a post-trade or batch process, it becomes live, adaptive, and fully embedded within the trading pipeline itself (see Wang et al. [2024]). As predictive models generate trading signals, each proposed trade is automatically passed through a **live exposure check** that evaluates sector concentration, liquidity availability, and counterparty risk before submission to the market.

For example, real-time VaR estimates update as each trade is proposed, ensuring that risk metrics are dynamically monitored alongside alpha generation, rather than reconciled afterward in batch processes. Additionally, a global asset manager might integrate reinforcement learning algorithms into its trade execution stack to dynamically adjust execution strategies in response to fragmented liquidity or volatility spikes (see Zhang et al. [2023], Macrì et al. [2024]). These algorithms can react instantaneously to deteriorating market conditions, rebalancing portfolios within predefined exposure corridors while minimising execution costs and market impact.

Yet, despite technological advancements, the financial markets remain characterised by non-stationarity, structural breaks, and unpredictable discontinuities (see Section (2.3)). The reality of systematic trading is that it represents a **high-dimensional, real-time decision-making problem under uncertainty**, where even the most sophisticated algorithms will inevitably encounter outliers, such as price jumps, stop-runs, or manipulation-driven fakeouts (see Section (8.1)).

Perfect detection of such events is, by nature, impossible. Instead, practical systems must respond probabilistically to symptoms of structural breaks using a combination of 'early-warning indicators, regime detection heuristics, and signal attenuation mechanisms' when uncertainty escalates.

Building a robust framework to handle these challenges requires solving three tightly coupled problems in real time:

1. Real-time detection of actionable market signals.
2. Live risk management validation of those signals against evolving portfolio exposures.
3. Dynamic execution using adaptive algorithms capable of responding to changing liquidity and volatility conditions.

To build a model capable of performing these three interrelated tasks in real time, we require a reliable **signal-to-position mapping** (see Section (8.1.3)), a systematic logic that transforms price observations and predictive indicators into actionable portfolio decisions (see Goluza et al. [2024]). This ensures that trading decisions are not based solely on raw signals but are contextually aligned with portfolio objectives, risk limits, and prevailing market conditions.

Beyond market data, the system must ingest external information from real-time news feeds, social media platforms, and other event-driven sources. This information needs to be analysed and scored using sentiment and behavioural models to generate structured metrics (see Konstantinidis et al. [2024], Zhou et al. [2025], Yang et al. [2025]).

Critically, these external metrics must be mapped coherently to both the detected trading signals and the existing portfolio exposures, ensuring that decisions are context-aware. All relevant data streams must be consolidated into a predefined, **live decision-making grid**, capable of translating diverse inputs into portfolio actions. Executing this process requires solving a complex, multi-dimensional optimisation problem under risk, liquidity, and regulatory constraints, all within fractions of a second, in the presence of noise, volatility, and potentially chaotic market conditions (see Ghosh et al. [2022]).

Consequently, attempting to perform these three interrelated tasks simultaneously in a live environment introduces inevitable trade-offs between speed and accuracy (see Qin et al. [2023]). False positives and false negatives are unavoidable, and without carefully designed controls, they can lead to significant financial losses or regulatory breaches. Therefore, the objective is not to eliminate uncertainty, but to build adaptive systems capable of mitigating its worst effects while preserving agility, transparency, and compliance.

3 Quantitative methods for dynamic market environments

We begin by examining the core methodologies underpinning quantitative trading in the context of dynamic and evolving financial markets. It begins with a foundational overview of quantitative trading strategies, focusing on momentum and mean-reversion models, the two dominant paradigms in systematic trading. These models leverage historical data and statistical techniques to make rules-based trading decisions, thereby reducing discretionary bias and enabling algorithmic implementation at scale.

The discussion then turns to risk management, a critical component of any robust trading system. Here, the objective is to manage downside exposure without stifling the strategy's performance potential. Key concepts include capital preservation, exposure control, and aligning risk-taking with predefined objectives through dynamic adjustment of position sizing, stop-loss mechanisms, and portfolio-level constraints.

Finally, we address the challenge of non-stationarity in financial time series. It explores the limitations of conventional estimation techniques in the face of structural breaks, jumps, and regime shifts. While rolling window estimators offer adaptability, they also introduce high variance and instability when local stationarity fails. The section emphasises the importance of balancing long-run and short-run estimation through shrinkage methods, exponentially weighted moving averages, and hybrid approaches. Ultimately, the goal is to align modelling techniques with the underlying market regime and intended use, be it long-term risk assessment or short-term signal triggering, thereby enhancing robustness in uncertain and discontinuous market environments.

3.1 Foundations of quantitative trading strategies

Quantitative trading strategies form the backbone of modern algorithmic trading systems. These strategies use mathematical models and statistical techniques to make systematic trading decisions, often based on the analysis of historical price data and market microstructure signals. Unlike discretionary approaches, quantitative methods aim to eliminate emotional bias by encoding decision rules into algorithms that can be rigorously tested, optimised, and deployed at scale. This section provides an overview of core quantitative strategies, focusing particularly on momentum and mean-reversion models, the two most widely adopted paradigms in the field.

3.1.1 Overview of quantitative trading strategies

Quantitative trading encompasses a broad class of methods that rely on empirical data and systematic logic to make investment decisions (see Narang [2009], Lien [2015]). At its core, a quantitative strategy involves four components: signal generation, risk management, execution, and performance evaluation. Signals are derived from statistical relationships in the data, which may include price trends, volume dynamics, volatility measures, or more complex factor models. Once a signal is generated, the strategy must assess the expected return and associated risk, size positions accordingly, and execute trades efficiently.

Key advantages of quantitative approaches include consistency, scalability, and the ability to backtest hypotheses on historical data. However, these systems are not without limitations. They rely heavily on model assumptions, which can break down in the presence of regime shifts, structural breaks, or unmodelled risks such as market jumps and fakeouts. As such, robust model design, continuous monitoring, and adaptive mechanisms are essential for long-term viability.

3.1.2 Momentum strategies

Momentum strategies exploit the empirical tendency of assets that have performed well (or poorly) in the recent past to continue performing in the same direction in the near future. These strategies are built on the assumption that asset prices exhibit serial correlation, driven by factors such as investor herding, delayed information diffusion, and trend-following behaviour in the market.

In practice, momentum signals are computed using past return measures over specified horizons, such as 20-day, 60-day, or 252-day lookbacks. In the case of intraday trading, the horizons range from seconds, to minutes and hours. When returns exceed a threshold, the strategy takes a long position (for upward momentum) or a short position (for downward momentum). Common implementations include:

- Time-series momentum: Trades based on the asset's own past performance.
- Cross-sectional momentum: Ranks assets relative to one another and takes long positions in the strongest performers and short positions in the weakest.

Despite their strong historical performance across asset classes, momentum strategies are vulnerable to sharp reversals and mean-reverting corrections. Their risk profile often includes large drawdowns during regime transitions or during episodes of heightened volatility.

3.1.3 Mean-reversion strategies

Mean-reversion strategies are predicated on the belief that asset prices fluctuate around a long-term equilibrium or fair value and tend to revert back after significant deviations. These strategies are supported by various economic and behavioural rationales, including overreaction, liquidity provision, and statistical noise.

Signal construction typically involves detecting extreme deviations from a reference mean using measures such as Z-scores, Bollinger Bands, or moving average convergence. When the price deviates beyond a set threshold, the strategy enters a contrarian position, expecting the price to revert.

Mean-reversion is most effective in range-bound or low-volatility regimes, and particularly in high-frequency settings where short-term inefficiencies are more pronounced. However, it faces critical challenges when trends emerge or when structural breaks cause the 'mean' to shift. Rigid adherence to outdated means can result in persistent losses, particularly during breakout phases or momentum-driven rallies.

To mitigate these risks, practitioners often enhance mean-reversion systems with regime filters, volatility-adjusted signals, or adaptive thresholds that account for time-varying distributions.

3.1.4 Core quantitative tools

Some of the core quantitative tools for momentum and mean-reversion strategies are:

- **Moving Average (MA):** A smoothed average of past prices over a fixed window, used to identify trends or mean levels.
- **Exponential Moving Average (EMA):** A weighted moving average that assigns more weight to recent prices, making it more responsive to changes.
- **Z-Score:** A standardised measure of how far a value deviates from the mean, used to detect statistical extremes or mean-reverting signals. See Appendix (14.1.3).
- **Bollinger Bands:** A volatility-based envelope around a moving average, where price touching the upper or lower band signals potential overbought or oversold conditions.
- **Relative Strength Index (RSI):** A momentum oscillator measuring the speed and change of price movements, typically used to detect overbought/oversold conditions.
- **MACD (Moving Average Convergence Divergence):** A trend-following momentum indicator based on the difference between two EMAs, used to detect shifts in momentum.
- **Rate of Change (ROC):** Measures the percentage change in price over a fixed lookback period; often used to capture momentum.
- **Autocorrelation:** Quantifies the correlation of a time series with its own past values; positive autocorrelation is often exploited in momentum strategies.
- **Mean Absolute Deviation (MAD):** The average of absolute deviations from the mean, offering a robust measure of dispersion.
- **Rolling Volatility:** Standard deviation of returns over a moving window; used for dynamic risk estimation and volatility filtering.
- **Sharpe Ratio:** Measures risk-adjusted return as the ratio of excess return to volatility; used for evaluating strategy performance.
- **Drawdown:** The peak-to-trough decline in equity or returns, used to quantify downside risk and assess strategy robustness.
- **Kalman Filter:** A recursive estimator used to track time-varying parameters or latent signals in noisy financial data.
- **Hurst Exponent:** A measure of long-term memory in time series, used to detect mean-reverting versus trending behaviour.
- **ADF Test (Augmented Dickey-Fuller):** A statistical test for stationarity by testing for the presence of a unit root in a time series.
- **Average True Range (ATR):** A volatility indicator that measures the average range between high and low prices (including gaps) over a given period, used to size positions or detect regime shifts. See Appendix (16)

3.1.5 Conclusion

Both momentum and mean-reversion represent foundational paradigms in quantitative trading, each grounded in empirical regularities and supported by extensive literature. Yet, they operate under fundamentally different assumptions about market behaviour, one trends, the other reverts. In practice, combining elements of both, alongside careful calibration, risk control, and adaptivity to changing market regimes, often leads to more robust and resilient trading systems.

3.2 Risk management

The goal of risk management is to preserve capital and prevent catastrophic losses while allowing the strategy to perform within expected bounds. This involves identifying, assessing, and prioritising potential risks, then applying appropriate controls to mitigate their impact. Effective risk management balances caution with opportunity, ensuring that risks taken are intentional and aligned with the overall objectives and risk tolerance of the investment or trading strategy.

3.2.1 Key risk management techniques

3.2.1.1 Position sizing

- **Core idea:** Adjust the size of each position to manage risk and return trade-offs, using information such as volatility, expected returns, signal strength, or capital at risk.
- **Objectives:**
 - Control downside risk and avoid overexposure to any single asset or trade.
 - Ensure consistency in portfolio contribution across trades or strategies.
 - Dynamically adapt to changing market conditions or signal confidence.
- **Volatility targeting:**
 - Scale position sizes inversely with asset volatility to target a fixed level of risk.
 - Example: Size each trade to contribute 10 basis points to overall portfolio volatility.
 - Can use rolling estimates of standard deviation or average true range (ATR) (see Equation (16.19)) for implementation.
- **Capital allocation methods:**
 - *Fixed allocation:* Assign a constant dollar amount or portfolio percentage to each position (simple but inflexible).
 - *Signal-weighted allocation:* Scale positions in proportion to signal strength or expected alpha.
 - *Risk parity:* Equalize the risk contribution of each asset or strategy, often using volatility estimates.
 - *Kelly criterion (fractional):* Optimize capital allocation to maximize expected logarithmic growth?use with caution due to high sensitivity to parameter estimates and drawdown risk.
- **Practical considerations:**
 - Impose minimum/maximum position sizes to control for liquidity and slippage.
 - Incorporate transaction costs and turnover when adjusting positions.
 - Rebalance periodically or when risk conditions change materially.

3.2.1.2 Stop losses and profit targets

- Hard stops are used less often in systematic trading but may still be applied:
 - *Intra-bar stop-loss*: Exit a position immediately if the unrealised loss exceeds a fixed threshold (e.g., $X\%$).
 - *Time-based stop*: Automatically close trades after a predetermined duration to limit exposure.
 - *Profit target*: Secure gains once a predefined return is achieved, though this may cap the long-term expectancy of the strategy.
- Soft stops are more commonly used in quantitative trading as they offer greater adaptability:
 - *Dynamic exit conditions*: Close a position based on changes in indicator values, such as momentum weakening or reversal in the Relative Moving Average (RMA).
 - *Trailing logic*: Adjust stop levels as the trade moves in the desired direction, locking in profits without prematurely exiting.
 - *Signal-based exit*: Rely on the same model that generated the entry signal to trigger an exit, e.g., crossing a threshold or detecting a trend reversal.
 - *Volatility-aware thresholds*: Adapt stop levels dynamically using recent volatility, e.g., via ATR (Average True Range) or standard deviation bands.

3.2.1.3 Exposure limits

- **Purpose**: Manage concentration risk and maintain portfolio diversification by capping exposure to individual positions, correlated assets, or directional market bets.
- **Single-asset limits**:
 - Prevent overconcentration in any one instrument.
 - Example: Cap position size at a maximum of 10% of total portfolio capital.
- **Sector and correlation limits**:
 - Group assets by industry, geography, or correlation clusters.
 - Limit cumulative exposure to a sector or correlated group to reduce systemic risk (e.g., max 25% in tech or energy).
 - Use statistical methods (e.g., PCA, correlation matrices) to detect hidden exposure overlaps.
- **Directional exposure constraints**:
 - Impose caps on net long or short exposure to maintain market neutrality or prevent over-leveraging in one direction.
 - Example: Limit net long exposure to 60% of capital; net short to 40%.
 - Useful for strategies with beta-neutral or relative value objectives.
- **Regime-aware exposure management**:
 - Adjust limits based on macro indicators, market stress levels, or volatility regimes.
 - Example: Tighten exposure during high-volatility environments (e.g., $VIX > 30$) to reduce drawdown risk.
 - Can be implemented using volatility filters, regime-switching models, or machine learning classifiers.
- **Implementation tips**:
 - Monitor exposures dynamically and enforce constraints during rebalancing or signal updates.
 - Integrate exposure limits into the portfolio optimization step to ensure compliance at the allocation level.
 - Maintain flexibility to adapt constraints based on strategy type (e.g., trend-following vs. mean-reversion).

3.2.1.4 Leverage controls

- **Purpose:** Prevent overexposure and manage risk by limiting the amount of capital borrowed relative to portfolio equity.
- **Leverage metrics:**
 - **Gross leverage** = Sum of absolute position sizes ($\text{long} + \text{short}$), divided by portfolio equity.
Example: A portfolio with \$150 long and \$50 short on \$100 equity has gross leverage = 2.0.
 - **Net leverage** = Net directional exposure ($\text{long} - \text{short}$), divided by portfolio equity.
Example: Same portfolio has net leverage = 1.0.
- **Control strategies:**
 - Set maximum allowable gross and net leverage levels (e.g., gross ≤ 3.0 , net ≤ 1.0).
 - Use margin requirements and volatility filters to adapt leverage dynamically.
 - Monitor leverage drift due to price movements or overlapping signals and rebalance when necessary.
- **Intraday trading considerations:**
 - Individual trades may use low leverage, but cumulative exposure across multiple simultaneous positions can escalate risk quickly.
 - Apply real-time leverage tracking and auto-throttling to avoid unintended exposure spikes.
 - Stress-test for scenarios where leverage compounds due to correlated signals or directional bias.
- **Risk mitigation:**
 - Integrate leverage caps with position sizing and exposure limits for coherent portfolio-level risk management.
 - Use simulated drawdowns and value-at-risk (VaR) measures to calibrate leverage tolerance under worst-case scenarios.
 - Be cautious with leverage in illiquid markets, where slippage and gap risk can amplify losses.

3.2.1.5 Slippage and transaction cost modelling We must model trading costs accurately:

- **Key components to model:**
 - **Bid-ask spread:** The cost of crossing the spread when using market orders. Spreads vary by liquidity, time of day, and volatility.
 - **Market impact:** The price movement caused by one's own order. More significant in large trades or illiquid assets.
 - **Slippage:** The difference between the expected execution price and the actual price received. Can occur due to latency, queue position, or partial fills.
 - **Latency and order execution lag:** Especially relevant for high-frequency or low-latency strategies where milliseconds matter.
- **Cost sensitivity by strategy type:**
 - **Intraday and high-frequency strategies:** Highly sensitive to transaction costs, small inefficiencies can become unprofitable after accounting for slippage and fees.

- **Swing and position trades:** Less affected by microstructure noise, but large positions can still incur market impact and spread costs.
- **Model integration:**
 - Embed cost models directly into backtests to avoid overestimation of strategy performance.
 - Include realistic assumptions on order type (market, limit, TWAP, VWAP), fill probability, and execution delay.
 - Use historical limit order book data when possible to calibrate slippage and impact models.
- **Other considerations:**
 - Account for exchange fees, commissions, taxes, and rebates depending on venue and broker.
 - Avoid 'free liquidity' assumptions, liquidity is dynamic and adversarial during market stress.
 - Model slippage asymmetrically; adverse slippage is more common than favourable execution.

3.2.1.6 Drawdown controls

- **Implement kill switches and circuit breakers** to preserve capital during adverse conditions:
 - **Daily/weekly/monthly drawdown limits:** Halt trading if cumulative losses exceed a pre-defined threshold (e.g., 2% daily or 5% weekly). This prevents emotional or model-driven overtrading after losses.
 - **Volatility triggers:** Pause trading during volatility spikes (e.g., sudden VIX jumps, liquidity drops, or abnormal intraday swings).
 - **Unusual behaviour detection:** Detect large deviations in strategy output or trade frequency compared to historical patterns.
- **Early warnings and auto-adjustments:**
 - Implement soft limits or warning zones (e.g., alert at 1.5% drawdown) to review strategy before hard limits are breached.
 - Adjust risk parameters dynamically when approaching drawdown thresholds (e.g., reduce position size or widen stop levels).
- **Beyond capital protection:**
 - Drawdown controls act as a proxy for detecting **model degradation, overfitting, or regime shifts**.
 - Useful for real-time monitoring of **strategy robustness and market compatibility**.
 - Helps mitigate tail risk and reduce psychological strain during turbulent periods.

3.2.1.7 Real-time monitoring

- **Live dashboards and metrics tracking:**
 - Continuously monitor **positions, P&L, order book activity, execution latency, and realised vs. expected slippage**.
 - Track key risk indicators: portfolio exposure, leverage, volatility, and drawdowns in real time.
- **Trigger alerts and fail-safes on anomalies:**
 - Flag unexpected deviations from historical norms (e.g., large trade sizes, sudden price gaps, unusual order rejections).

- Set automated alerts (email, Slack, terminal warnings) for latency spikes, execution failures, capital breaches, or strategy drift.
- Pause or throttle trading activity if system stress or irregularities are detected.

- **Health checks and infrastructure monitoring:**

- Monitor data feeds, broker connectivity, API response times, and internal computation lags.
- Implement redundancy and fallback mechanisms for critical trading components.

- **Human-in-the-loop oversight:**

- Ensure real-time human supervision, especially during market opens/closes, macro events, or outages.
- Support intervention tools for trade cancellation, manual shutdown, or parameter override.

3.2.2 Empirical validation and robustness analysis

3.2.2.1 Pre-trade risk controls Ensure strategy robustness before deployment (see Peterson [2017], Perumal et al. [2018]):

- **Out-of-sample testing:**

- Validate strategy performance on unseen data to detect overfitting.
- Use walk-forward or rolling-window approaches to assess consistency across time.

- **Monte Carlo simulations:**

- Resample trade sequences or return paths to estimate the distribution of potential outcomes.
- Quantify the impact of sequencing risk and performance variability.

- **Stress testing:**

- Simulate extreme but plausible events such as volatility spikes, flash crashes, or market halts.
- Assess drawdown resilience and capital preservation under tail scenarios.

- **Slippage and transaction cost modelling:**

- Incorporate realistic estimates of bid-ask spreads, market impact, latency slippage, and order book depth.
- Intraday strategies are especially sensitive to costs?underestimating them leads to misleading backtest results.

- **Robustness checks and baselines:**

- Test sensitivity to parameter changes and evaluate generalisation across markets or timeframes.
- Compare against simple baselines (e.g., buy-and-hold, random trades) to ensure true alpha.

3.2.2.2 Futuretesting platform: Confidence in every strategy Futuretesting (see Bloch [2023]) is a go-to infrastructure for confidently productionalising strategies. With a consistent interface at every stage, we can seamlessly transition code up and down the development ladder. Here are the key steps:

1. Synthetic Data Environment: Mimic real-world conditions with an identical interface to fine-tune hyperparameters and optimize strategies.
2. Backtest Environment: Fine-tune strategies and test hypotheses with a backtesting interface that mirrors real-world scenarios for ultimate accuracy.
3. Real-World Paper Account: Conduct live performance tests without financial risk?this is the final hurdle before trading with real money.
4. Live Low-Balance Account: Validate execution efficacy with a minimal investment. Once confirmed, strategies can be scaled up by adjusting position sizing configurations.

3.2.3 Evaluating strategy performance

3.2.3.1 Risk metrics to track Continuous assessment of strategy performance:

- **Sharpe Ratio (realised vs. expected):**
 - Compare the in-sample Sharpe Ratio with the live (out-of-sample) Sharpe to detect degradation. See Sharpe [1994].
 - Track rolling Sharpe over different time windows to assess stability.
- **Maximum Drawdown:**
 - Monitor both absolute and duration of drawdowns to assess recovery risk.
 - Use trailing windows to identify abnormal risk spikes or structural breaks.
- **Value at Risk (VaR) and Expected Shortfall (ES):**
 - Quantify tail risk at different confidence levels (e.g., 95%, 99%).
 - ES complements VaR by measuring the average loss beyond the VaR threshold.
- **Skewness and Kurtosis of Returns:**
 - Skew helps identify asymmetry (e.g., strategies with frequent small wins but rare large losses).
 - Kurtosis highlights fat-tailed behaviour, which may indicate higher tail risk.
- **Turnover and Execution Cost per Trade:**
 - High turnover can degrade performance via slippage and fees?track daily and monthly levels.
 - Measure average cost per execution vs. modeled assumptions to identify cost drift.
- **Additional metrics (optional):**
 - Calmar ratio, Sortino ratio, and Omega ratio for richer risk-adjusted performance views.
 - Rolling beta and correlation to benchmark or factor models for exposure analysis.

3.2.3.2 Success criteria The strategy's success depends on:

- Proper calibration of signal thresholds,
- Filtering noisy signals in low-momentum regimes,
- Tight risk control to avoid drawdowns from false signals
- Possibly ensemble filtering (e.g., using other indicators like volatility, volume, or trend strength).

3.3 Portfolio allocation in intraday systematic trading

In intraday systematic trading, portfolio allocation plays a critical role in balancing return opportunities with risk constraints under the specific dynamics of high-frequency and short-horizon trading. Unlike traditional portfolio theory, which typically assumes multi-day holding periods and relies on stable estimates of returns and covariances, intraday strategies must operate under intense time constraints, high transaction costs, and rapidly changing market microstructure.

3.3.1 Objectives and constraints

The main objective of portfolio allocation in this context is to distribute capital across a set of signals or strategies in a way that maximises risk-adjusted return, subject to execution feasibility, turnover constraints, and risk limits. These considerations lead to a more tactical, often data-driven approach, with frequent rebalancing and a greater emphasis on execution-aware portfolio construction.

Allocation decisions in intraday trading must simultaneously respond to fast-changing alpha forecasts, real-time market conditions, and operational constraints. The portfolio construction engine must, therefore, optimise capital deployment while maintaining sufficient flexibility to adapt to signal decay, liquidity fragmentation, and transaction costs. In many systems, this results in lightweight, regularised optimisation routines or rule-based allocations with embedded risk logic.

Key constraints include:

- Intraday risk budgets: $P\&L$ drawdown limits, realised or forecast volatility caps. Intraday strategies often operate under tight capital preservation mandates. Live drawdown monitoring is used to implement soft or hard risk stops, for example, halting trading if cumulative $P\&L$ falls below a predefined threshold within a trading session. Volatility targeting techniques are also common, whereby position sizing is dynamically adjusted to stabilise realised volatility. For instance, portfolio weights may be scaled as:

$$w_t \leftarrow w_t \cdot \frac{\sigma_{target}}{\hat{\sigma}_t}$$

where $\hat{\sigma}_t$ is the estimated portfolio volatility over a rolling window.

- Execution capacity: Slippage, impact, and available liquidity per instrument or venue. Because alpha signals are often small and fleeting, efficient execution is critical to capturing their value. The portfolio allocator must account for instrument-level constraints such as bid-ask spread, order book depth, and venue-specific fill probabilities. Trade sizes are frequently capped at a percentage of volume (e.g., 5 – 15% of 1-minute AIV) to mitigate market impact and avoid detection by predatory algorithms. Cross-asset liquidity correlations are also relevant, for example, an ETF arbitrage strategy may face execution risk due to the illiquidity of a subset of its components.

- **Latency and speed:** Computation must be fast enough to allow real-time decision-making. Intraday allocation pipelines must complete within milliseconds to seconds, depending on signal frequency. Optimisation solvers are often constrained to linear or quadratic objectives with diagonal or low-rank covariance approximations to ensure tractability. In low-latency environments, allocation logic may be reduced to precomputed lookup tables, cached model parameters, or even rule-based schemes to minimise runtime delays.
- **Position limits:** Regulatory or internal constraints on net/gross exposure, leverage, and sector or asset-class concentration. Allocation systems must ensure compliance with portfolio-level exposure rules, such as maximum gross exposure (e.g., $2\times$ capital), net neutrality (e.g., net dollar exposure $< 1\%$ of capital), or concentration limits (e.g., no single name exceeding 5% of NAV). Sector, industry, or region-based exposures may also be restricted depending on mandate or regulatory jurisdiction. These constraints are typically enforced via projections or penalties within the optimisation routine, or through rejection sampling and soft clipping in rule-based systems.

These constraints introduce a trade-off between alpha capture and execution practicality. The role of the portfolio allocator is not only to optimise expected utility, but to enforce feasibility and robustness under operational constraints, often requiring integration with real-time risk engines, execution systems, and monitoring infrastructure.

3.3.2 Common allocation approaches

1. **Signal-Based Allocation:** Positions are allocated proportionally to the strength of alpha signals, often standardised across assets:

$$w_{t,i} = \frac{s_{t,i}}{\sum_{j=1}^N |s_{t,j}|}$$

where $s_{t,i}$ is the signal for asset i at time t , and $w_{t,i}$ is the weight (or scaled notional position). This approach assumes independent signals and ignores cross-asset risk, but it is computationally efficient and suitable when execution costs dominate.

2. **Volatility Scaling and Risk Parity:** To avoid concentration in more volatile instruments, signals are often rescaled by recent intraday volatility estimates:

$$w_{t,i} \propto \frac{s_{t,i}}{\sigma_{t,i}}$$

where $\sigma_{t,i}$ is an intraday estimate of realised volatility for asset i . This ensures more stable P&L contributions across assets and simplifies risk targeting at the strategy level.

3. **Online Mean-Variance Optimisation:** A more sophisticated method uses online or adaptive optimisation to solve:

$$\max_{w_t} \left(\mu_t^\top w_t - \frac{1}{2} \lambda w_t^\top \Sigma_t w_t \right)$$

where μ_t is the vector of forecast returns from alpha models, Σ_t is the estimated intraday covariance matrix, and λ is a risk-aversion parameter. Given the challenges of estimating Σ_t intraday, practitioners often use shrinkage, diagonal approximations, or factor models to stabilise inputs.

4. **Cost-Aware Allocation:** Transaction costs, particularly slippage and market impact, are significant in intraday trading. A cost-aware objective includes a penalty term:

$$\max_{w_t} \left(\mu_t^\top w_t - \frac{1}{2} \lambda w_t^\top \Sigma_t w_t - \gamma C(w_t - w_{t-1}) \right)$$

where $C(\cdot)$ is a convex function modelling trading costs, and w_{t-1} is the previous position. Solutions typically involve quadratic programming or greedy approximations, depending on the dimensionality and latency requirements.

3.3.3 Execution considerations

Because of the short holding period, execution efficiency is often more important than signal accuracy. Small improvements in execution quality can outweigh marginal improvements in alpha forecasts, particularly in high-turnover intraday strategies. Therefore, portfolio allocation must be closely aligned with execution logic to ensure that theoretical positions can be translated into realised $P\&L$ without excessive slippage or impact.

Portfolio allocation decisions must:

- Respect liquidity limits (e.g., percentage of average volume). To avoid excessive market impact, intraday strategies often cap the notional traded per instrument to a fixed percentage of its average daily or intraday volume (ADV or AIV). A typical constraint might be 10% of 1-minute AIV. This ensures that trading does not distort prices or invite detection by other market participants. Liquidity-aware position sizing may also scale down trades during low-liquidity regimes (e.g., lunch hours or just after open/close auctions).
- Be synchronised with execution algorithms that incorporate order book dynamics. Execution engines must adaptively choose between passive and aggressive order types based on real-time order book state, spread, queue position, and short-term price momentum. For example, a predictive alpha signal with high urgency might trigger a marketable limit order, while a mean-reversion signal may prefer to provide liquidity. Portfolio allocations should be structured to support such microstructural decisions, for instance, through prioritisation of signals by urgency or expected alpha decay.
- Account for execution slippage and potential adverse selection. Slippage from predicted mid-price and market impact can erode signal profitability, especially for short-term trades. Moreover, aggressive executions may be picked off in adverse selection scenarios (e.g., buying before a large downward move). Allocation models can incorporate cost estimates in the form of convex functions $C(w_t - w_{t-1})$ where C includes both temporary and permanent market impact models. Adjusting weights dynamically based on predicted implementation shortfall or fill probabilities is common in latency-sensitive environments.

In practice, execution-aware portfolio allocation often involves a feedback loop between alpha models, allocation, and execution infrastructure. For example, slippage observed during live trading may be fed back into cost models to refine future allocation. Moreover, the latency profile of the strategy, whether co-located or cloud-based, can influence the optimal trade-off between execution speed and price improvement.

Execution considerations are not merely an implementation detail; they are integral to intraday strategy design. Many strategies that appear profitable in backtests fail in live environments due to unrealistic assumptions about fill prices, latency, and liquidity availability.

3.3.4 Intraday rebalancing and position lifecycle

Intraday portfolios are often rebalanced multiple times per day or continually in a streaming framework. Because alpha signals at this frequency typically decay within minutes to hours, maintaining alignment between portfolio positions and current model outputs is critical. The position lifecycle, from entry to exit, is tightly managed to control risk, reduce slippage, and comply with operational constraints.

Strategies may employ:

- Pre-specified rebalancing windows, e.g., every 15 minutes. This approach partitions the trading day into fixed intervals (e.g., 5-, 15-, or 30-minute bars), within which signals are aggregated and positions are recalculated. It offers a predictable execution schedule and is often easier to implement in backtesting and compliance systems. However, it may miss transient opportunities or lag behind fast-changing signals. Allocations within each window may use the latest available market data snapshot and assume execution within the next time slice, with slippage modeled accordingly.
- Event-driven updates, triggered by signal changes, volume thresholds, or volatility spikes. Instead of relying on a fixed clock, event-driven rebalancing reacts to significant changes in market state or model output. For example, a breakout in realized volatility, crossing a signal threshold, or a news-driven volume spike might prompt an immediate update. These systems often operate in a streaming architecture where signals are continuously monitored, and rebalancing logic is triggered by asynchronous events. This is particularly effective in alpha models sensitive to short-lived price dislocations or microstructure anomalies.
- End-of-day flattening, where all positions are closed before market close to avoid overnight risk. Many intraday strategies are designed to be *flat* at the end of each trading session to avoid overnight risk and regulatory capital charges. This flattening process typically begins within a closing window (e.g., last 5?15 minutes) and may involve prioritising the exit of less liquid or more volatile positions earlier. Flattening trades are sometimes executed more aggressively to ensure timely exit, accepting higher transaction costs in exchange for risk reduction. In high-turnover systems, forced flattening may also be used to reset state and avoid position drift across days.

In advanced systems, intraday rebalancing policies are dynamically configured based on market conditions. For example, during periods of high volatility or news events, the system may shorten rebalancing intervals or switch from time-based to event-driven logic. Additionally, rebalancing frequency is often optimised jointly with signal horizon and execution cost models to maximise net alpha retention.

The position lifecycle in intraday trading is therefore an orchestrated process that integrates forecasting, allocation, execution, and risk management in a tightly-coupled loop. Rebalancing decisions must be latency-aware, risk-adjusted, and cost-sensitive to maintain performance in production environments.

3.3.5 Risk management integration

Portfolio allocation is tightly integrated with risk management, especially in the context of intraday systematic trading where positions evolve rapidly and exposures can change materially within seconds. Real-time monitoring and automated control loops are essential to detect deviations from expected behavior, enforce constraints, and dynamically adjust positions in response to changing market conditions.

- Live $P\&L$ and position drift. Continuous tracking of live profit and loss is crucial for detecting unexpected slippage, execution errors, or divergence between model-predicted and actual trading outcomes. Position drift, the divergence between the intended allocation w_t and the realised position $\hat{w}_t$, can occur due to partial fills, delays in execution, or stale signal updates. Risk management systems often enforce intra-day stop-loss thresholds based on live $P\&L$, with actions ranging from reduced sizing to full strategy halts. Techniques such as real-time position reconciliation and shadow accounting are used to ensure alignment between trading intent and actual exposure.
- Factor exposures (beta, sector, macro). Even in short-horizon trading, latent exposure to systematic risk factors (e.g., equity market beta, duration, currency, or volatility) can accumulate unintentionally. Portfolio allocation engines may include real-time factor models that decompose positions into exposures relative to a risk model basis f_t , and enforce soft or hard bounds on these exposures. For instance, an intraday strategy might require dollar neutrality, sector neutrality, or low correlation with major indices (e.g., $S\&P$ 500, VIX). Constraints are often imposed through projections or penalisation terms during optimisation, such as:

$$\min_{w_t} (Loss(w_t) + \lambda \|B^\top w_t\|^2)$$

where B is the factor loading matrix.

- Limit breaches or anomaly detection (e.g., model drift or flash crash response). Real-time systems monitor for violations of operational limits, including gross/net exposure caps, turnover limits, leverage thresholds, and concentration rules. In addition, statistical and ML-based anomaly detectors may track deviations in input features, output signals, or execution patterns. For example, persistent divergence between predicted and realized $P\&L$, unusual signal persistence, or sudden breakdown in hit rates may indicate model drift or structural breaks. In extreme events such as flash crashes, automatic circuit breakers can trigger position reduction, forced flattening, or disconnection from execution venues to prevent catastrophic losses.

Risk management in intraday trading is not a post-hoc control function, but a real-time co-actor with portfolio allocation and execution. Effective integration allows for dynamic adaptation, such as down-weighting volatile instruments, applying signal throttles during market stress, or automatically reducing capital allocation to underperforming components. In many production systems, allocation decisions are gated through a risk manager module that provides real-time approval or rejection based on current exposure, market state, and capital utilisation.

3.4 Limitations of intraday portfolio allocation

3.4.1 Signal estimation for intraday allocation

The signal $s_{t,i}$ represents the model-generated expectation of return, price movement, or directional advantage for asset i at time t . In intraday systematic trading, these signals must be computed with high temporal resolution, adapt quickly to changing market conditions, and remain robust to noise and microstructural distortions.

Common approaches to estimating $s_{t,i}$ include:

- Statistical forecasting models. Many intraday signals are based on short-horizon return forecasts using autoregressive models, Kalman filters, or state-space representations. For example, a basic linear signal might take the form:

$$s_{t,i} = E_t[r_{t+\delta,i}] = \beta^\top x_{t,i}$$

where $x_{t,i}$ is a vector of features (e.g., lagged returns, volatility, volume imbalance) and β is estimated via regularised regression. Model parameters are typically fit on rolling windows with exponential decay to prioritise recent data.

- Order book and microstructure features. High-frequency signals often rely on limit order book dynamics, such as imbalance, queue position, or spread/volatility ratios, as predictors of short-term price moves. Machine learning models (e.g., gradient boosting, logistic regression, deep nets) are trained to predict mid-price changes over a fixed time horizon δ , with outputs calibrated into expected return or probability-of-direction scores. These are then standardized across assets and time to form $s_{t,i}$.
- Market-making or liquidity-provision models. For intraday strategies providing liquidity, $s_{t,i}$ may reflect the expected $P\&L$ from quoting or inventory management. For example, the signal might encode adverse selection risk or expected spread capture. In this case, $s_{t,i}$ often depends on both market state and the trader's current inventory, e.g.:

$$s_{t,i} = \alpha \cdot Edge_{t,i} - \gamma \cdot Inventory_{t,i}$$

- Event-driven or news-based alpha. For strategies reacting to scheduled announcements (e.g., earnings, economic releases), or unscheduled events (e.g., headline sentiment), signals can be derived from natural language processing models or real-time news scoring engines. Here, $s_{t,i}$ might be an estimated return impact over a fixed horizon post-event.
- Normalisation and scaling. Raw signals are often standardised to ensure comparability across assets and time:

$$\tilde{s}_{t,i} = \frac{s_{t,i} - \mu_{t,s,i}}{\sigma_{t,s,i}}$$

where $\mu_{t,s,i}$, $\sigma_{t,s,i}$ are exponentially weighted means and standard deviations of signal s_i seen at time t . This normalisation stabilises the mapping from signal to portfolio weight and allows for integration across heterogeneous alphas.

Signal estimation is typically performed in parallel across assets and updated at high frequency (e.g., every 1 – 5 seconds). Latency, robustness to microstructure noise, and feature drift detection are essential considerations in production.

3.4.2 Evaluating forecast accuracy and its impact on allocation

Statistical forecasting models produce intraday signals $s_{t,i}$ that aim to predict short-horizon returns $r_{t+\delta,i}$. However, since the true outcome is only known after the forecast horizon δ has elapsed, model evaluation lags real-time decision-making. This creates a structural problem: we allocate capital based on signals whose true quality we cannot verify at the moment of execution.

The implications are twofold:

1. Forecast error directly translates into allocation error. If $s_{t,i}$ is systematically miscalibrated or noisy, the resulting weights $w_{t,i} \propto s_{t,i}$ (or volatility-scaled variants) can lead to poor performance, either through overexposure to weak signals or excessive turnover in noisy directions.
2. Model miscalibration is often persistent but hard to detect in real time. Since forecast performance is evaluated on a delay, models may degrade or overfit without immediate feedback. This is particularly dangerous in non-stationary regimes or after market structure changes.

Practitioners address this problem through a combination of delayed evaluation, real-time diagnostics, and robust portfolio construction:

- Out-of-sample forecast evaluation: The ex post quality of $s_{t,i}$ is assessed via backtesting and live shadow models, using metrics such as:

$$IC = \text{corr}(s_{t,i}, r_{t+\delta,i}), \text{ Hit rate}, \text{ Sharpe of signal buckets}, \text{ Calibration plots}$$

However, these require many samples and are themselves noisy at intraday frequency.

- Online performance tracking: Implement rolling metrics over recent windows (e.g., past 1,000 forecasts) to monitor signal efficacy. Deterioration in signal performance may trigger reweighting, model retraining, or fallback to baseline models.
- Model confidence and weighting: Rather than treating all forecasts equally, weight signals by their recent predictive power or statistical confidence. For instance, assign lower weight to models with high recent forecast dispersion or poor calibration:

$$w_{t,i} \propto \frac{s_{t,i}}{\text{Var}(s_{t,i})}$$

- **Ensembling and regularisation:** Combine multiple signals or models to reduce the impact of individual forecast error. Regularisation techniques (e.g., signal shrinkage toward zero or cross-sectional normalisation) reduce overreaction to poor forecasts.
- **Fallback heuristics:** If signal quality cannot be reliably verified, strategies may default to uniform allocation, volatility-scaling, or minimum-risk weights until confidence is reestablished.

Ultimately, the inability to observe forecast accuracy contemporaneously is a fundamental limitation of all real-time predictive strategies. The role of robust allocation is to minimise the cost of forecast error, not eliminate it. This is achieved by smoothing, diversification, and treating signals as probabilistic indicators rather than deterministic bets.

3.4.3 On the stability and usefulness of signal-based allocation

The distinctive empirical features of financial time series, including heavy tails, regime-switching, volatility clustering, and non-stationarity, pose significant challenges to the direct estimation of return forecasts $s_{t,i}$ (see Section (3.4.1)). These features violate the assumptions of many classical models, resulting in unstable signals that are sensitive to estimation error and model specification. In the intraday context, this instability is magnified by noise, market microstructure effects, and rapidly shifting dynamics.

Despite these challenges, signal-based allocation remains widely used due to its ability to translate differentiated views into capital deployment. The key lies not in assuming precise forecasts, but in using $s_{t,i}$ as a relative ranking or confidence score, rather than as a literal estimate of expected return. In this interpretation, the signal serves as a directional or probabilistic cue, e.g., 'asset i is more attractive than asset j ', rather than a point forecast of return magnitude.

However, unregularised use of noisy, high-variance signals leads to chaotic portfolio weights. This raises two core issues:

- **Signal volatility induces weight volatility.** If raw signals fluctuate wildly, then weights $w_{t,i} \propto s_{t,i}$ will inherit this instability, leading to excessive turnover and transaction costs. This undermines the net profitability of the strategy, even if the signals have statistical edge. Moreover, frequent sign changes in weak signals can result in mean-reverting rather than profitable trades.
- **Estimation risk dominates alpha.** When the signal is unstable, the noise in $s_{t,i}$ may exceed the true underlying signal, making allocation decisions dominated by estimation error. In this regime, uniform allocation (or capital-weighted allocation) may outperform, as it avoids overreacting to spurious fluctuations.

To address these problems, practitioners use a variety of stabilisation techniques:

- **Signal smoothing and filtering:** Apply exponential moving averages or Kalman filters to $s_{t,i}$ to reduce high-frequency noise.
- **Score-based allocation:** Map raw signals through monotonic, bounded transformations such as:

$$w_{t,i} \propto \tanh(\gamma s_{t,i}) \text{ or } w_{t,i} \propto \text{rank}(s_{t,i})$$

which reduce sensitivity to outliers and suppress noise.

- **Shrinkage toward equal weighting:** Interpolate between signal-based and uniform allocation using a regularisation parameter $\alpha \in [0, 1]$:

$$w_{t,i} = \alpha \cdot \tilde{w}_{t,i} + (1 - \alpha) \cdot \frac{1}{N}$$

where $\tilde{w}_{t,i}$ is the normalised signal-based weight.

- Turnover penalisation: Apply explicit penalties for weight changes in the optimisation stage to avoid overreacting to transient signal shifts.

In this framework, Signal-Based Allocation is not about trusting the absolute value of $s_{t,i}$ but rather about exploiting relative, persistent structure in a noisy environment. When designed with proper regularisation and robustness, it can outperform naive equal-weight strategies, particularly when alphas are heterogeneous and partially orthogonal. However, in highly chaotic or adversarial regimes, simplicity and robustness (e.g., equal-weighting or volatility-scaling) may dominate.

Ultimately, the choice between signal-based and uniform allocation should be viewed as a bias-variance trade-off, where the goal is not to perfectly forecast returns, but to make better-than-random capital allocation decisions under structural uncertainty.

3.4.4 On the estimation of intraday volatility and its role in allocation

The empirical characteristics of financial time series, such as volatility clustering, regime-switching, heavy tails, and abrupt structural breaks, make the real-time estimation of asset-level volatility $\sigma_{t,i}$ especially challenging. Intraday volatility is not only highly time-varying, but also influenced by endogenous feedback effects, market microstructure noise, and order flow dynamics, all of which undermine the assumptions of classical estimators.

Despite these challenges, volatility scaling remains a foundational technique in intraday portfolio allocation, where the aim is to normalise position sizing across assets to achieve balanced risk contribution. In its simplest form, position weights are scaled inversely by recent volatility: $w_{t,i} \propto \frac{s_{t,i}}{\sigma_{t,i}}$. Here, even if $s_{t,i}$ is noisy, scaling by $\sigma_{t,i}$ aims to avoid over-allocating to inherently volatile or unstable assets. However, this only holds if $\sigma_{t,i}$ itself is reasonably stable and well-estimated, which, in intraday settings, is often not the case.

The problems with naive volatility estimation include:

- Nonstationarity and regime shifts. Volatility can jump abruptly due to news or order flow shocks. Estimators that assume smoothness or stationarity (e.g., fixed-window standard deviations) will lag or overreact.
- Microstructure noise and sampling bias. At high frequencies, bid-ask bounce and irregular spacing of trades introduce significant bias into realised volatility estimates. This noise inflates short-term volatility and can mislead allocation.
- Volatility clustering and path dependence. Volatility exhibits long memory and clustered periods of high/low activity. Estimators must adapt quickly during transitions while not overreacting to transient spikes.

To address these issues, practitioners use robust, adaptive methods for estimating $\sigma_{t,i}$ including:

- Exponentially weighted estimators:

$$\hat{\sigma}_{t,i}^2 = \lambda \hat{\sigma}_{t-1,i}^2 + (1 - \lambda) r_{t,i}^2$$

where $r_{t,i}$ is a short-horizon log-return and $\lambda \in [0.9, 0.99]$. This provides adaptive smoothing that downweights old data but buffers against jumpy updates.

- High-frequency realised volatility with filtering: Use 5-minute or 1-minute returns aggregated over rolling windows, possibly with microstructure filters (e.g., subsampling, kernel smoothing, or robust median-based methods).

- Robust volatility floors and shrinkage: To prevent division by near-zero volatility (which can cause excessive scaling), apply a minimum volatility floor or shrink toward a cross-sectional average:

$$\tilde{\sigma}_{t,i} = \alpha \cdot \sigma_{t,i} + (1 - \alpha) \bar{\sigma}_t$$

where $\bar{\sigma}_t$ is a cross-sectional volatility average.

Even with robust estimation, volatility-scaled weights may still exhibit excess sensitivity. To prevent this from destabilising the portfolio, further stabilisation techniques are used:

- Volatility smoothing: Apply a longer half-life to volatility estimates than to signals, reducing the feedback between short-term noise and allocation weights.
- Weight clipping or squashing: Impose bounds on max weight change or map $\frac{1}{\sigma_{t,i}}$ through bounded functions such as:

$$w_{t,i} \propto \frac{1}{1 + e^{a\sigma_{t,i} - b}}$$

- Turnover penalisation: Combine volatility-scaling with regularisation to reduce rebalancing cost due to short-term noise.

Alternatively, we can simply use uniform allocation (i.e., equal weighting across assets). It offers stability and simplicity, particularly when signals and volatilities are both noisy. However, it ignores:

- Cross-asset heterogeneity in volatility and liquidity.
- Risk budgeting objectives (e.g., equal risk contribution).
- The potential for volatility to encode information about signal quality or execution risk.

In practice, volatility scaling, when done robustly, serves as a risk normalisation layer, not a prediction engine. It helps make signals more comparable across instruments and mitigates overexposure to unstable assets. The goal is not to estimate true instantaneous volatility precisely, but to produce a stable proxy for relative risk that improves capital allocation over naive uniform rules.

4 Cross-functional leadership in systematic trading and research

In a modern systematic investment firm such as Two Sigma, Citadel, or Millennium, performance depends on seamless collaboration between specialised leadership domains. The Heads of Trading, Quantitative Research, and Machine Learning each oversee distinct yet tightly interconnected areas, from alpha signal discovery and model development to live execution and continuous feedback. Together, they form the operational and analytical backbone that connects market insight, data science, and real-world trading performance.

4.1 Head of Trading

The Head of Trading at a quantitative or systematic fund is responsible for overseeing all aspects of trade execution, market interaction, and trading infrastructure, bridging the gap between portfolio management, quant research, and operations.

4.1.1 Role overview

4.1.1.1 Morning routine and market preparation

1. Pre-market review:

- Check overnight PnL, risk exposures, and execution performance from the previous session.
- Review liquidity conditions, macro news, and market events that could impact trading today.
- Examine slippage and transaction cost analysis (TCA) reports to detect inefficiencies or anomalies in execution.

2. Coordination with teams:

- Meet with quant researchers to discuss model updates that may affect trade flow (e.g., signal turnover or risk budgets).
- Align with risk managers and portfolio managers (PMs) to ensure position limits and execution strategies remain within mandates.
- Coordinate with technology on any latency, connectivity, or exchange feed issues.

4.1.1.2 Execution and monitoring during market hours

1. Trade supervision:

- Oversee algorithmic and manual execution across asset classes (equities, futures, FX, fixed income, etc.).
- Monitor real-time order books, VWAP/TWAP execution, market impact, and fill quality.
- Intervene when necessary - for example, switching an algo from passive to aggressive mode during volatility spikes.

2. Market interaction:

- Maintain relationships with brokers, liquidity providers, and exchanges.
- Ensure best execution standards and manage counterparty performance.

3. Risk oversight:

- Track intra-day exposure to ensure compliance with fund-wide and regulatory limits.
- Manage execution risk - e.g., routing errors, stale data, or algo malfunctions, often via automated alerts.

4.1.1.3 Afternoon: Strategy evaluation and infrastructure

1. Post-trade analysis:

- Review execution cost breakdowns (implementation shortfall, spread capture, adverse selection).
- Work with quants to refine execution algorithms, improve smart order routing (SOR) logic, and reduce market impact.

2. Infrastructure and tooling:

- Oversee improvements to trading systems, latency pipelines, and connectivity with exchanges.
- Prioritise technology investments that improve execution performance or reduce operational risk.

3. Governance and compliance:

- Approve and document trade procedures in line with MiFID II, SEC, or ASIC requirements (depending on jurisdiction).
- Conduct trade surveillance to ensure no market abuse or algo misbehaviour.

4.1.1.4 Strategic and long-term functions

1. Strategy scaling and collaboration:

- Decide how to scale or decommission trading strategies based on liquidity, performance, and execution costs.
- Collaborate with data scientists and quant PMs on research into execution alpha - i.e., using microstructure data for short-term predictive signals.

2. Talent and team leadership:

- Manage traders, execution quants, and trading technologists.
- Ensure the trading team maintains high operational discipline and cross-departmental coordination.

4.1.2 Improving trading algorithms and market connectivity

This responsibility sits at the intersection of quant research, engineering, and market microstructure expertise. The Head of Trading's goal here is to minimise trading costs, improve execution performance, and ensure robust connectivity to markets, all while maintaining operational and regulatory integrity.

4.1.2.1 Trading algorithm improvement These algorithms determine how orders are executed, not what to trade (that's the signal/research side), but how to interact with the market efficiently.

1. Execution algorithm design and tuning

- Collaborate with execution quants and trading technologists to design and continuously refine algos such as:
 - VWAP/TWAP algos: time-based execution for low urgency trades.
 - Implementation Shortfall algos: adaptive to minimise deviation from model price.
 - Liquidity-seeking algos: probing hidden liquidity and dark pools.
 - Smart order routing (SOR): optimising which venues to hit and when.
- Adjust algorithmic parameters dynamically based on:
 - Market conditions (volatility, spread, depth, correlation regime).
 - Trade urgency (alpha decay, portfolio rebalancing needs).
 - Historical TCA feedback (slippage by venue, time of day, or order size).

2. Feedback loop and data-driven optimisation

- Collect high-frequency trade and quote data (TAQ) to analyse execution performance at a granular level.
- Run post-trade regression or reinforcement learning models to predict and minimise slippage.
- Use market microstructure models to quantify short-term impact and adverse selection risk.
- Example: Replace a fixed participation rate VWAP with an adaptive participation model that responds to volume bursts or liquidity fragmentation in real time.

3. Simulation and backtesting of execution logic

- Build or oversee market simulators that replicate limit order book (LOB) dynamics for safe testing of new algos before deployment.
- Simulate how new strategies would perform under different:

- latency regimes,
- liquidity conditions, and
- execution aggressiveness profiles.

4. Cross-asset and cross-venue execution strategy

- Harmonise execution behaviour across instruments (e.g., equities, futures, FX, options) to optimise multi-asset portfolios.
- Account for synthetic hedging relationships (e.g., index futures vs basket execution).

4.1.2.2 Improving market connectivity Connectivity is about how efficiently and robustly the fund interfaces with markets and counterparties. This is the infrastructure layer that enables execution algorithms to perform optimally.

1. Low-latency and resilient infrastructure

- Ensure ultra-low-latency connectivity to exchanges and liquidity providers, colocated servers, optimised network paths, and direct market access (DMA).
- Oversee real-time monitoring systems to detect dropped packets, lag spikes, or stale market data.
- Ensure failover and redundancy mechanisms so that no single point of failure (network, gateway, or broker) disrupts trading.

2. Smart Order Routing (SOR) infrastructure

- Develop routing systems that decide:
 - Which exchange, MTF, or dark pool to hit.
 - Whether to use passive or aggressive orders.
 - How to minimise information leakage (especially for large trades).
- Continuously calibrate SOR behaviour using venue analytics:
 - Fill ratios
 - Average queue times
 - Reversion statistics post-fill

3. Counterparty and venue management

- Evaluate and maintain relationships with:
 - Prime brokers
 - Electronic liquidity providers (ELPs)
 - Dark pool operators
 - Clearing counterparties
- Benchmark latency, fill quality, and toxicity across brokers and venues.
- Use transaction cost analysis (TCA) to feed broker and venue scorecards.

4. Exchange and API upgrades

- Stay ahead of exchange protocol changes (e.g., new matching engines, FIX upgrades, or throttling rules).
- Coordinate testing and rollout of new FIX/ITCH/OUCH APIs with engineering teams.
- Benchmark performance impact of different network providers and market access points.

Team	Contribution	Head of Trading's Role
Quant Research	Develops alpha signals and execution models	Ensures execution efficiency supports alpha realisation
Technology / Engineering	Builds execution systems, connectivity, and data pipelines	Prioritises feature development and performance upgrades
Risk / Compliance	Ensures trades conform to mandates and regulations	Oversees real-time controls and pre-trade risk checks
Operations	Handles reconciliation and post-trade reporting	Ensures smooth operational flow and low operational risk

Table 1: Cross-functional integration in trading algorithm and connectivity improvement

4.1.3 Cross-functional integration

The above tasks are not done in isolation - the Head of Trading integrates efforts across multiple teams:

4.2 Alpha signals generation

Alpha signals is primarily done by Quant Research, but not exclusively.

4.2.1 Defining alpha signals

At a high level:

- Research alpha = predictive of future returns over minutes, hours, or days.
- Execution alpha = predictive of near-term market microstructure dynamics (milliseconds to seconds).
- Portfolio alpha = predictive of relative performance across instruments, sectors, or risk factors.

For example

- A Quant Researcher might design a signal predicting short-term momentum decay based on volume imbalance.
- The Execution Quant or Head of Trading's team might then find a pattern in order book depth suggesting when that signal's execution is likely to incur less market impact - i.e., a microstructure-timed execution alpha.

4.2.2 Who develops alpha signals and how the roles overlap

See Table (2).

4.2.3 How the Head of Trading fits in

The Head of Trading sits in a pivotal position, not creating directional alpha per se, but amplifying realised alpha through better market interaction: see Table (3).

Hence, while Quant Research creates alpha, the Trading function realises it more efficiently, sometimes even generating incremental alpha through execution modelling and short-term predictive analytics.

Function	Primary Focus	Typical Contributions to Alpha Development
Quant Research	Designing and testing predictive models (signal generation).	Builds new alpha factors, statistical models, and machine learning predictors using historical data. Validates them for robustness, orthogonality, and turnover.
Portfolio Management (PMs)	Allocating capital and managing risk across signals and strategies.	Chooses which alphas to deploy, sizes them, and blends them into portfolios. Sometimes engineers cross-signal combinations or adds domain-level overrides.
Trading / Execution Quant	Minimising cost and impact of turning alpha into trades.	Builds execution alpha (short-term predictive signals in the order book), and microstructure-driven enhancements that improve realised PnL.
Data Science / Engineering	Maintaining and curating data pipelines.	Finds new data sources (alt data, NLP signals, satellite data, etc.) that feed alpha discovery.

Table 2: Who develops alpha signals and how the roles overlap

Type	Description	Who Typically Leads
Predictive Alpha	Forecasting asset returns before trading.	Quant Research / PMs
Execution Alpha	Forecasting short-term market microstructure to improve execution.	Head of Trading / Execution Quants
Liquidity Alpha	Capturing rebates, spreads, or flow predictability.	Trading / Market Microstructure Team
Cost Alpha	Reducing slippage or adverse selection beyond benchmarks.	Head of Trading (via execution optimisation)

Table 3: How the Head of Trading fits in

4.3 Head of Quantitative Research

The Head of Quantitative Research leads the development of systematic investment models, alpha signals, portfolio construction frameworks, and risk models that form the intellectual foundation of the fund's strategies.

4.3.1 Role overview

His team bridges financial economics, statistics, and machine learning, producing models that drive capital allocation decisions across asset classes and time horizons. He owns the research agenda: defining what ideas to test, how to test them, and how they become tradable systems.

4.3.1.1 Core research domains Here are the typical research domains under the Head of Quant Research, organised by functional area:

1. Alpha research and signal discovery.
Goal: Find statistically robust predictors of future returns.
 - Cross-sectional and time-series modeling across equities, futures, FX, rates, and credit.

- Factor discovery (momentum, value, quality, carry, volatility, flow-based factors).
- Alternative data modelling: NLP on filings/news, satellite data, supply chain analytics, etc.
- Regime-aware signals: conditioning alpha on volatility or macro regimes.
- Market microstructure-based signals: short-term signals derived from limit order book dynamics (often overlapping with the ML and Trading domains).

Key challenge: Balancing predictive strength (Sharpe) with scalability, liquidity, and decay profile.

2. Portfolio construction and optimisation.

Goal: turn signals into portfolios that balance alpha, risk, and transaction cost.

- Signal combination frameworks: Bayesian model averaging, meta-learning, or shrinkage-based ensemble methods.
- Risk budgeting and capital allocation: volatility targeting, dynamic leverage, or Kelly-based weighting.
- Optimisation under constraints: turnover limits, liquidity caps, leverage rules.
- Cross-asset allocation: building integrated multi-strategy portfolios (equities, macro, commodities, vol).
- Transaction cost modelling (TCM): integrating execution cost forecasts into optimization (often linking to Trading).

Key challenge: Ensuring theoretical portfolio efficiency translates into realised Sharpe ratio after costs.

3. Risk modelling and forecasting.

Goal: Understand and control the statistical and structural sources of risk.

- Covariance and correlation modelling: shrinkage estimators, factor models, random matrix methods.
- Volatility and tail risk modelling: GARCH, stochastic volatility, realised volatility forecasting, neural volatility models.
- Factor decomposition and stress testing: exposures to macroeconomic, liquidity, or sentiment factors.
- Scenario generation: Monte Carlo or generative ML for stress testing.

Key challenge: Maintaining stable yet adaptive risk models under regime shifts and data noise.

4. Backtesting, simulation, and research infrastructure.

Goal: Ensure scientific rigour and reproducibility.

- Backtest engines: high-performance simulation frameworks for strategies and execution.
- Data cleaning, adjustment, and survivorship-bias control.
- Statistical significance frameworks: out-of-sample testing, multiple hypothesis adjustment, decay analysis.
- Performance attribution: decomposing returns into alpha, beta, and cost contributions.
- Research pipelines: standardised workflows for feature generation, validation, and deployment.

Key challenge: Avoiding overfitting while maintaining a fast research-to-production cycle.

5. Cross-disciplinary model integration.

Goal: Fuse insights from ML, econometrics, and trading data.

- Collaborate with the Head of ML on integrating representation learning, reinforcement learning, and generative models into alpha research.
- Collaborate with the Head of Trading on using execution and order book data for execution-aware alpha.

- Develop meta-models that blend signals from diverse sources (fundamental, technical, and alternative).

Key challenge: Creating a unified signal architecture where each model contributes orthogonal, cost-adjusted information.

6. Research governance and validation.

Goal: Institutionalise scientific discipline and repeatable discovery.

- Maintain research standards (e.g., minimum out-of-sample period, data provenance requirements).
- Lead the Investment Research Committee, approving models for production.
- Oversee model retirement and decay monitoring.
- Define KPIs for research productivity and alpha sustainability.

4.3.1.2 Typical teams reporting to the Head of Quant Research See Table (4).

Team	Focus	Typical Example Output
Alpha Researchers	Predictive modelling and signal discovery.	Cross-sectional factor models
Quant Strategists	Portfolio construction, allocation frameworks.	Multi-factor optimisation engine
Risk Researchers	Covariance, volatility, and scenario modelling.	Dynamic factor risk model
Data Scientists	Feature engineering, alt data curation.	Cleaned and engineered datasets
Quant Developers	Research infrastructure and simulation tools.	Unified backtesting platform

Table 4: Typical teams reporting to the Head of Quant Research

4.3.1.3 Interfaces with other Heads See Table (5).

Counterpart	Collaboration Focus
Head of Trading	Integrate transaction cost and execution feedback into alpha models; align portfolio liquidity with trading capability.
Head of Machine Learning	Embed ML into alpha discovery; test ML-based risk and regime models; build shared data/feature stores.
Head of Technology	Build scalable research and production infrastructure.
Chief Investment Officer (CIO)	Define research priorities aligned with firm strategy and capital allocation goals.

Table 5: Interfaces with other Heads

4.3.1.4 Strategic mandate The Head of Quant Research is ultimately responsible for:

- The intellectual property of the fund's investment process.
- The research pipeline that continuously produces and refines alpha.
- The consistency and scientific credibility of all deployed models.

4.4 Head of Machine Learning

While the Head of Trading is responsible for execution, market connectivity, and trade realisation, the Head of Machine Learning (Head of ML) focuses on how the firm extracts, structures, and learns from data to generate predictive power and operational efficiency.

4.4.1 Role overview

The Head of Machine Learning is responsible for building and governing the entire ML ecosystem within the firm, from data acquisition and feature engineering, through model development and validation, to deployment and monitoring in both research and production environments.

His goal is to ensure that machine learning capabilities translate into alpha, risk efficiency, and execution intelligence - safely, systematically, and reproducibly.

4.4.1.1 Strategic responsibilities

1. ML vision and architecture

- Define the fund's machine learning strategy - e.g., reinforcement learning for execution, deep learning for signal extraction, or generative modelling for scenario simulation.
- Architect the ML platform: data pipelines, training infrastructure, model registry, and evaluation protocols.
- Set standards for model governance, reproducibility, and explainability (critical in regulated contexts).

2. Integration with core business functions

- Partner with:
 - Quant Research to integrate ML into alpha generation and feature discovery.
 - Trading to develop execution algos, microstructure models, and real-time signal adaptivity.
 - Risk and PMs to build forecasting models, stress simulators, and regime detectors.
- Ensure that ML is not siloed: it's embedded across the entire research and trading lifecycle.

4.4.1.2 Daily and weekly activities

1. Supervising model development

- Oversee multiple teams building models for:
 - Alpha research (cross-sectional prediction, factor combination, representation learning)
 - Execution prediction (short-term order book forecasting, optimal trade scheduling)
 - Market regime detection (unsupervised learning and clustering)
 - Risk forecasting (volatility, tail risk, drawdown probability)
 - Operational intelligence (anomaly detection in systems, flow analytics)
- Review model performance metrics (Sharpe, hit rate, calibration, feature importance).

2. Data pipeline and quality control

- Oversee ingestion of massive data sources:
 - Traditional market and fundamental data
 - Alternative datasets (news, sentiment, satellite, NLP, transaction, ESG, etc.)

- Real-time data feeds for online learning
- Enforce data validation, bias detection, and lineage tracking to avoid 'garbage in, garbage out'.

3. Cross-functional collaboration: Coordinate with:

- Engineering for compute and deployment architecture (Kubernetes clusters, GPU/TPU pipelines, cloud/on-prem hybrid systems).
- Trading for live model calibration, especially when models drive adaptive execution logic.
- Compliance to document model logic under AI governance and explainability rules (e.g., EU AI Act, MiFID II).

4.4.1.3 Core research domains The Head of ML ensures that the firm explores, validates, and scales ML methods across relevant timescales and functions. Typical domains include:

Domain	Objective	Typical Techniques
Alpha Discovery	Find predictive structure in noisy, high-dimensional data.	Deep learning (transformers, GNNs), autoencoders, self-supervised learning
Execution Optimisation	Predict liquidity, slippage, and short-term impact.	Reinforcement learning (Q-learning, policy gradients), imitation learning
Portfolio Construction	Optimise signal weighting, hedge ratios, and diversification.	Bayesian ML, meta-learning, Gaussian processes
Market Regime Detection	Identify shifts in volatility, correlation, or factor structure.	Clustering, hidden Markov models, variational inference
Risk Modelling	Forecast volatility, tail events, or covariance structures.	Probabilistic deep learning, normalising flows
Operational AI	Monitor and optimise systems (latency, order flow, outages).	Anomaly detection, time-series classification

Table 6: Core research domains under the Head of ML

4.4.1.4 Governance, standards, and tooling A senior ML head must institutionalise rigour, ensuring that models can be trusted, audited, and improved systematically.

1. Model lifecycle governance

- Define the model development lifecycle (MDLC): 1. Data sourcing → 2. Feature generation → 3. Model training → 4. Validation → 5. Deployment → 6. Monitoring and drift detection.
- Oversee model registry and versioning, ensuring every live model is reproducible.
- Establish model risk controls: bias audits, backtesting discipline, out-of-sample verification.

2. Tooling and infrastructure

- Manage:
 - Experiment tracking (e.g., MLflow, Weights and Biases)
 - Feature stores
 - Distributed training clusters
 - Low-latency inference pipelines integrated with trading systems
- Partner with IT to optimise GPU allocation, data caching, and latency.

4.4.1.5 Leadership and team management

1. Organisational scope: Leads a team that may include:

- ML Researchers: developing new architectures and approaches.
- ML Engineers: productionising and maintaining live models.
- Data Scientists: performing exploratory analysis and feature research.
- Execution Quants / ML Traders: embedding models into live trading systems.

2. Culture and talent

- Promote a culture of scientific rigour, reproducibility, and cross-team knowledge sharing.
- Mentor researchers in both financial domain understanding and ML best practices.

4.4.1.6 Collaboration with the Head of Trading

 These two roles are tightly coupled:

Topic	Head of ML	Head of Trading
Execution algos	Builds reinforcement learning models to optimise trade scheduling.	Tests and validates them in live markets.
Market microstructure	Models short-term order flow and impact.	Translates those predictions into routing and participation strategies.
Data feedback	Collects granular tick and TCA data for training.	Ensures quality and stability of execution data.
System latency and reliability	Designs ML inference architecture.	Ensures trading infrastructure supports it operationally.)

Table 7: Collaboration with the Head of Trading

Together, they close the loop between learning from the market and acting in it. In short: The Head of Trading optimises execution of decisions; the Head of Machine Learning optimises how those decisions are learned and adapted.

4.4.2 Structural overlap

There is a real and deliberate overlap between the Head of Quant Research and the Head of Machine Learning roles in many systematic hedge funds (such as Two Sigma, Jump, AQR, Citadel's Systematic business, or smaller technology-first asset managers). This overlap is structural, not accidental - it is what allows the firm to merge financial theory, data science, and machine learning into a coherent alpha engine.

4.4.2.1 Why the overlap exists

 Machine learning in finance doesn't exist in a vacuum, it is embedded in the quantitative research process:

- The Head of Quant Research is responsible for what problems the firm investigates and what signals or risk models it ultimately trades.
- The Head of Machine Learning is responsible for how the firm uses computational learning methods to investigate and model those problems.

As a result, both roles touch:

- Alpha generation (predicting returns)
- Risk forecasting (modelling volatility and correlation)
- Portfolio optimisation (structuring bets efficiently)
- Execution optimisation (minimizing costs and slippage)

Since machine learning is increasingly central to each of these, the Head of ML's domain inevitably overlaps with, and enables, the Head of Quant Research's.

4.4.2.2 The core distinction

Let's draw a clean conceptual separation.

Focus	Head of Quant Research	Head of Machine Learning
Primary Mandate	Build financially meaningful models of markets (economic intuition + empirical testing).	Build methodologically robust learning systems (algorithms, feature architectures, model governance).
Primary Question	What predicts returns, risk, or liquidity and why?	How can we learn those relationships most effectively and adaptively?
Core Skillset	Financial modelling, statistics, econometrics, portfolio theory, factor design.	Machine learning theory, model training/inference infrastructure, optimisation and software engineering.
Output	Alpha factors, portfolio models, risk models, backtesting frameworks.	Learning algorithms, ML frameworks, predictive model pipelines, deployment and drift control systems.
Timescale of Thinking	Investment horizons (hours to months).	Methodological and computational horizons (training cycles, feedback loops, system adaptability).

Table 8: The core distinction

While they both build models,

- The Head of Quant Research owns the economic and financial logic behind those models.
- The Head of ML owns the learning systems and architectures that make them scalable, adaptive, and data-efficient.

4.4.2.3 Overlapping domains and how they divide responsibilities

1. Alpha research and signal discovery

- Overlap: Both teams build predictive models.
- Quant Research: Designs financial hypotheses (e.g., momentum, earnings drift, flow imbalance) and evaluates their predictive power.
- ML: Provides model architectures (transformers, GNNs, embeddings, meta-learners) and develops feature-learning pipelines to extract nonlinear relationships.

Quant defines what is worth learning; ML defines how best to learn it.

2. Risk and regime modelling

- **Overlap:** Both forecast volatility, correlation, and tail behaviour.
- **Quant Research:** Focuses on interpretability and consistency with financial structure (factor exposures, volatility clustering, stress scenarios).
- **ML:** Focuses on model expressiveness and adaptability (probabilistic deep learning, regime-switching models, anomaly detection).

Quant explains the 'risk structure'; ML improves its predictive fidelity.

3. Portfolio optimisation

- **Overlap:** Both deal with weighting and allocation under uncertainty.
- **Quant Research:** Defines the optimization problem (objective, constraints, risk model, turnover control).
- **ML:** Provides adaptive or meta-learning methods for optimisation under nonstationary conditions (e.g., reinforcement learning for rebalancing, Bayesian optimisation for hyperparameter tuning).

Quant builds the optimiser; ML teaches it to learn from evolving data.

4. Execution and market microstructure

- **Overlap:** Both model short-term dynamics.
- **Quant Research:** Studies microstructure statistically (impact curves, spread vs. volatility relationships).
- **ML:** Builds predictive execution models (RL agents, limit order book prediction, short-term flow forecasting).

Quant models the economics of impact; ML learns to act optimally within it.

4.4.2.4 How they interface in the research pipeline See Table (9).

Stage	Ownership	Description
Data acquisition and cleaning	Head of ML + Quant Research	ML ensures quality and scale; Quant ensures financial relevance.
Feature generation	Shared.	Quant defines feature meaning (e.g., valuation ratios); ML engineers automate and embed them.
Model training	Head of ML	Oversees algorithms, hyperparameter tuning, and infrastructure.
Model validation	Quant Research	Tests for economic sense, robustness, and performance stability.
Portfolio integration	Quant Research	Converts model outputs into positions, capital weights, and constraints.
Execution and feedback	Head of Trading + ML	Executes trades, collects TCA data, and feeds learning loops.

Table 9: How they interface in the research pipeline

In mature firms, this pipeline is circular, the Head of ML and Head of Quant Research co-own the research-to-production feedback loop.

4.4.2.5 Why modern firms separate the two roles Historically, the Head of Quant Research covered everything: data, models, and trading logic. However, as ML grew more complex and computationally intensive, many funds separate the roles because:

1. Machine learning now demands its own specialised infrastructure, governance, and research culture.

2. Quant Research remains responsible for economic coherence ? ensuring signals are not just statistically predictive, but financially interpretable and risk-efficient.
3. Scale, hundreds of models and terabytes of data require dedicated ML systems leadership.

So the split creates complementarity:

- Quant Research ensures what the firm learns aligns with market structure.
- ML ensures the firm learns it well, fast, and continuously.

5 Limitations of classical statistical tools in chaotic financial systems

Classical statistical analysis relies heavily on the use of moments, mean, variance, correlation, skewness, and kurtosis, to summarise the behaviour of time series data (see Hamilton [1994]). However, the foundational assumptions behind these measures often fail in financial contexts (see Tsay [2010]). Empirical financial data exhibit path-dependence, jumps, volatility clustering, and structural breaks, features that violate the conditions under which moments are meaningful, stable, or informative (see Appendix (14.3)).

Since returns are not independently and identically distributed (i.i.d.), commonly used performance metrics such as the mean squared error (MSE) and mean absolute error (MAE) become highly misleading in financial contexts. These metrics are designed to capture average fit or linear association but are blind to the timing, direction, and economic impact of rare but consequential events, precisely the features that dominate real-world financial outcomes.

5.1 Navigating non-stationarity in financial modelling

Most of the core quantitative tools underlying momentum and mean-reversion strategies, as discussed in Section (3.1), rest on the assumption that market returns are Gaussian or, at a minimum, stationary. However, financial time series are inherently non-stationary, often exhibiting shifts in mean, variance, and distributional properties over time. This non-stationarity challenges the direct application of classical statistical techniques and necessitates more adaptive methodologies. This section outlines a comprehensive approach to handling such dynamics by combining statistical tests (e.g., unit root tests), transformation techniques (e.g., differencing), and adaptive estimation frameworks.

We explore the trade-off between long-run and short-run estimators: while global estimators offer reduced variance, they risk bias under regime changes; conversely, rolling window estimators adapt to evolving conditions but suffer from higher estimator variance. This apparent contradiction is (supposedly) resolved by acknowledging their complementary roles and managing the bias-variance trade-off through techniques such as exponential weighting, shrinkage estimators, and hybrid multi-timescale strategies.

The section also discusses practical implementation, including window size selection, cross-validation techniques, and the use of robust and adaptive models. Emphasis is placed on aligning estimation methodology with modelling objectives, whether structural risk assessment or short-term signal triggering, to ensure statistical soundness in dynamic and uncertain market environments.

Relying solely on rolling window techniques to address evolving market conditions is overly optimistic and may underestimate the complexity of structural shifts. While rolling window techniques offer adaptability to evolving market conditions, this adaptability comes at the cost of increased estimator variance, particularly when applied in environments characterised by structural breaks, price jumps, or fakeouts. In such discontinuous settings, the assumptions of local stationarity break down, leading to unstable and potentially misleading estimates. This section delves into the limitations of rolling window estimators under these conditions and highlights the risks of relying on them for precision-critical tasks such as signal generation. The analysis reinforces the need for hybrid approaches, smoothing techniques, and rigorous validation when deploying short-window estimators in volatile or non-stationary regimes.

5.1.1 Statistical estimation under non-stationarity

5.1.1.1 Addressing non-stationarity in financial time series Financial time series often exhibit non-stationary behaviour, characterised by changing means and variances over time (see Peters [1991-96], Mandelbrot et al. [1997]). To analyse such data we can perform the following tasks:

- **Differencing:** Applying differencing techniques can help stabilise the mean of a time series, making it more amenable to analysis.
- **Rolling Statistics:** Computing rolling means and variances allows for the assessment of local stationarity and the detection of structural breaks.
- **Unit Root Tests:** Statistical tests like the Augmented Dickey-Fuller (ADF) test can be employed to assess the presence of unit roots, indicating non-stationarity.

5.1.1.2 Practical implications for financial modelling

When modelling financial returns:

- **Robust Estimators:** Employ estimators that are robust to deviations from normality and can handle heavy tails and skewness.
- **Model Validation:** Use diagnostic tools and goodness-of-fit tests to validate model assumptions, ensuring that the chosen model adequately captures the data's characteristics.
- **Adaptive Models:** Consider models that can adapt to changing market conditions, such as regime-switching models or models with time-varying parameters.

5.1.1.3 Rolling windows vs. long-run estimation: A trade-off There is a tension between asymptotic consistency (which requires large samples) and rolling statistics (which operate on small, local windows). This is a classic trade-off in time series analysis under non-stationarity and limited data. The contradiction is only apparent, not fundamental. The trade-off is as follows:

- Global estimates reduce variance but may introduce bias if the regime has shifted.
- Local estimates reduce bias to regime changes but increase variance.

Purpose Determines the Method:

- **Long-run estimators** (large n) are used when we assume the underlying process is stationary, or at least piecewise stationary, over the entire data set. They provide precise estimates but can miss structural breaks or evolving dynamics.
- **Rolling window estimators** (small n) assume local stationarity: the statistical properties are stable over a short window but may change over time. They offer responsiveness, but come with higher estimator variance due to smaller sample size.

5.1.1.4 How practitioners handle this in practice?

- **Window Size Selection:** Window length is a hyperparameter, chosen based on a trade-off between bias and variance:
 - Use cross-validation or rolling backtests to find the optimal length.
 - Practically, common window lengths are 20 (1 month), 60 (1 quarter), or 252 (1 year) in daily data.

- **Shrinkage Estimators:** To stabilise high-variance estimates from small windows, practitioners often apply shrinkage techniques:
 - Shrink rolling variances toward a global variance estimate.
 - Use James-Stein shrinkage or Bayesian priors.
- **Exponential Weighting (EWMA):** Rather than using a hard window cutoff, EWMA (Exponentially Weighted Moving Average) smooths volatility estimates by assigning more weight to recent data while still using all past data. This mitigates the instability of small windows. Given Equation (16.18), we get

$$\sigma_t^2 = (1 - \alpha)r_t^2 + \alpha\sigma_{t-1}^2$$

where $\alpha \in [0, 1]$.

- **Hybrid Approaches:**
 - Use global estimates for structural risk metrics (e.g., position sizing).
 - Use rolling/local estimates for tactical decisions (e.g., signal thresholds or regime detection).
 - This multi-timescale approach mirrors how experienced traders and algorithms blend information.

The contradiction dissolves when you acknowledge:

- Long-run statistics are consistent but slow to adapt.
- Short-run (rolling) statistics are adaptive but noisy.
- In practice, you manage the trade-off with smart parameter tuning, smoothing, shrinkage, or by combining estimators at multiple scales.

5.1.2 The limitations of rolling window estimation

We saw in Section (5.1.1) that rolling statistics, such as moving averages and rolling variances, were standard tools for modelling time-varying behaviours in financial time series. The central assumption behind these tools is local stationarity, the idea that statistical properties of the process (mean, variance, autocorrelation structure) remain approximately constant within the window.

However, in modern electronic markets characterized by structural breaks, sharp reversals, and high-frequency discontinuities, this assumption often fails. Using rolling windows without accounting for such events can lead to unstable, misleading, and potentially dangerous estimates.

5.1.2.1 High estimator variance undermines precision In signal-triggering applications, such as identifying breakouts or reversals, precision is paramount. Decisions must be based on high-confidence statistical thresholds. Yet, small sample sizes inherent in rolling windows yield high estimator variance (see Equation (15.5)). In distributions with excess kurtosis (as market returns typically exhibit), this variance is exacerbated.

As a result, using short windows to capture 'local' behaviour introduces substantial noise in the variance estimate. For signal-sensitive systems, this noise increases the likelihood of both false positives (triggering on noise) and false negatives (missing real events).

5.1.2.2 Non-stationarity from structural breaks Rolling windows implicitly assume stationarity over their span. Yet, discontinuities such as news shocks, stop-runs, and algorithmic liquidity events violate this assumption. In these cases:

- Mean and variance change abruptly, invalidating local estimates.
- Even a small window includes non-homogeneous data: one half pre-event, one half post-event.

This creates internal inconsistency within the window, and statistical estimates reflect a blend of incompatible regimes, not a coherent local process.

5.1.2.3 Double counting discontinuities A subtle but critical flaw of rolling estimators is their treatment of discontinuities. Suppose a price jump occurs at time t_0 . Then

- The window $[t_0 - W + 1, t_0]$ captures the jump at the end.
- The window $[t_0, t_0 + W - 1]$ captures the same jump at the beginning.

Thus, the same shock is counted twice, artificially inflating estimates of volatility or skewness, despite being a single event. The second window incorrectly treats the past jump as a new volatility surge.

5.1.2.4 Implications for trading system design Given these limitations, the blind application of rolling window estimators for high-stakes decisions (e.g., signal generation, position entry/exit) is problematic. Instead:

- Regime-switching models or change-point detection algorithms should be used to identify structural breaks and adapt windowing accordingly.
- Outlier-robust estimators (e.g., MAD, robust volatility estimators) reduce the impact of jumps.
- Asymmetric weighting or state-space models (e.g., Kalman filters) allow for more controlled smoothing without artificial jumps.

5.1.2.5 Conclusion Rolling window techniques offer adaptability, but their use in non-stationary environments with high-frequency jumps introduces serious statistical flaws. In precision-sensitive applications, they may do more harm than good. Effective system design must incorporate mechanisms to detect and adapt to structural discontinuities, rather than assuming local smoothness where none exists.

5.2 Why correlation fails in real-world financial data

In recent years, financial research has increasingly aimed to quantify relationships between market variables, such as sentiment indicators and asset prices, often through the estimation of correlations. This approach seeks to identify predictive signals by measuring co-movement between variables. However, it relies on assumptions that are fundamentally at odds with the empirical nature of financial markets, particularly the notion that prices evolve smoothly over time.

In reality, financial markets display non-diffusive dynamics (see Appendix (14.3)). Prices do not follow the continuous trajectories assumed in classical diffusion models like geometric Brownian motion (GBM). Instead, they are marked by jumps, sudden reversals, fake-outs, and regime shifts, discontinuities that reflect the complex interplay of macroeconomic shocks, policy announcements, geopolitical surprises, and imbalances in market liquidity or order flow. These discontinuities are not mere noise: they are often informative and structural, driven by latent forces that reshape the risk landscape abruptly and unpredictably.

These jump-driven features fundamentally violate the assumptions of stationarity and time-invariance that correlation metrics depend on. In a stationary system, the statistical properties, including mean returns, variances, and correlations, are assumed to be stable over time. But in financial markets, these properties are structurally unstable, shifting in response to changes in monetary regimes, market microstructure, investor behaviour, or macroeconomic conditions. As a result, correlations estimated from historical windows are often ephemeral, capturing relationships that may vanish, invert, or become irrelevant in future states of the market.

Furthermore, the act of computing correlation assumes linearity and symmetry, that increases in one variable are proportionally and consistently related to changes in another. Yet market responses are highly nonlinear and context-dependent. A surge in positive sentiment during a bull market might reinforce momentum, while that same signal during a market crisis could be dismissed or even interpreted as contrarian noise. This regime sensitivity, combined with nonlinear reactions to volatility and liquidity conditions, renders static correlation metrics both fragile and misleading.

In sum, correlation-based inference breaks down in markets characterised by jumps, path dependency, and structural change. Models built on such measures risk identifying spurious relationships that perform poorly out-of-sample, a critical concern for both researchers and practitioners seeking robust signals in an inherently unstable environment.

5.3 Why MSE and MAE fail in financial time series forecasting

MSE and MAE, though statistically convenient, fail to capture the nonlinear, regime-switching, and risk-sensitive nature of financial prediction tasks. Their continued use in evaluating financial AI models not only distorts model selection, but contributes to the false sense of reliability that often precedes costly real-world failure. Finance demands metrics aligned with financial objectives, not just statistical smoothness.

5.3.1 The limits of traditional error metrics

Both Mean Squared Error (MSE) and Mean Absolute Error (MAE) are distance-based metrics, they compute how far off a model's prediction is from the actual outcome, on average. In machine learning and statistics, MSE and MAE are widely used to quantify predictive accuracy:

- MSE is the average of squared differences:

$$MSE = \frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2$$

where y_t is the true value, and $\hat{y}_t$ is the predicted value at time t . This penalises large errors more than small ones due to the squaring.

- MAE is the average of absolute differences:

$$MAE = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|$$

This penalises all deviations equally, regardless of size.

In practice, both metrics are minimised when predictions are close to the average value of the target variable, especially when:

- The target variable is stationary (i.e., has a stable mean and variance).
- The distribution of outcomes is symmetric and light-tailed (e.g., Gaussian).

Hence, if our target is a financial time series of daily returns, which mostly hover around 0%, then predicting 0% every day will often yield a low MSE or MAE, because:

- Most daily returns are close to 0%.
- Big movements are rare, so the model avoids being 'wrong' on average.

This results in what we can call 'mean-reverting behaviour' in model outputs: a model that 'plays it safe' by ignoring extreme or rare events (which may actually be economically important).

More formally, these metrics assume:

- A stationary distribution of errors,
- Homoscedasticity (constant variance),
- Symmetry in prediction errors,
- No structural breaks or heavy tails.

Financial time series, by contrast, violate all of these assumptions. Asset prices and returns often exhibit:

- Non-stationarity: Mean and variance change over time,
- Volatility clustering: Periods of high and low volatility alternate,
- Heavy-tailed distributions: Large deviations occur more frequently than Gaussian models suggest,
- Jumps and structural breaks: Shocks due to news, policy, or liquidity events.

As a result, using MSE or MAE on financial data often penalises the model for accurately capturing the inherent volatility (or spikes) of markets, rather than rewarding it.

5.3.2 Illustration: A stylised example

5.3.2.1 The alpha in the outliers: Exploiting spikes for profit Spikes in financial time series often reflect sudden, information-driven price movements, such as earnings surprises, policy shifts, or macroeconomic announcements, that temporarily disrupt the prevailing mean-reverting regime. These spikes represent tradable opportunities because they can be systematically exploited by event-driven or momentum-based strategies that seek to capture short-term drift or directional persistence.

For example, a momentum strategy might detect a sharp upward jump following an unexpected positive earnings release and initiate a long position, anticipating that other market participants will continue to reprice the asset in that direction in the short term. Conversely, a mean-reverting model that forecasts zero returns will ignore this deviation, assuming it to be noise, and thus fail to capitalise on the predictable short-term continuation caused by information diffusion or liquidity cascades.

Here, spikes are tradable because they often signal predictable short-term drift in price action, which strategies like momentum or event-driven trading are specifically designed to exploit, yet models minimising MSE typically smooth over these movements, missing the very alpha such strategies aim to capture.

5.3.2.2 Signal vs noise: The strategic importance of rare events Imagine now a return series that is mostly mean-reverting around 0% (long-run mean), but occasionally spikes due to policy shocks or earnings announcements. A model that forecasts the long-run mean (i.e., zero) will have low MSE because it avoids reacting to noisy fluctuations, but it will miss every tradable opportunity because it consistently underreacts to meaningful price movements, treating them as noise rather than signals, resulting in no actionable predictions when returns actually deviate in profitable ways.

Now consider a volatility-sensitive model that anticipates these spikes correctly. It may incur large residuals in absolute terms (raising the MSE), but its economic value is much higher, because it captures the events that matter to a trader or portfolio manager.

Thus, low statistical error does not imply economic usefulness.

5.3.2.3 Error metrics vs. economic value: A mismatch in financial modelling Metrics like Mean Squared Error (MSE) and Mean Absolute Error (MAE) are designed to reward models that make smooth, average predictions that stay close to the typical value of the target variable. In financial markets, however, the most economically significant movements, like price jumps, volatility spikes, or regime shifts, are rare and large.

A model that correctly anticipates such a spike (e.g., a sudden 5% drop after an earnings miss) will produce a prediction far from the mean, and even if it is correct, this prediction will be penalised by MSE or MAE because of the large residual (difference between predicted and actual, even if directionally accurate or valuable for trading). Conversely, a model that always predicts the long-run mean (e.g., 0% return every day) may minimise MSE simply by avoiding big errors, but it contributes nothing useful to a trader.

Hence, standard error metrics may reward 'safe' models that do nothing economically useful, and punish models that capture what actually matters, the high-impact, volatile, and non-stationary nature of financial markets.

5.4 Rethinking model evaluation in finance

In trajectory-based prediction for trading, where actionable events are rare but crucial, we should move away from general-purpose statistical loss functions and toward signal-aware, economically relevant, and asymmetry-sensitive metrics.

5.4.1 Why safe metrics are misleading in finance

In finance:

- The mean return is not very informative: While the mean return is often reported as a summary statistic, it provides limited insight into the true behaviour of financial assets. In most financial time series, especially over short horizons, returns exhibit high volatility, fat tails, and asymmetric distributions. This means the average return is frequently dwarfed by the variability around it, and may not reflect what investors actually experience. A strategy with a slightly positive mean return could still carry significant downside risk, while one with a near-zero mean might offer valuable upside in specific regimes. Moreover, in non-stationary markets, the mean itself can shift over time, making it a moving target rather than a stable anchor. Relying on it to assess model performance or asset behaviour is therefore misleading.
- Risk-adjusted gains from rare events (e.g., policy shifts, earnings surprises) often drive actual profitability: In real-world trading, the majority of meaningful returns often come from rare, high-impact events such as unexpected central bank announcements, geopolitical shocks, or corporate earnings surprises. These events break with historical patterns and drive large market moves. Models that are tuned only to minimize average error (e.g.,

via MSE) often ignore or underweight these tail events because they contribute disproportionately to short-term error, even when they carry significant economic value. Yet in practice, the ability to correctly anticipate and capitalise on these outliers is what drives superior risk-adjusted performance. Successful strategies often don't win by being right all the time, they win by being right when it matters most.

- Volatility, jumps, and tail events are where real economic value lies, not in minimising tiny day-to-day errors: Financial markets are not driven by smooth, continuous trends ? they are punctuated by bursts of volatility, abrupt jumps, and rare but consequential tail events. These discontinuities are not noise; they are signals of shifting risk regimes, liquidity imbalances, or informational shocks. While traditional error metrics like MSE or MAE disproportionately emphasise accuracy during tranquil periods, they ignore the fact that most meaningful profits, and losses, occur during moments of stress or surprise. The real economic value of a model lies in its ability to anticipate and respond to these irregular dynamics, not in making incremental improvements on low-variance, low-stakes days. Models that minimise day-to-day noise may appear statistically elegant but often fail when it matters most.

So, MSE and MAE implicitly reward models that ignore these rare but vital events, because they are optimised for average proximity, not for decision-making utility.

5.4.2 Consequences in practice

Using MSE or MAE in model selection can lead to:

- Underestimation of tail risks (low MSE = safe model illusion): MSE-driven models often appear safer than they are because they minimise average error by ignoring rare, high-impact events ? the very tail risks that dominate financial outcomes. By consistently predicting near the mean, such models avoid penalisation for everyday noise but systematically miss extreme moves, creating the illusion of robustness while leaving portfolios exposed to sharp drawdowns during crises or market dislocations.
- Overfitting to periods of low volatility (models do well in-sample but collapse in crises): When trained using MSE or MAE, models tend to overweight calm market periods where small errors dominate the loss function. This leads to overfitting on stable regimes and under-preparation for turbulent ones. As a result, these models may perform well in-sample but fail catastrophically out-of-sample, especially during crises?because they haven't learned to anticipate or adapt to the volatility spikes and structural breaks that define real financial stress events.
- Discarding predictive features that add value in turbulent markets but increase short-term error: Standard error metrics like MSE and MAE push models to minimise average deviations, leading them to ignore or down-weight features that may not help during calm periods but are highly informative during crises. For example, a volatility spike indicator or liquidity stress measure might slightly worsen short-term prediction accuracy under normal conditions, raising the error, but provide critical foresight during extreme events. Traditional loss functions treat this as noise, encouraging models to discard such features. This bias against rare-event predictors results in models that are smoother but blind to the very conditions where predictive power is most economically valuable.

In AI applications to finance, this misalignment becomes particularly dangerous. Models that are trained and selected purely on MSE/MAE may appear 'accurate' but fail to produce robust, tradable, or hedgable outputs in real-world settings.

5.4.3 What should be used instead?

To address these issues, alternative evaluation frameworks are needed. Some academics proposed the following metrics:

1. Risk-adjusted return metrics:

- Sharpe ratio: $\frac{E[R]}{\sigma(R)}$. The Sharpe ratio evaluates a model based on its risk-adjusted return, making it far more relevant for financial applications than MSE or MAE. It measures how much excess return a model or strategy delivers per unit of volatility, thus rewarding models that not only generate returns but do so consistently and with controlled risk. A strategy with occasional large profits and high variance may look impressive in raw return terms, but a low Sharpe ratio reveals its unreliability. In contrast, a model that captures fewer but more stable and repeatable opportunities, especially during volatile periods, will be favored, aligning evaluation with real investment objectives.
- Sortino ratio: $\frac{E[R]}{\sigma_{-}(R)}$, focusing on downside deviation. The Sortino ratio refines the Sharpe ratio by distinguishing between harmful and harmless volatility, penalising only downside deviations from a target return. This is particularly important in finance, where investors are more concerned with losses than with variability due to gains. Models that occasionally deliver large positive returns might be unfairly penalised by MSE or even the Sharpe ratio, but the Sortino ratio rewards models that avoid downside risk while still capturing upside potential. This makes it a more aligned performance measure in contexts where capital preservation and drawdown control are critical.
- Maximum drawdown: Largest peak-to-trough loss. Maximum drawdown measures the largest loss from a portfolio's peak to its subsequent trough before a new high is reached. It captures the worst-case capital erosion over a period and reflects the investor's exposure to severe downside risk. Unlike MSE or MAE, which average out errors, maximum drawdown focuses on extreme events that can trigger forced liquidation, margin calls, or psychological exit, making it a vital performance metric for evaluating a model's resilience under stress. Models that look statistically sound but suffer large drawdowns are often unusable in practice.

These directly measure economic performance.

2. Tail-sensitive scoring rules:

- Quantile loss (used in Value-at-Risk estimation): $L_{\alpha}(y, \hat{q}_{\alpha})(\alpha - 1_{\{y < \hat{q}_{\alpha}\}})(y - \hat{q}_{\alpha})$. Quantile loss, also known as pinball loss (comes from the shape of the loss function), evaluates how well a model predicts the tails of a distribution by penalizing underestimation and overestimation asymmetrically. It is particularly useful for estimating risk measures like Value-at-Risk (VaR), which focus on extreme quantiles of return distributions (e.g., the 5% worst-case loss). Unlike MSE or MAE, which treat all errors equally, quantile loss emphasises tail behaviour, where real financial risk resides. This makes it a more appropriate objective for models concerned with downside protection and capital preservation.
- Expected Shortfall (CVaR): Penalises underestimation of extreme losses. Expected Shortfall, also known as Conditional Value-at-Risk (CVaR), goes beyond Value-at-Risk by not only identifying the threshold of extreme losses but also quantifying the average loss in the worst-case tail (e.g., the worst 5% of outcomes). This makes CVaR especially valuable in evaluating models under conditions of market stress or crisis. Unlike standard metrics that may ignore rare but catastrophic events, CVaR explicitly penalises models that fail to account for fat tails and extreme downside risk, aligning more closely with the priorities of risk-averse investors and regulators.

3. Distributional and probabilistic metrics:

- Continuous Ranked Probability Score (CRPS): CRPS is a proper scoring rule used to evaluate the quality of probabilistic forecasts, particularly in time series and forecasting contexts. Unlike point-based metrics like MSE or MAE, CRPS compares the entire predicted cumulative distribution function (CDF) to the actual observed value, rewarding forecasts that are both sharp and well-calibrated. This makes it especially useful in finance, where capturing the full range of possible outcomes, including uncertainty and tail events, is

critical. CRPS is sensitive to both bias and dispersion, offering a more nuanced evaluation of models that predict not just a single value but a distribution over future returns.

- Logarithmic score (Log score): Evaluates the log-likelihood of the observed outcome under the predicted probability distribution; encourages sharp and accurate probabilistic forecasts.
- Kullback-Leibler divergence: If probabilistic forecasts are used. Quantifies how one probability distribution diverges from a reference distribution, often used to measure information loss in approximate models.

These metrics respect the structure and risk asymmetries of financial time series.

5.4.4 Recommended metrics for one-step-ahead trajectory forecasting in trading

Metrics like CRPS, Log score, and KL divergence are designed for evaluating probabilistic forecasts or full distributions, which isn't directly aligned with the typical objective in quantitative trading, where we're forecasting the next move (point prediction) along a single realised trajectory, not the entire distribution.

In financial markets most days might see small, routine fluctuations that don't offer meaningful trading opportunities. But occasionally, there are large jumps or breakouts, big moves caused by earnings surprises, policy changes, market shocks, etc., that create real opportunities (or risks) for traders. In quantitative trading, alpha is concentrated in jumps, breakouts, and regime shifts. Hence, we want to know if the model correctly predict the direction or magnitude of these rare but crucial large moves. We also need to know whether the model can identify that a significant event is about to occur, rather than just guessing. As a result, we need metrics that:

- Reward correct directional predictions around large moves,
- Penalise misses during volatility spikes, even if average error is low,
- Reflect economic value, not just statistical fit.

Some of these metrics are

- Directional accuracy (hit ratio): Measures the proportion of times the predicted direction (up/down) matches the actual direction. This captures trading signal value more directly than MSE. However, it doesn't distinguish between small vs. large moves, and needs complementing.
- Tail event precision / recall: Focused evaluation only during large absolute returns (above a percentile threshold, e.g., top 10% of moves). It specifically tests performance on breakouts or jumps, where trading value lies.
- Expected profit / PnL per trade: Evaluates model based on hypothetical trading strategy returns using predicted signals. It directly measures economic value, accounts for asymmetry, transaction costs, and risk. One can use Sharpe or Sortino ratio for risk-adjusted evaluation.
- Event-weighted error metrics: Modified MSE or MAE where errors during high-volatility or high-return periods are weighted more. It penalises underperformance during critical periods, unlike flat-weighted MSE.
- Thresholded accuracy / hit on jump events: It focuses evaluation only on those 'important' days when returns exceed a certain size threshold (like a big jump or crash). Conditional accuracy only on days when return magnitude $>$ threshold (e.g., 2σ).

6 Digital transformation in systematic trading

6.1 Foundations and challenges

Systematic trading refers to the execution of trades based on a predefined set of rules, with minimal or no discretionary intervention during live trading (see Treleaven et al. [2013], Pereira [2021]). As these strategies evolve, firms increasingly integrate real-time data with machine learning models to detect alpha-generating signals and anticipate market dynamics before they become apparent to others.

However, deploying real-time, adaptive systems requires a fundamental transformation of traditional model risk frameworks. These systems must not only deliver speed and agility, but also ensure that strategies are executed in a mechanical, disciplined, and repeatable manner. While this creates opportunities for superior execution, it simultaneously introduces new layers of risk, particularly concerning model explainability, operational robustness, and regulatory compliance.

Achieving this level of automation depends not only on advanced technological infrastructure, but also on fostering an organisational culture that aligns data-driven innovation with robust risk management, governance, and ethical accountability. This alignment is at the heart of digital transformation, where technology, process, and culture converge to enable sustainable, compliant, and adaptive systematic trading.

Digital transformation can be defined through the following five key dimensions:

1. The integration of digital technologies across all areas of an organisation, fundamentally changing how it operates, delivers value, and manages risk.
2. Beyond IT upgrades: it involves reengineering processes, transforming decision-making, and reshaping organisational culture to leverage data, automation, and emerging technologies such as artificial intelligence (AI).
3. Driving agility, efficiency, and innovation, enabling firms to respond rapidly to changing market dynamics, regulatory developments, and client expectations.
4. For risk leaders, embedding risk and compliance controls directly into the design of digital processes, rather than layering them on as afterthoughts.
5. In today's regulatory environment, successful digital transformation requires balancing innovation with robust governance and proactive regulatory alignment.

Applying this framework to systematic trading, we are going to examine three interrelated areas: (1) real-time signal detection for market opportunities, (2) the risk management frameworks necessary to support them, and (3) AI-enhanced trade execution algorithms. Integrating these components is far from trivial, it demands a structured, multi-stage mapping process that systematically translates external signals into actionable trading decisions. Crucially, this process must account for portfolio exposures, market constraints, and operational risks. In what follows, we present a framework that illustrates how such a mapping can be designed and implemented in practice.

6.2 The real-time trading decision stack

6.2.1 Real-time signal detection for market opportunities

Real-time signal detection transforms the alpha-generation process by shifting from periodic, batch-based model outputs to continuous, event-driven insights. Advanced data pipelines combined with machine learning models enable systematic strategies to capitalise on short-lived market dislocations, news events, or sentiment-driven flows. While this enhances agility and return potential, it also raises the stakes for data quality, model explainability, and ethical data usage, particularly when alternative or unstructured data sources are involved.

1. The integration of digital technologies into all areas of an organisation

- Example: A systematic hedge fund integrates real-time data ingestion pipelines (tick-by-tick market data, alternative data like social sentiment) with machine learning models that detect alpha-generating signals intraday.
- This requires collaboration between quant researchers, data engineers, risk teams, and compliance, because data itself becomes a regulated asset, especially under regimes like MiFID II or SEC Best Execution duties.

2. Goes beyond IT upgrades

- The firm moves from batch-based signal generation (daily or hourly) to real-time, event-driven trading signals that can trigger trades within milliseconds.
- Cultural change: Portfolio managers, who once relied on slower discretionary overlays, now trust signal-driven automation, demanding more sophisticated risk explainability before acting.
- Compliance teams must now assess whether certain data sources (e.g., alternative data) could create market conduct risks. This requires new joint working practices between legal, compliance, and data science functions.

3. Drives agility, efficiency, and innovation

- Result: The fund can exploit short-lived market dislocations or sentiment-driven anomalies that competitors relying on slower signals will miss.
- Real-time signal detection enables dynamic factor exposures, realigning portfolio tilts on the fly in response to geopolitical events, policy announcements, or even breaking news sentiment streams.
- Faster detection of market microstructure shifts improves execution quality by anticipating order book dynamics before they become obvious to others.

4. For risk leaders, it means embedding risk and compliance controls into the design of new digital processes

- Key risk control: Real-time signal systems must have kill switches and circuit breakers to prevent cascading trades on false positives (e.g., erroneous sentiment spikes due to misinformation).
- Pre-trade validation rules must check signal strength and integrity before triggering portfolio shifts.
- Compliance reviews ensure that alternative data sources comply with data privacy laws (e.g., GDPR) and market abuse rules, especially if signals are derived from web-scraped or social media content.
- Audit trails: Every real-time signal must generate logs for why a position was adjusted. It is critical for post-trade compliance reviews and potential regulatory scrutiny.

5. In today's regulatory landscape, successful digital transformation must balance speed of innovation with robust governance and regulatory alignment

- Governance includes:
 - Standing data ethics committees to approve new data sources used in signal generation.
 - Model risk frameworks adapted specifically for real-time adaptive systems, not just static factor models.
 - Engagement with regulators (e.g., FCA, SEC) to help define what constitutes 'material influence' of alternative data on trading decisions, shaping future market conduct expectations.
- Outcome: The firm leverages real-time alpha detection to enhance returns, while demonstrating robust governance and market integrity, positioning itself as a leader in responsible innovation.

6.2.2 Risk management for real-time signal detection

As systematic strategies embrace real-time data and adaptive algorithms, risk management frameworks must evolve in parallel. Traditional pre- and post-trade controls are no longer sufficient. Modern frameworks embed real-time exposure monitoring, pre-trade risk validation, circuit breakers, and explainability mechanisms directly into the trading infrastructure. These controls are essential not only for maintaining internal risk appetite but also for meeting heightened regulatory expectations on algorithmic trading, market abuse prevention, and responsible AI use.

1. The integration of digital technologies into all areas of an organisation
 - Example: A multi-asset quant fund deploys a real-time risk management engine alongside its real-time signal detection system.
 - The risk engine consumes the same live data feeds as the signal model, running parallel simulations on potential market impacts of signal-driven trades before execution.
 - Technology stack integration: Risk management is no longer an end-of-day or post-trade activity, it is live, adaptive, and embedded directly into the trading pipeline.
 - Cultural integration: Risk professionals, quants, and traders work together during signal model development, not after deployment.
2. Goes beyond IT upgrades
 - Shift from traditional VaR-based daily risk management to continuous real-time exposure management.
 - Decision-making evolves:
 - No trade is triggered purely on alpha strength; trades are gated by real-time exposure checks (sector, counterparty, liquidity).
 - Pre-trade risk filters prevent the execution of signals that might inadvertently breach internal or regulatory exposure limits.
 - Cultural impact: Traders must embrace the fact that some signals will be suppressed by real-time risk controls for the sake of long-term integrity and compliance.
3. Drives agility, efficiency, and innovation
 - Agility: Risk management evolves from being a brake to becoming an enabler of faster, smarter trades, because:
 - Real-time checks allow instant portfolio rebalancing within safe exposure corridors.
 - Faster detection of correlated positions or hidden exposures (e.g., to liquidity or credit risk) even when trading across multiple desks or strategies.
 - Example: Signal spikes during geopolitical news events are automatically capped to prevent portfolio over-concentration in sensitive sectors or regions.
 - Outcome: The firm is more responsive to alpha, while still preserving risk appetite discipline.
4. For risk leaders, it means embedding risk and compliance controls into the design of new digital processes
 - Risk controls embedded in signal detection pipelines:
 - Model governance frameworks classify signals by risk tier (e.g., low/medium/high uncertainty signals).
 - Pre-trade stress testing of signals: Before execution, the risk system simulates the impact of executing the trade under adverse market conditions.

- Explainability layers built in: Why was this signal generated? Why was this risk threshold tripped? Generates an audit trail for compliance and regulatory inspection.
 - Example risk scenario managed: Preventing momentum chasing during flash events that might later be classified as market abuse triggers by regulators.
5. In today's regulatory landscape, successful digital transformation must balance speed of innovation with robust governance and regulatory alignment
- Governance enhancements for real-time signal risk:
 - Establish signal validation committees: cross-functional groups that vet new signal sources/models before deployment.
 - Implement regulatory reporting interfaces that log both signals generated and risk overrides applied during live trading, e.g., for MiFID II best execution audits or Market Abuse Regulation (MAR) inquiries.
 - Engage proactively with regulators to share risk control methodologies around signal-driven strategies, especially in relation to algorithmic trading guidelines (e.g., FCA Algorithmic Trading Compliance Review expectations or ESMA guidelines).
 - Outcome: Risk management isn't a cost to innovation, it is a competitive differentiator, enabling the firm to pursue cutting-edge real-time strategies with regulatory confidence.

6.2.3 AI-enhanced trade execution algorithms

AI-enhanced trade execution algorithms represent a key area of digital transformation, where machine learning models dynamically adapt execution strategies in response to evolving market conditions. By leveraging real-time data, these systems aim to reduce transaction costs, minimise market impact, and optimise trade timing. However, the introduction of adaptive algorithms into execution workflows requires close coordination between trading, technology, risk, and compliance functions to ensure transparency, accountability, and regulatory alignment.

1. The integration of digital technologies into all areas of an organisation
 - Example: A global asset manager integrates reinforcement learning algorithms into its trade execution systems to dynamically adjust execution strategies based on real-time market conditions.
 - It isn't just an improvement to execution desks, it changes how portfolio managers, risk, and compliance teams interact with the entire trade lifecycle.
2. Goes beyond IT upgrades: it is about transforming processes, decision-making, and culture
 - This transformation shifts execution strategy from static VWAP/TWAP algorithms to adaptive AI-driven execution that continuously optimises for slippage, market impact, and latency.
 - Cultural change: Traders move from manually monitoring execution to trusting model-driven execution, requiring new skills in algorithm oversight and risk analytics.
 - Collaboration between quant teams, risk managers, and compliance becomes routine, not exceptional.
3. Drives agility, efficiency, and innovation
 - Result: The firm reduces transaction costs by several basis points per trade, improving fund performance.
 - Algorithms can react instantly to fragmented liquidity conditions or volatility spikes, providing an edge over competitors using static algorithms.
 - Enables faster entry into new markets or instruments without having to redesign execution workflows from scratch.

4. For risk leaders, it means embedding risk and compliance controls into the design of new digital processes
 - Regulatory expectations (e.g., MiFID II, SEC Reg SCI) require pre-trade risk controls like order size limits and kill-switch mechanisms.
 - Compliance, technology, and trading teams work together to embed:
 - Audit trails for every algorithmic adjustment.
 - Pre-trade and post-trade analytics dashboards to demonstrate adherence to best execution principles.
 - Explainability layers, providing rationale for why the AI recommended a particular execution path over another. It is critical for internal review and external regulatory inquiries.
5. In today's regulatory landscape, successful digital transformation must balance speed of innovation with robust governance and regulatory alignment
 - Governance includes:
 - Model validation frameworks adapted for real-time algorithmic execution.
 - Active participation in industry working groups on algorithmic trading governance (e.g., FIA PTG, AFME Algo Working Groups).
 - Continuous engagement with regulators to shape expectations around AI in trading algorithms, positioning the firm as a responsible innovator.
 - Outcome: The firm achieves competitive execution advantages without compromising on market conduct standards or regulatory scrutiny.

6.3 Towards modular architectures for real-time systematic trading

As trading systems grow increasingly sophisticated, integrating external information, such as news sentiment, social media signals, and behavioural metrics, into trading decisions becomes critical for achieving a contextual edge. However, this integration is not trivial. It requires a structured, multi-layered mapping process that systematically links external signals to actionable trading decisions, while accounting for portfolio exposures, market constraints, and operational risks. The following framework outlines how this mapping can be structured in practice.

6.3.1 Mapping external information to trading actions

The mapping of external information to trading actions is done by defining a structured framework or model that links external metrics to trading actions. Here is how the mapping is typically done in practice:

1. Feature engineering and transformation: External data (e.g., news, sentiment scores, social media signals) is processed and converted into quantitative features that can be used by the trading system. For example, a positive sentiment score about a particular sector might be transformed into a numeric boost to the weight of related signals.
2. Signal conditioning layer: The external features are then combined with the detected trading signals using statistical models, machine learning models, or rule-based systems. This step adjusts the raw trading signal upward or downward depending on the external input.
3. Portfolio alignment logic: Next, the combined signal is adjusted in light of the existing portfolio exposures:
 - If the portfolio is already heavily exposed to a sector associated with positive external sentiment, the adjustment might be attenuated to avoid concentration risk.
 - Conversely, if exposure is low, the signal's position size recommendation may be increased.

4. Decision grid or scoring matrix: These interactions (between external metrics, trading signals, and portfolio exposures) are codified in a predefined decision matrix or scoring grid, often implemented as:
 - Rules (e.g., Boost long signals by 20% if sentiment $>$ threshold and exposure $<$ limit),
 - Probabilistic models, or
 - Optimisation algorithms that maximise expected utility under constraints.
5. Real-time optimisation: Finally, the adjusted signals are fed into a real-time optimiser, subject to constraints (risk, liquidity, regulation), to determine position sizing and execution strategy.

6.3.2 Challenges in implementing multi-stage decision frameworks

Konstantinidis et al. [2024], Zhao et al. [2024], Nawathe et al. [2024], among others, propose a deep learning or reinforcement learning framework that ingests multimodal data, including historical prices, sentiment scores, and topic embeddings from news, and utilise them to construct a comprehensive state representation for the agent. The system demonstrates how these diverse inputs are scored, fused, and optimised in real time to adjust portfolio allocations under constraints like transaction costs, liquidity, and risk limits. This exemplifies scoring and integrating external information into state representations and optimising in real time under multiple constraints.

While these articles leverage multimodal data, typically passed into a single model (often reinforcement learning or deep learning architectures) for end-to-end portfolio optimisation, the framework described in the five bullet points represents a fundamentally different design. It envisions a modular, multi-stage architecture, where distinct specialised networks process individual components (e.g., external data scoring, sentiment mapping, signal validation, position sizing), and their outputs are aggregated by a final decision layer.

Each individual module performs its own constrained optimisation in a non-stationary, chaotic environment, making this not just a portfolio optimisation problem, but a complex multi-task learning challenge under real-time operational and regulatory constraints. To date, no comprehensive, practical implementation of such a framework exists in the academic or practitioner literature.

Consequently, while theoretical progress has been made on individual elements, attempting to perform these three interrelated tasks simultaneously in a live, production-grade trading environment remains aspirational rather than a present reality.

6.4 Leveraging deep learning for intraday systematic trading

Intraday systematic trading using deep learning models, particularly deep neural networks (DNNs) and large language models (LLMs), offers a compelling avenue for capturing short-term market inefficiencies. However, translating predictive signals into sustained profitability requires careful attention to model calibration, strategy design, and execution efficiency.

6.4.1 Model calibration and validation

The predictive power of deep learning models depends critically on how they are trained and calibrated. Calibration involves not only minimising prediction errors but also ensuring that outputs correspond to meaningful, probabilistic assessments of future market direction. This process typically includes the following steps:

1. Feature selection and engineering:
 - Price-based features: OHLCV data, momentum indicators, order book imbalances.

- News and sentiment features: Extracted using LLMs from real-time news feeds, social media, or analyst reports.
 - Latent representations: Embeddings derived from LLMs for qualitative data.
2. Model selection:
 - Convolutional Neural Networks (CNNs) for spatial relationships in order books or volatility surfaces.
 - Recurrent Neural Networks (RNNs) or Transformers for temporal dependencies in time-series price data.
 - Fine-tuned LLMs (e.g., Llama, GPT variants) for interpreting unstructured news data and generating sentiment scores.
 3. Calibration techniques:
 - Temperature scaling or Platt scaling to adjust model confidence.
 - Regularisation techniques to prevent overfitting in noisy, non-stationary environments.
 - Walk-forward validation or rolling cross-validation to ensure robustness in the face of regime shifts.
 4. Error analysis:
 - False positives (predicting upward moves that fail to materialise) are particularly costly in intraday settings due to transaction costs.
 - Distribution of prediction confidence is as important as directional accuracy for controlling risk exposure.

6.4.2 Strategy design: From signals to positions

The predictive output of the deep model, typically a probability of upward or downward movement within a specific horizon ($P(up)$ and $P(down)$), must be translated into concrete trading actions through a signal-to-position mapping. Several strategy types are commonly employed:

1. Threshold-based execution:
 - Only act when the predicted probability exceeds a predefined confidence threshold (e.g., 60% probability of upward move).
 - Adjust thresholds dynamically depending on volatility or liquidity conditions.
2. Volatility-adjusted sizing: Use predicted volatility to adjust position size: larger positions in low-volatility environments, smaller in high-volatility periods.
3. Ensemble models:
 - Combine signals from multiple models: one for price action, another for sentiment, and another for macro news.
 - Weight each signal's contribution based on recent predictive performance.
4. Hybrid approaches: Blend systematic signals with traditional market microstructure signals (e.g., order flow imbalance, liquidity gaps).

6.4.3 Generating alpha: Beating the market

Generating sustainable profits, or alpha, requires moving beyond raw predictive accuracy to building a complete trading system. Key components include:

1. Execution algorithms:
 - Implementation Shortfall algorithms (e.g., VWAP, TWAP) to minimise slippage.
 - Smart Order Routing (SOR) to split orders across venues based on real-time liquidity analysis.
2. Dynamic risk management:
 - Position-level stop-losses calibrated in real time based on volatility forecasts.
 - Portfolio-level exposure controls to avoid concentration risk, sector risk, and unintended macro factor bets.
 - Real-time kill switches and circuit breakers embedded in execution logic.
3. Feedback and adaptation:
 - Continuous monitoring of signal quality.
 - Adaptive learning loops where poor model performance triggers recalibration or down-weighting of that signal in ensemble strategies.
4. Market adaptivity:
 - Regime detection models (e.g., Hidden Markov Models) to identify periods of model degradation.
 - Switching between strategies depending on detected market regime (e.g., trend versus mean-reversion environments).
5. Cost-aware optimisation:
 - Explicit inclusion of transaction costs, market impact models, and latency considerations in the decision-making framework.
 - Using reinforcement learning where the reward function incorporates realised PnL adjusted for transaction costs.

6.4.4 Limitations and practical challenges

Despite sophisticated modelling:

- Sharp market jumps or unexpected news events can lead to substantial prediction error.
- In high-volatility regimes, the model may maintain an outdated directional bias due to its dependency on the preceding window's data, causing temporal distortion.
- Probability Stickiness: Following sharp moves, rolling-window-based models tend to 'remember' large jumps, skewing subsequent predictions until enough new data flows into the window to recalibrate.

6.4.5 Benchmarking and performance assessment

To validate that the strategy is genuinely adding value:

- Compare against relevant benchmarks: e.g., market-neutral returns versus intraday volatility benchmarks, or Sharpe ratio versus alternative systematic strategies.
- Attribution analysis: Break down returns into components attributable to signal quality, execution skill, and market conditions.
- Out-of-sample testing: Demonstrating consistent performance across different market conditions is critical for credibility.

7 Forecasting short-term market directions: Precision or illusion?

Intraday systematic trading using deep learning models, including deep neural networks (DNNs) and large language models (LLMs), has opened new avenues for capturing short-term market inefficiencies. Among these, threshold-based execution strategies have emerged as a practical framework, triggering trades when model-estimated probabilities of near-term market direction cross predefined thresholds. For instance, a strategy might initiate a long position when the probability of an upward move over the next ten minutes exceeds 0.6, with predictions continuously updated using rolling windows of real-time data.

While conceptually appealing, this approach faces well-documented challenges in live, volatile markets. Sudden price jumps can distort the rolling inputs, causing prediction lags and leading to premature exits, whipsaw reversals, or overreactions once past shocks roll out of the forecast window. With realistic limits on trading frequency due to execution costs and slippage, these effects can erode performance and amplify drawdowns precisely during unstable conditions.

At the heart of these strategies is the challenge of estimating the probability of market direction over short horizons, typically ranging from a few minutes to an hour, depending on market structure and strategy design. We explore two fundamentally distinct approaches to this task: a data-driven, AI-enhanced methodology, and a stochastic modelling framework rooted in option pricing. While AI models rely on structured real-time features and advanced calibration techniques to generate actionable probability forecasts, stochastic models link directional probability to the spread between expected market risk premia and expected variance. Both approaches ultimately highlight the difficulty of accurate short-term forecasting in environments dominated by noise, regime shifts, and asymmetric information.

Although probabilistic forecasting provides a structured foundation for risk-aware trade selection, the non-stationarity of financial markets ultimately breaks the classical assumption that repeated trades will converge to positive expected returns. The informational edge provided by probability estimates must be understood as local and transient, requiring continuous adaptation to evolving market conditions.

7.1 Threshold-based execution in intraday trading

Threshold-based execution strategies rely on predictive models to estimate the probability of near-term market direction, with trades triggered when these probabilities cross predefined thresholds (see Lovas et al. [2021]). For example, a systematic strategy might initiate a long position when the probability of an upward move over the next ten minutes exceeds 0.6. This decision-making process typically operates on a rolling-window basis, where both the input features and the predictive horizon are continuously updated as new market data arrives.

While conceptually appealing, this approach faces significant challenges in live, chaotic environments. Sharp price jumps can distort the rolling input, causing prediction lags and leading to directional misalignment. After a sudden

market move, the influence of that event persists in the rolling window, biasing the model's output and often resulting in premature exits or erroneous position reversals. Moreover, because prior information is counted multiple times across overlapping windows, subsequent predictions may overreact when the jump event finally rolls out of the window, compounding the risk of whipsaw losses.

In practice, with realistic constraints on trading frequency due to transaction costs, slippage, and execution latency, these distortions can rapidly degrade performance. Systems constrained to two or three trades per day risk exhausting their trading capacity on reactive, poorly timed entries and exits, often amplifying drawdowns precisely during volatile conditions. Addressing these limitations requires either more sophisticated temporal modelling, such as regime-switching layers, or adaptive mechanisms that suppress overreaction to transient market shocks (see Goluz et al. [2024], Chattopadhyay et al. [2024], Genet [2025]).

7.1.1 Description

In threshold-based execution, we only execute trades when the model's predicted probability of a specific market move (up or down) exceeds a certain predefined **confidence threshold**. This reduces the risk of overtrading on weak or noisy signals, thereby improving the signal-to-noise ratio and expected return per trade. For example, trade only if $P(up) \geq 0.6$, meaning at least a 60% probability that the next price move will be upward. This way, we can reduce transaction costs and slippage, and focus trading capital on higher-confidence opportunities.

The predictive model operates on a rolling forecast horizon, generating updated probabilities for the market direction over the next fixed interval, typically 10 minutes. At each new observation point, e.g., every minute, a fresh probability estimate is produced for the subsequent 10-minute window. For example, at $t = 10:01$, the model forecasts the likelihood of the price increasing by $t = 10:11$; at $t = 10:02$, it recalculates the probability for $t = 10:12$, and so forth. This sliding window structure ensures that trading decisions continuously adapt to the most recent market information, enabling the strategy to remain responsive to evolving conditions. While the prediction intervals overlap, positions are reviewed and adjusted at each step based on the model's most current assessment of short-term directional risk and opportunity.

To support this rolling forecast process, the data inputs to the predictive model are continuously refreshed, typically on a minute-by-minute basis. As each new market observation arrives, whether it is price ticks, order book dynamics, or external signals such as sentiment indicators, the model ingests this updated information to generate a new probability forecast. This rolling data feed ensures that the model's internal representation of market conditions evolves in real time, allowing it to adjust its predictions as new patterns, trends, or disruptions emerge. The continuous refresh of both data and predictions is essential for maintaining responsiveness in fast-moving intraday markets.

This strategy can be classified as a momentum-based approach (see Section (3.1.2)), as it systematically increases exposure in the direction predicted by the model, aiming to capture trending price movements. Position sizing is adaptive, scaling with the model's confidence in directional continuation. Unlike mean-reversion strategies, which seek to profit from price reversals, this framework explicitly targets periods of directional strength, aligning with the core principles of trend-following.

7.1.2 A practical example

The operational pipeline is as follows:

1. Prediction generation: model predicts the probability of the next price movement being up or down over a short horizon, e.g., next 10 minutes. Example output from the model every few seconds/minutes:

Timestamp	$P(up)$	$P(down)$
10:01:00	0.53	0.47
10:02:00	0.64	0.36
10:03:00	0.58	0.42
10:04:00	0.72	0.28
10:05:00	0.35	0.65

Table 10: Probability of the next price movement being up or down

2. Apply confidence threshold:

- Define confidence threshold T (e.g., 0.6).
- Trading action only taken if $P(up) > T$ or $P(down) > T$
- No action if model confidence is low (close to 0.5 = random guess).

3. Trade execution:

- Long position if $P(up) \geq T$
- Short position if $P(down) \geq T$
- No position / Hold if both $< T$
- The size of the position can be made proportional to the confidence above the threshold, e.g., Position Size $\propto P(up) - T$ or adjusted by volatility forecasts.

This strategy can be formalised as follows:

- The trading rule is

$$\Psi_t = \begin{cases} +1 & \text{if } P_t(up) \geq T \\ -1 & \text{if } P_t(down) \geq T \\ 0 & \text{otherwise} \end{cases}$$

- Position Sizing (example linear rule):

$$\omega_t = \kappa \cdot (P_t(up) - T)$$

where κ is a risk adjustment factor.

The pseudo code implementation is

```
threshold = 0.6
for each_time in market_time:
    p_up = model.predict(market_state)

    if p_up >= threshold:
        go_long(size = risk_factor * (p_up - threshold))
    elif (1 - p_up) >= threshold:
        go_short(size = risk_factor * ((1 - p_up) - threshold))
    else:
        hold_position()
```

In practice, model-driven intraday trading requires clear protocols for position management based on the evolving probability outputs of the predictive model. For instance, if at $t = 10:02$ the model estimates a $P(up) = 64\%$ probability of the market moving up, the strategy would initiate a long position, with sizing proportional to the signal strength above a predefined threshold. In our example, the position size is proportional to $\omega_t = (0.64 - 0.6) = 0.04$. If, at the following timestamp $t = 10:03$, the probability remains close to but below the threshold ($P(up) = 0.58$), the position is maintained but not increased. Should the model subsequently strengthen its conviction, for example, rising to $P(up) = 72\%$ at $t = 10:04$, the position is incrementally increased in proportion to the updated signal confidence. Here, it is proportional to $\omega_t = (0.72 - 0.6) = 0.12$.

Conversely, if the model flips to predicting a downward move with high probability (e.g., $P(down) = 65\%$ at $t = 10:05$), the protocol is to first unwind the existing long position before establishing a new short exposure. This two-step process ensures that positions are cleanly reversed, avoiding overlapping exposures that could introduce unnecessary risk or confusion in execution. Position sizing, entry, and exit logic are thus systematically tied to the dynamic evolution of model confidence, maintaining mechanical discipline while adapting to market changes in real time.

7.1.3 Benefits and extensions

The benefits from this strategy are as follows:

- **Filters noise:** Models will often make low-confidence predictions. Thresholds eliminate weak signals that are likely indistinguishable from random noise.
- **Transaction Cost Mitigation:** Reduces unnecessary trades, meaning you save on commissions and slippage.
- **Alpha Concentration:** By trading only when the model is relatively sure, you concentrate your risk-taking in situations with higher expected return per unit of risk.

Some practical extensions are

- **Dynamic Thresholds:** Adapt the threshold based on market volatility. Example: higher volatility $\rightarrow$ higher threshold to compensate for uncertainty.
- **Bid-Ask Spread Adjustment:** If spread costs are high, increase threshold further to ensure trades overcome transaction costs.
- **Regime-Dependent Thresholds:** Use lower thresholds in trending markets, higher thresholds in choppy markets.
- **Trade Direction Filtering:** Combine with filters on order flow imbalance, liquidity, or news sentiment to confirm trades.

7.1.4 Limitations of rolling predictions in chaotic markets

7.1.4.1 Prediction lag following sharp market movements When we have a sharp event, say a sudden price drop at $t = 10:04:30$, the model's rolling window remembers that event, and that dominates the feature set feeding into predictions for several subsequent minutes. So even if the price starts recovering at $t = 10:06$, the model is still biased by that earlier sharp movement because its statistical or learned features (momentum indicators, volatility spikes, short-term sentiment features, etc.) are still 'contaminated' by that sudden move.

As a result:

- We can stay short longer than is optimal.
- We can miss the recovery.

- Worse, we can rack up losses because of that lag.

Three possible standard mitigation strategies are:

1. Decay weighting: Give exponentially less weight to older observations in the window, so that sharp but quickly-reversed jumps lose influence faster.
2. Event detection filters: Introduce explicit features or filters for jump detection. If a jump is identified as an anomaly or event-driven (e.g., from news or macro data), we can flag the predictions as unreliable for a short period or reduce position sizes temporarily.
3. Position throttling: It requires a large change in probability to flip positions immediately after a sharp move.

7.1.4.2 Temporal challenges in real-time predictive trading Further, this approach suffers from temporal distortion due to double-counting and abrupt forgetting. Here is how it works:

1. Double-counting during the window: Right after a sharp move (e.g., the jump at $t = 10:04:30$), the event heavily influences the input features for several minutes as the rolling window keeps sliding forward. This leads to the model repeatedly 'seeing' the same shock, artificially amplifying its perceived importance in the near-term predictions.
2. Abrupt forgetting: Then, when the jump finally falls out of the 10-minute window (e.g., at $t = 10:14:30$), the model suddenly forgets about it, even if the market is still reacting or if secondary effects are in play. This can cause **prediction discontinuities**, probabilities might jump or flip direction unexpectedly.
3. Risk of wrong reversal signals: Because of this, we can see $P(up)$ suddenly jump after the event exits the window, even if the market is stabilising or about to reverse. This risks triggering trades that are out of sync with actual market dynamics, especially in noisy or mean-reverting conditions.
4. Solution approaches: Practitioners often mitigate this with:
 - Exponential decay weighting rather than flat windows.
 - Shock detection layers that tag anomalous moves separately.
 - Feature engineering that incorporates time since last shock as an input.
 - Hybrid models combining short- and long-memory components.

7.1.4.3 Trading constraints and compounding execution risk In volatile or chaotic market conditions, a rolling-window predictive strategy can suffer from reactive misalignment, where the model persistently lags behind sudden directional shifts. Given transaction costs, slippage, and execution frictions, this lag increases the likelihood of entering and exiting positions just as the market reverses, compounding realised losses. With a constraint of only two or three trades per day, the risk of clustering poor trades rises significantly, often leading to premature stop-loss triggers and systematic underperformance relative to a naïve passive benchmark. This highlights the need for adaptive filtering mechanisms or regime detection layers to prevent overreacting to transient shocks.

7.2 Estimating the probability of near-term market direction

At the core of threshold-based execution strategies lies the challenge of accurately estimating the probability of upward or downward market movements over a short forecast horizon, typically ranging from a few minutes to an hour, depending on the trading frequency, strategy design, and underlying market microstructure. Accurate estimation of this probability is essential for making rational, data-driven trade entry and exit decisions. We present two fundamentally

distinct approaches for estimating the probability of near-term market direction: a data-driven, AI-enhanced approach and a model-based, stochastic process framework.

Estimating the probability of near-term market direction in AI-enhanced trading involves processing rolling, real-time market data into structured features, such as price action, volatility, order book dynamics, and optionally sentiment signals. These features feed probabilistic or sequential models (e.g., logistic regression, gradient boosting, or deep learning architectures like LSTMs or Transformers) (see Section (2.3.2)) to generate rolling forecasts over short horizons (typically 5 – 15 minutes) (see Ferreira et al. [2021]). Since raw model outputs often misalign with real-world frequencies, calibration techniques like Platt scaling or isotonic regression are applied to improve probability accuracy, with continuous real-time adjustments during live trading.

In the model-based approach, the probability of near-term market direction is linked to the price of a Digital Bond option when the spot price is close to the forward at-the-money strike. This probability depends on the spread between the expected mean market price of risk (EMPR) and the expected mean variance. While the variance component is observable through instruments like variance swaps, the EMPR is not directly tradable and remains difficult to estimate reliably (see Ben Alaya et al. [2024]). As a result, forecasting short-term price direction is inherently subjective, favouring market participants with superior or faster access to information.

7.2.1 Data-driven AI-enhanced approach

Assuming an AI-enhanced approach, the steps for estimating the probability of near-term market direction are as follows:

1. Data inputs and feature construction: The predictive model relies on rolling, real-time streams of market data, commonly structured as overlapping time windows. These input features typically include:
 - Market microstructure features: Last trade price, bid-ask spread, order book imbalance, volume at price, VWAP deviation.
 - Technical Indicators: Short-term momentum, RSI, moving averages, Bollinger Bands.
 - Volatility measures: Realised volatility, implied volatility (when available), intraday variance estimators.
 - Event-based features: Detection of large price jumps or abrupt changes in order book depth.
 - Sentiment or exogenous signals (optional): For sophisticated models, real-time sentiment scores derived from news, social media, or analyst reports can be included as external features.

All features are normalised or transformed to ensure stationarity over the estimation horizon, especially important in noisy, chaotic market regimes.

2. Model choice and training: Two main approaches dominate probability estimation in this context:

2.1 Probabilistic Classifiers:

- Logistic Regression: Simple, interpretable, often used for baseline models.
- Gradient Boosted Decision Trees (e.g., XGBoost, LightGBM): More capable of capturing non-linear interactions, popular for structured financial data.
- Deep Learning Models (e.g., LSTMs, Temporal CNNs): Applied when using richer feature sets or for capturing longer temporal dependencies.
- The model output is typically $P(\text{up} \mid \text{data})$, the probability that the price will be higher at the end of the forecast window than it is at the moment of prediction.

2.2 Sequence Models for Time Dependency:

- For sequential dependencies, Recurrent Neural Networks (RNNs) or Transformers can model the evolving market dynamics over time, especially in intraday applications.
 - These models generate rolling forecasts updated every minute (or faster), providing a continuous stream of probability estimates.
3. Forecast horizon definition: The forecast window defines the near-term horizon for the predicted movement. For intraday threshold-based execution strategies, this might be:
- 5 to 15 minutes: Short enough to capture local patterns, but long enough to smooth noise.
 - Rolling forecasts: Updated at fixed intervals (e.g., every minute) to reflect the latest market developments.

Each probability estimate $P(up)$ corresponds to the likelihood that the market price at $t + h$, where $h = \text{forecast_window}$, will be greater than the price at time t .

4. Calibration and probability adjustment: Raw probability estimates from models may not correspond well to observed frequencies in real markets due to regime shifts, volatility clustering, or overfitting on historical data. Therefore, calibration techniques are used:
- Platt Scaling or Isotonic Regression to correct systematic bias.
 - Brier Scores or Logarithmic Loss minimisation during validation to ensure the predicted probabilities reflect real outcome frequencies.

Models can also include real-time feedback mechanisms, adjusting probabilities based on realised prediction errors during live trading.

7.2.2 Model-based stochastic processes approach

We let the spot price, $(S_t)_{t \in [0, T]}$, follows a general diffusion process with dynamics specified in Equation (17.20). In this framework, the probability of near-term market direction corresponds to the value of a Digital Bond option in the special case where the initial spot price $S_t = x$ is close to the at-the-money forward strike, $K \approx xe^{(r-q)(T-t)}$. We derived the expression for this digital bond price in Appendix (17.2.2), showing that it depends explicitly on the spread between the expected cumulative market price of risk and the expected cumulative variance. This spread is defined by

$$\phi(t, T) = (T - t)(H^\lambda(t, T) - \frac{1}{2}I^2(t, T))$$

where $H^\lambda(t, T)$ is the mean market price of risk and $I^2(t, T)$ is the mean variance. Taking conditional expectation with respect to the market filtration $\mathcal{F}_t$, we obtain the expected mean spread

$$E[\phi(t, T) | \mathcal{F}_t] = (T - t)(H_0^\lambda(t, T) - \frac{1}{2}I_0^2(t, T))$$

where $H_0^\lambda(t, T)$ is the expected mean market price of risk (EMPR) and $I_0^2(t, T)$ is the expected mean variance.

Consequently, estimating the probability of near-term market direction reduces to estimating these two quantities: the expected mean market price of risk and the expected mean variance. The relationship between the probability and the expected mean spread is direct:

- $P(up) > \frac{1}{2}$ if $H_0^\lambda(t, T) > \frac{1}{2}I_0^2(t, T)$, and
- $P(up) < \frac{1}{2}$ if $H_0^\lambda(t, T) < \frac{1}{2}I_0^2(t, T)$.

While $(T-t)I_0^2(t, T)$ corresponds to the payoff of variance swap (observable and tradable in practice), $(T-t)H_0^\lambda(t, T)$ does not correspond to any traded contract. Although several attempts have been made to estimate the EMPR from market data, these efforts have proven challenging and largely inconclusive. Estimating $H_0^\lambda(t, T)$ remains inherently subjective and heavily dependent on the quality and timeliness of information available at any given moment.

In summary, forecasting the direction of spot price movements, on average, requires estimating a quantity that is both difficult to observe and inherently subjective. This places a structural advantage on trading firms or funds with access to superior, faster, or exclusive information, highlighting the role of information asymmetry in directional forecasting.

7.3 Limits of probabilistic reasoning in financial forecasting

While probabilistic forecasting provides a valuable framework for risk-informed trading decisions, the non-stationary nature of financial markets fundamentally undermines the classical argument of 'positive expected returns over many trades'. The informational edge provided by probability estimates must be understood as local and transient, requiring continuous adaptation to evolving market conditions.

7.3.1 Bridging probabilistic forecasts and single realisations

Forecasting the near-term direction of asset prices inherently involves dealing with uncertainty. As we have discussed above, models, whether AI-enhanced or grounded in stochastic processes, can only provide probabilities associated with various possible future scenarios. In the context of diffusion-based models, this uncertainty is commonly visualised as a cone of diffusion, representing a spread of potential price outcomes over the forecast horizon (see Section (2.3.2)). Yet in live trading, we are ultimately confronted with a single realisation of this probabilistic process: the realised market price path.

This apparent mismatch, between a probabilistic view of the future and the reality of a single realised trajectory, raises an important question: how can estimated probabilities meaningfully guide execution decisions if only one path will ultimately unfold?

The key lies in interpreting probability not as a guarantee of outcome, but as a risk-weighted informational edge. Threshold-based execution strategies leverage these probabilistic estimates to frame trade decisions in terms of expected value and risk management. For example, if a model estimates a 65% probability of an upward move within the next ten minutes, this suggests that over many repetitions of similar situations, upward moves would materialise 65% of the time, even though for this single instance, the market will either go up or down. The probabilistic signal, therefore, does not claim certainty over the immediate outcome, but provides a statistical advantage when applied consistently over time.

In practical terms, traders can:

1. Use probabilistic thresholds to filter noise: By only triggering trades when the estimated probability materially deviates from 50%, the strategy avoids acting on weak or ambiguous signals, focusing instead on scenarios where the informational edge is strongest.
2. Align position sizing with probability strength: Larger estimated deviations from neutrality (e.g., $P(\text{up}) = 0.7$) could justify larger position sizes or more aggressive execution, while marginal signals (e.g., $P(\text{up}) = 0.52$) may be ignored or treated with caution.
3. Incorporate risk controls based on uncertainty: Even with favourable probabilities, adverse realisations are inevitable. Integrating stop-losses, volatility-adjusted thresholds, or adaptive execution windows helps contain drawdowns when the realised outcome contradicts the model's probabilistic forecast.

4. Exploit long-term statistical edge: While the outcome of any single trade is uncertain, over a sufficiently large sample of trades, adherence to well-calibrated probabilistic signals should translate into positive expected returns, assuming transaction costs, slippage, and execution risks are adequately managed.

Ultimately, while each trade decision corresponds to one realised path, probabilistic forecasting enables traders to make consistent, risk-informed choices across many trading opportunities. The challenge of bridging probabilistic models with single-path realisations underscores the importance of calibration, scenario analysis, and disciplined execution frameworks within intraday systematic trading.

7.3.2 The challenge of accumulating statistical edge in non-stationary markets

A common justification for probabilistic trading strategies is the notion that, while the outcome of any single trade is uncertain, positive expected returns can be achieved over a sufficiently large sample of trades, provided the probabilities are well-calibrated. This reasoning implicitly relies on a key assumption: that the underlying probability distribution governing market behaviour remains sufficiently stable over time for past data to inform future outcomes.

However, this assumption does not hold in financial markets. As we explored in Section (2.3.1), markets are fundamentally non-stationary systems. The distributions that describe their future evolution, their finite-dimensional distributions (FDDs), depend on absolute time points rather than relative differences. In other words, the probabilistic structure itself is moving, often abruptly, in response to shifts in monetary policy, geopolitical events, technological developments, or investor sentiment. As a result, **the estimated probability of an upward or downward move at any given moment is itself a highly stochastic quantity**, vulnerable to rapid reconfiguration as new information arrives or structural breaks occur.

This reality introduces a critical limitation to the naive application of probabilistic reasoning in trading:

1. No stable long-term frequency: In stationary systems, repeating a probabilistic experiment many times allows the empirical frequency of outcomes to converge to the true underlying probability (via the Law of Large Numbers). In financial markets, however, **there is no guarantee that the conditions governing a trade today will resemble those for trades tomorrow or next week**. The very act of accumulating a 'large sample of trades' becomes problematic because the statistical environment keeps shifting.
2. Probability drift and miscalibration: Even if a model is well-calibrated over a recent training window, non-stationarity ensures that its predictive validity can deteriorate without warning. Regime shifts may render previous probability estimates irrelevant, leading to systematic mispricing of risk and the breakdown of expected performance.
3. Finite sample paradox: Intraday trading, especially when constrained by transaction costs, slippage, and operational risk, typically allows for only a modest number of trades over realistic time horizons. Since the underlying distributions are non-stationary, **we cannot rely on long-run averaging to 'smooth out' errors**, and poor calibration during a volatile regime can result in disproportionate drawdowns.
4. The risk of false informational edges: A probability estimate of, say, 65% for an upward move may reflect a genuine informational advantage in one market regime, but devolve into noise, or worse, structural bias, in another. Without active adaptation, traders risk mistaking transitory, regime-specific relationships for universal statistical edges.

Addressing this challenge requires moving beyond static probability estimates toward adaptive modelling frameworks. These might include regime-switching models, online learning algorithms, or probabilistic forecasts that explicitly account for parameter uncertainty and structural breaks. Furthermore, trading systems need robust feedback mechanisms to detect model degradation and recalibrate or deactivate strategies during periods of heightened non-stationarity.

8 System design under uncertainty

In financial markets, uncertainty is not a transient inconvenience but a structural reality. This section addresses the design of quantitative trading systems in environments marked by non-stationarity, structural breaks, and unpredictable discontinuities. Financial time series frequently exhibit evolving statistical properties due to macroeconomic shifts, policy changes, and investor behaviour, rendering traditional static models inadequate. As such, systematic strategies must be built not to eliminate uncertainty, but to operate effectively within it.

We begin with a case study of real-time decision-making under uncertainty, illustrating how quantitative systems can respond to statistical deviations and local extrema without relying on perfect foresight. The discussion emphasises the probabilistic nature of market dynamics and highlights the strategic focus on maximising expected returns over many trades, rather than predicting individual outcomes. The case study reinforces the principle that adverse events, such as price jumps, stop-runs, or fakeouts, are inevitable outliers within a complex and evolving distribution.

Next, we confront the specific challenge of managing discontinuities in systematic trading. While such events are rare and difficult to model, they cannot be ignored: their impact on risk and performance is disproportionately large. Practitioners argue that discontinuities are not modelling failures but statistical certainties, and that robust strategies must embed protective mechanisms, such as volatility throttles, fallback states, and residual-based disbelief logic, to handle them prudently and adaptively.

Finally, we examine the question of detecting structural changes in real time, highlighting both its necessity and its inherent limitations. While perfect jump detection is impossible, practical systems can respond to symptoms of structural breaks through probabilistic indicators, regime detection heuristics, and signal attenuation when uncertainty is high. In this light, adaptive design is not about anticipating the future, but about recognising when the past no longer applies and adjusting accordingly.

Through these three perspectives, this section provides a cohesive framework for building quantitative strategies that are resilient to the unpredictable nature of financial markets. It reinforces a core principle: success in systematic trading does not come from eliminating uncertainty, but from systematically navigating it.

8.1 The decision process under uncertainty: A case study

As discussed in Section (2), financial markets are fundamentally non-stationary. Their dynamics evolve continuously in response to a complex and ever-changing landscape shaped by economic policies, geopolitical developments, regulatory adjustments, and shifts in investor sentiment. Consequently, statistical properties such as volatility, correlation, and return distributions are not stable over time. This structural instability presents a core challenge for modelling and prediction, requiring adaptive approaches capable of responding to regime shifts and structural breaks.

Sudden price movements, such as jumps, fakeouts, or reversals, are often driven by latent variables (e.g., macroeconomic shocks, large hidden orders, or unexpected news) that are not observable at high frequency, such as in 1-minute price data. In what follows, we define and analyse the formation of such anomalies and explain why quantitative strategies must base their decisions on recurring statistical patterns in historical data. However, every market event is a new realization drawn from a complex, evolving distribution. Thus, even with the best data and models, we operate in a world of probabilities, not certainties.

Quantitative strategies are not tasked with predicting rare price events, rather, they are designed to respond intelligently to observable price structure and shifting probabilistic regimes. Their objective is to maximise expected returns over many trades, while ensuring robust risk management so that adverse outcomes, such as a sudden price jump, do not compromise long-term performance.

To clarify this decision-making process, we present a representative example of a real-time, window-based systematic trading system that reacts to local price extrema and statistical deviations. This includes reactions to events like price jumps, stop-runs or fakeouts, reversals, and other outliers that are inherently unpredictable. We demonstrate that no model or dataset can fully anticipate these anomalies. For every trading decision made along a price series, a plausible counter-example can always be found.

This unpredictability is not a flaw of quantitative strategies but a fundamental feature of market behaviour. The role of a robust strategy is to make probabilistic decisions informed by context and to manage risk prudently around those decisions. Because outcomes are inherently uncertain, strategy performance can only be evaluated in expectation, which requires a sufficiently large dataset for empirical averages to meaningfully approximate long-run returns.

8.1.1 Price formation anomalies

Price formation anomalies arise from the complex and often imperfect mechanisms through which financial markets aggregate information, liquidity, and trader behavior. Unlike the assumptions of idealised models, such as continuous price paths, perfect information, and rational agents, real markets are shaped by heterogeneous participants with varying objectives, time horizons, and access to information. Anomalies often reflect liquidity constraints, behavioural biases, or strategic interactions among market participants, such as stop orders clustering around technical levels or algorithmic strategies reacting to short-term price signals. Additionally, market microstructure effects, such as order book dynamics, execution latency, and fragmented liquidity, can lead to temporary distortions in price that deviate from fundamental value. These anomalies are not merely noise but can contain structural information exploitable by quantitative strategies that model the underlying mechanisms generating them.

First, we briefly describe the double bottom pattern, which will serve as the basis for analysing W-shaped reversal traps. This classic formation in technical analysis resembles the letter 'W' and typically signals a major trend reversal following a preceding downtrend. The twice-touched low is considered a significant support level. Practically, it indicates a period of selling, causing an asset's price to drop below a level of support. It will then rise to a level of resistance, before dropping again. Finally, the trend will reverse and begin an upward motion as the market becomes more bullish. It is a bullish reversal pattern, because it signifies the end of a downtrend and a shift towards an uptrend.

Next, we present several well-known price formation anomalies and provide their definitions:

- **Price Jumps:** Price jumps refer to sudden, large movements in asset prices that occur over very short time intervals. In quantitative finance, they are commonly modeled using jump diffusion processes, where returns follow a combination of continuous Brownian motion and discrete Poisson-driven jumps. These discontinuities break the assumption of smooth price paths found in classical models and are essential for capturing extreme events, gap risks, and the effects of unexpected news. In high-frequency contexts, jumps often reflect imbalances in order flow, sudden liquidity withdrawal, or the rapid assimilation of new information by market participants.
- **Stop-run or Fakeout:** Stop-runs, also known as fakeouts, occur when price temporarily breaches a key technical level, such as a recent high or low, only to reverse sharply afterwards. These events are often the result of liquidity-seeking behaviour, where large players intentionally push price through stop-loss clusters to trigger forced exits and gain access to hidden liquidity. Commonly referred to as stop-loss hunting, liquidity sweeps, or false breakouts, such movements exploit predictable order placement by retail or systematic participants. In quantitative models, these behaviors may be captured through order flow imbalance, market impact modelling, or the identification of liquidity voids around key price levels.
- **W-shaped Reversal Trap:** A W-shaped reversal trap occurs when price forms a double bottom pattern, giving the illusion of a strong reversal, only to fail and resume its prior trend. This behaviour is commonly referred to as a double bottom trap, reversal trap, or whipsaw pattern, and is characterized by rapid reversals that mislead both momentum and mean-reversion strategies. These traps often arise in volatile or low-liquidity environments,

where short-term noise overrides genuine structural shifts. Quantitative models may treat such events as pattern-based regime misclassifications, or relate them to volatility clustering and transient shifts in market sentiment.

- **Healthy Pullback:** A healthy pullback refers to a temporary retracement in price that occurs within the context of a prevailing trend. Also known as a corrective move, retracement within trend, or trend-conforming pullback, this pattern reflects a controlled pause or consolidation before the trend resumes. Unlike reversal traps or fakeouts, these pullbacks typically exhibit bounded volatility, reduced momentum, and respect for key technical levels such as moving averages or support zones. In quantitative frameworks, such behaviour is often modelled as mean-reversion within a trending regime, or captured by trend-following models with volatility constraints.

8.1.2 The price series

We are going to consider a representative example of a real-time, window-based systematic trading system reacting to local price extrema and statistical deviations. The price series must include jumps, fakeout and reversals since they are often fundamentally unpredictable. We illustrate these events by considering the GSPC index on the third of February 2025. We display prices in Figure (6). The x-axis is time expressed in minutes.

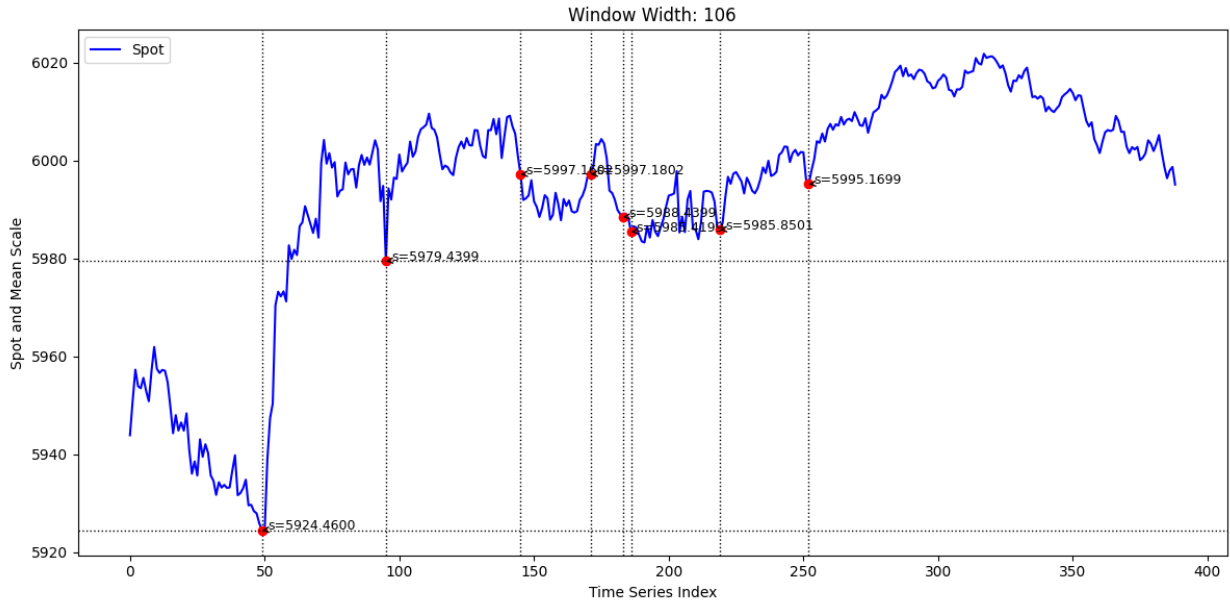


Figure 6: Prices for GSPC on the 03/02/25.

We consider eight decision points corresponding to price anomalies, each marked by a red dot on the price series: just before a price jump at $t = 49$ with value $S_0 = 5924.46$, a stop-run or fakeout at $t = 95$ with value $S_1 = 5979.44$, a fake W-shaped starting at $t = 145$ with value $S_{2,0} = 5997.1$ and ending at $t = 171$ with value $S_2 = 5997.18$, a W-shaped reversal trap starting after $t = 171$ and ending at $t = 183$ with value $S_3 = 5988.44$, an outlier at $t = 186$ with value $S_4 = 5985.42$, just before a fakeout at $t = 219$ with value $S_5 = 5985.85$, and just before a mild fakeout or a healthy pullback at $t = 252$ with value $S_6 = 5995.17$. The optimum window size for the sliding moving average is $window_size = 106$.

8.1.3 Signal-to-position mapping

The decision-making process under uncertainty is a core component of systematic strategy design. At its foundation lies the task of translating observed market signals into trading decisions, based not on deterministic rules, but on probabilistic inferences drawn from historical context.

We are now going to illustrate the decision-making process under uncertainty with the case study presented in Section (8.1.2). We will focus on the price jump at time $t = 49$ with price S_0 , the stop-run at time $t = 95$ with price S_1 , and the W-shaped reversal trap at time $t = 183$.

- A strategy may learn that a sharp drop (jump condition) observed at time $t = 95$ with spot price S_1 has historically been followed by profitable short positions.
- Likewise, a W-shape² starting at $t = 145$ with spot price $S_{2,0} = 5997.1$ and ending $t = 171$ with spot price $S_2 = 5997.18$ may signal a high-probability opportunity to go long.
 - In our example, there are more than two distinct lows forming a strong support level. While those lows hold, the upside has significant potential.
 - Traders don't buy at the second bottom, instead they wait for the price to reach the confirmation point of another buy, denoted S_2 , to be certain that this is a W-shaped pattern.
 - Hence, the trading volume is lower on the second bottom than on the first one. If this isn't the case, there is a risk of falling to even lower lows. Then, volume increases again as the pattern completes, to further push the price upwards.
- However, the inability of the algorithm to anticipate a stop-run or fakeout at $t = 96$, or, a W-shaped reversal trap at $t = 183$ with spot price S_3 reflects a limitation of the strategy's forecasting ability, not a failure of its risk management system.
- Risk management does not attempt to predict market direction; its role is to preserve capital, control drawdowns, and ensure robustness under adverse scenarios.

This framework is known more broadly as a sequential decision process under uncertainty, and it underpins fields such as online learning, control theory, and adaptive trading system design. In quantitative trading, this process is often referred to as:

- Signal interpretation or signal-to-position mapping,
- Position management logic, or
- Tactical execution logic within the overall strategy.

It is important to distinguish this decision logic from risk management. While the former drives trade initiation based on statistical context, the latter operates downstream, monitoring, constraining, and adjusting exposure to ensure resilience in the face of market uncertainty.

8.1.4 Strategy versus risk management

The decision to stay in, reverse, or exit a position, based solely on information available up to the current time, is a central function of the trading strategy. However, this decision-making process operates in close coordination with risk management, which serves as a protective layer to ensure robustness and capital preservation.

²A double bottom signifies the end of a downtrend and a shift towards an uptrend.

8.1.4.1 Strategy component The strategy is responsible for generating trade signals based on historical and real-time data. This includes determining not just whether to enter or exit, but also the direction and size of a position, using a set of predefined rules.

- Trade decisions are made using only information available at the current time step (e.g., $t = 95$, $t = 171$), with no foresight.
- Key elements of the strategy include:
 - Signal generation: deriving signals from features like momentum, mean reversion, or statistical anomalies.
 - Signal interpretation: applying thresholds, crossovers, or other decision criteria to convert raw signals into actionable trades.
 - Position logic: deciding whether to maintain, close, or reverse a position based on model outputs.
- These decisions are made because the model infers a structural change or continuation, not because of access to future information. The logic is inherently forward-looking but constrained by historical context.

8.1.4.2 Risk management component Risk management acts as a real-time evaluator and constraint system, assessing whether the strategy's proposed actions are permissible within defined risk parameters. It does not initiate trades but can restrict or override them when necessary.

- Before executing a trade, risk management evaluates factors such as:
 - Position size: does the trade exceed exposure limits?
 - Volatility environment: is current market risk within acceptable bounds?
 - Drawdown status: has the strategy reached predefined loss thresholds?
 - Liquidity constraints: is there sufficient market depth to execute the order safely?
- In adverse conditions, risk controls may override or suppress the strategy's decision. For example, a reversal may be blocked if volatility has surged unexpectedly, even if the model suggests a directional shift.

8.1.4.3 Summary In summary, the strategy proposes actions based on observed data and learned behaviour, while risk management ensures that these actions are aligned with broader capital preservation and robustness objectives. The two components are distinct yet interdependent, forming the core of any sound quantitative trading system.

8.2 Managing discontinuities in systematic trading

8.2.1 The unpredictability of jumps and reversals

Price jumps, reversals, and sharp discontinuities are often cited as the most difficult elements to model in financial markets. Their unpredictability is not merely a technical challenge, it reflects deeper limitations tied to the stochastic nature of markets and the inherent constraints of any modelling framework.

8.2.1.1 Markets are stochastic, not deterministic

- Even with perfect data and the most sophisticated mathematical model, trading decisions are based on probabilities, not certainties.
- Jumps are frequently triggered by latent variables, such as macroeconomic announcements, geopolitical news, or large hidden institutional orders, that are not observable in standard high-frequency data (e.g., 1-minute bars).
- As a result, a sudden move may appear random from the model's perspective, even though it is entirely rational from a broader informational context unknown to the strategy.

8.2.1.2 Path dependency and branching futures

- At any decision point, such as $\tau = \{49, 95, 171, 183\}$, the market can branch into multiple futures: breakouts, reversals, consolidations, or false moves (fakeouts).
- Quantitative strategies rely on historical analogues to estimate what typically happens in similar contexts, but each new instance is a fresh draw from a complex, high-dimensional distribution.
- Because of this path dependency, two apparently identical setups may lead to very different outcomes depending on subtle, unobservable conditions.

8.2.1.3 Model risk versus irreducible risk

- Model risk arises when a model fails to capture existing structure or misrepresents statistical relationships, this can, in principle, be reduced with better design or more data.
- Irreducible risk, however, stems from randomness, regime shifts, or the arrival of new information not present in the historical record. No amount of modelling can eliminate this uncertainty.
- Sudden price jumps and sharp reversals often fall into this second category: they are not predictable, not because we haven't found the right model, but because they reflect genuine uncertainty in a complex, evolving system.

8.2.2 Designing robust strategies

8.2.2.1 The role of the strategy Given that sharp moves like jumps, reversals, or stop-runs are fundamentally unpredictable, what, then, is the role of a trading strategy?

Crucially, the strategy's purpose is not to forecast every price anomaly, but to respond intelligently and probabilistically to evolving price dynamics. Its primary objectives are:

- To interpret price structure in real time and adapt to shifting statistical patterns.
- To maximise expected return over a large number of trades by exploiting small, repeatable edges.
- To manage risk effectively, ensuring that an unexpected adverse move doesn't jeopardise long-term performance.

In short, quantitative trading is not a game of certainties, it is a game of edges, where success comes from executing a sound process consistently under uncertainty.

8.2.2.2 What should a strategy do when anomalies occur? At any given decision time $t \in \tau$, especially when a large price move occurs, the model does not need to 'know' with certainty what caused the move. Instead, it should:

- Use its training and historical experience to probabilistically classify the event, as a breakout, reversal, noise, or anomaly.
- Weigh the likelihood of different outcomes:
 - The move could be the start of a trend (e.g., a news-driven breakout),
 - Or it could be a short-term reversal (e.g., a liquidity spike or stop-run).
- Act based on the most likely scenario, while acknowledging and preparing for the possibility that it may be wrong.

Robust strategies accept this probabilistic framework. They are designed not to be right every time, but to be right often enough and to cut losses when they are not.

8.2.2.3 Signal-to-position mapping: A human-like interpretation To build a model that responds in this way, we need a reliable signal-to-position mapping: a logic that transforms price observations and indicator signals into portfolio actions.

In particular, this requires tools that describe the behaviour of indicators like the Relative Moving Average (RMA) in a manner similar to how a seasoned trader would interpret charts with their eyes and brain. These tools should:

- Be objective, interpretable, and not sensitive to arbitrary features of the time series (such as scale or granularity).
- Capture structural elements like local extrema, mean-reversion patterns, or breakout acceleration.
- Translate these features into actionable signals that reflect probabilistic expectations, not certainties, about future price movement.

In effect, this bridges the gap between statistical learning and market intuition, allowing the strategy to act intelligently without relying on prediction alone.

8.3 Detecting structural discontinuities in real time

Price jumps, characterised by large discontinuities in price evolution, are an essential part of the price time series generated on financial markets (see Jiang et al. [2005] [2013]). The detection of structural discontinuities in real time remains one of the hardest problems in systematic trading. It is not solvable in a deterministic sense but must be addressed probabilistically and defensively. Rather than viewing jumps as a failure of signal precision, robust strategy design reframes them as statistical outliers that are inevitable, and thus must be managed, not predicted.

8.3.1 The challenge of jumps

Jumps, fakeouts, and abrupt reversals in financial markets are by nature ex-post observable events. They break the assumption of continuity implicit in most statistical models and are often driven by latent, unobservable factors, macro shocks, order flow imbalances, or unexpected news releases. Detecting such structural discontinuities in real time, before their full impact unfolds, is statistically problematic and often impossible in the classical sense.

Yet, systematic strategies must still respond to such events, ideally, with minimal delay and controlled risk exposure. The core issue can be summarised as follows: Discontinuities cannot be reliably predicted or identified in real time using only past data; they can only be inferred probabilistically after-the-fact, through their statistical imprint on recent price action.

Some of the explanation as to why real-time detection is a hard problem are:

1. Jumps are non-infinitesimal: Unlike diffusive moves, jumps are characterised by large changes over a very short time, violating assumptions like Lipschitz continuity or Brownian motion used in many models. See Bellamy et al. [2000].
2. Backward-looking statistics fail: Rolling estimators assume local stationarity. When a jump occurs:
 - Variance spikes, contaminating estimates.
 - The sliding window registers the event twice: once on entry, once on exit.
 - Signal triggers built on thresholds become unstable.
3. Noisy short-term volatility: A jump is statistically indistinguishable from noise in real time until it exceeds certain thresholds, by which point it may be too late to react profitably.

8.3.1.1 Detecting discontinuities Despite these limitations, practitioners have developed several approximate mechanisms to infer discontinuities and adapt dynamically:

- Jump tests on high-frequency data:
 - Barndorff-Nielsen and Shephard jump test: Decomposes realised variance into continuous and jump components using high-frequency intraday returns. See Barndorff-Nielsen et al. [2006].
 - Lee-Mykland test: Based on bipower variation to statistically identify discontinuities. See Lee et al. [2008].

Limitation: These are lagged diagnostics. Useful for backtesting and regime classification, but not for predictive control.

- Volatility-of-Volatility heuristics:
 - Sudden spikes in short-term realized volatility (e.g., over the last 5 bars) may indicate latent structural changes.
 - Heuristic triggers: If σ_t increases by more than k standard deviations within a short window, flag the environment as unstable.

Tradeoff: Responsive but noisy; leads to more false positives.

- Regime-Switching models:
 - Use Hidden Markov Models (HMMs) or Switching VARs to detect latent state changes. See Srivastava et al. [2018], Aydinhan et al. [2023], Bloch et al. [2024].
 - Once a jump is detected probabilistically, the system transitions to a different rule set (e.g., conservative risk parameters).

Limitation: HMMs assume slowly changing regimes and require large sample sizes to estimate transition probabilities reliably.

- Entropy and information measures:
 - Sudden increases in entropy or Kullback-Leibler divergence between the current distribution and recent history can signal structural change.
 - Useful in online learning frameworks where the system continuously re-estimates its belief about the market's statistical structure.
- Model Residual Monitoring
 - Maintain an ensemble of predictive models.
 - Large residuals between expected and observed returns trigger 'model disbelief' flags, prompting the strategy to either reduce exposure or switch to capital preservation mode.

8.3.1.2 Design principles for handling jumps Given the inherent limitations in detecting discontinuities:

- Accept irreducible uncertainty: Don't try to forecast every jump. Instead, design systems that remain robust when surprised.
- Incorporate Bayesian updating: Strategies should down-weight prior beliefs quickly after observing statistically inconsistent behaviour.

- Apply control theory: Use dynamic risk buffers, position scaling, and exposure throttling to prevent overreaction to noise or underreaction to true regime shifts.
- Use ensemble signals: No single metric can detect jumps reliably. Combine volatility measures, trend breaks, order book shifts, and residuals for more stable inference.
- Design with optionality: In jump-prone environments, hold optionality in portfolio structure (e.g., flexible sizing, stop logic, conditional entry/exit).

8.3.2 The role of ex-post jump estimation in financial modelling

The usefulness of ex-post jump estimation critically depends on the intended objective. If the goal is real-time signal execution, such as avoiding a stop-run or capitalising on a fakeout, then estimating a jump after it has occurred offers no benefit for immediate decision-making. By the time the jump is identified, the market has already responded, and the trade has already been affected.

However, this does not render ex-post jump detection irrelevant. Its value lies not in real-time intervention, but in its utility for downstream tasks such as regime identification, model recalibration, and performance diagnostics. Recognising jumps after they occur allows models to adapt to structural changes in the market, update assumptions about volatility or return distributions, and inform future strategy adjustments. Thus, ex-post estimation serves as a critical tool in the ongoing refinement and robustness of quantitative trading systems.

8.3.2.1 The real value of ex-post jump estimation Since analysis of market jumps can only be performed ex-post, it cannot safeguard an individual trade in real time. However, over a large number of trades, ex-post jump detection can be instrumental in protecting the integrity of the overall strategy through model adaptation and improved risk management. This includes:

- Model credibility and adaptive control:
 - Suppose our model expected a smooth mean-reverting behaviour, but suddenly gets hit with a sharp loss.
 - Ex-post jump detection helps us flag when our model's assumptions have failed (e.g., continuity, normality, stationarity).
 - This lets us deactivate or shift our model to a different regime, reducing its influence on trading decisions. (we are not shifting regime in presence of jumps)

Think of it as a safety interlock in engineering: we don't predict the failure, but we shut things down safely once it has occurred.

- Post-mortem risk attribution:
 - After a sequence of bad trades, we want to know: was this bad luck, or was the model structurally wrong?
 - Jump detection helps separate model risk from irreducible market shocks.
 - This supports model auditability, particularly in institutional or regulatory contexts.

Real-time feedback with delay

- While we can't act before the jump, we can act quickly after it (if we are lucky).
- For instance, if jump indicators spike, the strategy could:
 - * Pause new entries,
 - * Reduce risk limits for the next N bars,
 - * Wait for statistical stability to return.

This is not prediction, it is controlled re-engagement after surprise.

- Recalibrating Expectations and statistics
 - Post-jump, our volatility, skew, and other moments are no longer valid.
 - Rolling estimates become contaminated.
 - We need to segment our dataset, exclude jump-contaminated periods, or re-estimate parameters for stability.

Without jump estimation, our entire signal infrastructure remains polluted.

8.3.2.2 Real-time safeguards in extreme market conditions To mitigate the risk of significant losses during abrupt market events such as fakeouts or stop-runs, real-time trading systems incorporate robust safeguards. These mechanisms do not rely on ex-post analysis but are designed to respond adaptively as new data arrives. Common components include:

- **Position Sizing and Stop-Loss Protocols:** Trades are constrained in size such that no single position can cause disproportionate damage to the portfolio, even in the event of model failure or adverse price movement.
- **Volatility Filters:** Signal activation may be temporarily suspended when short-term volatility exceeds a predefined threshold, reducing exposure to noise or regime transitions.
- **Model Deviation Controls:** If observed returns diverge materially from model forecasts, signal strength may be attenuated to reflect reduced confidence in the model's relevance under current conditions.
- **Fallback States:** During periods of elevated uncertainty, evidenced by increased signal variance or conflicting indicators, the system can default to a conservative operational mode, reducing position sizes or halting trade initiation altogether.

These safeguards represent practical adaptations that enhance the robustness of quantitative strategies in real-time, especially when underlying statistical assumptions are momentarily invalidated by market discontinuities.

9 The RMA framework

Building on the quantitative foundations outlined in Section (3), we now introduce the Relative Moving Average (RMA) framework (see Bloch [2024a]), an innovative indicator that integrates the strengths of traditional momentum and mean-reversion models while addressing their limitations in non-stationary market environments. Designed to operate robustly under evolving structural conditions, RMA provides a dynamic and adaptive lens for characterising short-term market behaviour, enabling real-time decision-making in systematic trading.

The RMA framework combines the statistical rigour of moving averages with the contextual awareness of price action analysis. By constructing a normalised, sliding-window distribution around a local moving average, RMA captures key features of the underlying price dynamics, specifically, the expansion and contraction phases of the distribution that signal potential opportunities for trade entry and exit. This structure allows RMA to adaptively profile local price behaviour, identify probabilistic disequilibria, and support robust signal generation without relying on fixed assumptions about stationarity or specific distributional forms.

Unlike conventional technical indicators that are either purely reactive or excessively noisy, RMA provides a calibrated balance between responsiveness and stability. It supports both trend-following (momentum) and counter-trend

(mean-reversion) strategies by exploiting localised deviations from the dynamically evolving center of mass. The indicator generalises naturally to multi-asset strategies such as pairs trading, where co-movement and divergence offer exploitable signals.

In this section, we present a formal exposition of the RMA methodology, detailing its mathematical formulation, implementation parameters, and associated risk management protocols. Central to this approach is the introduction of Relative Extremum Ratios, a technique for detecting market regimes and their transition phases by comparing the current distribution range to its historical minimum and maximum values. We then demonstrate how the RMA framework can be employed to design actionable trading strategies specifically adapted to varying market conditions. This methodology provides a probabilistic framework for modelling short-term structural changes in price dynamics, thereby improving the robustness and adaptability of systematic strategies operating in uncertain and discontinuous financial environments.

9.1 Description

Forecasting financial time series remains a challenging task due to the complex nature of market behaviour. Financial data often display multifractal characteristics, heavy-tailed distributions, abrupt extremes, and long-range dependencies, features that hinder the identification of stable, exploitable relationships. Traditional approaches, including Technical Analysis, stochastic modelling, and derivative pricing frameworks, rely on mathematical indicators to estimate alpha and define systematic trading rules. While these methods offer probabilistic insights, they are constrained by their assumptions and often fail in the presence of structural breaks or regime changes.

Standard technical indicators tend to be lagging, as they are derived from historical data. Excessive use of such tools can introduce signal noise and conflicting inputs, complicating decision-making. Price Action Trading (PAT), though more flexible and responsive to market context, is inherently subjective; identical price patterns may yield divergent interpretations depending on the trader, reducing the repeatability of results.

To address these limitations, Bloch [2024a] introduced the Relative Moving Average (RMA), a method that integrates Moving Averages with elements of price action analysis to dynamically profile time series data. This approach seeks to enhance intraday strategy design by capturing evolving price distributions and identifying periods of potential disequilibrium. RMA operates on the principle that normalised deviations from a sliding mean can reveal localised patterns indicative of market expansion or contraction. This methodology also generalises to use cases such as pairs and correlation trading, where price divergences and convergences are of interest.

In practical terms, RMA constructs a rolling, normalised distribution over a sliding window applied to the observed price path. This framework enables the development of an adaptive trading strategy that is responsive to short-term structural changes in market dynamics. The main components of the method are as follows:

- **Market Dynamics:** Prices are assumed to oscillate around a local moving average, defining a dynamic center for a distribution whose spread reflects market volatility. Phases of distributional expansion and contraction correspond to potential trading opportunities.
- **Entry and Exit Logic:** Trades are initiated at the onset of an expansion phase and exited during contraction. The window length determines the resolution of these shifts and implicitly governs trading frequency.
- **Parameter Sensitivity:** The window size provides a mechanism for adjusting the temporal granularity of the strategy, balancing responsiveness and noise reduction.
- **Risk Management:** The framework includes provisions for position sizing, stop-loss placement, and target setting, based on real-time estimations of return-to-risk ratios derived from the local distribution.

By aligning a distributional view of price action with classical moving average structures, the RMA approach offers a consistent mechanism for modelling short-term market behaviour in a non-stationary environment. It does not attempt to forecast precise price levels, but rather characterizes probabilistic tendencies within a local, evolving structure, offering a robust foundation for real-time strategy deployment under uncertainty.

9.2 Construction and application

9.2.1 Formulation

We consider the time series $X = \{x_1, \dots, x_n\}$, $x_t \in \mathbb{R}$, and generate a set of overlapping segments $V = \{v_1, \dots, v_{n^*}\}$, where $n^* = n - W + 1$, by applying a sliding window of size W on X . In finance, we cannot look forward and should start the sliding window from x_W with k th segment being $v_k = (x_{k-W+1}, \dots, x_k)$, $k \in [W, n]$. However, for simplicity, we choose to go forward and let the k th segment be given by $v_k = (x_k, \dots, x_{k+W-1})$, $k \in [1, n^*]$. To look backward, simply change the index k to $k + W - 1$ in the segment v_k .

We normalise that k th segment with respect to any element of that segment, which we call the anchor. We denote $v_k^{(i)}$, $i = 1, \dots, W$, the normalised k th segment with respect to the i th element. This leads to rewriting each element of the segment in terms of some kind of returns as follows:

$$v_k^{(i)} = \frac{1}{x_{k+i-1}} \cdot v_k = \frac{1}{x_{k+i-1}} \cdot (x_k, x_{k+1}, \dots, x_{k+W-1}), i = 1, \dots, W$$

We compute the sample average (SA) of the normalised segment $v_k^{(i)}$, getting

$$\bar{v}_k^{(i)} = \frac{1}{x_{k+i-1}} \cdot \bar{v}_k = \sum_{l=0}^{W-1} \frac{x_{k+l}}{x_{k+i-1}}$$

Given a segment v_k , for every normalised segment $v_k^{(i)}$ with respect to the i th element, the RMA is

$$\bar{v}_k^{(i,-1)} = \bar{v}_k^{(i)} - 1 = \frac{1}{x_{k+i-1}} \cdot \bar{v}_k - 1, i = 1, \dots, W \quad (9.1)$$

Anchors can be specific values within the k th segment, such as the first, middle, or last element, or statistical measures like the minimum, maximum, or selected quantiles of the segment. For example, for the last element x_{k+W-1} , we set $i = W$ and get $\bar{v}_k^{(W,-1)} = \bar{v}_k^{(W)} - 1$, where $v_k^{(W)} = \frac{1}{x_{k+W-1}} \cdot (x_k, x_{k+1}, \dots, x_{k+W-1})$.

9.2.2 The trading problem

The RMA model the distance between the sliding moving average (SMA) and any specific point in the sliding segment. Given the k th segment v_k , we can rewrite the RMA as

$$\bar{v}_k^{(i,-1)} = \frac{d_k^{(i,-1)}}{x_{k+i-1}}, i = 1, \dots, W$$

where

$$d_k^{(i,-1)} = \bar{v}_k - x_{k+i-1} \quad (9.2)$$

is the mean-reverting drift of the i th process. We can therefore measure the deviation of the stock price from its dynamic equilibrium level. When $i = W$, the distance $d_k^{(W,-1)}$ corresponds to the mean-reverting drift of the current stock price. This distance determines the profitability of the trade when using the RMA method.

For example, we can assume that the stock price S_t follows the dynamics

$$dS_t = \kappa(Z_t - S_t)dt + \dots$$

where κ is the speed of mean-reversion and $Z_t = E_t[S_T]$ for some maturity $T > t$. In our setting, Z_t is stochastic and statistically estimated. It corresponds to the forecasted price of the trading opportunity. Hence, the distance $d_k^{(W,-1)}$ corresponds to the expected profit for the trade given by

$$E_t[\pi_T] = \omega(E_t[S_T] - S_t)$$

where ω is the portfolio weight.

9.3 Implementation of the RMA

We briefly discuss the implementation of the Relative Moving Average (RMA) indicator and present a use case which we will use through out this article to illustrate our purpose.

9.3.1 The RMA

At the start of the trading day, the first 100 minutes of data for each asset are used to determine the optimal window size, denoted *window_size*, which best captures the time series' underlying dynamics. This window is then applied to compute the sliding moving average (SMA), as illustrated in Figure (7). As detailed in Section (9.2.1), the SMA subsequently serves as the basis for computing the Relative Moving Average (RMA) indicator throughout the remainder of the trading day.

We create a 2D array of overlapping sliding windows from a 1D spot time series:

```
windows = np.lib.stride_tricks.sliding_window_view(spot_time_series, window_width)
```

This results in an array of shape $(n_{\text{windows}}, \text{window_size})$ where each row corresponds to a sliding window from the original time series. We then normalise each window to obtain the RMA indicator (see Equation (9.1)), which provides a compact summary of the distribution of values within each window.

```
normalized_returns = windows / means[:, None] - 1.0
```

The RMA indicator (normalisation with respect to first, middle, last, minimum, maximum) is mapped into curves as follows:

```
first_elements = windows[:, 0]
mid_elements = windows[:, mid]
last_elements = windows[:, -1]
max_values = windows.max(axis=1)
min_values = windows.min(axis=1)
```

This is illustrated with the dotted curves in Figure (8), where the dotted lime green curve represents *w* (normalisation with respect to last element), the dotted light pink curve represents *mid* (normalisation with respect to middle element), the dotted gold curve represents 1 (normalisation with respect to first element), the dotted dodger blue curve represents *min* (normalisation with respect to minimum), and the dotted aqua blue curve represents *max* (normalisation with respect to maximum element).

9.3.2 Compute the qth quantile of the given data

Next, we transform the RMA indicator into quantiles by dividing the value distribution within each window into intervals of equal probability. Each window (i.e., each row in the array) (of size *window_size*) is associated with a quantile value at level, q , the value below which a proportion q of the window's elements fall. By computing a

sufficient number of quantiles (e.g., deciles or percentiles), we can reconstruct the empirical cumulative distribution function (ECDF) for the values within each window.

Let qS be a list (or iterable) of quantile values, for example, $[0.1, 0.25, 0.5, 0.75, 0.9]$. Each $q \in qS$ represents the quantile level (i.e., the probability threshold) to compute. The function `np.quantile` calculates the q -th quantile along each row (window) of the 2D array `windows`. The result is stored in a dictionary, denoted by `quantiles`, which maps each quantile level q to its corresponding computed values:

- **Keys:** Each element q from the list or array qS , representing the quantile levels (e.g., $[0.1, 0.25, 0.5, 0.75, 0.9]$).
- **Values:** For each q , `np.quantile` returns a 1D array of shape $(n_{\text{windows}},)$, where each entry is the quantile value at level q computed across the corresponding row (or 'window') of the 2D array `windows`.

9.3.3 Normalise the quantile values

Although computing quantiles provides rich information about the shape, spread, and skewness of a distribution, we still need to normalise them. Normalisation does not add new information - rather, it ensures that the existing information is comparable across different windows or time periods. Quantile values, in their raw form, reflect absolute levels. By normalising, we express each quantile as a relative deviation from the mean, making the representation scale-invariant and suitable for comparison across windows. Subtracting 1 centers this proportion around zero. If a quantile is equal to the mean then $result = 0$, if it is above the mean then $result > 0$, and if it is below the mean then $result < 0$. To summarise, quantiles normalisation highlights shape independently of scale, allows comparison across windows, and it is useful for machine learning features.

Following the normalisation used in the RMA indicator (see Equation (9.1)), we convert each quantile value into a relative deviation from the mean of its corresponding window which we center around zero. This process removes scale effects, highlights asymmetry, and makes the features more suitable for modelling.

```
for q, quantile_values in quantiles.items():
    normalized_by_quantiles[q] = quantile_values / means - 1.0
```

9.3.4 Map curves into fractiles

Like quantiles, a fractile highlights the proportion (or fraction) of the distribution that lies below a certain value. In addition to computing the RMA indicator, we also map its curves - (w, l, mid, min, max) - to corresponding fractile values, resulting in the set $(f_w, f_l, f_{mid}, f_{min}, f_{max})$. This transforms the RMA indicator into a representation on a $[0, 1]$ scale, making it directly comparable across different assets. To achieve this, we compute the fractile of each selected normalized return value - (w, l, mid, min, max) - relative to the normalised return distribution within each sliding window. These fractile values are then stored in new columns of the dataframe.

```
new_columns = {}
columns_to_fractile = ['l', 'mid', 'w', 'max', 'min']
for column_name in columns_to_fractile:
    fractile_column_name = f'f_{column_name}'
    column_values = normalized_df[column_name].values
    new_columns[fractile_column_name] = compute_fractile(column_values,
                                                         normalized_returns)
```

For each row:

- Take the value in the column (e.g., a specific normalized return like ' w ').
- Compares it against the corresponding row in `normalized_returns`, which is a distribution of all normalized returns in the window.

- Uses `compute_fractile(...)` (see Appendix (14.1)) to compute how high that value ranks in the window (its empirical cumulative distribution or percentile).
- Stores the result as a new column.

This is illustrated with the plain curve in Figure (10), where the plain dark green curve represents f_w , the plain dodger blue curve represents f_{min} , and the plain aqua blue curve represents f_{max} .

9.3.5 Compute lower bound quantile values of the RMA curves

Finally, we compute the lower and upper bound quantile values of the RMA curves ($w, l, mid, etc.$) along with their corresponding quantile levels (alpha values). The lower bound alpha values are curves that take values in the interval $[0, 1]$, indicating the current position within the distribution, expressed as probabilities.

```
normalized_df = compute_lower_higher_quantiles(normalized_df, quantiles, "w")
normalized_df = compute_lower_higher_quantiles(normalized_df, quantiles, "l")
normalized_df = compute_lower_higher_quantiles(normalized_df, quantiles, "mid")
```

For instance, w_lower_alphas is the lower bound of alpha values for quantiles of w , and $w_lower_alphas_means$ is the resulting smoothed curved on a sliding window of n -minutes. This is illustrated with the plain curves in Figure (8), where the plain green curve represents $w_lower_alpha_means$, the plain magenta curve represents $mid_lower_alpha_means$, and the plain orange curve represents $l_lower_alpha_means$.

9.3.6 Stop losses and profit targets

To manage stop-loss and profit-target conditions, we define a simple function that determines whether trading should be halted. This function, displayed below, evaluates whether the net PnL has reached dynamic thresholds for profit or loss.

The logic is as follows:

- We monitor the maximum historical PnL throughout the trading session and stop trading only once that peak exceeds a dynamic profit target. Crucially, we focus on drawdowns from this peak, allowing for larger tolerances early in the session, which then progressively shrink to zero as the session progresses to the end.
- In parallel, we apply a time-decaying loss threshold that starts, for example, at 0.007 and also decreases to 0.0 as the session progresses.

This asymmetric design separates absolute loss from drawdown to avoid premature exits during profitable phases. It allows the strategy to realistically capture upside potential while becoming progressively more risk-averse. The profit cap is dynamic, enabling trades to run when in strong trends, while the loss limit tightens over time to curtail exposure in deteriorating conditions.

The advantages of this approach are threefold: it avoids premature exits during profitable runs, it mimics a trailing-stop mechanism by triggering only after significant drawdowns from the peak, and it imposes a decaying loss limit to ensure growing conservatism as the day unfolds.

```
def check_stop_trading(net_pnl, max_net_pnl, drawdown_limit, loss_limit):
    drawdown = max(max_net_pnl - net_pnl, 0)
    condition_drawdown = drawdown >= drawdown_limit
    condition_loss = net_pnl <= -loss_limit
    stop_trading_bool = condition_drawdown or condition_loss
    return stop_trading_bool
```

9.4 The RMA in action

Following the implementation of the RMA in Section (9.3), we present the indicator curves along with their associated fractile curves for the GSPC index on third of February 2025. Given that the window size is set to $window_size = 106$, and additional data points are required for smoothing, we display the curves starting from $t = 120$ onward. We display prices and rolling MA in Figure (7), prices and rolling RMA in Figure (8), prices and rolling RMA Fractiles in Figure (10), prices and rolling PnL with Extremum-Revert-Trigger in Figure (11), and prices and rolling PnL with Full-Distribution-Crossover in Figure (12).

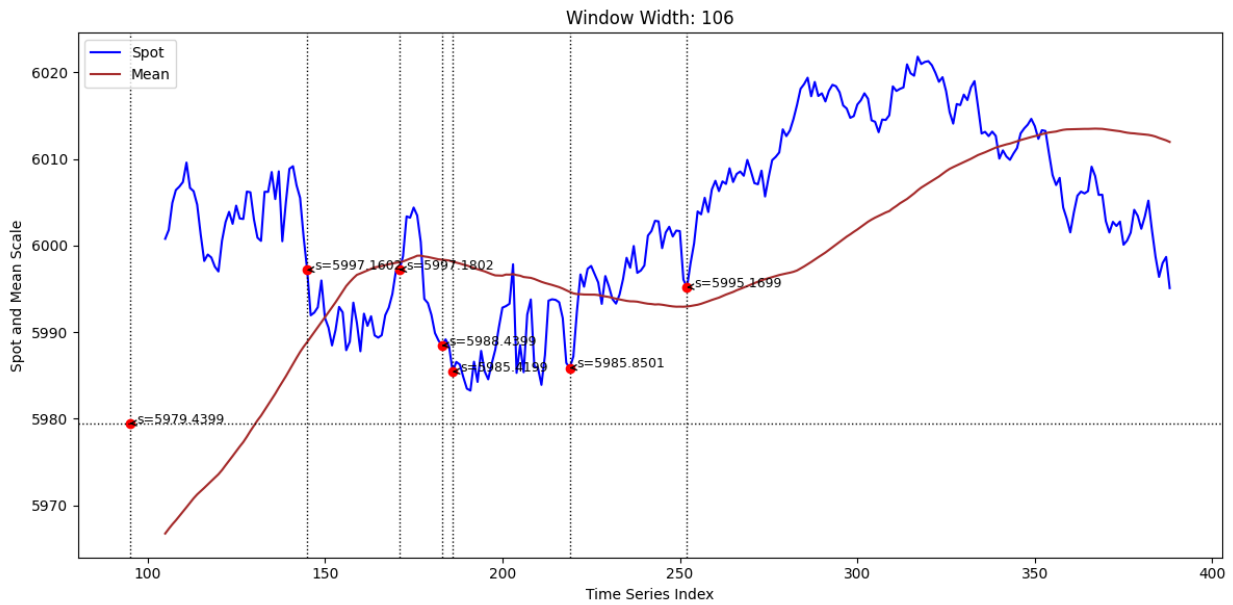


Figure 7: Prices and rolling MA for GSPC on the 03/02/25.

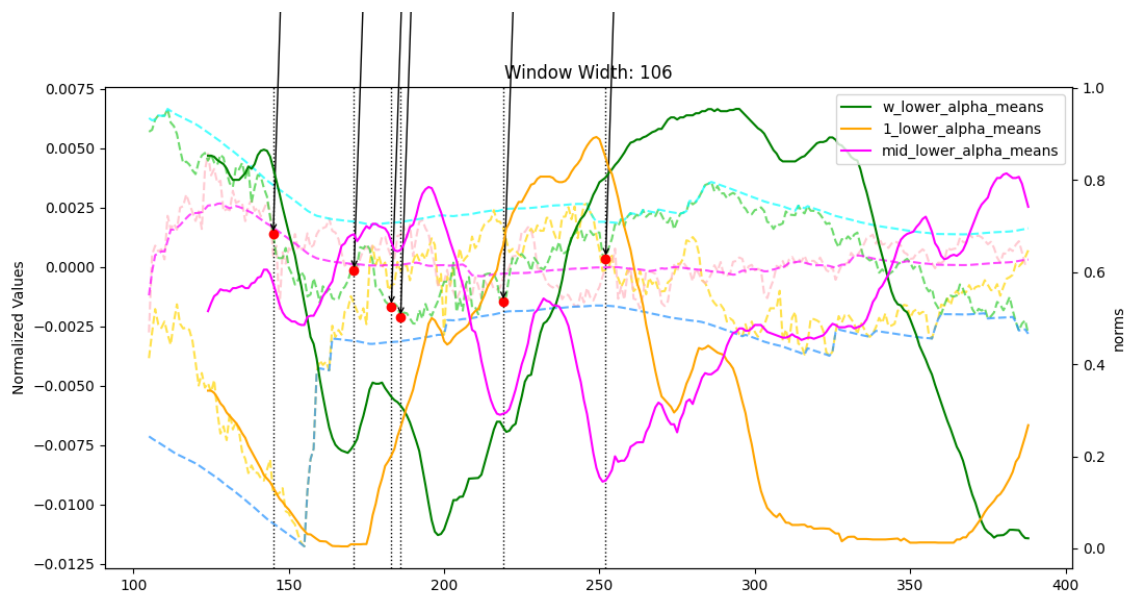


Figure 8: Rolling RMA for GSPC on the 03/02/25.

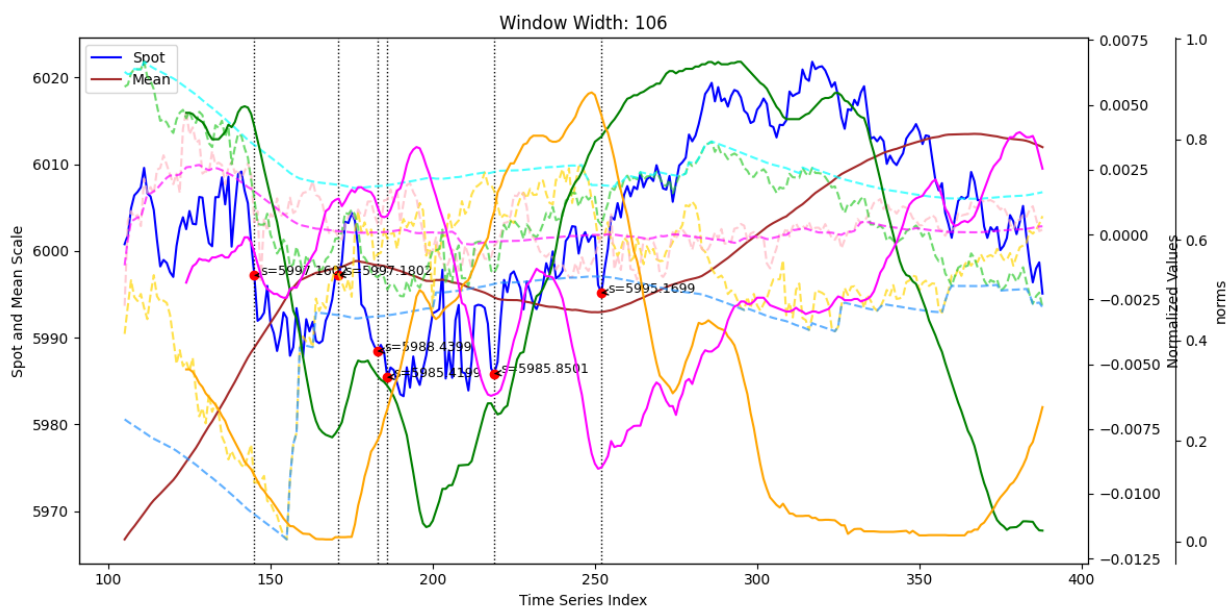


Figure 9: Prices and rolling RMA for GSPC on the 03/02/25.

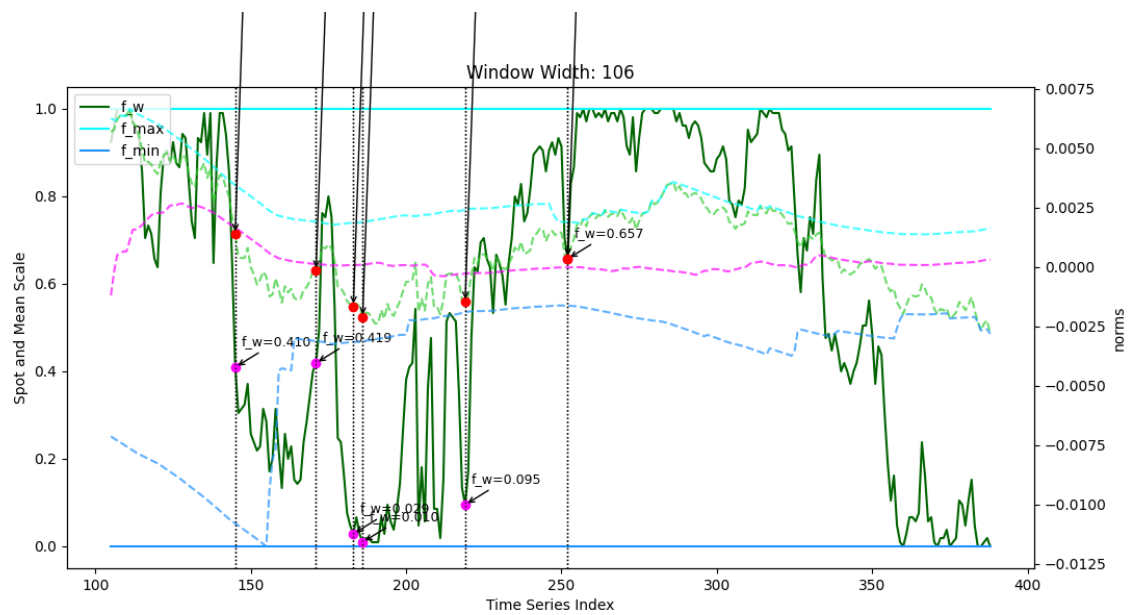


Figure 10: Rolling RMA Fractiles for GSPC on the 03/02/25.

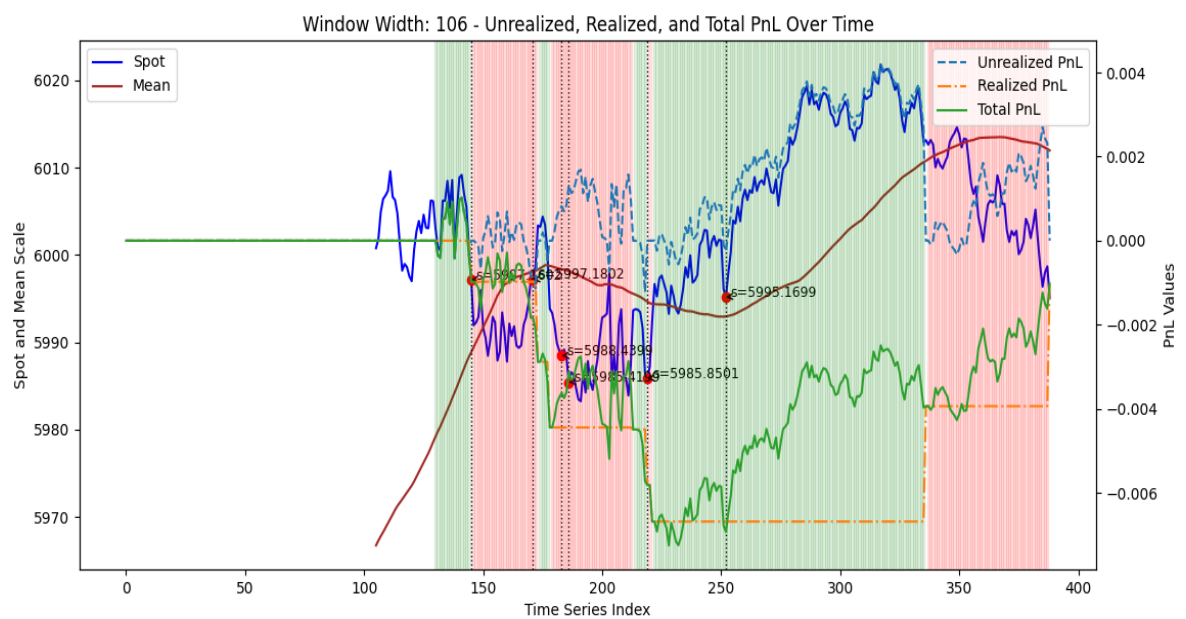


Figure 11: Prices and rolling PnL with Extremum-Revert-Trigger for GSPC on the 03/02/25.

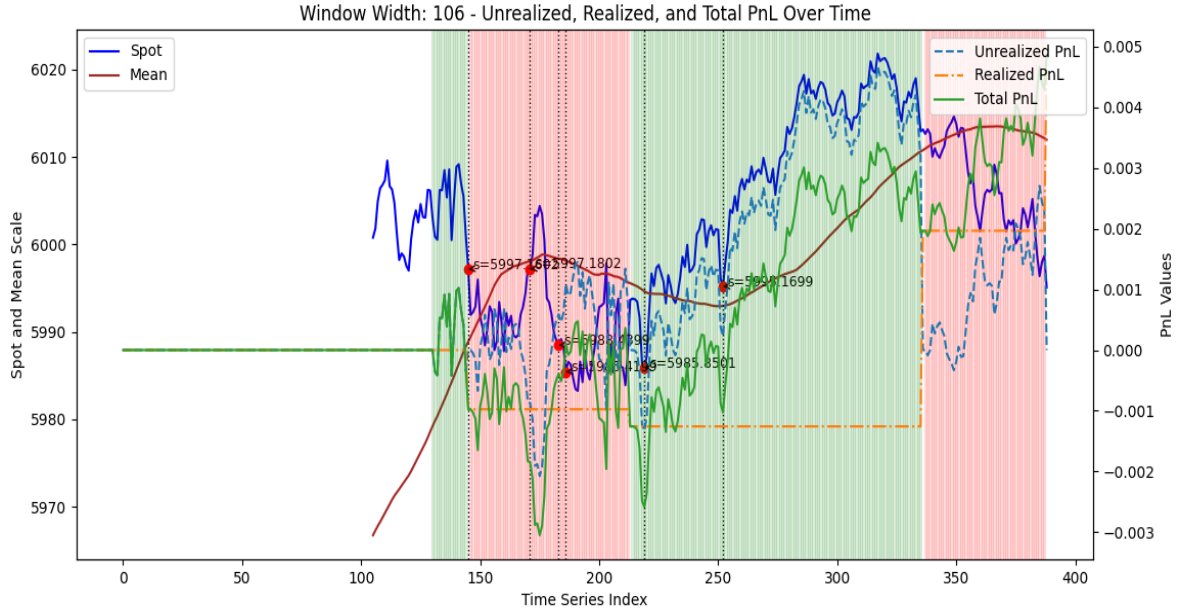


Figure 12: Prices and rolling PnL with Full-Distribution-Crossover for GSPC on the 03/02/25.

9.5 Accounting for outliers

As discussed in Section (9.3.4)) mapping the RMA indicator to fractiles on a $[0, 1]$ scale allows for direct comparison across different assets. However, this comparability breaks down when the underlying distribution is heavily skewed due to the presence of outliers. In this section, we illustrate this issue with an example, explain how outliers distort the interpretation of fractiles, and propose using a measure of closeness as a more robust alternative.

9.5.1 Description

In Figure (10), the dotted lime green curve represents the current normalised return values, denoted by w , and the solid dark green curve shows their corresponding quantile positions, f_w . The dotted blue lines indicate the minimum and maximum normalised return values, while the solid blue lines represent their quantile positions f_{min} and f_{max} at 0 and 1, respectively.

In general, when w reaches the minimum or maximum, its quantile position, f_w , typically reaches 0 or 1. This expected behaviour is observed throughout most of the graph. However, in the time range between $t = 185$ and $t = 190$, we notice a deviation from this pattern: although f_w approaches 0, the value of w , associated to S_4 in Figure (7), does not coincide with the minimum. In other words, during this interval, the last normalised value in the sliding window is not the lowest, yet its quantile position is close to the lower extreme.

9.5.2 Explanation

Quantiles are determined by rank: if a value is the lowest among all elements in its window, its quantile should be 0. However, even if $w[k]$, where k is the time index, lies numerically halfway between the median and minimum, it may still fall in the bottom 1% rank-wise if the distribution is heavily skewed. This highlights that when the window's distribution is non-uniform, the fractile reflects the rank rather than the visual height of the curve.

This can be verified by examining the ranked values within the window. The first two sorted values are $-3.05970696 \times 10^{-3}$ (the blue curve representing the minimum) and $-2.16769568 \times 10^{-3}$ (the dotted lime green curve representing w). We observe that the minimum value (approximately -0.00306) is substantially lower than the second-lowest (approximately -0.00217), creating a significant gap relative to the rest of the values. This gap skews the distribution, which in turn affects the computed quantiles.

In general, a large gap between the minimum and the second-smallest value within a window indicates that the minimum is likely an outlier, a value unusually low relative to the rest of the data. This is clearly visible in Figures (6) and (7). Recall that $window_size = 106$, so that the normalised value of S_1 , resulting from a sharp downward jump and thus considered an outlier, still remains within the sliding window at time $t = 185$ and $t = 190$.

9.5.3 Solution

Such outliers can be detected statistically (e.g., using z-scores or signatures), visually (e.g., via histograms), or handled through techniques such as clipping or Winsorizing. Whether to retain or adjust these values depends on the intended downstream application - for example, detecting anomalies versus producing a smoothed signal.

To ensure that the value w truly matches the minimum or maximum value within a window, we measure their proximity when both are of very small magnitude. We use the function `are_values_close`, which checks whether two values a and b are sufficiently close based on their relative difference (see Appendix 14.1). This function uses a threshold parameter to define the maximum allowed relative deviation between the two values. Then, the condition to satisfy to be at a minimum or maximum is that both f_w is close to 0 and w is close to the minimum or both f_w is close to 1 and w is close to the maximum.

```
condition_is_close_to_max, difference_to_max_in_value = are_values_close(w_value,
    max_value, threshold=0.1)
condition_is_close_to_min, difference_to_min_in_value = are_values_close(w_value,
    min_value, threshold=0.1)
condition_at_max_range = at_max_range and condition_is_close_to_max
condition_at_min_range = at_min_range and condition_is_close_to_min
```

9.6 Identifying expansion and contraction regimes

As discussed in Section (9.1), the RMA effectively captures key features of the underlying price dynamics, particularly the expansion and contraction phases of the distribution, which provide potential signals for trade entry and exit. In this framework, trades are typically initiated at the onset of an expansion phase and closed during contraction.

A central challenge, therefore, is the real-time identification of whether the market is currently in a regime of expansion, contraction, or transitioning between the two. Several quantitative measures can assist in this task, including the percentiles of the distribution's range, the Z-score of the range, the variation ratio, and the slope of the range. However, none of these measures individually offers consistently high predictive accuracy.

A more robust approach is to combine these complementary indicators into a unified framework. Alternatively, we introduce the concept of Relative Extremum Ratios to quantify the position of the current range relative to its historical minimum and maximum, denoted as $\mathcal{R}_k^m$ and $\mathcal{R}_k^M$ respectively.

In the following, we illustrate this methodology through a representative example.

9.6.1 A representative example

While Figure (8) illustrates an expansion regime followed by a contraction regime, it does not capture the more complex scenario of an expansion transitioning into a contraction phase, followed by a breakout. To address this, we

consider a more representative example: the FTSE index on seventh of February 2025. The corresponding price series is shown in Figure (13), and the combined price and RMA representation is provided in Figure (14).

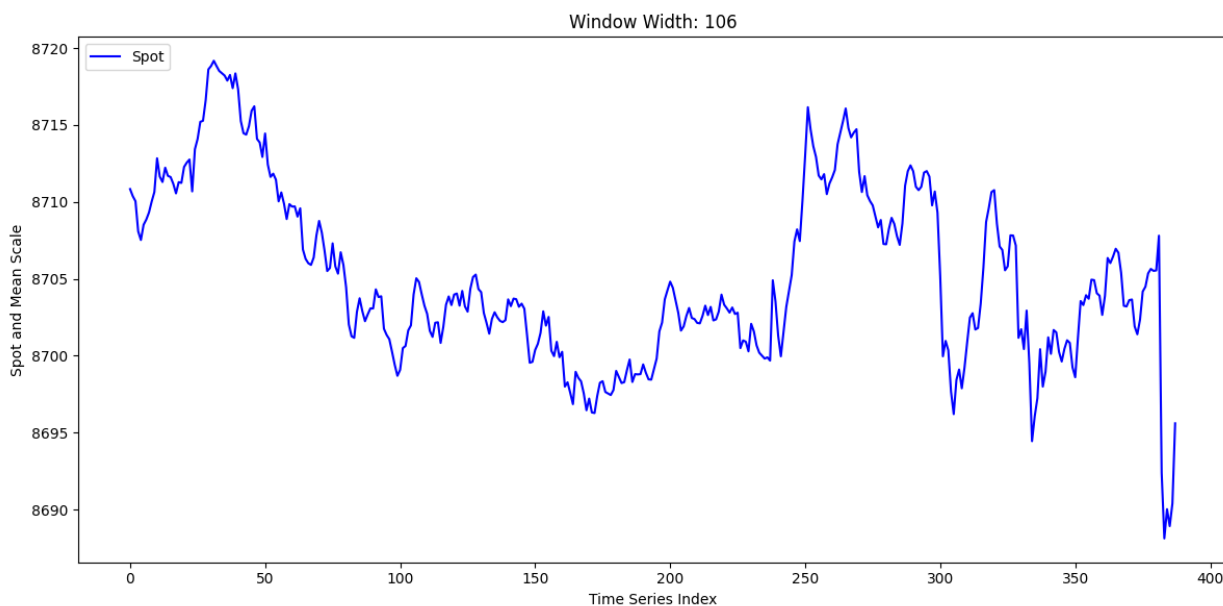


Figure 13: Prices for FTSE on the 07/02/25.

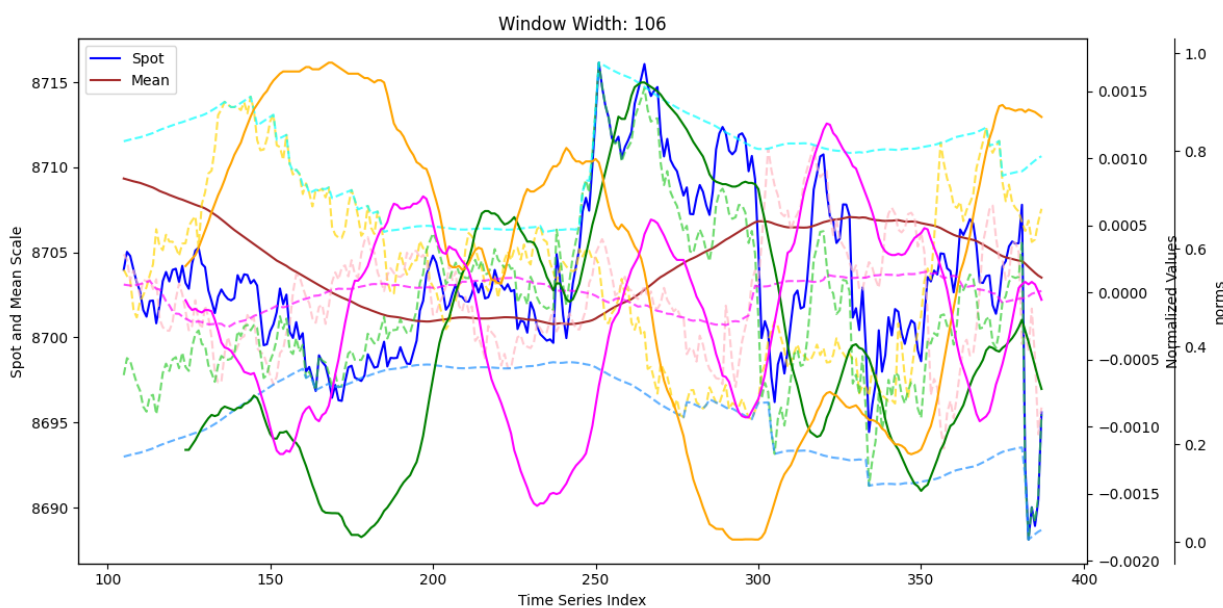


Figure 14: Prices and rolling RMA for FTSE on the 07/02/25.

We observe in Figure (13) that the price reaches a local high of 8719 at $t = 31$, before decreasing sharply to a minimum

of 8696 at $t = 170$. It then steadily rises to 8700 at $t = 241$, followed by a breakout to 8716 at $t = 251$.

As a consequence, the Moving Average in Figure (14) begins to decline from $t = 106$, reaching a plateau around $t = 193$, which lasts until approximately $t = 246$, after which it resumes a steady increase.

Regarding the normalised distribution (minimum and maximum of the RMA), it enters an expansion regime from $t = 106$ to $t = 144$, then steadily contracts, entering a transition phase, to reach a contraction regime from $t = 185$ to $t = 244$. Finally, it surges into a new expansion regime, reaching a maximum at $t = 251$.

9.6.2 Detecting contraction and expansion

Given the example presented in Section (9.6.1), at each time k we compute the minimum and maximum of the normalised distribution to determine its range, denoted by R_k . As time evolves, our objective is to detect in real time the transition phase between the expansion and contraction regimes, as well as periods of sustained contraction.

9.6.2.1 Some measures At every time k , we observe the set of historical absolute ranges $\mathcal{R}_k = \{R_0, \dots, R_k\}$. There are several techniques that we can use to measure contraction (see Appendix (14.1)). This includes

- The percentile: We can compute the percentile P_k on the distribution ranges and use it to identify if we are in contraction or in expansion. That is, $P_k < \epsilon_{p,c}$, where $\epsilon_{p,c} = 0.1$, and $P_k > \epsilon_{p,e}$, where $\epsilon_{p,e} = 0.9$, respectively. We let `d_abs_max_min_series` be a series of absolute range values, and `start_k` be the index from which to begin comparison. For global measure we use `past_ranges` and for rolling measure we use `recent_values`. Then,

```
past_ranges = d_abs_max_min_series[start_k:k]
current_range = d_abs_max_min_series[k]
historical_percentile = np.sum(past_ranges <= current_range) / len(past_ranges)
Contraction = historical_percentile <= threshold_down
Expansion = historical_percentile >= threshold_up
```

- The Z-score: We get contraction if $Z_k < \epsilon_{z,c}$, where $\epsilon_{z,c} = -1.0$, and expansion if $Z_k > \epsilon_{z,e}$, where $\epsilon_{z,e} = 1.0$.

```
z_score = (current_range - np.mean(past_ranges)) / (np.std(past_ranges) + eps)
Contraction = z_score < -threshold_z_c
Expansion = z_score > threshold_z_e
```

- The rolling standard deviation: Track the rolling standard deviation of the ranges over time. Long-lasting low-volatility contractions will have persistently low rolling standard deviation. We get contraction if $\sigma_k < \epsilon_{s,c}$, and expansion if $\sigma_k > \epsilon_{s,e}$.

```
rolling_std = pd.Series(past_ranges).rolling(window=lookback_for_trend).std().iloc[-1]
Contraction = rolling_std < std_threshold_down
Expansion = rolling_std > std_threshold_up
```

- The variation ratio: Track the variation ratio of the ranges over time. We get contraction if $V_k < \epsilon_{v,c}$ and expansion if $V_k > \epsilon_{v,e}$.

```
variation_ratio = np.std(past_ranges) / (np.mean(past_ranges) + eps)
Contraction = variation_ratio < low_variation_threshold
Expansion = variation_ratio > high_variation_threshold
```

- The rolling variation ratio: Track the rolling variation ratio of the ranges over time. We get contraction if $V_k < \epsilon_{v,c}$ and expansion if $V_k > \epsilon_{v,c}$.

```
recent_values = d_abs_max_min_series[k - lookback_for_trend:k]
variation_ratio = np.std(recent_values) / (np.mean(recent_values) + eps)
Contraction = variation_ratio < low_variation_threshold
Expansion = variation_ratio > high_variation_threshold
```

- The rolling slope: Track the rolling slope of the ranges over time. We get contraction if $S_k < \epsilon_{s,c}$ and expansion if $S_k > \epsilon_{s,e}$.

```
slope_recent = np.polyfit(np.arange(lookback_for_trend), recent_values, deg=1)[0]
Contraction = slope_recent < threshold_down
Expansion = slope_recent > threshold_up
```

- The rolling directional change ratio: Track the rolling imbalance ratio of the ranges over time. We get contraction if $D_k < \epsilon_{d,c}$, where $\epsilon_d = 0.45$, and expansion if $D_k > \epsilon_{d,e}$, where $\epsilon_{d,e} = 0.55$.

```
diff = np.diff(recent_values)
min_step = 0.02 # example threshold to reduce false triggers from noise
num_drops = np.sum(diff < -min_step)
num_increases = np.sum(diff > min_step)
Contraction = num_drops >= lookback_for_trend // 2
Expansion = num_increases >= lookback_for_trend // 2
```

- Quantile of recent values: Computes the 30th/70th percentile of those *recent_values*, denoted *quantile_recent*. That is, 30% of *recent_values* are $\leq$ to *quantile_recent* and 70% of *recent_values* are $\geq$ to *quantile_recent*. In contraction, we want the current range to be lower than the former value, and in expansion higher than the latter value.

```
quantile_recent_down = np.quantile(recent_values, 0.3)
quantile_recent_up = np.quantile(recent_values, 0.7)
Contraction = current_range < quantile_recentt_down
Expansion = current_range > quantile_recentt_up
```

- Quantile smoothing: Smooth percentiles with exponential moving averages (EMA) or weighted rolling averages to prevent noise.

```
smoothed_percentile = EMA(percentile_series, alpha=0.1)
contraction = smoothed_percentile < threshold_down
Expansion = smoothed_percentile > threshold_up
```

Alternatively, we can require persistence across multiple samples, like:

```
condition_recent_contraction = all(p <= threshold_down for p in last_N_percentiles
)
condition_recent_expansion = all(p >= threshold_up for p in last_N_percentiles)
```

- Combine Quantile smoothing with rolling standard deviation: We get

```
Contraction = (smoothed_percentile < threshold_down and rolling_std <
std_threshold_down)
Expansion = (smoothed_percentile > threshold_up and rolling_std > std_threshold_up
)
```

- State Machine / Regime Detection: Treat contraction/expansion as regimes. Model the transitions explicitly using Hidden Markov Models (HMMs) or Bayesian Filters.

9.6.2.2 Problems However, none of these measures individually provides consistently high predictive accuracy. Several challenges and limitations are associated with their use:

- **Percentile:**

- As soon as the distribution enters a transition phase - here, decreasing from $t = 145$ till $t = 184$ - each new current value R_k pulls the percentile closer to zero. This leads to a false positive, where the condition suggests that we are already in contraction, even though we are merely transitioning from an expansion regime.
- *Drift over time:* As new data accumulates, the position of the current range relative to the historical distribution fluctuates, even if the underlying range remains relatively stable.
 - * In flat ranges (during contraction), percentiles tend toward ≈ 0.5 regardless of the actual contraction state. In such cases, percentiles become statistically uninformative, as each new value remains close to the central tendency.
 - * Long-lasting contractions increase the likelihood of false negatives.

- **Z-score:**

- Similarly, when the distribution enters a decreasing transition phase, each new value R_k progressively drives the Z-score toward extreme negative values (e.g., $Z_k \approx -3$), incorrectly indicating a strong contraction. This again results in a false positive, signalling contraction when the system is still transitioning.
- When the distribution breaks from a contraction phase into a strong expansion, the Z-score exhibits a lagging response. Initially, it remains around 0 (at the mean) and only gradually increases, eventually exceeding 1 and then 2 standard deviations. This delayed reaction can cause missed or late signals for the onset of expansion.
- As the time index transitions back into an expansion regime following a contraction, the mean converges toward the range maximum, and the Z-score shifts slightly above zero. This leads to poor detection of prolonged expansion regimes and increases the likelihood of false negatives.

- **Rolling standard deviation:**

- Double counting discontinuities: The rolling standard deviation is highly sensitive to new values entering or exiting the window, often producing abrupt jumps in its estimate (see Section (5.1.2)). These fluctuations can lead to both false positives and false negatives when identifying regime changes.
- Delayed adaptation in transitional phases: When entering a contraction regime after transitioning from an expansion regime, the rolling window incorrectly interprets the past oscillations from the transition phase as a new surge in volatility. This artificially inflates estimates of volatility or skewness, even though these fluctuations are part of a completed event. Conversely, when transitioning into an expansion phase, it takes time for the older, non-event elements within the window to be fully replaced by the more volatile events, delaying the responsiveness of volatility estimates.
- Tuning parameters: Selecting an appropriate threshold to reliably distinguish between contraction and expansion phases is particularly challenging, as the optimal value tends to vary with market conditions.

- **Rolling slope:** As illustrated in Figure (14), although the overall slope of the distribution range decreases during the transition phase from an expansion regime to a contraction regime, the local behaviour of the slope is highly oscillatory. Consequently, the rolling estimate of the slope fluctuates, intermittently indicating periods of increasing or decreasing slope. This leads to false positives in regime detection. Similarly, during sustained phases of contraction or expansion, the slope is not perfectly constant but exhibits small fluctuations, which can generate spurious signals of regime change when none is actually occurring.

9.6.2.3 Integrating multiple measures We have seen above that conditions based on percentiles and Z-scores are reasonably effective for detecting contraction or expansion regimes but struggle to reliably capture transition phases. Additionally, we observed that the standard deviation is highly sensitive to changes in the distribution range, making it useful primarily for detecting flat ranges and, by extension, contraction regimes. The same limitation applies to the rolling variation ratio, as it directly depends on the standard deviation. Similarly, the slope of the distribution range is sensitive to variations in the range itself, making it effective for identifying flat ranges (contraction) but unreliable during transition phases due to the inherent oscillations in the range.

As a result, none of these measures individually provides consistently high predictive accuracy. A natural solution is to combine several of these indicators, with the expectation that some false positives and false negatives will offset one another.

Assuming that the contraction and expansion regimes are relatively stable over time, we employ percentiles and Z-scores as primary signals for regime identification. These are combined with additional conditions on the standard deviation, rolling variation ratio, and slope of the distribution range. Specifically, in the case of contraction, we require negligible standard deviation, low variation ratio, and a near-zero slope. This approach is illustrated in the code snippet below:

```
condition_contraction_variation_ratio = variation_ratio < eps_stability
condition_contraction_slope = abs(slope_recent_range) < eps_slope
condition_contraction_measures = condition_contraction_variation_ratio or
    condition_contraction_slope
condition_contraction_percentile_or_z_score = condition_contraction_percentile or
    condition_contraction_z_score
condition_contraction = condition_contraction_percentile_or_z_score and
    condition_contraction_measures
```

In contrast, when detecting a transition phase (e.g., from expansion to contraction), we expect to observe a decreasing slope, elevated volatility, and a high variation ratio. We again utilise the percentiles and Z-score, but combine them with conditions on high standard deviation, high rolling variation ratio, and a negative slope. This is illustrated in the code snippet below for the case of a transition toward contraction:

```
condition_toward_contraction_variation_ratio = not
    condition_contraction_variation_ratio
condition_toward_contraction_slope = slope_recent_range < -eps_slope
condition_toward_contraction_num_drops = num_drops >= lookback_for_trend // 2
condition_toward_contraction_imbalance = imbalance_ratio < 0.45
condition_toward_contraction_measures = condition_toward_contraction_variation_ratio
    and condition_contraction_variation_ratio and condition_toward_contraction_slope
    and condition_toward_contraction_num_drops and
    condition_toward_contraction_imbalance
condition_toward_contraction = condition_contraction_percentile_or_z_score and
    condition_toward_contraction_measures
```

9.6.3 Pattern-based regime detection using Relative Extremum Ratios

Another approach for detecting regimes in real time is to interpret the observed sequence of expansion followed by contraction, and then another expansion, illustrated in Figure (14), as a recurring pattern. Geometrically, this sequence resembles a bicone tube with a cylindrical waist: a funnel-like shape featuring two conical ends connected by a straight cylindrical segment. More precisely, it consists of a horizontal cone tapering inward, followed by a cylindrical section of constant diameter, and then another horizontal cone flaring outward in the opposite direction.

We detect regimes and their transition phases by comparing the current range of the distribution to its historical minimum and maximum values. We refer to these measures as the Relative Extremum Ratios, representing normalised

range ratios relative to historical extremes. This approach is straightforward to implement and avoids many of the pitfalls associated with rolling statistical measures, such as double-counting discontinuities and sensitivity to non-stationary data (see Section (5.1)).

At any given time k , we consider the set of historical absolute ranges $\mathcal{R}_k = \{R_0, \dots, R_k\}$, where R_k is the current range. From this set, we compute the minimum and maximum range values, denoted by $m_k = \min(\mathcal{R}_k)$ and $M_k = \max(\mathcal{R}_k)$, respectively. We then define the relative positions of the current range with respect to these historical extremes by

- Minimum Extremes Ratio: $\mathcal{R}_k^m = \frac{R_k}{m_k}$
- Maximum Extremes Ratio: $\mathcal{R}_k^M = \frac{R_k}{M_k}$

These Extremes Ratios allow us to distinguish between contraction and expansion regimes, as well as their transition phases, according to the following conditions:

- Contraction: $|\mathcal{R}_k^m| \approx 1$ and $\mathcal{R}_k^M < \epsilon_{low}$
- Expansion: $|\mathcal{R}_k^M| \approx 1$ and $\mathcal{R}_k^m > \epsilon_{high}$
- Transition to contraction: $\epsilon_{low} < \mathcal{R}_k^M < 1$ and $1 < \mathcal{R}_k^m < \epsilon_{high}$
- Transition to expansion: $\epsilon_{low} < \mathcal{R}_k^M < 1$ and $1 < \mathcal{R}_k^m < \epsilon_{high}$

Since the mathematical conditions for transitions to contraction and expansion are identical, we disambiguate them using the rolling slope S_k of the target range, computed as described previously. A positive slope indicates an increasing range (i.e., a transition toward expansion), while a negative slope indicates a decreasing range (i.e., a transition toward contraction). Incorporating the slope leads to the refined conditions for transitions:

- Transition to contraction: $\epsilon_{low} < \mathcal{R}_k^M < 1$ and $1 < \mathcal{R}_k^m < \epsilon_{high}$ and $S_k < \epsilon_{s,c}$
- Transition to expansion: $\epsilon_{low} < \mathcal{R}_k^M < 1$ and $1 < \mathcal{R}_k^m < \epsilon_{high}$ and $S_k > \epsilon_{s,e}$

9.7 Trading strategy

9.7.1 Description

As discussed in Section (9.2.2), the Relative Moving Average (RMA) indicator quantifies the deviation of a stock's price from its dynamic equilibrium level, represented by the sliding moving average (SMA). This deviation is projected onto a dynamic price distribution that evolves within a sliding window. The window size is calibrated so that prices oscillate naturally around the SMA, forming cycles of expansion and contraction (see Figure (7)).

These oscillations generate a time-varying distribution of deviations, which serves as the basis for a responsive trading strategy. Within this framework, the normalised price levels, denoted by $(w, mid, 1)$, fluctuate between the minimum and maximum of the distribution (see Figure (8)). The strategy aims to 'ride' these oscillations: entering long positions when prices move from the minimum toward the maximum, and short positions when they reverse from the maximum toward the minimum.

9.7.1.1 Detection of long-cross and short-cross reversals To systematically monitor trading opportunities, we track both the position and directional movement of the normalised current price, w , within its evolving distribution. We define a **long-cross-reverse** as a downward movement of the price, originating near the maximum of the distribution and traversing toward the minimum. Conversely, a **short-cross-reverse** refers to an upward movement starting near the minimum and rising toward the maximum.

A long-cross-reverse is identified when either (1) the normalised price w is located in the upper quadrant, above the median, or (2) when w is in the lower quadrant, between the median and the minimum. Similarly, a short-cross-reverse is detected either (1) when w lies in the lower quadrant, below the median, or (2) when it is positioned in the upper quadrant between the median and the maximum.

9.7.1.2 Exit conditions for the Relative Moving Average strategy We consider two distinct exit conditions for the RMA strategy, each designed to respond differently to anomalies in price formation:

1. **Full Distribution Crossover Exit:** The first exit condition assumes that the price series fully traverses the distribution, moving from its maximum to its minimum or vice versa. Under this scenario, the exit condition for a short-cross-reverse coincides with the entry condition for a long-cross-reverse, and similarly in the opposite direction. This symmetric behaviour enables a consistent, rule-based mechanism for switching between long and short positions.
2. **Extremum Revert Trigger:** The second exit condition allows for partial traversals. The price may begin to move across the distribution from maximum to minimum (or the reverse), but then revert back toward the originating extremum, either before reaching the median, or, shortly after crossing it. In such cases, the strategy exits the short-cross or long-cross-reverse position when the normalised price, w , is detected to revert direction and return toward the distribution's extremum. When we exit a **long-cross-reverse**, we immediately enter a **long-cross-revert**, and conversely, exiting a **short-cross-reverse** leads to entering a **short-cross-revert**. In other words, the strategy transitions directly from a short to a long position, or from a long to a short position, maintaining continuous market exposure through systematic position reversal.

9.7.1.3 Adaptive exit conditions using regime detection We can clearly observe in Figure (14) that during a contraction regime, the trajectory of the normalised price w is tightly constrained within the cylindrical section of constant diameter. This confinement forces w to traverse from the minimum to the maximum of the distribution, or vice versa, before breaking out during a transition phase toward an expansion regime. Furthermore, in Section (9.6), we introduced the Relative Extremum Ratios as a real-time indicator to distinguish whether the market is currently in a regime of expansion, contraction, or transitioning between the two.

Building on this framework, we propose a third exit condition that blends the two previously defined mechanisms. Specifically, we employ the Extremum Revert Trigger during expansion or transition phases, where partial traversals and reversals are likely. However, once a contraction regime is identified, we switch to the Full Distribution Crossover Exit, which is better suited to the constrained dynamics of the contraction phase. This combined approach, called Adaptive Crossover Exit, allows the strategy to adapt dynamically to prevailing market conditions, improving robustness across different volatility regimes.

9.7.2 Quantitative tools for the RMA

As discussed in Section (8.1), the price series in Figure (6) displays a combination of momentum and mean-reversion regimes, punctuated by extreme events such as jumps, stop-runs, fakeouts, and W-shaped reversal traps. To make effective trading decisions in such a complex and evolving environment, it is essential to incorporate quantitative tools that capture both the local behaviour of the asset and the broader structural context of the time series. This dual

perspective helps mitigate the risk of overreacting to short-term noise or falling into local traps. To address this, we now turn to the development of quantitative tools based on the Relative Moving Average (RMA) indicator.

As detailed in Section (9.2.1), the RMA is constructed relative to the sliding moving average (SMA), which serves as a dynamic equilibrium point and is fixed at zero within the RMA framework (see Figure (8)). Consequently, the RMA does not directly convey the value of the SMA itself but instead encodes the deviation of key price statistics, such as the first, middle, last, minimum, and maximum values of a segment, from this equilibrium. To overcome this limitation, the trading strategy must integrate the local, relative insights provided by the RMA with the absolute, global perspective offered by the SMA (see Figure (9)). This combined approach enables a more robust characterization of the time series, improving the reliability of trading decisions across varying market conditions.

To analyse the position of the normalised current price, w , within the RMA distribution and to detect barrier crossings, we begin by estimating its relative location. Using the median as a reference point, we partition the upper half of the distribution (from the median to the maximum) into ascending quantile bins, ranging from 0.05 to 0.95. The lower half of the distribution (from the median to the minimum) is partitioned similarly.

At each time step k , we compute the normalised distance between w and the median, denoted $d_{med,w}$. This value is used to determine the bin in which w resides at both time steps k and $k - 1$, as well as whether the bin index has increased, decreased, or remained the same. For each bin, we track whether $d_{med,w}$ has entered, exited, or stayed within that interval, and whether the movement direction is ascending or descending. Based on this information, we define threshold levels (e.g., 20th, 25th, ..., 95th percentiles) and monitor whether $d_{med,w}$ crosses these thresholds. This allows us to determine whether w is crossing the median from below and moving toward the maximum, or conversely, descending from the maximum back toward the median. The same logic applies symmetrically for movements in the lower half of the distribution.

This framework also extends naturally to the analysis of f_w , enabling the detection of quantile crossings for additional dimensions of the RMA indicator.

We also monitor instances when f_w approaches or reaches the lower and upper bounds of its range within the RMA, denoted f_{min} and f_{max} , respectively.

```
set_f_w_around_max = condition_at_max_range and (w_q_lower_mid_up_trend or f_w_diff >
0)
set_f_w_around_min = condition_at_min_range and (w_q_lower_mid_down_trend or f_w_diff
< 0)

if not set_f_w_around_min: # f_w is reaching max
    f_w_around_max, f_w_around_max_index = set_cross_base_k(f_w_around_max,
        set_f_w_around_max, k, f_w_around_max_index)
else:
    f_w_around_max = False

if not set_f_w_around_max: # f_w is reaching min
    f_w_around_min, f_w_around_min_index = set_cross_base_k(f_w_around_min,
        set_f_w_around_min, k, f_w_around_min_index)
else:
    f_w_around_min = False
```

Using the same logic, we detect when f_w is crossing the median upward or downward.

We identify long-term SMA trends and their turning points. Initially, we detect local turning points and count their occurrences over the past n minutes. This information is then used to infer broader SMA patterns as the trading day progresses. Additionally, we continuously monitor and update the most recent local trend of the SMA in real time.

```

ma_min_vec = df['mean_min'].iloc[cum_mean_norm_start_index:k+1].values
ma_max_vec = df['mean_max'].iloc[cum_mean_norm_start_index:k+1].values
ma_local_turning_points_vec = detect_ma_local_turning_points(ma_min_vec, ma_max_vec)
tuple_ma_count_turning_points = count_ma_local_turning_points(
    ma_local_turning_points_vec, 10)
list_of_ma_detected_patterns = detect_ma_patterns(tuple_ma_count_turning_points,
    min_percent_count=min_percent_count)
list_ma_last_trends, global_dominant_trend = detect_ma_last_trends(
    tuple_ma_count_turning_points, min_percent_count=min_percent_count)

```

9.7.3 Implementation of the strategy

We briefly outline parts of the implementation of the RMA trading strategy described in Section (9.7.1) and introduce a set of basic algorithms governing the entry and exit conditions.

9.7.3.1 Key conditions We list below a few key conditions of the strategy and give their definition:

- Enter long-cross revert, denoted *selected_cross_up_for_revert_event*: long-cross reversal movement reverts before reaching the minimum. The normalised price w go back up towards the maximum.
- Enter short-cross revert, denoted *selected_cross_down_for_revert_event*: short-cross reversal movement reverts before reaching the maximum. The normalised price w go back down towards minimum.
- *set_crossing_quantiles_down*: Set crossing quantiles downwards
- *set_crossing_quantiles_up*: Set crossing quantiles upwards
- *set_long_cross_reverse*: Set the long-cross-reverse entry.
- *set_short_cross_reverse*: Set the short-cross-reverse entry.
- *set_long_cross_revert*: Set the long-cross-revert entry.
- *set_short_cross_revert*: Set the short-cross-revert entry.
- Enter short position, long-cross-reverse *w_q_lower_long_cross_reverse*: The normalised price w comes from the maximum and goes to the minimum. (1) above the median, (2) from the median to the minimum.
- Enter long position, short-cross-reverse *w_q_lower_short_cross_reverse*: The normalised price w comes from the minimum and goes to the maximum. (1) below the median, (2) from the median to the maximum.
- Enter long position, long-cross-revert, *w_q_lower_long_cross_revert*: The normalised price w goes up, f_w is reverting back to the maximum.
- Enter short position, short-cross-revert, *w_q_lower_short_cross_revert*: The normalised price w goes down, f_w is reverting back to the minimum.
- Exit long-cross position *long_cross_reach_min_and_up*: long-cross reversal comes from the minimum and goes up towards the maximum.
- Exit short-cross position *short_cross_reach_max_and_down*: short-cross reversal comes from the maximum and goes down towards the minimum.
- Exit long position / enter short position *long_cross_exit*: (1) the normalised price w reaches the minimum and goes up, (2) the normalised price w reverts to the maximum.

- Exit short position / enter long position *short_cross_exit*: (1) the normalised price w reaches the maximum and goes down, (2) the normalised price w reverts to the minimum.

```

if long_cross_exit:
    w_q_lower_long_cross_reverse, w_q_lower_long_cross_reverse_before_med = False,
    False
else:
    w_q_lower_long_cross_reverse, w_q_lower_long_cross_reverse_before_med =
    set_double_cross_base(w_q_lower_long_cross_reverse,
    w_q_lower_long_cross_reverse_before_med, set_long_cross_reverse,
    crossing_quantiles_down_above_med)

if short_cross_exit:
    w_q_lower_short_cross_reverse, w_q_lower_short_cross_reverse_before_med =
    False, False
else:
    w_q_lower_short_cross_reverse, w_q_lower_short_cross_reverse_before_med =
    set_double_cross_base(w_q_lower_short_cross_reverse,
    w_q_lower_short_cross_reverse_before_med, set_short_cross_reverse,
    crossing_quantiles_up_below_med)

```

9.7.3.2 Basic algorithms We present the pseudo-code for the entry and exit conditions for the long-cross-reverse, short-cross-reverse, long-cross-revert, and short-cross-revert components of the strategy:

Algorithm 1 Entry long-cross-reverse / short-cross-reverse conditions.

Require: The current time t_k , the normalised price levels $(w, mid, 1)$.

- 1: Set *set_long_cross_reverse* and *set_short_cross_reverse*
 - 2: **if** $w_q_lower_long_cross_reverse$ **then**
 - 3: Go short
 - 4: Set $t_{entry} = t_k$
 - 5: **else if** $w_q_lower_short_cross_reverse$ **then**
 - 6: Go long
 - 7: Set $t_{entry} = t_k$
 - 8: **else**
 - 9: Do nothing
 - 10: **end if**
-

Algorithm 2 Exit long-cross-reverse / short-cross-reverse conditions.

Require: The current time t_k , the normalised price levels $(w, mid, 1)$.

- 1: Set *long_cross_reach_min_and_up* and *long_cross_revert_to_max_before_min*
 - 2: Set *short_cross_reach_max_and_down* and *short_cross_revert_before_max*
 - 3: **if** *long_cross_exit* **then**
 - 4: Exit short position
 - 5: Set $t_{exit} = t_k$
 - 6: **else if** *short_cross_exit* **then**
 - 7: Exit long position
 - 8: Set $t_{exit} = t_k$
 - 9: **end if**
-

Algorithm 3 Entry long-cross-revert / short-cross-revert conditions.

Require: The current time t_k , the normalised price levels $(w, mid, 1)$.

```

1: Set set_long_cross_revert and set_short_cross_revert
2: if w_q_lower_short_cross_revert then
3:   Go short
4:   Set  $t_{entry} = t_k$ 
5: else if w_q_lower_long_cross_revert then
6:   Go long
7:   Set  $t_{entry} = t_k$ 
8: else
9:   Do nothing
10: end if

```

Algorithm 4 Exit long-cross-revert / short-cross-revert conditions.

Require: The current time t_k , the normalised price levels $(w, mid, 1)$.

```

1: if short_cross_revert_exit then
2:   Exit short position
3:   Set  $t_{exit} = t_k$ 
4: else if long_cross_revert_exit then
5:   Exit long position
6:   Set  $t_{exit} = t_k$ 
7: end if

```

10 Limits of systematic trading under uncertainty

We saw in Section (8) that price anomalies such as price jumps, fakeouts, reversals, and sharp discontinuities were fundamentally unpredictable. As a result, systematic trading decisions must be based on probabilities rather than certainties. The goal is not to forecast every anomaly, but to respond intelligently and adaptively to evolving market dynamics. This involves interpreting price structure in real time and adjusting to shifting statistical patterns. With sufficient historical data, a model can be trained to probabilistically classify events, estimate the likelihood of various outcomes, and act based on the most probable scenario.

However, even when strategies are built on strong statistical foundations and robust historical analogues, they remain vulnerable to the intrinsic uncertainty of financial markets. Each event is a new draw from a complex, high-dimensional, and partially observable distribution. For instance, two seemingly identical setups may lead to vastly different outcomes due to latent variables or unobservable market forces. This irreducible risk, uncertainty that no model or amount of data can eliminate, sets a fundamental limit on the performance of systematic strategies.

Using the case study presented in Section (8.1.2), we will analyse eight key decision points $\mathcal{S} = \{S_0, \dots, S_6\}$ occurring at times $\tau = \{49, \dots, 252\}$, through the lens of two canonical strategy types introduced in Section (3.1): momentum and mean reversion. We are going to show that in the presence of fakeouts or reversal traps, both strategies may initially capture gains during brief directional moves, but then suffer losses when the market sharply reverses. These losses often outweigh the short-term profits, exposing the vulnerability of such models to deceptive and transient market events.

These cases illustrate the structural limitations of purely reactive strategies. Whether based on trend-following or mean-reverting signals, systematic approaches can fail in the face of rapid regime shifts or manipulative price action.

Such events invalidate simple technical signals and highlight a common blind spot: while quantitative strategies can respond to market conditions, they do not inherently anticipate structural breaks or discontinuities. This underscores the importance of incorporating robust risk management and adaptive mechanisms into strategy design.

10.1 Price jumps

In Figure (6), we observe the following market behaviour:

- From time $t = 0$ to $t = 49$, the price declines steadily from above 5940 to a local minimum of $S_0 = 5924.46$ at $t = 49$ (indicated by the first red dot).
- At $t = 50$, the price experiences a sharp upward move and enters a strong rally phase, jumping to 5971 at $t = 54$ and reaching 6002 at $t = 71$.

Assuming that only data up to $t = 49$ is available (i.e., excluding any knowledge of future movements), the trading system observes a sustained downtrend and has likely initiated a short position earlier. At this decision point, it must evaluate whether to maintain the short exposure, close the position, or reverse into a long trade.

A momentum-based strategy, which relies on trend persistence, would typically opt to remain short, expecting further downside continuation. Conversely, a mean-reversion strategy might interpret the recent decline as an overshoot and choose to exit the short or even initiate a long position, anticipating a reversion toward the mean.

The following analysis explores how these two strategic approaches would interpret and respond to the observed price structure under uncertainty.

- **Momentum Strategy:** Under a momentum or trend-following strategy, the most likely expectation is that the price will continue to decline, and the appropriate decision is to remain in the short position. From a purely momentum-based perspective, initiating a long position at $t = 49$ would be unjustified, as all local information up to that point supports maintaining the short exposure.
- **Mean-Reversion Strategy:** In a mean-reversion or contrarian framework, the system might consider the following real-time actions at $t = 49$: if the price has reached a statistically significant low (e.g., Z-score < -2), closing the short position to realise profits could be a prudent choice. Some aggressive strategies even interpret strong downside extremes as buy signals, initiating long positions without requiring confirmation. Such actions represent pure contrarian mean-reversion signals, not because the model anticipates an upcoming rebound, but because it is programmed to reduce risk or fade statistically extreme moves.

Mean reversion systems are designed to fade large deviations rather than follow prevailing trends. Without foreknowledge of the sharp rebound that occurs shortly afterwards, reversing to a long position at $t = 49$ is risky and would typically be viewed as premature, unless it is supported by additional reversal indicators, such as a spike in volume or signs of volatility compression. In this particular case, however, the price movement does not constitute a pronounced downward extreme, which limits the justification for a contrarian entry.

10.2 Stop-run or fakeout

10.2.1 Description

In Figure (6), we observe that following the local minimum at $t = 49$, where the price reached $S_0 = 5924.46$, the market entered a strong upward rally. This move culminated in a local high of approximately 6002 at $t = 71$. Subsequently, the price action transitions into a consolidation phase, characterised by sideways movement. This period of indecision ends with a pullback to a local low of $S_1 = 5979.44$ at $t = 95$.

This sequence illustrates a common pattern in intraday dynamics: an initial breakout followed by mean-reverting behaviour or profit-taking, raising a new decision point for any strategy managing an open long position. The following analysis explores how momentum and mean reversion approaches would interpret and respond to this observed price structure under uncertainty.

- **Momentum Strategy:** Before $t = 95$, momentum indicators (e.g., return over window, moving average slope) are likely strongly positive. Enter long during the early rally (perhaps between $t = 60$ and $t = 70$), and hold the position if the strength continues. At $t = 95$, the price has plateaued and might be starting to stall. Without confirmed reversal signals (e.g., negative returns, MA crossover, loss of trend strength), momentum systems often stay long. Exiting here would typically require evidence of trend weakening (like RSI divergence, decreasing volume, volatility contraction), or take-profit threshold. A pure momentum system would likely still be long.
- **Mean-Reversion Strategy:** Before $t = 95$, the price has deviated significantly above recent averages. Mean-reversion indicators (e.g., Z-score of return vs moving average, Bollinger Bands) are likely flashing overbought. At $t = 95$, exit long to lock in profits if already long from the bottom, or even initiate a short position assuming the move is unsustainable. This is because the move may be interpreted as an overshoot, especially if the volatility-adjusted deviation from the mean is high (e.g., > 2 standard deviations).

10.2.2 The challenge of false signals and stop-runs

Limitations inherent with Risk Management System:

- In the case of a momentum strategy, a well-designed trading system would exit the long position at $t = 49$ either due to a stop-loss being triggered or through the detection of a trend reversal, signalling the onset of a downward move.
- Conversely, given that the price movement from $t = 49$ to $t = 95$ is steep and extended, conditions often flagged by mean-reversion indicators such as Z-score thresholds or significant deviations from a moving average, a mean-reversion system would likely exit the long position and enter short, anticipating a reversion toward the recent mean.

However, as evident from the graph, this move does not represent a sustained downward trend, but rather a sharp stop-run or fakeout. The price rapidly recovers and resumes the upward trajectory shortly after. As a result, both momentum and mean-reversion strategies may initially capture some gains during the brief down move but subsequently incur losses when the sharp rebound occurs. The loss on the short position typically outweighs any temporary profit from the pullback, highlighting the challenge of reacting to such transient, deceptive market events.

10.2.3 Strategic failures in the face of market anomalies

10.2.3.1 Why momentum and mean-reversion strategies misfire We now outline the core reasons why both momentum and mean-reversion strategies often fail to capture stop-runs or fakeouts:

- **False Signals:**
 - The local low at $t = 95$ appears to be a turning point (with a maximum at 6002), but it is misleading.
 - The price dynamics reflect a trend-with-volatility regime rather than genuine mean-reversion.
- **Market Microstructure Effects:** Such sharp, abrupt moves frequently stem from liquidity imbalances or news-related events, which invalidate the statistical assumptions underlying mean-reversion strategies.

From this price sequence, the strategic takeaway is clear: it serves as a textbook example of the risks involved in applying mean-reversion logic within a trending market:

- Mean-reversion strategies become highly vulnerable when volatility coexists with directional momentum.
- These models are especially exposed when the price briefly corrects before resuming its original trend, misinterpreting a pause as a reversal.

A more robust, hybrid strategy might instead:

- Wait for clearer confirmation of a trend breakdown (e.g., a lower high, or a momentum crossover).
- Avoid shorting into strength, particularly when upward volatility is elevated.

10.2.3.2 Understanding stop-runs and liquidity voids In trading, the event observed at $t = 95$, a sharp downward spike, followed by a rapid recovery and continuation of the prior uptrend, is commonly described as a liquidity vacuum, stop-run, or flash dip/fakeout, depending on the context and time scale.

- Liquidity vacuum: A sudden price drop due to the absence of resting buy orders, quickly corrected by an influx of buying.
- Stop-run: A deliberate sweep of stop-loss levels just below recent highs/lows, triggering exits before reversing.
- Fakeout / Flash dip: A brief and aggressive move beyond a key technical level that quickly reverses, trapping counter-trend traders.

On lower timeframes such as the 1-minute chart, such moves are typically caused by:

- Stop-runs targeting intraday trailing stops, or
- Large institutional sell orders that encounter thin liquidity, followed by market absorption and reversal.

These events carry important strategic implications:

- They often invalidate conventional technical signals.
- They tend to coincide with algorithmic trading behaviour, order book imbalances, or unexpected news releases.
- For systematic strategies, both trend-following and mean-reverting, these patterns present a fundamental challenge, as they resemble a trend break but ultimately are not.

10.2.4 Systematically filter or react to fakeout

To systematically filter or react to fakeout or stop-run events like the one at $t = 95$, we need to:

1. Detect the Signature Pattern: These events typically have a V-shape in price with a sharp directional move followed by a reversal. We can characterise them with:

- Sudden Large Move Down (or Up): Detect large 1-minute returns

```
returns = price_series.pct_change()
threshold = returns.std() * k # e.g. k = 2.5 or 3
large_moves = abs(returns) > threshold
```

- Followed by Reversal Within N Minutes: Check if price returns close to the previous level within a short time window

```
window = 5 # 5-minute window
for t in large_moves[large_moves].index:
    recovery = price_series[t+1:t+1+window]
    if abs(recovery.iloc[-1] - price_series[t-1]) < epsilon:
        mark_as_fakeout(t)
```

Set ϵ on average volatility or use Z-score.

2. Filter in Strategy Logic

- Avoid Trading During/Right After Fakeouts: If a fakeout is detected, pause strategy execution for a cooling-off period

```
if fakeout_detected:
    skip_trade = True # For N minutes
```

- Tag Trades for Post-Analysis: Track trades that happen around these events to evaluate slippage, draw-down, and alpha decay.
- Add a 'Volatility Trap' Filter: If realized volatility over last few minutes spikes unusually

```
rolling_vol = returns.rolling(5).std()
if rolling_vol[-1] > rolling_vol.mean() + 2 * rolling_vol.std():
    suppress_signals()
```

3. Adapt Strategy Response

- For Momentum Strategies
 - Add confirmation delay: Don't enter after the first signal - wait for a second bar to confirm trend continuation.
 - Use volume filters: Require rising volume to validate breakout.
- For Mean-Reversion Strategies
 - Tighter stop-loss during spikes: Assume you're fading temporary panic moves.
 - Check for V-structure: Only enter reversion trades after clear price reversal.

We can also tag past events (fakeouts vs real breaks) and train a simple ML model (like logistic regression or XGBoost) using features such as: price return, time since last local extrema, volume spike, order book imbalance (if available), volatility.

10.3 W-shaped reversal trap

10.3.1 Description

Formation dynamics of a W-bottom pattern:

- Following the stop-run or fakeout dip at $t = 95$ observed in Figure (6), the price enters a sustained uptrend, peaking above 6000 around $t = 140$. This is followed by a sharp downward move from approximately 6009 to 5992, and then a steady decline that bottoms out near 5989 at $t = 164$, forming a short-term local downtrend.
- The market then rebounds sharply, climbing to around 6003 at $t = 175$, nearly retracing the entire drop from the earlier high.

- This sequence forms what appears to be a 'W-bottom' pattern, a classic double-dip recovery shape. For simplicity, we have defined the W-shape starting before $t = 145$ with value $S_{2,0} = 5997.1$ and ending after $t = 171$ with value $S_2 = 5997.18$.

Formation dynamics of a W-bottom fakeout pattern:

- However, rather than confirming a bullish continuation (from the definition of a W-bottom pattern), the price abruptly reverses downward again just after $t = 175$, falling to 5983 by $t = 189$.
- This failed recovery is best described as a bull trap within a W-bottom fakeout, a deceptive setup that initially mimics a reversal, but ultimately resumes the prior move, punishing both momentum and mean-reversion positions.

10.3.2 A comparative analysis of strategy responses

At time $t = 171$, the price reaches a local high of $S_2 = 5997.18$, following a modest rally from the recent trough. This level coincides with the previous high observed at $t = 145$. A price action algorithm would likely identify this formation as a W-bottom pattern, typically interpreted as either a continuation of the broader upward trend or a successful mean-reversion recovery. Both momentum- and mean-reversion-based systems would typically respond affirmatively at this point:

- Momentum strategy:
 - The price has recovered from the preceding local low and is climbing in a way that suggests the prior downtrend is over.
 - Indicators such as short-term moving averages or MACD crossovers would likely generate bullish signals around this point.
 - In accordance with standard momentum logic, the system would either initiate or maintain a long position at $t = 171$ in anticipation of further upward continuation.
- Mean-reversion strategy:
 - The drop from $t = 140$ to $t = 164$ has likely stretched Z-score or deviation metrics, triggering a long entry somewhere between $t = 164$ and $t = 171$.
 - By $t = 171$, the rebound appears validated, particularly if accompanied by contracting volatility or stabilising volume, common confirmation signals in mean-reversion frameworks.
 - As such, the strategy would maintain its long position, expecting a return to recent highs or at least a stabilisation around current levels.

This point marks a clear confluence between two otherwise divergent approaches, each recognising a statistical or structural signal of upward momentum.

10.3.3 The trap: Unanticipated reversal

What happens next breaks both models:

- Instead of continuing upward after $t = 171$, the market reverses again sharply just after $t = 175$.
- By $t = 183$, the price drops to 5988.44, effectively erasing the entire bounce.
- Neither momentum nor mean-reversion strategies have mechanisms to preempt this second reversal:

- There is no strong signal of exhaustion at $t = 171$.
- The move up was clean and looked trend-confirming.
- The reversal is not preceded by excessive overbought conditions or volatility spikes.

The outcome by $t = 183$ is losses for both strategies:

- Momentum: Caught in false breakout leading to small loss.
- Mean-Reversion: Reversal resumes leading to loss and confusion.

To summarise:

- At $t = 171$, both momentum and mean-reversion strategies have legitimate reasons to be long: trend strength, upward momentum, and mean reversion are all aligned.
- The drop by $t = 183$ is unforeseeable from a rules-based perspective; no standard indicator would signal an imminent downturn.
- This is an example of a local maximum in strategic agreement, followed by a structural failure in signal robustness.

This situation exposes a blind spot shared by both strategies: they are reactive, not predictive. Once again, we see a structural fakeout, where the market lures in long trades by presenting a textbook recovery, then reverses abruptly, invalidating both trend and reversion models.

10.3.4 RMA strategy response to distribution dynamics

Figure (8) illustrates that by time $t = 171$, the normalised price, w , is traversing the global distribution: it has declined from its maximum, passed through the median, and partially retraced toward the upper range. At this point, w is approximately back at the median level (indicated by the dotted pink curve).

Depending on the specific strategy and its risk management framework (see Section (9.7)), this configuration could support either (1) exiting the short position, or, (2) holding it until w crosses the full distribution and approaches the minimum. Furthermore, the smoothed lower bound alpha values, $w_lower_alphas_means$, are trending upward toward 1, signalling the possible emergence of a local bullish trend.

- **Exiting long-reversal at the median:** The strategy exits the short position when the normalised price, w , crosses the median from below, reverting to the maximum, and the smoothed lower bound alpha values rise toward 1. This is demonstrated in Figure (11), where the algorithm exits the short position (red marker) at $t = 171$ and enters a long position (green marker). However, with the benefit of hindsight, this move is revealed to be part of a reversal trap: the price soon reverses again, and based on the updated state S_3 , the strategy reverts to a short position at $t = 183$.
- **Exiting long-reversal at the minimum:** Alternatively, the strategy can opt to remain short until the normalised price, w , reaches the minimum of the distribution and begins to ascend again. In this approach, the short position is maintained beyond $t = 171$ and only reconsidered after $t = 183$, which in this case would have aligned more closely with the realised market behaviour. This is demonstrated in Figure (12), where the algorithm remains short (red marker) at $t = 171$.

This scenario highlights a core principle of real-time trading: a successful strategy depends less on perfectly predicting market movements and more on the consistency and robustness of the decision-making process. Even with sound analysis, outcomes can lead to losses due to the inherent uncertainty of price action. For this reason, strategies must not be built on a single market realisation but calibrated across large datasets. The response must be probabilistic, embracing uncertainty through systematic, repeatable processes that are validated over time.

10.4 Fakeouts in sideways market

10.4.1 Description

We saw in the previous section that the window leading up to time $t = 183$ exhibited a W-shaped reversal trap. Following this, as shown in Figure (6), the price action transitions into a sideways market lasting until $t = 219$, oscillating between a support level near 5985 and a resistance level around 5995. Within this range, the price briefly pushes to 5998 at $t = 203$ (breakout resistance), then sharply drops to 5985.7 at $t = 204$; it climbs to 5994 at $t = 207$, falls to 5984.3 at $t = 210$, and rises again to 5994 at $t = 214$, before ultimately reaching $S_5 = 5986$ at $t = 219$.

In sideways trading, a fakeout occurs when price briefly breaks through a key support or resistance level but quickly reverts back within the original range, often trapping traders who act on the initial breakout. The repeated penetrations and sharp reversals observed in this interval exhibit precisely this dynamic, suggesting a sequence of fakeouts within a bounded, non-trending market.

10.4.2 A comparative analysis of strategy responses

Following the W-shaped reversal trap up to $t = 183$, the market transitions into a sideways regime marked by sharp fluctuations between support (≈ 5985) and resistance (≈ 5995), including multiple fakeouts, brief breaches of key levels quickly reversed.

- Momentum strategy response: A momentum-based strategy seeks to exploit directional trends by entering positions aligned with price continuation. In this context:
 - At the initial breakouts (e.g., at $t = 203$, $t = 207$, $t = 214$), the strategy is likely to interpret the price push beyond resistance as emerging bullish momentum and initiate long positions.
 - However, due to the fakeout nature, these entries quickly turn unprofitable as price reverts into the range.
 - Similarly, sharp dips toward support (e.g., at $t = 204$, $t = 210$) might trigger short entries, again vulnerable to reversal.
 - Net effect: the momentum strategy suffers from whipsaw losses in this non-trending environment, where false signals are frequent and trend confirmation is weak or absent.
- Mean-Reversion strategy response: A mean-reversion strategy assumes that price tends to revert to a central value (e.g., the range midpoint or a moving average) after deviating toward extremes. In this setting:
 - The repeated approaches to support and resistance without sustained breaks are treated as overextended moves ripe for reversion.
 - Long entries near support (e.g., $t = 204$, $t = 210$) and short entries near resistance (e.g., $t = 203$, $t = 207$, $t = 214$) align well with the price action.
 - As fakeouts revert quickly, the strategy capitalises on mean-reverting pullbacks with tighter stop-losses and favourable risk-reward.
 - Net effect: the mean-reversion strategy performs more robustly, exploiting the range-bound structure with fewer false signals.

In this phase of low directional conviction and repeated fakeouts, the momentum strategy underperforms due to frequent trap entries, while the mean-reversion strategy is better suited, effectively leveraging the lack of trend and profiting from price oscillations between established support and resistance levels.

10.4.3 RMA strategy response to distribution dynamics

Figure (8) shows that by time $t = 211$, the normalised price, w , is approaching the minimum of the global distribution before sharply rising to reach the median at $t = 213$ (indicated by the dotted pink curve). The RMA strategy is built on the assumption that normalised prices tend to oscillate around the local moving average, and thus, w would typically continue traversing the distribution toward its maximum. However, at $t = 217$, the normalised price, w , reverses direction and begins to revert toward the minimum, ultimately reaching a local low at point S_5 at $t = 219$.

Depending on the specific configuration of the strategy and its associated risk management parameters (see Section (9.7)), this price action could trigger either (1) an exit from the long position, or (2) continuation of the long position in anticipation of a full traversal toward the maximum. Notably, the smoothed lower bound alpha values, $w_lower_alphas_means$, also revert toward 0, indicating the possible onset of a local bearish phase.

- **Exiting short-reversal at the median:** The strategy exits the long position when the normalised price, w , crosses the median from above and reverts toward the minimum, while the smoothed lower bound alpha values fall toward 0. This behaviour is shown in Figure (11), where the algorithm exits the long position (green marker) at $t = 219$ and enters a short position (red marker). In hindsight, however, this move corresponds to a fakeout, as the price soon reverses again and begins an upward trend.
- **Exiting short-reversal at the maximum:** Alternatively, the strategy may choose to remain long until the normalised price, w , reaches the maximum of the distribution and begins to decline. This approach would have kept the long position open beyond $t = 219$, better aligning with the realised market trajectory. Figure (12) illustrates this scenario, where the algorithm maintains the long position (green marker) at $t = 219$.

10.5 Mild fakeout or a healthy pullback

10.5.1 Description

In Figure (6), the window leading up to $t = 252$ exhibits a generally upward price trajectory, following a recovery from the consolidation phase observed between $t = 170$ and $t = 240$, reaching 6003 at $t = 243$. However, at $t = 252$, the price experiences a sharp drop to $S_6 = 5995.17$.

Interpretation of market behaviour:

- The drop to $S_6 = 5995.17$ at $t = 252$ does not represent a collapse, but rather a local retracement within a bullish regime.
- From a risk management perspective, this move could be interpreted as either a mild fakeout or a healthy technical pullback, depending on the surrounding volatility and market context.
- A mean-reversion system would typically expect price to revert toward a local equilibrium. However, it must be sensitive to regime shifts. Anchoring to outdated statistical baselines risks misclassification, treating a trend continuation as a reversion opportunity.

This scenario underscores the importance of adaptive models that not only respond to local price anomalies but also account for the evolving macrostructure in which those anomalies occur. That is, they must be capable of distinguishing between structural regime changes and routine noise.

10.5.2 A comparative analysis of strategy responses

- **Momentum Strategy:** At $t = 252$, a momentum-based strategy would likely interpret this point as either a local dip in a continuing uptrend or a potential breakdown if price action appears to be flattening or turning.

- If trend-following filters (e.g., moving average slope or breakouts) confirm the continuation, the strategy may stay long or add to the position.
- If price breaches a dynamic support level (e.g., a trailing stop or EMA), it might exit long, but not short yet, as the prevailing trend is still up.

Action: Stay long or reduce exposure depending on the strength of the pullback. No reversal to short unless trend breaks.

- Mean-Reversion Strategy: Here, the key point is that we must not use a long-term mean centered around $S_0 = 5924$. That would make no sense given the current regime. A proper mean-reversion model would recompute local statistics based on a rolling window (e.g., last 50–100 minutes). Let's assume: the local mean in the recent window is around 6002 – 6005, and the Z-score of the drop to 5995.17 is moderately negative (say, between -1.0 and -1.5), suggesting a local downside deviation, but not extreme. At $t = 252$, the system may detect a mild dislocation from the local mean and see this as a buy opportunity (fade the dip). However, because this dip is shallow and not statistically extreme, a conservative system may do nothing.

10.5.3 Momentum confirmation in the RMA framework

In the RMA configuration using a fixed moving average computed over the initial 100 minutes, we set *window_size* = 106 to account for the strong rally following $t = 49$. As a result, the MA curve at $t = 252$ in Figure (7) exhibits a slight decline and begins to plateau, reflecting the lagging influence of the earlier uptrend.

In contrast, Figure (8) illustrates the normalised current price, w , moving across the global distribution from minimum to maximum, with a downward spike toward the median at $t = 252$. Depending on the adopted risk management framework, this transient dip could trigger an exit from the long position to avoid short-term downside. Alternatively, the strategy may opt to remain in the trade, anticipating further gains. Notably, the smoothed lower-bound alpha values, $w_lower_alphas_means$, are rising sharply from 0 toward 1, indicating the development of a sustained local uptrend. This strengthens the underlying momentum signal and supports the decision to maintain the long position. This is reflected in Figures (11) and (12), where the algorithm holds the long position (green marker) at $t = 252$.

11 Results

To Do

11.1 The data

For simplicity, we consider the process-driven methods presented in Section (17.3), where we need to generate scenarios. To illustrate our approach, we let the stock prices follow a modified GBM with a stochastic market price of risk (MPR). The dynamics of the stock price, under the $\mathbb{P}$ -measure, are given by

$$\frac{dS_t}{S_t} = (r - q + \sigma_S \lambda_t)dt + \sigma_S d\widehat{W}_S(t) \quad (11.3)$$

where $\widehat{W}_S$ is a Brownian motion under the historical measure. Here, the repo rate q and the volatility σ_S are positive constants. We let the market price of risk, λ_t , follows an Ornstein-Uhlenbeck process (see Equation (17.24)). The Brownian motions are correlated as follows: $d < \widehat{W}_S, W_\lambda >_t = \rho_{S\lambda} dt$.

The settings for the model generating the returns, the option contracts, and the strategy are as follows:

- Market parameters taking values in $\mathcal{P}$:
 - GBM: $r = \{-0.1, 0, 0.1\}$, $q = \{0\}$, $\sigma_S = \{0.1, 0.2, 0.3\}$

- MPR: $\lambda_0 = \{0.1\}$, $a_\lambda = \{0.1, 0.5\}$, $b_\lambda = \{0.1, 1.0\}$, $\sigma_\lambda = \{0.1, 0.5\}$
- Simulation parameters: sample frequency 15 minutes, time steps $N = 780$ which gives 10 days, number of simulation 1,000
- Option parameters taking value in $\mathcal{P}$: maturity $T_O = \{1, 5, 10\}$ days, strike $K = S_0$
- Strategy parameters taking value in $\mathcal{M}$:
 - Entry level: $\epsilon_L = \{0.02, 0.03, 0.04, 0.05\}$
 - Global distance level: $\epsilon_{GSL} = \{0.1, 0.3, 0.5, 0.7, 1.0\}$
 - Local distance level: $\epsilon_{LSL} = \{0.1, 0.3, 0.5, 0.7, 1.0\}$

12 Conclusion

Systematic trading is not a quest for perfect foresight but a discipline of navigating uncertainty with structure, adaptability, and precision. In this article, we have advanced a framework that embraces the non-stationary, discontinuous nature of financial markets, where price patterns are fragmented, and traditional statistical assumptions often fail. By introducing the Relative Moving Average (RMA) as a dynamic and context-aware tool, we have shown how strategies can be designed to respond intelligently to local disequilibria and structural breaks without relying on rigid predictive models. Empirical results confirm that the RMA approach provides enhanced robustness and responsiveness across market regimes, outperforming static momentum and mean-reversion systems.

Ultimately, the resilience of a trading strategy lies in its ability to operate effectively when markets are most chaotic, when signals conflict, regimes shift, and uncertainty dominates. This requires not only statistical innovation but also disciplined execution and a continuous commitment to risk-aware design. In confronting the limits of predictability, we reaffirm the central principle of systematic trading: the value of decisions lies not in forecasting the future, but in preparing to act when it defies expectation.

13 Appendix

14 Miscellaneous

14.1 Some useful functions

14.1.1 The empirical cumulative distribution function

14.1.1.1 The percentiles Let $\mathcal{R} = \{R_0, \dots, R_n\}$ be the set of historical absolute ranges of the distribution, and let R_k be the current value at time k . That is, $R_k = \max(X_k) - \min(X_k)$ is the range of the normalised distribution X_k at time k . Then, the historical percentile is

$$P_k = \frac{1}{k} \sum_{i=0}^{k-1} I(R_i \leq R_k)$$

where $I(\cdot)$ is the indicator function, equal to 1 when the condition is true, and 0 otherwise. This gives us the proportion of past range values that are less than or equal to the current range. Hence, if this value is low, say $P_k \leq 0.2$, it means that the current range is among the smallest observed, suggesting a contraction in the distribution.

Note, there is an inherent bias in the empirical percentile computation when the index k is close to the first element of the set $\mathcal{R}$. The estimate becomes noisy, discrete, and sensitive to small fluctuations in the absolute ranges. Thus, it requires enough historical samples to give a stable estimate. As $k \rightarrow \infty$, the empirical distribution:

- converges to the true underlying distribution of the ranges (assuming stationarity),
- and becomes a smooth, unbiased estimator.

14.1.1.2 The fractiles Adding fractile-based features to the dataset is very useful in time series modelling, particularly for:

- regime detection, feature scaling, anomaly detection, momentum analysis.

The function `compute_fractile(...)` computes for each value v_i the fraction of values in the corresponding window W_i that are $\leq v_i$. That is,

$$f_i = \frac{1}{|W_i|} \sum_{j=1}^{|W_i|} I(W_{ij} \leq v_i)$$

where $I(\cdot)$ is the indicator function. This gives the fractile or quantile position of the value within its window.

14.1.2 Measures of proximity of two values

To measure how close two small values a and b are, we can use

- Relative Difference: $f(a, b) = \frac{|a-b|}{\max(|a|, |b|)}$ Threshold rule of thumb: $< 5\%$ almost identical, in $[5, 10]\%$ close, in $[10, 20]\%$ borderline, $> 20\%$ not close. For financial time series, $\approx [5, 10]\%$ is a good proximity cutoff when comparing normalised indicators.

14.1.3 The Z-score

The Z-score is a statistical measure that indicates how many standard deviations a data point is from the mean of a distribution. The formula for the Z-score of a data point x is:

$$Z = \frac{x - \mu}{\sigma}$$

where x is the value of the data point, μ is the mean of the population or sample, and σ is the standard deviation of the population or sample. This formula assumes that we are working with known population parameters. If we are working with a sample, we may use the sample mean $\hat{\mu}$ and sample standard deviation S , getting

$$Z = \frac{x - \hat{\mu}}{S}$$

The Z-score measures how many standard deviations the current point x is from the historical mean. For instance, for $Z = \pm 1$, the current point x is one standard deviation above/below the historical mean. It signifies that x is meaningfully above/below average, but not so extreme. Depending on the volatility of the data and the shape of distribution, we might need to consider two standard deviations. In general, ± 0.5 is slightly above/below average, ± 1 is moderately above/below average, ± 2 is significantly above/below average, and ± 3 is extreme event.

As an example, the action in a mean-reversion strategy is as follows:

- If $Z < -2$ (strong deviation): likely buy signal (long).
- If $Z \approx -1.0$: no trade or weak buy.
- If volatility is increasing or momentum is turning, mean-reversion models may avoid trading to reduce whipsaw risk.

Note, we can let the threshold for the Z-score depend on the percentiles of its past values. For instance, choose the 20th percentile of past Z-scores. We can also let the threshold be dynamic or adaptive.

14.1.4 The variation ratio

The variation ratio is a simple statistical measure used to quantify the dispersion or variability of a dataset relative to its central tendency. In the context of time series analysis, it can be used to track the relative spread of a distribution over time. For a given time k , the variation ratio V_k is defined as the ratio of the range of the distribution to its central value, typically the median or mean. Using the median, the formula is given by

$$V_k = \frac{R_k}{M_k}$$

where $R_k = \max(X_k) - \min(X_k)$ is the range of the normalised distribution X_k at time k , and M_k is the median of X_k at time k .

A higher variation ratio indicates greater dispersion relative to the central value, often signalling an expansion phase in the distribution, while a lower ratio suggests contraction.

Note, the coefficient of variation (CV) is typically defined as the standard deviation over the mean of a dataset. But when applied specifically to the range of a distribution evolving over time, we can define our own version of a variation ratio for this application. In this context, the variation ratio at time k refers to the relative dispersion of the range of the distribution. It is defined as the ratio of the standard deviation of the range to its mean, computed over a sliding window or over time. Its purpose is to quantify how much the range of the distribution fluctuates relative to its average value. The formula is given by

$$V_k = \frac{\sigma(R)}{\mu(R)}$$

where $\sigma(R)$ is the standard deviation of the range R over a defined window or over time up to k , and $\mu(R)$ is the mean of the range R over the same period.

This formulation provides a dimensionless measure of variability, allowing for the comparison of fluctuations regardless of scale. A high variation ratio indicates significant fluctuations in the spread of the distribution, which may correspond to transitions between expansion and contraction regimes.

14.1.5 The slope of the range

In order to capture dynamic changes in the spread of the distribution, we introduce the slope of the range as an indicator of significant fluctuations. Given a sliding window of size W applied to the time series of the range R_k of the normalised distribution, the slope provides a local linear approximation of how the range evolves within that window.

For a window of W consecutive time points, the slope S_k at time k is computed by performing a linear regression of R against time within the window, yielding:

$$S_k = \frac{\sum_{i=1}^W (t_i - \bar{t})(R_{k-W+i} - \bar{R})}{\sum_{i=1}^W (t_i - \bar{t})^2}$$

where t_i are the time indices within the window (e.g., $t_i = k - w + i$), $\bar{t}$ is the mean of the time indices in the window, R_{k-W+i} are the corresponding range values, and $\bar{R}$ is the mean of the range values in the window.

A positive slope ($S_k > 0$) indicates that the spread of the distribution is expanding over the window, whereas a negative slope ($S_k < 0$) signals contraction. Rapid increases or decreases in S_k can serve as early indicators of transitions between expansion and contraction regimes in the distribution.

This slope-based measure complements other dispersion metrics by capturing trend-like behaviour in the spread, rather than just instantaneous variability.

14.1.6 The directional change ratio

To further characterise the dynamics of the range, we introduce the directional change ratio (or imbalance measure), which quantifies the imbalance between increases and decreases in recent range values. Given a sequence of recent range values $R_{k-w+1}, \dots, R_k$, we compute the sequence of first differences:

$$\Delta R_i = R_i - R_{i-1}, i = k - W + 2, \dots, k$$

We then define:

- number of increases: $N_{inc} = \text{Number of times } \Delta R_i > 0$
- number of decreases: $N_{dec} = \text{Number of times } \Delta R_i < 0$

From these counts, we can define the directional imbalance ratio as:

$$D_k = \frac{N_{inc}}{N_{inc} + N_{dec}}$$

where $D_k \in [0, 1]$ and

- $D_k = 0.5$ corresponds to a balanced number of increases and decreases,

- $D_k > 0.5$ indicates that increases dominate,
- $D_k < 0.5$ indicates that decreases dominate.

This measure is useful for detecting directional bias in the recent evolution of the range. For example, sustained expansion phases typically correspond to $D_k \gg 0.5$, while contraction phases correspond to $D_k \ll 0.5$.

```
diff = np.diff(recent_values)
num_increases = np.sum(diff > 0)
num_drops = np.sum(diff < 0)
D_k = num_increases / (num_increases + num_drops) if (num_increases + num_drops) > 0
      else 0.5
```

14.1.7 Time in low state

When capturing dynamic changes in the spread of the distribution, we can introduce time-based confirmation (like hysteresis in physics). In the case of contraction detection, only exit contraction after X consecutive points are above a threshold. This prevents flipping in and out of contraction on noise.

```
if contraction_detected:
    contraction_counter += 1
else:
    contraction_counter = 0
condition_contraction = contraction_counter >= min_duration
```

14.2 Jumps and their role in financial time series

14.2.1 Description

In the study of stochastic processes, jumps refer to discontinuous changes in the value of a process over infinitesimal time intervals (see Jeanblanc et al. [2009]). Formally, for a càdlàg (right-continuous with left limits) process X_t , a jump at time t is defined as:

$$\Delta X_t = X_t - X_{t-}$$

where $X_{t-} = \lim_{s \rightarrow t} X_s$ denotes the left-hand limit of the process at time t . A process exhibits jumps if there exist $t \in [0, T]$ such that $\Delta X_t \neq 0$.

In continuous-time modelling, jumps are typically represented using Poisson random measures or Levy processes. A canonical example is the jump-diffusion model, which augments Brownian motion with a compound Poisson jump process:

$$dX_t = \mu dt + \sigma dW_t + J_t dN_t$$

where W_t is a standard Brownian motion, N_t is a Poisson process with intensity λ , and J_t is a random jump size, often modeled as i.i.d. from a specified distribution.

Jumps are a critical empirical feature of financial time series, capturing abrupt price movements arising from macroeconomic news, earnings reports, policy announcements, or sudden shifts in market sentiment. Unlike the continuous paths of Brownian motion, which imply small, frequent fluctuations, jumps introduce rare but significant discontinuities that dominate risk in high-frequency or crisis-period data.

Key implications include:

- Risk modelling: Jumps induce fat tails in return distributions, invalidating Gaussian assumptions and affecting metrics such as Value-at-Risk (VaR) and Expected Shortfall.
- Option pricing: Jump risk significantly impacts implied volatility surfaces and requires models like Merton's jump-diffusion or exponential Levy models for accurate pricing.
- Forecasting and control: Jumps undermine the Markovian and smoothness assumptions common in time-series forecasting, requiring non-local models or regime-switching frameworks.

14.2.2 Apparent continuity in discretely sampled financial data

Conceptually, when drawing a path of a Levy process that includes jumps, we do 'lift the pen' at each jump time. This reflects the fact that the sample paths of a general Levy process are càdlàg (continue à droite, limites à gauche): they are right-continuous with left limits, and can have discontinuities (jumps) at isolated points. This differs fundamentally from standard Brownian motion, whose paths are almost surely continuous and can be drawn without lifting the pen.

In theory, many financial time series include discontinuities. That means at certain times, prices change abruptly, not smoothly. Mathematically, the process has jumps, so we would 'lift the pen' when plotting the path. However, in practical data analysis, especially at lower frequencies (e.g., 1-minute or 10-minute bars), we don't observe the path continuously. Instead, we sample it discretely. That changes how we visualise and interpret jumps.

Even if the true underlying process includes jumps, at coarse time resolutions, we may not detect them. So we appear to draw the trajectory without lifting the pen, not because the process is continuous, but because our data doesn't capture the discontinuities.

Let's break this down:

1. High-Frequency Reality: At nanosecond or microsecond resolution (e.g., 10^{-6} seconds):
 - Modern electronic markets generate price ticks continuously (limit orders, market orders, cancellations).
 - Data at this frequency may reveal jumps: e.g., price gaps caused by a sudden large trade.
 - If we plotted every tick, we would lift the pen at jumps.
2. Low-Frequency Observation: When we sample data every 1 minute, 5 minutes, or 10 minutes:
 - We only see snapshots of the price at discrete time points.
 - We don't observe what happens between those points.
 - So even if a jump occurred, we just connect the two prices with a straight line, as if the path were continuous.

This is why, in practical charts or simulations we draw paths without lifting the pen. But that's an artifact of the sampling, not the true nature of the process.

In summary, when working with low-frequency data, financial paths appear continuous, but that doesn't mean the underlying process is continuous. The apparent smoothness is a consequence of coarse sampling, not the absence of jumps.

14.2.3 Example in the BS-model

Corporate events such as discrete dividends and earnings per share (EPS) lead to stock price jumps. This is modelled as $S(t_i) = S(t_i) + \text{sgn}(d_i)d_i$, with amount d_i , if corporate event. As an example, given the time set $\mathcal{T} = \{t_0 = 0, t_1, \dots, t_N = T\}$, we let $S_{t_0} = 50$ be the spot price at inception time t_0 . We consider two cases: (1) a single proportional dividend $\tilde{d}_i = \alpha S_{t_i}$, $\alpha = 5\%$, (2) a single cash dividend $d_i = 2.5$, both at time $t_i = 60\delta$ days, where the step δ is one day.

In the BS-model the governing equation is:

$$\frac{dS_t}{S_{t-}} = \mu dt + \sigma dW_S^P(t) + \frac{\bar{D}_t}{S_{t-}} dt$$

where $\mu = r - q - \lambda\sigma$, and $\bar{D}_t = \sum_{i=0}^{\infty} \delta(t - t_{d_i})$ and δ is the Dirac delta function. The model parameters tested are:

#	μ	σ
	0.05	0.3

Table 11: BS parameters

We display a single proportional dividend $\tilde{d}_i = \alpha S_{t_i}$, $\alpha = 5\%$ at time $t_i = 60\delta$ days.

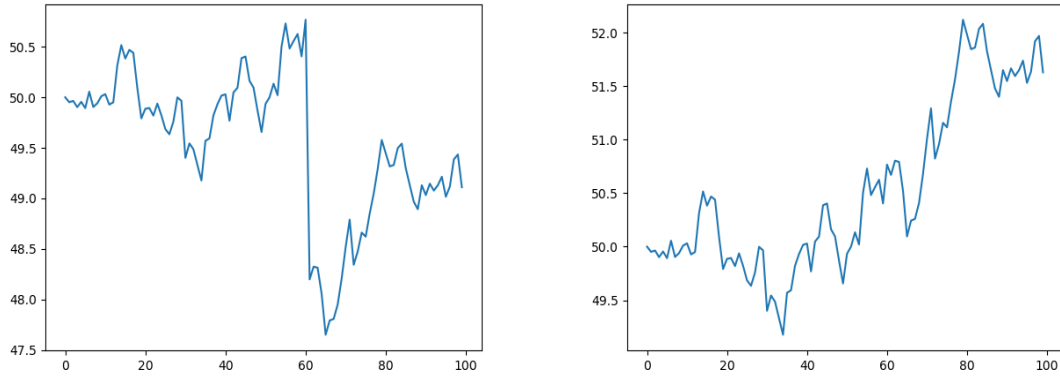


Figure 15: Stock price in the BS-model with and without proportional dividend.

We display a single cash dividend $d_i = 2.5$ at time $t_i = 60\delta$ days.

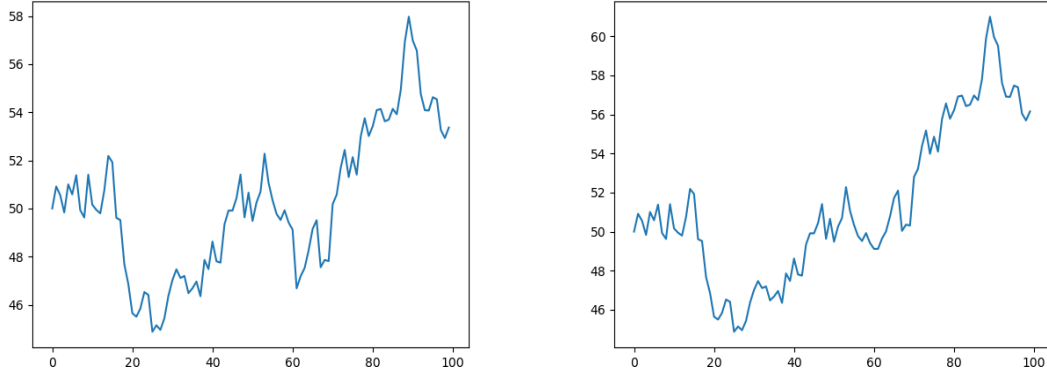


Figure 16: Stock price in the BS-model with and without cash dividend.

14.3 Characteristics of financial time series

Financial time series display a rich and complex set of behaviours that deviate significantly from the assumptions underpinning classical statistical models. These empirical regularities, often referred to as 'stylised facts', have been documented across asset classes, markets, and time horizons (see Cont [2001], McNeil et al. [2015]). Understanding these characteristics is essential for constructing realistic forecasting models, designing robust risk metrics, and interpreting econometric results in a financially meaningful way.

Below, we summarise and expand on the principal characteristics:

14.3.1 Heavy tails and non-Gaussianity

A foundational empirical characteristic of financial return series is their departure from normality. Unlike the Gaussian distribution assumed in many classical models, financial returns exhibit heavy tails, or fat-tailed distributions, meaning that extreme events, both large losses and gains, occur far more frequently than predicted by a normal distribution (see Mandelbrot [1963], Mandelbrot et al. [1997], Peters [1994]). Empirical return distributions often show excess kurtosis and power-law tails:

$$P(|X_t| > x) \sim x^{-\alpha} \text{ with } \alpha \in (2, 4)$$

This property implies that higher-order moments (such as variance or even the mean) may be unstable or undefined, depending on the tail index α . The Gaussian assumption significantly underestimates the probability of large losses or gains, leading to mispricing of risk and failure of traditional risk metrics (e.g., Value-at-Risk under normality).

14.3.2 Volatility clustering and persistence

Periods of high volatility tend to be followed by high volatility, and periods of calm by calm, an effect known as volatility clustering. This temporal dependence is well captured by GARCH-type models but violates the assumption of i.i.d. returns. Formally, returns r_t may be serially uncorrelated, but the squared returns r_t^2 exhibit strong autocorrelation:

$$\text{Cov}(r_t^2, r_{t+h}^2) > 0 \text{ for small } h$$

This feature reflects the endogenous amplification of shocks and the impact of investor herding behaviour, order flow dynamics, and changes in information flow.

14.3.3 Non-stationarity and structural breaks

Financial time series are inherently non-stationary. Statistical properties such as the mean, variance, and higher moments evolve over time in response to macroeconomic cycles, policy changes, liquidity regimes, or exogenous shocks. This undermines methods that rely on the assumption of time-invariant distributions, such as standard linear regressions or moving-average based models.

Structural breaks, sudden and lasting shifts in market dynamics, are particularly important. They may arise from central bank interventions, regime shifts in monetary or fiscal policy, or even changes in market microstructure.

14.3.4 Jumps and discontinuities

Asset prices frequently exhibit discrete jumps that are too abrupt to be explained by continuous diffusion processes. These may be caused by earnings announcements, geopolitical events, regulatory changes, or systemic liquidity crises. A more realistic model of returns takes the form of a jump-diffusion process:

$$dX_t = \mu dt + \sigma dW_t + J_t dN_t$$

where W_t is Brownian motion, N_t is a Poisson process, and J_t is the jump size. In many markets, jump components dominate the risk and return profile, particularly over short time horizons.

14.3.5 Leverage effects and asymmetric volatility

- Volatility reacts asymmetrically to returns: Increases more after negative returns than positive ones.
- Known as the leverage effect: Declining asset prices increase debt-to-equity ratios, amplifying risk.
- Violates assumptions of symmetric volatility models: Standard GARCH-type models may miss key dynamics.
- Negative shocks trigger stronger risk responses: Higher perceived risk leads to rising implied and realised volatility.
- Psychological biases amplify the asymmetry: Loss aversion makes investors more sensitive to drawdowns.
- Institutional factors play a role: Risk limits, margin calls, and deleveraging reinforce volatility spikes.

14.3.6 Autocorrelation in higher moments

- Raw returns show little autocorrelation: Markets quickly absorb information; arbitrage erases return patterns.
- But higher moments tell a different story: Volatility, skewness, and kurtosis show strong time-dependence
- Volatility clusters over time: High-volatility periods tend to follow high-volatility periods (and vice versa).
- Nonlinear dynamics in skewness and kurtosis: Tail risk and asymmetry persist across time.
- Foundation for stochastic volatility and GARCH-type models: Captures time-varying uncertainty more accurately.
- Enables volatility forecasting and risk targeting: Core to modern portfolio risk management and option pricing.

14.3.7 Path dependence and memory effects

- Financial markets are path-dependent: Outcomes depend not just on the current state, but how that state was reached.
- Impact of shocks varies with market context: A -2% drop feels different in a bull run vs. after a recent crash.
- Recent return sequences shape investor behaviour: Momentum, fear, or complacency emerge based on prior paths.
- Time since last major event matters: Market sensitivity changes after volatility spikes or drawdowns.
- Violates the Markov property: Future dynamics depend on more than just the present state.
- Limits of state-based models: Hidden Markov Models and simple regime-switching approaches may miss key dependencies.

14.3.8 Asymmetry and nonlinear dynamics

- Market reactions are not symmetric: A -1% move often triggers stronger reactions than a $+1\%$ move.
- Price changes lead to nonlinear responses: Small shocks can escalate unpredictably, especially on the downside.
- Investor behaviour is convex in losses: Risk aversion leads to exaggerated reactions during drawdowns.
- Nonlinear feedback loops drive volatility spikes: Panic selling, margin calls, and stop-loss cascades amplify shocks
- Volatility is state- and direction-dependent: Markets behave differently in uptrends vs. downtrends.
- Failure to model asymmetry leads to risk underestimation: Distorts derivative pricing, especially options and volatility products.

14.3.9 Market microstructure noise and frictions

- High-frequency prices are noisy, not clean signals: True value is obscured by trading mechanics.
- Bid-ask bounce distorts short-interval returns: Price flips between quoted bid and ask, not reflecting real movement.
- Latency and execution delays impact observed prices: Timing mismatches create artificial volatility.
- Order book dynamics influence price evolution: Hidden liquidity, queue priority, and order flow shape short-term moves.
- Noise dominates signal at microsecond to second scales: Makes raw returns unreliable for inference at high frequency.
- Requires specialised models: Techniques like filtering, aggregation, or noise-robust estimators are essential.

14.3.10 Cross-asset spillovers and contagion

- Shocks don't stay contained: Volatility and losses spill across assets, sectors, and geographies.
- Crisis periods break independence assumption: Assets that are normally uncorrelated can move together violently.
- Correlation structures are not stable: 'Diversification' often fails when it's needed most.
- Contagion effects drive systemic risk: Stress in one market can trigger chain reactions elsewhere.
- Tail dependence becomes dominant: Extreme co-movements aren't captured by linear correlation.
- Needs dynamic models to capture risk: Use DCC-GARCH (Dynamic Conditional Correlation), copulas, or regime-switching models to track evolving dependencies.

14.3.11 Implications

These features pose serious challenges to classical econometric tools based on assumptions of normality, linearity, and stationarity. Standard error metrics such as MSE or correlation coefficients often fail to recognize meaningful signals embedded in volatility spikes, jumps, or regime changes. Moreover, financial models must align with these empirical realities to be both statistically coherent and economically meaningful. In this context, alternative models, ranging from jump-diffusions and Levy processes to machine learning methods with memory and regime-awareness, are necessary to accurately model, forecast, and respond to the complex behaviour of financial markets.

15 Sample moments estimation

The accurate estimation of sample moments, such as the mean, variance, and derived performance metrics like the Sharpe ratio, is fundamental to quantitative finance. These quantities underpin not only the design of systematic trading strategies but also the assessment of their risk-return profiles. In practice, however, financial return series are often short, noisy, and exhibit departures from the idealized assumptions of normality and stationarity. Small sample sizes, in particular, can lead to substantial estimation error, inflating apparent signal strength or underestimating risk. Moreover, real-world complexities such as jumps, heavy tails, and temporal dependence further degrade the reliability of classical estimators. This section examines the theoretical and practical challenges in estimating population moments under both idealized and realistic market conditions, quantifies the sample sizes required for statistical reliability, and highlights the implications for commonly used measures like the Sharpe ratio.

15.1 The population mean and volatility**15.1.1 Description**

We let X_t be a Gaussian random variable with drift μ and volatility σ . We observe values uniformly allocated in time with time difference δ . That variable seen at time $t = j\delta$ for the time $t + 1 = (j + 1)\delta$ is given by

$$X(j, j + 1) = \mu\delta - \sigma(W_{(j+1)\delta} - W_{j\delta})$$

Given N period of time, and $X_{j\delta} = X(j, j + 1)$, the first two sample moments are

$$\hat{\mu}_N = \frac{1}{N} \sum_{j=0}^{N-1} X_{j\delta} \text{ and } \hat{\sigma}_N^2 = \frac{1}{N} \sum_{j=0}^{N-1} (X_{j\delta} - \hat{\mu}_N)^2 \quad (15.4)$$

Since the observations are selected randomly, both $\hat{\mu}_N$ and $\hat{\sigma}_N^2$ are random variables. Their expected values can be evaluated by averaging over the ensemble of all possible samples of size N from the population. We get

$$E[\hat{\sigma}_N^2] = \frac{n-1}{n} \sigma^2$$

with population variance $\sigma^2 = E[X_j^2] - \mu^2$ where $\mu^2 = E[X_i X_j] = E[X_i]E[X_j]$ due to independence of X_i and X_j . Note, $\hat{\sigma}_N^2$ gives an estimate of the population variance that is biased by a factor of $\frac{n-1}{n}$. Hence, it is a biased sample variance. Correcting for this bias yields the unbiased sample variance

$$S^2 = \frac{n-1}{n} \hat{\sigma}_N^2$$

These two estimators are simply referred to as the sample variance. Since the sample variance is a function of random variables, it is itself a random variable. Hence, we must study its distribution. Assuming independent observations from a normal distribution, given the population variance σ^2 , the unbiased sample variance S^2 follows a scaled chi-squared distribution

$$(n-1) \frac{S^2}{\sigma^2} \sim \chi_{n-1}^2$$

As a direct consequence, it follows that

$$E[S^2] = E\left[\frac{\hat{\sigma}^2}{n-1} \chi_{n-1}^2\right] = \hat{\sigma}^2 \text{ and } Var(S^2) = \frac{\hat{\sigma}^4}{(n-1)^2} Var(\chi_{n-1}^2) = \frac{2\hat{\sigma}^4}{n-1}$$

For non-normal samples, the variance of the sample variance depends on the kurtosis. If the observations are independent and identically distributed, but not necessarily normally distributed, then

$$E[S^2] = \hat{\sigma}^2 \text{ and } Var(S^2) = \frac{\hat{\sigma}^4}{n} \left(K - 1 + \frac{2}{n-1}\right) = \frac{1}{n} \left(\mu_4 - \frac{n-3}{n-1} \hat{\sigma}^4\right) \quad (15.5)$$

where K is the kurtosis of the distribution and μ_4 is the fourth central moment. Note that the variance of the estimator tends asymptotically to zero (when n goes to infinity).

In order to use statistical techniques on market data, one must make sure that the data is stationary, and test the assumption of log-normality on the observed values of the underlying price. Note, the underlying prices rarely satisfy the Black-Scholes assumption. However, we are not trying to compute option prices in the risk-neutral measure, but we are estimating the moments of the stock returns under the historical measure.

15.1.2 Implementation

Given the sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, we compute the sample mean and sample variance recursively as follows: Starting from the identities

$$(n+1)\bar{X}_{n+1} = n\bar{X}_n + X_{n+1}$$

and

$$(n+1)(\sigma_{n+1}^2 + (\bar{X}_{n+1})^2) = n(\sigma_n^2 + (\bar{X}_n)^2) + X_{n+1}^2$$

we get the recursive sample mean

$$\bar{X}_{n+1} = \bar{X}_n + \frac{X_{n+1} - \bar{X}_n}{n+1} \quad (15.6)$$

and the recursive sample variance

$$\sigma_{n+1}^2 = \sigma_n^2 + (\bar{X}_n)^2 - (\bar{X}_{n+1})^2 + \frac{X_{n+1}^2 - \sigma_n^2 - (\bar{X}_n)^2}{n+1} \quad (15.7)$$

such that σ_{n+1}^2 only depends on σ_n^2 , $\bar{X}_n$, $\bar{X}_{n+1}$ and X_{n+1} . If we normalise the data with a constant K such that the i th value is $\frac{X_i}{K}$, we let $\tilde{X} = \frac{1}{n} \sum_{i=1}^n \frac{X_i}{K}$ be the sample mean and $\tilde{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n (\frac{X_i}{K} - \tilde{X})^2$ be the sample variance. Then, the recursive sample mean becomes

$$\tilde{X}_{n+1} = \tilde{X}_n + \frac{1}{n+1} \left(\frac{X_{n+1}}{K} - \tilde{X}_n \right) \quad (15.8)$$

and the recursive sample variance becomes

$$\tilde{\sigma}_{n+1}^2 = \tilde{\sigma}_n^2 + (\tilde{X}_n)^2 - (\tilde{X}_{n+1})^2 + \frac{1}{n+1} \left(\frac{X_{n+1}^2}{K^2} - \tilde{\sigma}_n^2 - (\tilde{X}_n)^2 \right) \quad (15.9)$$

15.2 The case of non-normal and non-stationary conditions

Estimating the first two sample moments, mean and variance, under non-normal and non-stationary conditions is a nuanced task, especially in financial contexts where return distributions often exhibit heavy tails, skewness, and time-varying properties. Below is a comprehensive overview of the theoretical frameworks and methodologies pertinent to such scenarios.

15.2.1 Variance of the sample variance in non-normal distributions

In non-normal distributions, the variance of the sample variance estimator is influenced by the distribution's kurtosis. Specifically, for a sample of size n drawn from a distribution with mean μ , variance σ^2 , and fourth central moment μ_4 , the variance of the sample variance s^2 is given in Equation (15.5). This expression highlights that higher kurtosis increases the variability of the sample variance estimator. As n approaches infinity, the variance of S^2 tends to zero, but the convergence rate is affected by the kurtosis of the underlying distribution.

15.2.2 Asymptotic behaviour and the Central Limit Theorem

The Central Limit Theorem (CLT) asserts that, under certain conditions, the distribution of the sample mean approximates a normal distribution as the sample size increases, regardless of the original distribution's shape. However, the CLT's applicability hinges on the finiteness of the variance. In cases where the underlying distribution has infinite variance, such as certain heavy-tailed distributions, the CLT does not hold, and alternative limit theorems must be considered.

15.2.3 Alternative estimation techniques for non-normal data

15.2.3.1 L-moments L-moments are linear combinations of order statistics and serve as robust alternatives to conventional moments. They are particularly advantageous in estimating parameters of distributions with heavy tails or skewness, as they require only the existence of the mean and are less sensitive to outliers.

15.2.3.2 Generalised method of moments The Generalised Method of Moments (GMM) is a flexible estimation technique that relies on moment conditions derived from the data. GMM is especially useful when dealing with non-normal and heteroskedastic data, as it does not necessitate full specification of the distribution and can accommodate time-varying volatility structures.

15.3 Sample size requirements for reliable estimation of mean, variance, and Sharpe ratio

We extend Appendix (15.1) to estimate the required sample size, both for the mean and variance, for statistical significance (see van Belle [2008]). We then use this result to estimate the required sample size for the Sharpe ratio.

15.3.1 Measuring statistical significance

Assuming Gaussian rates, we briefly explain why we measure statistical significance differently for the mean and the variance (or volatility). In the case of the mean, we distinguish between hypothesis testing and estimation. We consider the case of (1) detecting a mean shift (computing n so that a small deviation in the sample mean is statistically significant) and (2) testing a null hypothesis (computing n required to reject $H_0 : \mu = \mu_0$ if the true mean is different from μ_0 with high probability). On the other hand, when estimating variance we care about the relative uncertainty of the estimate.

15.3.1.1 Measuring statistical significance When we talk about detecting a mean shift (e.g. in trading), we are typically trying to answer: 'Is the true mean return different from 0 (or some baseline μ_0) by at least $\epsilon_\mu = |\mu - \mu_0|$?' This is a signal detection problem. We are interested in whether the mean is different enough from a reference point to be economically or statistically meaningful. Hence, we work with the absolute shift ϵ_μ and test whether this exceeds the sampling error: standard error $= \frac{\sigma}{\sqrt{n}}$. In empirical work, the quantity $\epsilon_\mu = |\hat{\mu}_n - \mu_0|$ is used where $\hat{\mu}_n$ is our sample estimate of the true mean. We search for n that makes $\hat{\mu}_n$ statistically significant at level α . This is backward-looking: we observe a shift and compute how much data would be needed to make it significant.

However, in theoretical derivations, we treat $\hat{\mu}_n$ as a random variable, and instead ask: 'If the true mean is μ , how large must n be so that $\hat{\mu}_n$ is sufficiently far from a null hypothesis mean μ_0 to be detected as statistically significant?'. While we observe the sample mean $\hat{\mu}_n$, to design the test (or compute required sample size), we must imagine what the sampling distribution of $\hat{\mu}_n$ would look like under both: (1) the null hypothesis $H_0 : \mu = \mu_0$, (2) the alternative hypothesis $H_1 : \mu = \mu_0 + \epsilon_\mu$. In that case, $\hat{\mu}_n$ is the random variable we compute, μ is the hypothesised true mean under the alternative, and $\epsilon_\mu = |\mu - \mu_0|$ is the minimum effect size we want to detect. We then compute how large n must be to detect that shift ϵ_μ with high probability (e.g. 95%). This is forward-looking: we decide the smallest shift ϵ_μ we want to reliably detect and solve for n .

In the analysis that we are going to perform, especially in the examples, we compute the required n such that the observed sample mean becomes statistically significant at a given confidence level. This is closely related to, but not exactly the same as, computing the power of a hypothesis test (which involves assumptions about the true mean μ , not the sample mean). While they are identical in the form, in meaning one uses ϵ_μ as an observed quantity (significance of sample estimate), the other treats it as a theoretical true shift (hypothesis testing). They match when we define the smallest effect size we care about (say, 1 bps/minute) and ask: 'How large must n be so that, if the true mean shift is ϵ_μ , we would likely detect it as significant?'. Then, we are using ϵ_μ as a minimum detectable effect size, and computing n such that $\hat{\mu}_n \sim N(\mu, \frac{\sigma^2}{n})$ is outside the confidence band under H_0 . This is equivalent to asking 'What is the smallest true shift we can reliably detect using a test at level α with n observations?'.

15.3.1.2 Variance estimation: Use relative confidence interval width In contrast, when estimating variance (or volatility), we rarely test: 'Is the variance different from some fixed number?'. Instead, we ask: 'How precise is our estimate of the population variance σ^2 ?'. We care about the relative uncertainty of the estimate, not detecting a shift from a benchmark. Hence, this leads to a relative error question:

$$\text{Relative CI width} = \frac{\text{CI upper} - \text{CI lower}}{2\sigma^2}$$

where CI is the confidence interval. If we let $L = \sigma^2(1 - \epsilon_{\sigma^2})$ and $U = \sigma^2(1 + \epsilon_{\sigma^2})$, then $\text{Relative CI width} = \epsilon_{\sigma^2}$. We want this width (say, $\pm 10\%$) to be small enough for our application.

15.3.2 Estimating the required sample size for statistical significance

Given that both the sample mean $\hat{\mu}_N$ and the unbiased sample variance S^2 are random variables, we wish to determine the minimum number of observations n required for these estimators to be statistically significant at a prescribed confidence level $1 - \alpha$.

15.3.2.1 Sample mean significance To get a confidence bound on the mean estimation error, want the sample mean $\hat{\mu}_N$ to be within a certain absolute error ϵ_μ of the true mean μ with probability at least $1 - \alpha$. That is,

$$P(|\hat{\mu}_N - \mu| < \epsilon_\mu) \geq 1 - \alpha$$

From the central limit theorem (CLT), for independent Gaussian observations with variance σ^2 , the distribution of the sample mean is

$$\hat{\mu}_N \sim N(\mu, \frac{\sigma^2}{n})$$

Hence,

$$P(|\hat{\mu}_N - \mu| < \epsilon_\mu) = P(|Z| < \frac{\epsilon_\mu \sqrt{n}}{\sigma})$$

where $Z \sim N(0, 1)$. It gives

$$P(|\hat{\mu}_N - \mu| < \epsilon_\mu) = 2\Phi(\frac{\epsilon_\mu \sqrt{n}}{\sigma}) - 1$$

where $\Phi(x) = P(Z \leq x)$ is the cumulative distribution function (CDF) of the standard normal distribution. When $\Phi(x) = 0.975$, the corresponding x is $z_{0.975} \approx 1.96$. In the sample size derivation, $\Phi^{-1}(p)$ is used to get the z-score that corresponds to a given tail probability.

We now want this to be at least $1 - \alpha$. So, we get

$$2\Phi(\frac{\epsilon_\mu \sqrt{n}}{\sigma}) - 1 \geq 1 - \alpha \implies \Phi(\frac{\epsilon_\mu \sqrt{n}}{\sigma}) \geq 1 - \frac{\alpha}{2}$$

We can take the inverse of Φ (quantile function) and estimate n .

If σ^2 is not known (common in practice), the estimate $\hat{\mu}_N$ follows a Student's t-distribution. That is, to ensure that the estimated mean is significantly different from zero (or another reference value μ_0), we perform a two-sided t-test:

$$T = \frac{\hat{\mu}_N - \mu_0}{\frac{S}{\sqrt{n}}} \sim t_{n-1}$$

where t_{n-1} is the t-distribution and S is the unbiased sample volatility. For a significance level α , the test rejects the null hypothesis $H_0 : \mu = \mu_0$ if

$$|T| > t_{1-\frac{\alpha}{2}, n-1}$$

Rearranging, the required sample size is approximately

$$n \geq \left(\frac{t_{1-\frac{\alpha}{2}, n-1} \sigma}{\epsilon_\mu} \right)^2$$

where $\epsilon_\mu = |\hat{\mu}_N - \mu_0|$ is the mean shift. For large n , we have $t_{1-\frac{\alpha}{2}, n-1} \approx z_{1-\frac{\alpha}{2}}$ from the standard normal distribution. Hence, we obtain

$$n \gtrsim \left(\frac{z_{1-\frac{\alpha}{2}} \sigma}{\epsilon_\mu} \right)^2 \tag{15.10}$$

This scales as $n \propto \frac{1}{\epsilon_\mu^2}$. Knowing the sample size n , we can estimate the mean shift as

$$\epsilon_\mu \gtrsim \frac{z_{1-\frac{\alpha}{2}} \cdot \sigma}{\sqrt{n}}$$

15.3.2.2 Sample variance significance We want to estimate the population variance σ^2 with a relative error bound of ϵ_{σ^2} with $1 - \alpha$ confidence. That is, we want a confidence interval $[L, U]$ such that:

$$P\left(\left|\frac{\hat{\sigma}^2 - \sigma^2}{\sigma^2}\right| < \epsilon_{\sigma^2}\right) \geq 1 - \alpha$$

and we want to find the minimum sample size n required to guarantee this. The relative error is related to the width of this interval relative to σ^2 . We want this width to be within a factor of $1 \pm \epsilon_{\sigma^2}$. This is equivalent to requiring that the confidence interval for σ^2 lies within the range:

$$[(1 - \epsilon_{\sigma^2})\sigma^2, (1 + \epsilon_{\sigma^2})\sigma^2]$$

Hence, we are aiming to find n such that the confidence interval has a relative half-width (compared to σ^2) no greater than ϵ_{σ^2} . That is,

$$\text{Relative half-width} \approx \frac{U - L}{2\sigma^2}$$

From the chi-squared distribution property, we have

$$(n - 1)\frac{S^2}{\sigma^2} \sim \chi_{n-1}^2$$

For Gaussian returns, a $100(1 - \alpha)\%$ confidence interval for the variance σ^2 is

$$(n - 1)\frac{S^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2} \leq \sigma^2 \leq (n - 1)\frac{S^2}{\chi_{\frac{\alpha}{2}, n-1}^2} \quad (15.11)$$

where $L = \sigma_{lower}^2 = (n - 1)\frac{S^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2}$ and $U = \sigma_{upper}^2 = (n - 1)\frac{S^2}{\chi_{\frac{\alpha}{2}, n-1}^2}$.

To ensure statistical significance, the relative width of this confidence interval should be below the desired tolerance ϵ_{σ^2} :

$$\frac{\sigma_{upper}^2 - \sigma_{lower}^2}{2\sigma^2} \leq \epsilon_{\sigma^2} \quad (15.12)$$

which we can rewrite as

$$\frac{(n - 1)}{2\sigma^2} S^2 \left(\frac{1}{\chi_{\frac{\alpha}{2}, n-1}^2} - \frac{1}{\chi_{1-\frac{\alpha}{2}, n-1}^2} \right) \leq \epsilon_{\sigma^2}$$

Assuming $S^2 \approx \sigma^2$, we get the approximation

$$\frac{(n - 1)}{2} \left(\frac{1}{\chi_{\frac{\alpha}{2}, n-1}^2} - \frac{1}{\chi_{1-\frac{\alpha}{2}, n-1}^2} \right) \leq \epsilon_{\sigma^2}$$

For large n , the chi-squared distribution with $(n - 1)$ degrees of freedom is approximately normal:

$$\chi_{n-1}^2 \approx N(n - 1, 2(n - 1))$$

Setting $k = n - 1$, we get

$$\chi_{p,k}^2 \approx k \pm z_p \sqrt{2k} \quad (15.13)$$

where $\chi_{p,k}^2$ is the p th quantile of the chi-squared distribution with k degrees of freedom, and $z_p = \Phi^{-1}(p)$ is the p th quantile of the standard normal distribution ($P(Z \leq z_p) = p$).

Since the standard normal distribution is symmetric about 0, we have

$$\chi_{p,k}^2 \approx k + z_p \cdot \sqrt{2k} = k - z_{1-p} \cdot \sqrt{2k}$$

Thus, we reuse $z_{1-\frac{\alpha}{2}}$ as a positive quantity in both approximations for notational convenience. As a result, the quantiles can be approximated as:

$$\chi_{1-\frac{\alpha}{2},n-1}^2 \approx (n-1) + z_{1-\frac{\alpha}{2}} \sqrt{2(n-1)} \text{ and } \chi_{\frac{\alpha}{2},n-1}^2 \approx (n-1) - z_{1-\frac{\alpha}{2}} \sqrt{2(n-1)}$$

Setting $z = z_{1-\frac{\alpha}{2}}$ and $k = n-1$, we get

$$\frac{1}{\chi_{\frac{\alpha}{2},n-1}^2} - \frac{1}{\chi_{1-\frac{\alpha}{2},n-1}^2} \approx \frac{1}{k - z\sqrt{2k}} - \frac{1}{k + z\sqrt{2k}}$$

Let $a = k$ and $b = z\sqrt{2k}$ in the identity ³, we get

$$\frac{1}{k - z\sqrt{2k}} - \frac{1}{k + z\sqrt{2k}} = \frac{2z\sqrt{2k}}{k^2 - 2kz^2} \approx \frac{2z\sqrt{2k}}{k^2} \text{ neglecting small term in denominator}$$

So,

$$\epsilon_{\sigma^2} \approx \frac{k}{2} \cdot \frac{2z\sqrt{2k}}{k^2} = \frac{z\sqrt{2}}{\sqrt{k}} = \frac{z_{1-\frac{\alpha}{2}}\sqrt{2}}{\sqrt{n-1}}$$

Squaring, we get

$$\epsilon_{\sigma^2}^2 \approx \frac{2z_{1-\frac{\alpha}{2}}^2}{n-1}$$

and rearranging, we obtain

$$n \gtrsim 1 + \frac{2z_{1-\frac{\alpha}{2}}^2}{\epsilon_{\sigma^2}^2} \quad (15.14)$$

15.3.3 Large-n approximation and practical meaning

The bounds for n in the section above rely on large-n asymptotics, where:

1. For the mean: the t-distribution t_{n-1} is replaced by the standard normal z-distribution.
2. For the variance: the chi-squared distribution χ_{n-1}^2 is replaced by a normal distribution with mean $n-1$ and variance $2(n-1)$.

In practical terms, 'large' means that the approximation error from these substitutions is negligible compared to the target significance level α and tolerance ϵ .

- For the mean, the normal approximation is typically adequate when $n \gtrsim 30$, though heavier-tailed data may require more observations.
- For the variance, the normal approximation to χ_{n-1}^2 becomes reasonably accurate when $n \gtrsim 50$ but higher accuracy demands larger n .

³We have the identity $\frac{1}{a-b} - \frac{1}{a+b} = \frac{2b}{a^2-b^2}$

15.3.3.1 Example Recall, the quantiles of the chi-squared distribution at probabilities 0.975 and 0.025 are $z_{0.975} = \Phi^{-1}(0.975) \approx 1.96$ and $z_{0.025} = \Phi^{-1}(0.025) \approx -1.96$, respectively. Suppose we want a two-sided 95% confidence ($\alpha = 0.05$):

- Mean significance: If $\sigma = 1$ and we want to detect a mean shift of $\epsilon_\mu = 0.1$, from Equation (15.10), the large-n bound is

$$n \gtrsim \left(\frac{z_{0.975} \cdot 1}{0.1} \right)^2 = \left(\frac{1.96}{0.1} \right)^2 \approx 384.16$$

If we instead used the exact t_{n-1} critical value, the required n would be 399.5, a difference of only 2%, confirming that $n \approx 384$ is already 'large' here.

- Variance significance: If we want a relative confidence interval width of $\epsilon_{\sigma^2} = 0.1$ at 95% confidence, From Equation (15.14), the large-n bound gives

$$n \gtrsim 1 + \frac{2z_{0.975}^2}{(0.1)^2} \approx 1 + \frac{2 \cdot (1.96)^2}{0.01} \approx 769.3$$

The exact chi-squared bound yields $n \approx 772$, again showing close agreement.

15.3.4 Quadratic relation for the sample size

15.3.4.1 The mean From the large-n bound for the mean:

$$n \geq \left(\frac{z_{1-\frac{\alpha}{2}} \sigma}{\epsilon_\mu} \right)^2$$

where $\epsilon_\mu = |\hat{\mu}_N - \mu_0|$. If we fix the confidence level α and σ , then the sample size becomes

$$n \propto \frac{1}{\epsilon_\mu^2}$$

That means:

- Halving the target detectable shift quadruples the required sample size.
- Detecting $10x$ smaller shifts requires $100x$ more data.

In our example above for 95% confidence ($z_{0.975} \approx 1.96$, $\sigma = 1$)

- For $\epsilon_\mu = 0.2$, we get $n \geq 96$
- For $\epsilon_\mu = 0.1$, we get $n \geq 384$
- For $\epsilon_\mu = 0.05$, we get $n \geq 1,537$
- For $\epsilon_\mu = 0.01$, we get $n \geq 38,416$

To detect a 1% mean shift with high statistical confidence, we would need on the order of tens of thousands of independent trades or periods. Hence, in practice, small alpha signals require either:

- Very long observation horizons, or
- Aggregating many independent strategies to increase statistical power.

Now, if we fix the detectable shift ϵ_μ and confidence level α , the large-n bound for the mean is

$$n \gtrsim \left(\frac{z_{1-\frac{\alpha}{2}} \sigma}{\epsilon_\mu} \right)^2$$

and n is quadratic in σ

$$n \propto \sigma^2$$

Interpretation:

- Doubling σ (higher volatility) quadruples the required sample size.
- Halving σ (lower volatility) reduces the required sample size by a factor of 4.

When volatility is high relative to the signal size ϵ_μ , we need far more observations to detect the edge.

15.3.4.2 The variance From the large-n bound for the variance:

$$n \gtrsim 1 + \frac{2z_{1-\frac{\alpha}{2}}^2}{\epsilon_{\sigma^2}^2}$$

where n is the number of observations, α is the significance level, and ϵ_{σ^2} is relative confidence interval (CI) width for the variance σ^2 (see Equation (15.12)).

The scaling with ϵ_{σ^2} (precision target) is as follows:

- Relation: $n \propto \frac{1}{\epsilon_{\sigma^2}^2}$
- Interpretation: halving the allowed relative CI width requires 4x more observations.

In our example above ($\alpha = 0.05$, $z_{0.975} \approx 1.96$), we get

- For $\epsilon_{\sigma^2} = 0.7$, we get $n \approx 17$
- For $\epsilon_{\sigma^2} = 0.5$, we get $n \approx 32$
- For $\epsilon_{\sigma^2} = 0.2$, we get $n \approx 193$
- For $\epsilon_{\sigma^2} = 0.1$, we get $n \approx 769$
- For $\epsilon_{\sigma^2} = 0.05$, we get $n \approx 3,074$
- For $\epsilon_{\sigma^2} = 0.04$, we get $n \approx 4,803$

Scaling with σ (population standard deviation): Unlike the mean case, in relative terms (ϵ fixed), the required n does not depend on σ at all, because the variance estimator's CI scales with σ^2 both in numerator and denominator. However, if instead you fix an absolute tolerance δ_{σ^2} (absolute CI half-width in variance units), then:

$$\epsilon_{\sigma^2} = \frac{\delta_{\sigma^2}}{\sigma^2} \Rightarrow n \propto \frac{\sigma^4}{\delta_{\sigma^2}}$$

So with an absolute accuracy goal, n scales like σ^4 . Hence, volatility makes the variance estimation problem much harder.

15.3.5 Sample Sharpe Ratio significance and sample size

The sample Sharpe ratio combines the two estimation problems we just discussed (mean and variance), so the error propagation depends on both σ and the signal μ .

15.3.5.1 Definition Given returns $X_1, \dots, X_n$ with sample mean $\hat{\mu}$ and unbiased sample variance S^2 , the sample Sharpe ratio is $\hat{R} = \frac{\hat{\mu}}{S}$ and the population Sharpe ratio is $R_\theta = \frac{\mu}{\sigma}$. We can propagate the relative errors from both the mean and the volatility into the Sharpe ratio. To claim that $\hat{R}$ is statistically significant, both the numerator $\hat{\mu}$ and denominator S must be estimated with sufficient precision.

15.3.5.2 Mean and variance estimation requirement From the large-n bound for the sample mean, to detect a mean shift ϵ_μ at confidence level $1 - \alpha$, we require Equation (15.10). Similarly, if we want to detect the relative confidence interval width ϵ_{σ^2} at confidence level $1 - \alpha$, we require Equation (15.14). Hence,

- For small mean shift $\epsilon_\mu \rightarrow 0$, we get $n_{SR} \sim \frac{1}{\epsilon_\mu^2}$. The mean estimation dominates.
- For small relative CI width $\epsilon_{\sigma^2} \rightarrow 0$, we get $n_{SR} \sim \frac{1}{\epsilon_{\sigma^2}^2}$. The variance estimation dominates.

The two effects are independent, so the larger of the two required n dominates.

15.3.5.3 Required n for the Sharpe ratio The sample Sharpe ratio inherits the slower-converging of the two estimation problems. Thus, the effective sample size requirement is

$$n_{SR} \gtrsim \max\left(\left(\frac{z_{1-\frac{\alpha}{2}}\sigma}{\epsilon_\mu}\right)^2, 1 + \frac{2z_{1-\frac{\alpha}{2}}^2}{\epsilon_{\sigma^2}^2}\right)$$

15.3.6 Relation between variance and volatility confidence interval widths

The relationship between the relative CI width for variance (ϵ_{σ^2}) and the relative CI width for volatility (ϵ_σ) is not a simple halving in general, but for small widths it can be approximated as

$$\epsilon_\sigma \approx \frac{\epsilon_{\sigma^2}}{2}$$

Proof: If (from the linearity of variance)

$$\sigma_{upper}^2 = \sigma^2(1 + \delta) \text{ and } \sigma_{lower}^2 = \sigma^2(1 - \delta)$$

then

$$\sigma_{upper} = \sigma\sqrt{1 + \delta} \text{ and } \sigma_{lower} = \sigma\sqrt{1 - \delta}$$

For small δ , applying a first order Taylor expansion, we get

$$\sqrt{1 + \delta} \approx 1 + \frac{\delta}{2} \text{ and } \sqrt{1 - \delta} \approx 1 - \frac{\delta}{2}$$

So the relative CI width for volatility is:

$$\epsilon_\sigma = \frac{\sigma_{upper} - \sigma_{lower}}{c \cdot \sigma} \approx \frac{\delta}{c}$$

for some constant $c > 0$, and the relative CI width for variance is:

$$\epsilon_{\sigma^2} = \frac{c \cdot \sigma_{upper}^2 - \sigma_{lower}^2}{c \cdot \sigma^2} \approx \frac{2\delta}{c}$$

Hence, $\epsilon_{\sigma} \approx \frac{\epsilon_{\sigma^2}}{2}$.

15.3.7 Some examples

For Gaussian returns, Equation (15.11) applies. We consider a forward-looking approach and decide the smallest shift ϵ_{μ} and the relative CI width target for variance ϵ_{σ^2} we want to reliably detect and solve for n . While in the first example we consider a forward-looking approach, we will focus on backward-looking approach for the two trading examples.

15.3.7.1 Example Following on our previous example, let the detectable mean shift be $\epsilon_{\mu} = 0.01$ and the relative CI width target for variance be $\epsilon_{\sigma^2} = 0.1$. The mean and variance requirements are

$$n_{\mu} \approx \left(\frac{1.96}{0.01}\right)^2 \approx 38,416 \text{ and } n_{\sigma^2} \approx 1 + \frac{2 \cdot (1.96)^2}{(0.1)^2} \approx 769.32$$

So, the Sharpe ratio requirement is

$$n_{SR} \approx \max(38,416, 769.32) = 38,416$$

So here, the mean estimation is the bottleneck.

15.3.7.2 Example: Minute-level trading Suppose we trade a liquid equity index future (e.g., S&P 500 E-mini) with 1-minute returns (that is 390 min/day). Assuming an annualised volatility of 20%, the per-minute volatility is

$$\sigma_{min} \approx \frac{0.2}{\sqrt{250 \times 390}} \approx 0.00064$$

That's about 0.064% per minute. Following on the above example where $\alpha = 0.05$, let $n = 5000$ minutes (about 35 trading days), and assume $S^2 \approx \sigma^2$. From Equation (15.13), we have $\chi_{0.025}^2 \approx 5000 - 1 - 1.96\sqrt{2 \times 4999} \approx 4,802$ and $\chi_{0.975}^2 \approx 5000 - 1 + 1.96\sqrt{2 \times 4999} \approx 5,198$. Then,

$$\begin{aligned} \sigma_{lower} &\approx \sqrt{\frac{(n-1)S^2}{5198}} = \sigma \times \sqrt{\frac{4999}{5198}} \approx 0.00064 \times 0.981 \approx 0.000628 \\ \sigma_{higher} &\approx \sqrt{\frac{(n-1)S^2}{4802}} = \sigma \times \sqrt{\frac{4999}{4802}} \approx 0.00064 \times 1.02 \approx 0.000653 \end{aligned}$$

The relative CI width for variance is

$$\begin{aligned} \epsilon_{\sigma^2} &\approx \frac{\sigma_{higher}^2 - \sigma_{lower}^2}{2\sigma^2} \approx \frac{(0.000653)^2 - (0.000628)^2}{2 \cdot (0.00064)^2} \\ &\approx \frac{4.267 \times 10^{-7} - 3.944 \times 10^{-7}}{2 \times 4.096 \times 10^{-7}} \approx 0.0394 \end{aligned}$$

From Appendix (15.3.6), the relative CI width for volatility is

$$\epsilon_{\sigma} \approx \frac{0.0394}{2} \approx 0.0197, 2\% \text{ relative width for volatility}$$

Thus, with ~ 5000 minutes of data, we can estimate volatility to within about $\pm 2\%$ precision (half-width) at 95% confidence, assuming no serial correlation.

In a forward-looking approach, a $\delta = 10\%$ shift on the variance would give

$$\begin{aligned}\sigma_{higher}^2 &= \sigma^2(1 + \delta) = 4.096 \times 10^{-7} \times 1.1 = 4.505 \times 10^{-7} \\ \sigma_{lower}^2 &= \sigma^2(1 - \delta) = 4.096 \times 10^{-7} \times 0.9 = 3.686 \times 10^{-7}\end{aligned}$$

so that the relative CI width for variance is

$$\epsilon_{\sigma^2} \approx \frac{4.505 \times 10^{-7} - 3.68 \times 10^{-7}}{2 \times 4.096 \times 10^{-7}} \approx 0.1$$

as expected.

For the mean shift, we consider three scenarios:

- Unrealistically gigantic edge: $\epsilon_\mu = 1 \times \sigma_{min}$, which is 0.064% return per minute ($\sim 25\%$ return per day). A $1x$ volatility shift is detectable in only a few minutes, $n \geq (\frac{1.96 \cdot 0.00064}{0.00064})^2 \approx 3.84$ minutes, but such a signal is unrealistic in live trading.
- Unrealistically huge edge: $\epsilon_\mu = 0.5 \times \sigma_{min}$, which is 0.032% return per minute ($\sim 12.5\%$ return per day). A $0.5x$ volatility shift is detectable in only a few minutes, $n \geq (\frac{1.96 \cdot 0.00064}{0.00032})^2 \approx 15.36$ minutes, but such a signal is unrealistic in live trading.
- Big by trading standards: $\epsilon_\mu = 0.1 \times \sigma_{min}$, which is 0.0064% return per minute ($\sim 2.5\%$ return per day). A $0.1x$ volatility shift (mean return = 0.0064% per minute) requires $n \geq (\frac{1.96 \cdot 0.00064}{0.000064})^2 \approx 384$ independent minutes (~ 1 trading day).
- Realistic alpha signal: $\epsilon_\mu = 0.01 \times \sigma_{min}$, which is 0.00064% return per minute ($\sim 0.25\%$ return per day or 63% annualised). A $0.01x$ volatility shift (mean return = 0.00064% per minute), more in line with realistic short-horizon alpha, requires $\sim 38,000$ minutes (100 trading days) of independent observations.

Hence, for the 'realistic alpha' case, we set $\epsilon_\mu = 0.0000064$, which is $0.01x$ volatility or about 0.64 basis points per minute.

We observe that

- In minute trading, volatility can be measured reasonably precisely with a few weeks of continuous data.
- However, mean returns are much smaller so that detecting them requires orders of magnitude more data than volatility estimation.

Thus, we assume

- Detectable mean shift: $\epsilon_\mu = 0.0000064$ (0.00064% return per minute)
- Relative CI width target for variance: $\epsilon_{\sigma^2}^2 = 0.0394$ (3.9%)

In that setting, we find:

$$n_\mu \approx \left(\frac{1.96 \times 0.00064}{6.4 \times 10^{-6}}\right)^2 = (196)^2 \approx 38,416 \text{ and } n_{\sigma^2} \approx 1 + \frac{2 \cdot (1,96)^2}{(0.0394)^2} \approx 1 + 4,949.36 \approx 4,950.36$$

so that $n_{SR} \approx \max(38,416, 4,950.36) \approx 38,416$. Here, the mean estimation dominates.

15.3.7.3 Scaling takeaway for trading

- Small alpha signals (small ϵ_μ): mean estimation dominates, $n_{SR} \sim \frac{1}{\epsilon_\mu^2}$, requiring very long histories of independent observations. Here, 5000 minutes would detect only mean shifts $\epsilon_\mu \gtrsim \frac{z_{1-\frac{\alpha}{2}} \cdot \sigma}{\sqrt{n}} \approx 0.000018 \approx 0.028 \times \sigma$. Even with a relatively loose volatility precision target ($\epsilon_{\sigma^2} \sim 2\%$), detecting a mean shift of $0.01 \times \sigma$ requires $\sim 38,416$ independent minutes of data. That's about 99 trading days.
- Tight volatility precision (small ϵ_{σ^2}): variance estimation dominates, $n_{SR} \sim \frac{1}{\epsilon_{\sigma^2}^2}$, even for large mean shifts.
- In systematic trading, serial correlation reduces the effective n , making both requirements harder to satisfy.

15.3.7.4 Example: Daily trading Again, we assume an annualised volatility of 20% so that the daily volatility is:

$$\sigma_{day} \approx \frac{0.2}{\sqrt{252}} \approx 0.0126$$

which is $\sim 1.26\%$ per day. We assume $S^2 \approx \sigma_{day}^2$ and pick $n = 252$ (about 1 year of daily data). For large n , from Equation (15.13), we get

- Lower chi-sq bound: $\chi_{0.025}^2 \approx 251 - 1.96\sqrt{502} \approx 207.15$
- Upper chi-sq bound: $\chi_{0.975}^2 \approx 251 + 1.96\sqrt{502} \approx 294.85$

Then, we get

$$\begin{aligned} \sigma_{lower} &\approx \sqrt{\frac{251 \cdot \sigma^2}{294.85}} \approx \sigma \times 0.922 \approx 0.01161 \text{ (1.161\%)} \\ \sigma_{upper} &\approx \sqrt{\frac{251 \cdot \sigma^2}{207.15}} \approx \sigma \times 1.082 \approx 0.01363 \text{ (1.363\%)} \end{aligned}$$

Squaring these volatilities, we get

$$\sigma_{lower}^2 \approx 1.348 \times 10^{-4} \text{ and } \sigma_{upper}^2 \approx 1.857 \times 10^{-4} \text{ and } \sigma^2 \approx 1.587 \times 10^{-4}$$

Putting the terms together, the relative CI width for the variance is

$$\epsilon_{\sigma^2} \approx \frac{1.857 \times 10^{-4} - 1.348 \times 10^{-4}}{2 \times 1.587 \times 10^{-4}} \approx 0.16 \text{ (16\%)}$$

From Appendix (15.3.6), the relative CI width for volatility is

$$\epsilon_\sigma \approx \frac{0.16}{2} \approx 0.08, 8\% \text{ relative width for volatility}$$

Hence, with one year of daily data, the volatility is estimated to within $\pm 8\%$ (half-width) at 95% confidence.

For the mean, with $\sigma_{day} \approx 0.0126$, we assume the mean shift to be $\epsilon_\mu = 0.01 \times \sigma_{day} \approx 0.000126$ (0.0126%). Hence, in that setting, we get

$$\begin{aligned} n_\mu &\gtrsim \left(\frac{1.96 \times 0.0126}{1.26 \times 10^{-4}} \right)^2 = (196)^2 \approx 38,416 \\ n_{\sigma^2} &\gtrsim 1 + \frac{2 \cdot (1.96)^2}{(0.16)^2} \approx 1 + 300.125 \approx 301.125 \end{aligned}$$

Thus, the Sharpe ratio requirement is

$$n_{SR} \gtrsim \max(n_\mu, n_{\sigma^2}) \approx \max(38, 416, 301.125) = 38, 416$$

The above results says that with actual data $n = 252$, we are well below $n_{\sigma^2} = 301.125$, which is the minimum effective sample size needed to get $\sim 16\%$ variance-relative CI width.

15.3.7.5 Final interpretation for daily trading

- Volatility estimation: About 1.2 year of data gives $\sim 16\%$ relative CI width for variance ($\pm 8\%$ for volatility). Since $n_{current} \approx \frac{2 \times (1.96)^2}{(0.16)^2} \approx 300$, to get 8% RCI for variance we have $n_{target} \approx \frac{2 \times (1.96)^2}{(0.08)^2} \approx 1200$. The ratio of sample sizes is $\frac{n_{target}}{n_{current}} \approx 4$, so that to go from 16% to 8% variance-relative CI width, we need $n_{target} \approx 4 \times n_{current}$. That is, $n_{target} \approx 4 \times 300 \approx 1200$ days, which is about 4.76 years. So, to get 8% variance-relative width, we would need over 4.5 years of daily data.
- Mean estimation: Detecting a mean shift of $0.01 \times \sigma_{day}$ requires $\sim 38,000$ days.
- So for small alpha daily strategies, mean estimation is vastly more data-hungry than volatility estimation.

15.4 Extension to non-Gaussian variables, jumps, and dependence

The large- n scaling laws for mean and variance estimation discussed earlier assume Gaussian i.i.d. returns. In practice, financial returns often deviate from this ideal in three ways:

- Heavy tails (high kurtosis) even without jumps,
- Jumps or discontinuities,
- Serial dependence (autocorrelation).

These deviations change the constants in the scaling laws and, in some cases, reduce the effective number of observations.

15.4.1 Heavy tailed returns (no jumps)

For independent observations with finite variance σ^2 the mean estimator still satisfies:

$$\hat{\mu}_N \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

so that $n_\mu \propto \frac{1}{\epsilon_\mu^2}$ remains valid.

For variance estimation, the Gaussian-based variance formula $Var(S^2) = \frac{2\sigma^4}{n-1}$ is replaced by:

$$Var(S^2) = \frac{\sigma^4}{n} \left(K - 1 + \frac{2}{n-1} \right)$$

where K is the kurtosis. If $K > 3$ (heavy tails), the sampling variance increases linearly with $K - 1$ so the required sample size is multiplied by $\frac{K-1}{2}$ relative to the Gaussian case.

15.4.2 Impact of jumps

Suppose the return process is:

$$r_t = \mu\delta t + \sigma\sqrt{\delta t}Z_t + J_t$$

where $Z_t \sim N(0, 1)$ and $J_t = \sum_{k=1}^{N_t} Y_k$ is a jump term with $N_t \sim Poisson(\lambda\delta t)$ and $Y_k \sim N(0, \sigma_J^2)$. The unconditional variance and kurtosis are:

$$\begin{aligned} Var(r_t) &= \sigma^2\delta t + \lambda\delta t\sigma_J^2 \\ K &= 3 \frac{\sigma^4(\delta t)^2 + 6\sigma^2\delta t\lambda\delta t\sigma_J^2 + \lambda\delta t\mu_{4,J}}{(\sigma^2\delta t + \lambda\delta t\sigma_J^2)^2} \end{aligned}$$

where $\mu_{4,J}$ is the 4th central moment of jump sizes.

As an example, assume a sampling of 1-min returns ($\delta t = \frac{1}{390}$), annualised volatility is $\sigma = 0.2$, jump intensity is $\lambda = 5$, jump volatility is $\sigma_J = 0.005$. This yields a kurtosis $K \approx 9.1$ so that $K - 1 \approx 8.1$. Relative to the Gaussian case ($K - 1 = 2$), the required sample size for variance estimation is:

$$\text{Multiplier} \approx \frac{8.1}{2} \approx 4.05$$

That is, we need $4x$ more data to achieve the same variance CI width.

15.4.3 Non-independent observations

If returns are serially correlated (e.g., due to microstructure effects or persistence in volatility), the effective sample size is reduced. For an estimator $\hat{\theta}$ based on n autocorrelated observations, the variance is inflated by:

$$n_{eff} \approx \frac{n}{1 + 2 \sum_{k=1}^{\infty} \rho_k}$$

where ρ_k are the autocorrelations.

- For the mean: If $\rho_k > 0$ and $n_{eff} < n$, more observations are needed to detect a given ϵ_μ .
- For the variance: Serial correlation in squared returns (volatility clustering) reduces the number of independent variance estimates, increasing ϵ_{σ^2} .

As an example, If the lag-1 autocorrelation of returns is $\rho_1 = 0.2$ and all higher lags negligible:

$$n_{eff} \approx \frac{n}{1 + 2 \times 0.2} = \frac{n}{1.4}$$

We need $1.4x$ more raw observations to get the same statistical precision as in the i.i.d. case.

15.4.4 Practical implications

- Heavy tails: use the kurtosis-adjusted $Var(S^2)$ to scale n_{σ^2} .
- Jumps: even small jump intensities inflate K and can multiply required n_{σ^2} by factors of 2 – 5. Dependence: correct n to n_{eff} before applying Gaussian formulas for n_μ or n_{σ^2} .

Large- n scaling ($n \propto \frac{1}{\epsilon_\mu^2}$ for the mean, $n \propto \frac{1}{\epsilon_{\sigma^2}^2}$ for the variance) still holds in form, but the constants can be much larger, and n_{eff} replaces n when observations are dependent.

16 Stable indicator levels

Technical Analysis tries to derive profitable buy and sell signals by isolating upward and downward price trends from oscillations around a stable level (see Schulmeister [2005]). We let C_t be the closing price at time t , O_t is the opening price, L_t is the low price, H_t the high price, and V_t is the volume of the trade at time t . LL_n is the lowest low and HH_n is the highest high in the look back period with n discrete events. Some of these levels are as follows:

- Range: $R_t = H_t - L_t$. Large or increasing ranges suggest traders are prepared to continue to bid up or sell down a stock through the course of the day.

- The Typical Price

$$M_t = \frac{H_t + L_t + C_t}{3} \quad (16.15)$$

- Momentum: $C_t - C_{t-n}$, where n is the number of discrete events in the look back period.
- Support and resistance: the former is a price level that a share has reached but not fallen below, the latter is a price level that the market has reached but has not risen above. A breach of either of these levels is sometimes considered to indicate a significant change in a share's price, with the level breached then forming a new support or resistance level respectively.
- Retracing: it assumes that prices trends will eventually tend to reverse to visit a support or resistance level. Thus, attempting to predict a future price.
- UpDown Length: It is the number of consecutive days that a security price has either closed up (higher than previous day) or closed down (lower than previous days). Closing up values represented in positive numbers and closing down is represented with negative numbers. If a security closes at the same price on back to back days, the UpDown Length is 0.
- Accumulation / Distribution (A/D) oscillator: $\frac{(H_t - O_t) + (C_t - L_t)}{2(H_t - L_t)} \times 100$
- Moving average indicator: the simple moving average (SMA), or just MA, is

$$MA_t(n) = \frac{1}{n} \sum_{i=0}^{n-1} C_{t-i} \quad (16.16)$$

A buy signal is generated when a share's price exceeds the moving average and a sell signal when the moving average exceeds the share price. A variation of this indicator is the moving average convergence divergence (MACD) oscillator calculated by taking the difference of a short run and a long run moving averages of differing length. If the result is positive, this is taken as a signal that the market is trending upward.

- Exponential moving average (EMA) indicator (also known as an exponentially weighted moving average (EWMA)): It gives more weight to the most recent periods, making it more reliable than the SMA and a better representation of the recent performance of the security. Hence, it can be used to create a better moving average strategy. At time t , given a window of k periods, it is given by the formula

$$EMA(k)_t = EMA(k)_{t-1} + \alpha \times (C_t - EMA(k)_{t-1}) \quad (16.17)$$

where $EMA(k)_{t-k+1} = C_{t-k+1}$, $\alpha = \frac{2}{k+1}$ is a smoothing factor, and k is the time period of k -day EMA. Note, it can be rewritten as

$$EMA(k)_t = \alpha \times C_t + (1 - \alpha)EMA(k)_{t-1} \quad (16.18)$$

The weightage to the most recent data is greater for a shorter period EMA than for a longer period. Each term in the moving average period has an exponentially greater weightage than its preceding term, making the EMA faster to react than the SMA.

- The average true range (ATR): It measures market volatility by decomposing the entire range of an asset price for that period (see Welles Wilder [1978]). The true range indicator is taken as the greatest of the following: current high less the current low; the absolute value of the current high less the previous close; and the absolute value of the current low less the previous close. The ATR is then a moving average, generally using 14 days, of the true ranges. The true range is given by

$$TR_t = \max\left((H_t - L_t), |H_t - C_{t-1}|, |L_t - C_{t-1}|\right)$$

Note, the range $H_t - L_t$ measures the range/size of one candlestick. The larger the candle, the higher the ATR is, while the smaller the candle, the lower the ATR is. Hence, large or increasing ranges suggest traders are prepared to continue to bid up or sell down a stock through the course of the day. Then, we compute the arithmetic mean of the TR over n periods (excluding TR_t), getting $\overline{TR}_t(n) = \frac{1}{n} \sum_{i=1}^n TR_{t-i}$. Finally the ATR is given by

$$ATR_t(n) = \frac{1}{n} \sum_{i=0}^{n-1} TR_{t-i} = \frac{\overline{TR}_t(n-1) \times (n-1) + TR_t}{n} = \frac{ATR_{t-1}(n-1) \times (n-1) + TR_t}{n} \quad (16.19)$$

Thus, the ATR is roughly an average measure of the range/size of the past n candlesticks. A stock experiencing a high level of volatility has a higher ATR, and a lower ATR indicates lower volatility for the period evaluated.

17 Stochastic modelling and applications in finance

17.1 Modelling returns

We let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space where $(\mathcal{F}_t)$ is a right-continuous filtration including all $\mathbb{P}$ negligible sets in $\mathcal{F}$. We just assume that the stock price $(S_t)_{t \geq 0}$ is a semimartingale with a unique representation. For simplicity, we let the stock price $\{S_u; u \in [t, T]\}$, over the life of the option price, be a diffusion process (see Oksendal [1998]).

17.1.1 A diffusion model

We write the dynamics of the stock price, under the historical measure $\mathbb{P}$, as

$$\frac{dS_t}{S_t} = b_t dt + dM_{P,S}(t) \quad (17.20)$$

with drift and martingale

$$b_t = r_t - q_t + \sigma_t \lambda_t \text{ and } dM_{P,S}(t) = \sigma_t d\widehat{W}_S(t), \sigma_t = \sqrt{V_t}$$

where $\widehat{W}_S$ is a Brownian motion in the $\mathbb{P}$ -measure, V_t is a stochastic variance process, and λ_t is a market price of risk (MPR) process. The stock price S_t is given by

$$S_t = x e^{\int_0^t (r_s - q_s + \sigma_s \lambda_s) ds} \mathcal{E}(\sigma \widehat{W}_S)(t), S_0 = x$$

where the Doleans-Dade martingale is

$$\mathcal{E}(\sigma \widehat{W})(t) = e^{-\frac{1}{2} \int_0^t \sigma_s^2 ds + \int_0^t \sigma_s d\widehat{W}_S(s)}$$

We define the cumulative mean drift, under the historical measure $\mathbb{P}$, as

$$\begin{aligned}
H_P(t, T) &= \frac{1}{T-t} \int_t^T (r_s - q_s + \sigma_s \lambda_s) ds \\
&= \frac{1}{T-t} \int_t^T r_s ds - \frac{1}{T-t} \int_t^T q_s ds + \frac{1}{T-t} \int_t^T \sigma_s \lambda_s ds = H^r(t, T) + H^q(t, T) + H^\lambda(t, T)
\end{aligned}$$

where $H^r(t, T)$ is the cumulative mean rate, $H^q(t, T)$ is the cumulative mean repo, and $H^\lambda(t, T)$ is the cumulative mean market price of risk. The expected cumulative mean drift, under the historical measure $\mathbb{P}$, is

$$H_{P,0}(t, T) = E[H_P(t, T) | \mathcal{F}_t]$$

We define the mean variance as

$$I^2(t, T) = \frac{1}{T-t} Y_t \text{ for } Y_t = \int_t^T V(X_s) ds$$

where the cumulative variance process Y_t is not an adapted process. We denote $\tilde{I}(t, T) = (T-t)I^2(t, T)$ the total variance over the period $[t, T]$. The expected mean variance, under the historical measure, is

$$I_0^2(t, T) = E[I^2(t, T) | \mathcal{F}_t]$$

From Equation (17.20), we observe that the cumulative return, under the historical measure $\mathbb{P}$, is

$$\begin{aligned}
R_P(t, T) &= \int_t^T \frac{dS_u}{S_u} = (T-t)H_P(t, T) + \int_t^T dM_{P,S}(u) \\
&= (T-t)H^r(t, T) + (T-t)H^q(t, T) + (T-t)H^\lambda(t, T) + \int_t^T dM_{P,S}(u)
\end{aligned}$$

Taking conditional expectation, we get the expected cumulative return

$$\begin{aligned}
R_{P,0}(t, T) &= E[R_P(t, T) | \mathcal{F}_t] = (T-t)H_{P,0}(t, T) \\
&= (T-t)H_{Q,0}(t, T) + (T-t)H_0^\lambda(t, T)
\end{aligned} \tag{17.21}$$

17.1.2 The forward price

Given the dynamics of the stock price in Equation (17.20), the spot price at time T seen at time t is

$$S_{P,T} = S_t C_P(t, T) \mathcal{E}_P^\sigma(t, T)$$

where

$$C_P(t, T) = e^{(T-t)H_P(t, T)} \text{ and } \mathcal{E}_P^\sigma(t, T) = e^{-\frac{1}{2}(T-t)I^2(t, T) + \int_t^T \sigma_s dW(s)}$$

Taking expectation, we get

$$C_{P,0}(t, T) = E[C_P(t, T) | \mathcal{F}_t], \quad E[\mathcal{E}_P^\sigma(t, T) | \mathcal{F}_t] = 1$$

where $C_{P,0}(t, T)$ is the normalised forward price seen at time t given by

$$\begin{aligned}
C_{P,0}(t, T) &= E[e^{(T-t)H_P(t, T)} | \mathcal{F}_t] = E[e^{(T-t)H^r(t, T) + (T-t)H^q(t, T) + (T-t)H^\lambda(t, T)} | \mathcal{F}_t] \\
&= E[e^{(T-t)H^r(t, T)} | \mathcal{F}_t] E[e^{(T-t)H^q(t, T)} | \mathcal{F}_t] E[e^{(T-t)H^\lambda(t, T)} | \mathcal{F}_t] \Psi(t, T) \Psi^\lambda(t, T) \\
&= \frac{D_q(t, T)}{P(t, T)} \Lambda^\sigma(t, T) \Psi(t, T) \Psi^\lambda(t, T)
\end{aligned}$$

and

- $P(t, T) = E[e^{-(T-t)H^r(t, T)} | \mathcal{F}_t]$ is the discount factor
- $D_q(t, T) = E[e^{(T-t)H^q(t, T)} | \mathcal{F}_t]$ is the repo factor
- $\Lambda^\sigma(t, T) = E[e^{(T-t)H^\lambda(t, T)} | \mathcal{F}_t]$ is the market price of risk factor.

Note, $\Psi(t, T)$ is an adjustment term accounting for the joint variation between the returns of all the stochastic processes in the drift and S , except for the market price of risk λ . It is a direct application of the semimartingale exponential. The quadratic variation can be composed of several covariances depending on the number of stochastic processes. Further, we let $\Psi^\lambda(t, T)$ denote the specific adjustment term accounting for the covariance between the market price of risk, λ , and the spot price S . We denote $\Lambda^{\Psi^\lambda}(t, T)$ the modified MRP factor

$$\Lambda^{\Psi^\lambda}(t, T) = E^{S^{\tilde{S}}} [e^{(T-t)H^\lambda(t, T)} | \mathcal{F}_t] = \Lambda^\sigma(t, T) \Psi^\lambda(t, T)$$

which corresponds to the MPR factor, under the $P^{\tilde{S}}$ measure. We can then rewrite the normalised forward price as

$$C_{P,0}(t, T) = \frac{D_q(t, T)}{P(t, T)} \Lambda^{\Psi^\lambda}(t, T) \Psi(t, T)$$

17.1.3 Defining returns

The returns over the period $[t, T]$ are estimated as follows:

- Simple return

$$R_{t,T} = \frac{S_{P,T}}{S_t} - 1 = C_P(t, T) \mathcal{E}_P^\sigma(t, T) - 1$$

with expected value

$$\begin{aligned}
E[R_{t,T} | \mathcal{F}_t] &= C_{P,0}(t, T) - 1 = \frac{F_P(t, T; S_t)}{S_t} - 1 \\
&\approx (T-t)y_P(t, T) \\
&\approx (T-t)H_{P,0}(t, T)
\end{aligned}$$

Hence, we get

$$E[R_{t,T} | \mathcal{F}_t] \approx E[R_P(t, T) | \mathcal{F}_t] = R_{P,0}(t, T)$$

- Log-return

$$\begin{aligned}
r_L(t, T) = \log \frac{S_{P,T}}{S_t} &= \log C_P(t, T) + \log \mathcal{E}_P^\sigma(t, T) \\
&= (T-t)H_P(t, T) - \frac{1}{2}(T-t)I^2(t, T) + \int_t^T \sigma_s dW(s)
\end{aligned}$$

with expected value

$$\begin{aligned}
E[r_L(t, T)|\mathcal{F}_t] &= (T-t)H_{P,0}(t, T) - \frac{1}{2}(T-t)I_0^2(t, T) \\
&\approx (C_{P,0}(t, T) - 1) - \frac{1}{2}(T-t)I_0^2(t, T) \\
&= E[R_{t,T}|\mathcal{F}_t] - \frac{1}{2}(T-t)I_0^2(t, T)
\end{aligned} \tag{17.22}$$

Thus, we can deduce that

$$(T-t)I_0^2(t, T) \approx 2\left\{E[R_{t,T}|\mathcal{F}_t] - E[r_L(t, T)|\mathcal{F}_t]\right\}$$

17.1.4 Example: the GBM

So far we have presented results based on a general diffusion model. We are now going to specify the approach by considering the simple case of a GBM. The dynamics of the stock price, under the $\mathbb{P}$ -measure, are

$$\frac{dS_t}{S_t} = (r - q + \sigma\lambda)dt + \sigma dW_t$$

where $\sigma > 0$ is the volatility and λ is a constant MPR. Integrating that dynamics in the range $[t, T]$ the lo stock price is given by

$$S_{P,T} = S_t C_P(t, T) \mathcal{E}_P^\sigma(t, T) \text{ where } C_P(t, T) = e^{(r-q+\sigma\lambda)(T-t)}, \mathcal{E}_P^\sigma(t, T) = e^{-\frac{1}{2}(T-t)I^2(t, T) + \sigma W_{T-t}}$$

where $I^2(t, T) = I_0^2(t, T) = \sigma^2$. Thus,

$$(T-t)H_P(t, T) = (r - q + \sigma\lambda)(T-t)$$

Taking the expectation, we get

$$(T-t)H_{P,0}(t, T) = (r - q + \sigma\lambda)(T-t) \text{ and } \mathcal{E}_{P,0}^\sigma(t, T) = 1$$

so that $H_P(t, T) = H_{P,0}(t, T)$. Similarly,

$$C_{P,0}(t, T) = E[e^{(r-q+\sigma\lambda)(T-t)}|\mathcal{F}_t] = e^{(r-q+\sigma\lambda)(T-t)}$$

so that

$$C_P(t, T) = C_{P,0}(t, T) \approx 1 + (r - q + \sigma\lambda)(T-t) = 1 + (T-t)H_{P,0}(t, T)$$

It can be computed with option prices as

$$C_{P,0}(t, T) - 1 = \frac{\bar{C}(t; K^*, T) - \bar{P}(t; K^*, T)}{K^*}$$

where $K^* = S_t$. The forward price becomes

$$F_P(t, T; S_t) = S_t C_{P,0}(t, T)$$

The yield curve of the stock price is

$$y_P(t, T) = \frac{1}{T-t} \log C_{P,0}(t, T) = H_{P,0}(t, T) = r - q + \sigma\lambda$$

Similarly, the logarithm of the stock price $S_{P,T}$ is given by

$$\log \frac{S_{P,T}}{S_t} = (r - q + \sigma\lambda)(T - t) + \log \mathcal{E}_P^\sigma(t, T)$$

and taking the expectation, we get

$$E[\log \frac{S_{P,T}}{S_t} | \mathcal{F}_t] = (r - q + \sigma\lambda - \frac{1}{2}\sigma^2)(T - t) = (T - t)H_{P,0}(t, T) - \frac{1}{2}I_0(t, T)(T - t)$$

17.2 Options pricing

17.2.1 Pricing options with digital products

Following Ingersoll [1998] [2000], we call $D_B(S, t, T; \xi)$ the Digital Bond and $D_S(S, t, T; \xi)$ the Digital Stock for the event ξ .

- Digital Bond: We let $D(S, t, T; A)$ be the value at time t of receiving \$1 at the maturity T if and only if the event A occurs. Such a contract is called a binary option or a cash-or-nothing option.
- Digital Stock: we let $S(S, t, T; A)$ be the value at time t of receiving one share of the stock at the maturity T (excluding any intervening dividends) if and only if the event A occurs. Such a contract is called a all-or-nothing share.

In the special case where the rate, repo and volatility are constants, Cox et al. [1976] showed that the Digital Bond and Digital Share with event $\xi = \{S_T > K\}$, where K is the strike price, could be valued under the risk-neutral measure $\mathbb{Q}$ with the Black-Scholes formula (see Black et al. [1973]). That is,

$$\begin{aligned} D(S, t, T; \xi) &= E^Q[e^{-\int_t^T r_s ds} I_{S_T > K} | \mathcal{F}_t] = e^{-r(T-t)} N(d_2(T-t, S_t e^{(r-q)(T-t)}, K)) \\ S(S, t, T; \xi) &= E^Q[e^{-\int_t^T r_s ds} S_T I_{S_T > K} | \mathcal{F}_t] = S_t e^{-q(T-t)} N(d_1(T-t, S_t e^{(r-q)(T-t)}, K)) \end{aligned}$$

European contracts are piecewise linear in the spot price on the maturity date. The value of any of these contracts can be represented

$$\sum_i a_i D(S, t, T; \xi_i) + \sum_j b_j S(S, t, T; \xi_j)$$

for some weights a_i and b_j .

In order to compute the Digital Bond for some event ξ we need to compute some probabilities under the cash numeraire, while to compute the Digital Stock we need to compute some probabilities under the stock numeraire. Combining the Digital Bond and Digital Stock, we can recover many financial products.

17.2.2 The digital bond

17.2.2.1 A diffusion model Assuming the price dynamics in Equation (17.20), the digital bond, under the historical measure, becomes

$$D(S, t, T; \xi) = E^P[e^{-(T-t)H^r(t, T)} I_{\{S_{P,T} > K\}} | \mathcal{F}_t]$$

We take the zero-coupon bond price $P(t, T)$ as Numeraire with density $Z_P(u) = \frac{P(u, T)}{P(t, T)} e^{-(T-u)H^r(t, u)}$, for $u \in [t, T]$. We can then change measure and rewrite the expectation as

$$\begin{aligned} E^P[e^{-(T-t)H^r(t, T)} I_{\{S_{P, T} > K\}} | \mathcal{F}_t] &= P(t, T) E^P[Z_P(T) I_{\{S_{P, T} > K\}} | \mathcal{F}_t] \\ &= P(t, T) E^T[I_{\{S_{P, T} > K\}} | \mathcal{F}_t] \end{aligned}$$

where $E^T[\cdot]$ is the conditional expectation under the forward measure. Hence, we can rewrite the digital bond as

$$D(S, t, T; \xi) = P(t, T) E^T[I_{\{S_{P, T} > K\}} | \mathcal{F}_t] = P(t, T) P^T(S_T^T \geq K)$$

where S_T^T is the spot price in the forward measure. In that setting, the Brownian motion of the spot price has been translated to the new Brownian motion W^T and an extra term is added to the cumulative mean drift H_P . Replacing the spot price with its components, we get

$$D(S, t, T; \xi) = P(t, T) P^T(C_T(t, T) \mathcal{E}_T^\sigma(t, T) \geq \frac{K}{S_t})$$

Setting $X_t^T = \int_t^T \sigma_s dW_s^T$ and taking the logarithm on the terms in the probability, we get

$$\begin{aligned} D(S, t, T; \xi) &= P(t, T) P^T((T-t)H_q(t, T) + (T-t)H^\lambda(t, T) - \frac{1}{2}(T-t)I^2(t, T) + X_t^T \geq \log \frac{K}{S_t}) \\ &= P(t, T) P^T(X_t^T \leq \log \frac{S_t}{K} + (T-t)H_q(t, T) + (T-t)H^\lambda(t, T) - \frac{1}{2}(T-t)I^2(t, T)) \end{aligned}$$

In the special case where the initial spot price $S_t = x$ is close to the at-the-money forward strike, $K \approx x e^{(r-q)(T-t)}$, the value of the digital bond depends on the spread

$$\phi(t, T) = (T-t)(H^\lambda(t, T) - \frac{1}{2}I^2(t, T))$$

17.2.2.2 The BS-assumptions In the special case where we consider the BS-assumptions, the digital bond, under the historical measure, simplifies to

$$D(S, t, T; \xi) = e^{-r(T-t)} P(X_t \leq \log \frac{S_t}{K} + (r-q)(T-t) + \lambda \sigma(T-t) - \frac{1}{2} \sigma^2(T-t))$$

where $X_t = \int_t^T \sigma dW_s = \sigma W_{T-t} = \sigma \sqrt{T-t}x$, and $x \sim N(0, 1)$. Hence,

$$D(S, t, T; \xi) = e^{-r(T-t)} N(d_2^\lambda)$$

where

$$d_2^\lambda = \frac{1}{\sigma \sqrt{T-t}} (\log \frac{S_t}{K} + (r-q)(T-t) + \lambda \sigma(T-t) - \frac{1}{2} \sigma^2(T-t))$$

For $\lambda = 0$, we get $d_2^0 = d_2$, and we recover the well know formula

$$D(S, t, T; \xi) = e^{-r(T-t)} N(d_2)$$

17.2.2.3 Some approximations Considering the BS-assumptions, in the special case where the initial spot price $S_t = x$ is close to the at-the-money forward strike, $K \approx xe^{(r-q)(T-t)}$, some approximations can be made. For x small enough, expanding both the CDF and the exponential term, we get

$$N(x) \approx \frac{1}{2} + \frac{1}{\sqrt{2\pi}}(1 - \frac{1}{2}x^2)x$$

If x is small, so is the quantity x^3 , and the above expression further simplifies to

$$N(x) \approx \frac{1}{2} + \frac{1}{\sqrt{2\pi}}x$$

Given $\frac{1}{\sqrt{2\pi}} \approx \frac{2}{5}$, we get

$$N(x) \approx \frac{1}{2} + \frac{2}{5}x$$

Since $K \approx xe^{(r-q)(T-t)}$, we get

$$d_2^\lambda \approx \frac{1}{\sigma\sqrt{T-t}}(\lambda\sigma(T-t) - \frac{1}{2}\sigma^2(T-t)) = (\lambda - \frac{1}{2}\sigma)\sqrt{T-t}$$

Replace in the CDF, we get

$$N(d_2^\lambda) \approx \frac{1}{2} + \frac{2}{5}(\lambda - \frac{1}{2}\sigma)\sqrt{T-t}$$

We see that

- for $N(d_2^\lambda) > \frac{1}{2}$, we must have $\lambda > \frac{1}{2}\sigma$,
- and for $N(d_2^\lambda) < \frac{1}{2}$, we must have $\lambda < \frac{1}{2}\sigma$.

Further, $N(d_2^\lambda) \approx \frac{1}{2}$ if either $\lambda \approx \frac{1}{2}\sigma$ or $\sqrt{T-t} < \epsilon$.

17.3 Generating data with stochastic processes

Synthetic data generation can roughly be categorised into two distinct classes (see Bloch [2023]):

1. **Process-driven methods:** It derives synthetic data from a computational or mathematical model of an underlying psychical process. Define a stochastic model of the underlying asset processes (see Bakshi [1997], Bates [1996] [2000]), calibrate it to the observed option prices, and use it to simulate many trajectories.
2. **Data-driven methods:** It has historically relied on modelling multivariate probability models such as Gaussian Mixture Models (GMM), Gaussian copulas, and Bayesian networks (see Russell et al. [2009]). In the case of deep learning, several model architectures have been investigated, with selforganising maps (SOMs), Variational Autoencoders (VAs) (see Kingma et al. [2014]), and Generative Adversarial Networks (GANs) (see Goodfellow et al. [2014]) being the most promising research trajectories. These methods are nonparametric, nonlinear estimators of latent variables. VAs learn how to represent high-dimensional observations in a low-dimensional space by maximising the likelihood of observing the given (original) data samples under the generated samples. SOMs apply competitive learning and use a neighbourhood function to preserve the topological properties of the input space. GANs train two competing neural networks, where one network (the generator) is tasked with generating simulated observations from a distribution function, and the other network (the discriminator) is tasked with predicting the probability that the simulated observations are false given the true observed data.

For simplicity we only consider the process-driven methods. We start by specifying a particular stochastic differential equation (SDE), such as the diffusion process given in Equation (17.20), and then we select the model parameters. We can then simulate as many trajectories as required with a Monte Carlo engine.

- In the case (1), where the data must be statistically indistinguishable from the original dataset, we estimate statistically the model parameters on the original dataset (see Bakshi [1997], Bates [1996] [2000]), which we can then perturbate.
- In the case (2), where we need to generate scenarios, we assume a (discrete) range for the model parameters and consider all the combinations which we simulate in turn with a Monte Carlo engine.

Focussing on case (1), the simplest approach is to assume that returns are normally distributed, fit the model parameters on the original dataset, and perturbate these parameters. Given a realisation of the stock price $S = (S_1, \dots, S_N)$, where $T = N\Delta t$, we estimate statistically its first two sample moments (see Appendix (15.1)). We denote $\hat{\mu}$ and $\hat{\sigma}$ the sample mean and sample standard deviation, respectively.

First, we let the returns be Gaussian $N(\hat{\mu}, \hat{\sigma})$ with dynamics given by

$$\frac{dS_t}{S_t} = \hat{\mu}dt + \hat{\sigma}dW_t \quad (17.23)$$

In order to generate scenarios for the stock price, we perturbate the model parameters by assuming they follow a Gaussian distribution centred on their values. We modify the sample mean $\bar{\mu} \sim N(\hat{\mu}, \sigma^*)$ and the sample standard deviation $\bar{\sigma} \sim N(\hat{\sigma}, \sigma^*)$, where σ^* is a chosen volatility. The dynamics of returns become

$$\frac{dS_t}{S_t} = \bar{\mu}dt + \bar{\sigma}dW_t$$

We can also complexify the dynamics of returns by adding jumps and stochastic volatility (see Bates [2000]). That is,

$$\frac{dS_t}{S_t} = b_t dt + dM_{P,S}(t)$$

with drift and martingale

$$b_t = \hat{\mu} + \sigma_t \lambda_t \text{ and } dM_{P,S}(t) = \sigma_t d\widehat{W}_S(t) + dM_S^{J,P^\gamma}(t)$$

where the drift is centred around the sample mean $\hat{\mu}$, $W_S^{P^\gamma}$ is a $\mathbb{P}^\gamma$ -Brownian motion and $M_S^{J,P^\gamma}(t)$ is an adjusted random measure for jump processes. As an example of a jump process, we can follow the approach by Merton (see Merton [?]) and consider a Poisson process with positive jump size log-normally distributed. In addition, $\sigma_t = \sqrt{V_t}$ is a stochastic volatility and λ_t is a stochastic market price of risk (MPR). We can further complexify the dynamics of returns by replacing the Brownian motions with fractal (or multi-fractal) Brownian motions.

Let x_t be a OU-process with dynamics

$$dx_t = a_x(b_x - x_t)dt + \sigma_x dW_x(t), \quad x_0 = x \quad (17.24)$$

where a_x is the speed of mean reversion, b_x is the infinite value of x , and $\sigma_x > 0$ is the instantaneous volatility. Let v_t be a square-root process with dynamics

$$dv_t = (b_v - a_v v_t)dt + \sigma_v \sqrt{v_t} dW_v(t), \quad v_0 = v$$

where b_v, σ_v are positive and $a_v \in \mathbb{R}$. The infinite variance is $\bar{v} = \frac{b_v}{a_v}$. For $b_v > \frac{1}{2}\sigma_v$ the process v_t never reaches zero. In our setting, x can represent the spot rate and the market price of risk, that is, $x = r, \lambda$.

As an example, we can let the MPR be an Ornstein-Uhlenbeck (OU)-process and the variance be a square-root process. Hence, we get

$$d\lambda_t = a_\lambda(b_\lambda - \lambda_t)dt + \sigma_\lambda dW_\lambda(t), \lambda_0 = \lambda$$

We can then define a (discrete) range for the model parameters: a_λ , b_λ , σ_λ , λ , a_v , b_v , σ_v , and v . At last, we simulate the trajectories with a Monte Carlo engine.

References

- [2025] Ataei M., Multiscale causal analysis of market efficiency via news uncertainty networks and the financial chaos index. Working Paper, arXiv:2505.01543.
- [2023] Aydinhan A.O., Kolm P.N., Mulvey J.M., Shu Y., Identifying patterns in financial markets: Extending the statistical jump model for regime identification. Working Paper.
- [1997] Bakshi G., Cao C., Chen Z., Empirical performance of alternative option pricing models. *Journal of Finance*, **52**, (5), pp 2003–2049.
- [2006] Barndorff-Nielsen O.E., Shephard N., Econometrics of testing for jumps in financial economics using bipower variation. *Journal of Financial Econometrics*, **4**, pp 1–30.
- [1996] Bates D., Jumps and stochastic volatility: Exchange rate processes implicit in Deutschemark options. *Review of Financial Studies*, **9**, pp 69–108.
- [2000] Bates D., Post-87 crash fears in the S&P 500 futures option market. *Journal of Econometrics*, **94**, (1-2), pp 181–238.
- [2019] Becker S., Cheridito P., Jentzen A., Deep optimal stopping. *Journal of Machine Learning Research*, **20**, (74), pp 1–25. Technical report in 2018.
- [2000] Bellamy N., Jeanblanc M., Incompleteness of markets driven by a mixed diffusion. *Journal of Finance and Stochastics*, **4**, (2), pp 209–222.
- [2024] Ben Alaya M., Kebaier A., Ren Z., Sarr D., Financial stochastic models diffusion: From risk-neutral to real-world measure. Working Paper, arXiv:2409.12783.
- [2020] Benidis K., Rangapuram S.S., Flunkert V., Gasthaus J., Wang Y., Neural forecasting: Introduction and literature overview. Working Paper, arXiv:2004.10240.
- [2021] Berdondini A., The theory of quantitative trading. Working Paper.
- [2009] Bertram W., Analytic solutions for optimal statistical arbitrage trading. Working paper. Available at <http://ssrn.com/abstract=1505073>
- In *Physica A: Statistical Mechanics and its Applications*, **389**, (11), pp 2234–2243.
- [1973] Black F., Scholes M., The pricing of options and corporate liabilities. *Journal of Political Economics*, **81**, pp 637–659.
- [2023] Bloch D., Futuretesting quantitative strategies. Working Paper, University of Paris 6 Pierre et Marie Curie.
- [2024a] Bloch D., Dynamically characterising time series with Relative Moving Moments. Working Paper, University of Paris 6 Pierre et Marie Curie. Published in *Wilmott Magazine*, Feb 26, 2025.
- [2024b] Bloch D., Pricing American options with OST-TDBP. Working Paper, University of Paris 6 Pierre et Marie Curie.
- [2024c] Bloch D., Optimal Trading With OST-TDBP. Working Paper, University of Paris 6 Pierre et Marie Curie.
- [2024] Bloch D., et al. Detecting and predicting price jumps with consecutive signature distance. *Wilmott Magazine*, Oct 25, 2024.

- [2025] Bloch D., Why AI will never predict financial markets. *Risk*, Apr 16, 2025.
- [1986] Bollerslev T., Generalized autoregressive conditional heteroskedasticity. *Journal of econometrics*, **31**, (3), pp 307–327.
- [1992] Bollinger J., Using Bollinger bands. *Technical Analysis of Stocks and Commodities*, **10**, February.
- [2025] Buehler H., Horvath B., Limmer Y., Schmidt T., Uncertainty-aware strategies: A model-agnostic framework for robust financial optimization through subsampling. Working Paper, arXiv:2506.07299.
- [2021] Chan E.P., Quantitative trading: How to build your own algorithmic trading business. Wiley Trading, 2nd edition.
- [2024] Chattopadhyay R., Malichkar A., Ren Z., Zhang X., Volatility-volume order Slicing via statistical analysis. Working Paper, arXiv:2412.12482.
- [2021] Cit A., AI-driven algorithmic trading with real-time risk management: Integrating control systems for optimized portfolio management. Working Paper.
- [2001] Cont R., Empirical properties of asset returns: Stylized facts and statistical issues. *Quantitative Finance*, **1**, (2), pp 223–236.
- [2025] Cortes A., Rehm R., Letzelter V., Winner-takes-all for multivariate probabilistic time series forecasting. Working Paper, arXiv:2506.05515.
- [1976] Cox J.C., Ross S., The valuation of options for alternative stochastic processes. *Journal of Financial Economics*, **3** (January-March), pp 145–166.
- [2024] Dolfín M., Kapetanios G., Leonida L., De Leon Miranda J., Investor behavior and multiscale cross-correlations: Unveiling regime shifts in global financial markets. Working Paper, arXiv:2408.17200.
- [2024] Duan L., Exploring Efficient Quantitative Trading Strategies: A Comprehensive Comparison of Momentum, SMAs and Machine Learning. *Advances in Economics Management and Political Sciences*, **86**, (1), pp 43–48.
- [2024] Dung P.T., Giudici P., SAFE machine learning in quantitative trading. Working Paper, University of Pavia
- [1982] Engle R.F., Autoregressive conditional heteroskedasticity with estimates of the variance of U.K. inflation. *Econometrica*, **50**, pp 987–1008.
- [2021] Ferreira I.H., Medeiros M.C., Modeling and forecasting intraday market returns: A machine learning approach. Working Paper, arXiv:2112.15108.
- [2025] Genet R., VWAP execution with signature-enhanced transformers: A multi-asset learning approach. Working Paper, arXiv:2503.02680.
- [2022] Ghosh S., Wynter L., Lim S.H., Nguyen D.T., Neural-progressive hedging: Enforcing constraints in reinforcement learning with stochastic programming. Working Paper, arXiv:2202.13436.
- [2024] Goluzá S., Kovacevic T., Bauman T., Kostanjcar Z., Deep reinforcement learning with positional context for intraday trading. Working Paper, arXiv:2406.08013.
- [2014] Goodfellow I.J., Pouget-Abadie J., Mirza M., Xu B., Warde-Farley D., Ozair S., Courville A., Bengio Y., Generative adversarial nets. In *Proceedings of the International Conference on Neural Information Processing Systems*, pp 2672–2680.
- [2025] Guo W., Wang Y., Huang Z., Zhang C., Ma S., Trading under uncertainty: A distribution-based strategy for futures markets using FutureQuant Transformer. Working Paper, arXiv:2505.05595.

- [1994] Hamilton J.D., Time series analysis. Princeton University Press.
- [2002] Harris L., Trading and exchanges market microstructure for practitioners. OUP Oxford.
- [2020] Ho J., Jain A., Abbeel P., Denoising diffusion probabilistic models. In NeurIPS, 2020.
- [1998] Ingersoll, Jr J.E., Approximating American options and other financial contracts using barrier derivatives. *Journal of Computational Finance*, **2** (Fall), pp 85–112.
- [2000] Ingersoll J.E., Digital contracts: Simple tools for pricing complex derivatives. *The Journal of Business*, **73**, (1), pp 67–88.
- [2009] Jeanblanc M., Yor M., Chesney M., Mathematical methods for financial markets. Springer.
- [2005] Jiang G.J., Oomen R., A new test for jumps in asset prices. Working Paper, Eller College of Management, University of Arizona.
- [2013] Jiang G.J., Yao T., Stock price jumps and cross-sectional return predictability. *The Journal of Financial and Quantitative Analysis*, **48**, (5), pp 1519–1544.
- [2014] Kingma D.P., Welling M., Auto-encoding variational bayes. In *Proceedings of International Conference on Learning Representations*, (ICLR).
- [2024] Konstantinidis T., Iacovides G., Xu M., Constantinides T.G., Mandic D., FinLlama: Financial sentiment classification for algorithmic trading applications. Working Paper, arXiv:2403.12285.
- [2008] Lee S.S., Mykland P., Jumps in real-time financial markets: A new nonparametric test and jump dynamics. *Review of Financial Studies*, (Univ. of Chicago), **21**, (6), pp 2535–2563. Technical Report No. 566, University of Chicago, 2006.
- [2019] Leung T., Zhang H., Optimal trading with a trailing stop. arXiv:1701.03960. Published in 2021 in *Applied Math & Optimization*, **83**, pages 669–698.
- [2015] Lien K., Day trading and swing trading the currency market: Technical and fundamental strategies to profit from market moves Wiley, (Third edition).
- [2000] Lo A.W., Mamaysky H., Wang J., Foundations of technical analysis: Computational algorithms, statistical inference, and empirical implementation. *The Journal of Finance*
- [2021] Lovas A., Rasonyi M., Ergodic aspects of trading with threshold strategies. Working Paper, arXiv:2111.14708.
- [1963] Mandelbrot B.B., The variation of certain speculative prices. *Journal of Business*, **36**, (4), pp 394–419.
- [1967] Mandelbrot B.B., Forecasts of future prices, unbiased markets and martingale models. *Journal of Business*, **39**, pp 242–255.
- [1968] Mandelbrot B.B., van Ness J., Fractional Brownian motions, fractional noises and applications. *SIAM Review*, **10**, (4), pp 422–437.
- [1997] Mandelbrot B.B., Fisher A., Calvet L., A multifractal model of asset returns. Cowles Foundation Discussion Paper No. 1164.
- [2024] Maneesoonthorn W., Frazier D.T., Martin G.M., Probabilistic predictions of option prices using multiple sources of data. Working Paper, arXiv:2412.00658.

- [2024] Macrì A., Lillo F., Reinforcement learning for optimal execution when liquidity is time-varying. Working Paper, arXiv:2402.12049.
- [2023] Marcondes D., Simonis A., Metastable financial markets. Working Paper, arXiv:2310.13081.
- [2022] Marti G., From data to trade: A machine learning approach to quantitative trading. Working Paper, Ecole Polytechnique.
- [2015] McNeil A.J., Frey R., Embrechts P., Quantitative risk management. Princeton University Press.
- [1999] Murphy J.J., Technical analysis of the financial markets: A comprehensive guide to trading methods and applications. NYIF.
- [2023] Nagy P., Calliess J-P., Zohren S., Asynchronous deep double dueling Q-learning for trading-signal execution in limit order book markets. Working Paper, arXiv:2301.08688.
- [2009] Narang R.K., Inside the black box: The simple truth about quantitative trading. John Wiley & Sons, (1st ed.).
- [2024] Nawathe S., Panguluri R., Zhang J., Venkatesh S., Multimodal deep reinforcement learning for portfolio optimization. Working Paper, arXiv:2412.17293.
- [2024] Ndikum P., Ndikum S., Advancing investment frontiers: Industry-grade deep reinforcement learning for portfolio optimization. Working Paper, arXiv:2403.07916.
- [2014] Neely C.J., Rapach D.E., Tu J., Guofu Z., Forecasting the equity risk premium: The role of technical indicators. *Management Science*, **60**, (7), pp 1617–1859.
- [1998] Oksendal B., Stochastic differential equation: An introduction with applications. Springer, Fifth Edition.
- [2021] Pereira O.S., Algorithmic trading strategies: Automation and back-testing the perfect order strategy. NOVA Information Management School, Universidade Nova de Lisboa.
- [1994] Peters E.E., Fractal market analysis: Applying chaos theory to investment and economics. John Wiley & Sons.
- [1991-96] Peters E.E., Chaos and order in the capital markets. second edition, John Wiley & Sons.
- [2017] Peterson B.G., Developing and backtesting systematic trading strategies. Working Paper, University of Illinois Urbana-Champaign.
- [2018] Perumal K., Flint E., Systematic testing of systematic trading strategies. *The Journal of Investment Strategies*, **7**, (3), pp 29–49.
- [2023] Qin M., Sun S., Zhang W., Xia H., Wang X., An B., EarnHFT: A three-stage reinforcement learning framework for high-frequency trading adaptation. Working Paper, arXiv:2309.12891.
- [2021] Rasul K., Seward C., Schuster I., Vollgraf R., Autoregressive denoising diffusion models for multivariate probabilistic time series forecasting. In ICML, 2021.
- [2009] Russell S.J., Norvig P., Artificial intelligence: A modern approach. Prentice Hall, (3rd ed.), (original version 1995, Prentice hall Englewood Cliffs), Upper Saddle River, New Jersey.
- [2005] Schulmeister S., Components of the profitability of technical currency trading. WIFO Working Papers, No. 263, December.
- [1994] Sharpe W.F., The Sharpe ratio. *Journal of Portfolio Management*, **21**, pp 49–58.

- [2018] Srivastava S., Bhattacharya R., Evaluating the building blocks of a dynamically adaptive systematic trading strategy. Working Paper, arXiv:1812.02527.
- [2024] Liu P., Wu B., Hu Y., Li N., Dai T., Bao J., Xia S-T., TimeBridge: Non-stationarity matters for long-term time series forecasting. Working Paper, arXiv:2410.04442.
- [2000] Thomas N.B., Encyclopedia of chart patterns. John Wiley & Sons.
- [2013] Treleaven P., Galas M., Lalchand V., Algorithmic trading review. *Communications of the ACM*, **56**, (11), pp 76–85.
- [2010] Tsay R.S., Analysis of financial time series. Wiley, (3rd ed.).
- [2008] van Belle G., Statistical rules of thumb. Wiley, (2nd ed.).
- [2020] Velu R., Hardy M., Nehren D., Algorithmic trading and quantitative strategies. CRC Press, Chapman & Hall.
- [2025] Vytelingum P., Baggott R., Stillman N., Zhang J., Zhu D., Chen T., Lyon J., Agent-based liquidity risk modelling for financial markets. Working Paper, arXiv:2505.15296.
- [2007] Wang J-L., Chan S-H., Stock market trading rule discovery using pattern recognition and technical analysis. *Expert Systems with Applications*, 33.2.
- [2024] Wang L., Cheng Y., Gu X., Wu Z., Design and optimization of big data and machine learning-based risk monitoring system in financial markets. Working Paper, arXiv:2407.19352.
- [1978] Welles Wilder Jr. J., New concepts in technical trading systems. Trend Research.
- [2025] Zhang Z., Tang Y., Li Y., Zhang L., Zhang H., Dynamic hedging strategies in derivatives markets with LLM-driven sentiment and news analytics. Working Paper, arXiv:2504.04295.
- [2023] Zhang C., Duan Y., Chen X., Chen J., Li J., Zhao L., Towards generalizable reinforcement learning for trade execution. Working Paper, arXiv:2307.11685.
- [2024] Zhang Z., Chen B., Zhu S., Langren N., Quantformer: From attention to profit with a quantitative transformer trading strategy. Working Paper, arXiv:2404.00424v2.
- [2024] Zhao Z., Welsch R.E., Hierarchical reinforced trader: A bi-level approach for optimizing stock selection and execution. Working Paper, arXiv:2410.14927.
- [2025] Zhou Z., Mehra R., An end-to-end LLM enhanced trading system. Working Paper, arXiv:2502.01574.